OIL INDUSTRY SAFETY DIRECTORATE

तरलीकृत पेट्रोलियम गैस (एल पी जी) प्रतिष्ठान
ओ आई एस डी-मानक-144

LIQUIFIED PETROLEUM GAS (LPG) INSTALLATIONS
OISD-STANDARD-144

Inception	1994
Revised Edition	October 2005
Amended Edition	July 2008
Amended Edition	October 2017
Errata and Updation	March 2022
Revised Edition	MMYYYY

FOR RESTRICTED CIRCULATION ONLY

OIL INDUSTRY SAFETY DIRECTORATE
Government of India
Ministry of Petroleum & Natural Gas
8th Floor, OIDB Bhavan, Plot No. 2, Sector – 73, Noida – 201301 (U.P.)
Website: https://www.oisd.gov.in Tele: 0120-2593833

This page is intentionally left blank.

PREFACE

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for prosperity of the country. As the industry deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January, 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (MoP&NG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders, drawing inputs from national and international standards and adapting them by leveraging the experience of the functional committee. These standards cover inbuilt design safety, asset integrity and best operating practices in the field of production, processing, storage and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD, are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

Executive Director
Oil Industry Safety Directorate

NOTE

Oil Industry Safety Directorate (OISD) publications are prepared by the Oil and Gas industry under the Ministry of Petroleum & Natural Gas. This standard is a property of Oil Industry Safety Directorate (OISD) and shall not be reproduced or copied and loaned or exhibited to others without written consent from OISD.

OISD standards are published to facilitate sound engineering, operating practices and aimed to supplement the information to the oil and gas industry, however this may be used by anyone desiring to do so and are not intended to obviate the need for applying sound engineering judgement regarding when and where these standards should be utilized. The formulation and publication of OISD standard is not intended in any way to inhibit anyone from using any other practices and not to replace the prevailing statutory requirements, which must be followed as applicable.

Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents. Whenever there is any difference between the statutory norm and OISD standard, more stringent requirement should be followed. However, in case of any difference in the requirement between OISD standards, the requirement as specified in the standard released at a later stage shall prevail.

The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed or withdrawn at least every ten years.

Suggestions for revision are invited from the oil and gas industry and should be submitted to OISD at **ps-ed.oisd@gov.in**.

CURRENT FUNCTIONAL COMMITTEE

NAME	DESIGNATION	ORGANISATION
TEAM LEADER		
Gopakumar Balakrishnan	General Manager	Hindustan Petroleum Corporation Ltd.
MEMBERS		
Ranjan Mehrotra	Retd. ED (HS&E) Corp. Office, IOCL	Subject Expert
Vineet Singh	General Manager	Bharat Petroleum Corporation Ltd
Rakesh Agrawal	General Manager	Hindustan Petroleum Corporation Ltd.
Manish Shandilya	Sr. General Manager	Engineers India Limited
Mahesh Goyal	Dy. General Manager	Bharat Petroleum Corporation Ltd. (since July 2024)
Perumal A	Dy General Manager	Bharat Petroleum Corporation Ltd.
Jaimin Kumar Shah	Team Lead – HSSE, LPG Business	Reliance Industry Limited.
M Vamshi Krishna	Sr. Manager (LPG-M&I)	Indian Oil Corporation Ltd.
Manpreet Singh	Chief Manager	Indian Oil Corporation Ltd.
Md. Mustaque Ahmed	Dy. Chief engineer (O)	Oil India Ltd.
Gagan Aggarwal	Dy. Director	PNGRB
Ashendra Singh	Dy. Chief Controller of Explosives	CCE, Ranchi (up to July 2024)
Amit Sharma	Joint Director (P&E)	Oil Industry Safety Directorate
MEMBER COORDINATOR		
Mahesh Goyal	Additional Director (MO-LPG)	Oil Industry Safety Directorate (up to July 2024)
Manish Kumar Lenpale	Additional Director (MO-LPG)	Oil Industry Safety Directorate (since July 2024)

OISD recognizes the support and cooperation of several other experts from industry, who contributed in the preparation, review and finalization of this document.

TABLE OF CONTENTS

Clause. No.	Content/Description	Page No.
1.0	Introduction	01
2.0	Scope	02
3.0	Definition	02
4.0	Statutory Rules/ Regulations	06
5.0	Installation Layout	06
5.1	General	06
5.2	Location and separation distance	06
5.3	Layout	06
5.4	LPG Storage Facilities	07
5.5	LPG Bulk Loading/ Unloading Facilities	08
5.6	LPF Bottling Facilities	10
5.7	Protection of Facilities	11
5.8	Utilities	12
6.0	Design Considerations	17
6.1	General	17
6.2	LPG Storage Vessels	17
6.3	LPG Loading/ Unloading Facilities	19
6.4	Cylinder Filling Facilities	20
6.5	Utilities	21
6.6	LPG Pumps	22
6.7	LPG compressors	23
6.8	Piping	23
6.9	Thermal Pressure Relief System	24
6.10	Valves	24
6.11	Fittings	24
6.12	Evacuation Facilities for Defective Cylinders	25
6.13	Purging of new cylinders/ Tankers	25
6.14	Equipment for Ethyl Mercaptan Service	26
6.15	Selection of Electrical Equipment	27
6.16	Installation Lighting	27
6.17	Installation Earthing	29
6.18	Earth Pits	31
6.19	Electrical independent Earth electrodes	31
6.20	Protection	31
6.21	Cable	31
6.22	PMCC Panels	32
6.23	Others	32
7.0	Storage and Handling of Bulk LPG	32
7.1	General	32
7.2	Bulk Handling for movement of Bulk LPG Tank Trucks	32
7.3	Bulk Handling for movement by rail	37
7.4	Bulk Handling for movement by Pipeline Transfer	43
7.5	Bulk Handling for movement by Sea	43
7.6	Storage of Bulk LPG	46

7.7	Sampling and Draining	46
8.0	Bottling Operations	47
8.1	General	47
8.2	LPG Bottling	47
8.2.1	Receipt of LPG Cylinder	47
8.2.2	Filling of LPG Cylinder	48
8.2.3	Handling and storage of LPG Cylinder	51
8.2.4	Auxiliary Operation	52
8.2.5	Operation and handling of 425 kg cylinder	53
8.2.6	Operation and handling of composite cylinder	53
8.2.7	Mercaptan receipt, storage, dosing and testing	58
8.2.8	Quality control	60
8.2.9	Personal Protective Equipment	60
9.0	Maintenance and Inspection of Equipment	60
10.0	Statutory Testing of LPG Cylinders	68
10.1	Testing Procedure	68
10.1.1	Degassing of LPG cylinders	68
10.1.2	Surface cleaning	69
10.1.3	Checking tare weight	69
10.1.4	Visual inspection	69
10.1.5	Hydrostatic testing	69
10.1.6	Valve fixing	70
10.1.7	Pneumatic test	70
10.1.8	Painting operations	70
10.1.9	Checking and marking the tare weights	71
10.1.10	Punching of marks	71
10.1.11	Route card	71
10.1.12	Colour coding of the cylinder	71
10.1.13	Certification	71
10.1.14	Others	72
11.0	Fire Protection Facilities	72
11.1	General	72
11.2	Design criteria	72
11.3	Fire water system	72
11.4	Flow rate design	72
11.5	Fire water system design	74
11.6	Fire water storage	74
11.7	Fire water pumps	74
11.8	Fire hydrant network	75
11.9	Material specifications	77
11.10	Medium velocity sprinkler system	78
11.11	Automatic fire protection system	78
11.12	Firefighting equipment	80
11.13	Emergency control/ alarm system	82
11.14	Special fire protection	83
11.15	Inspection and testing	84
11.16	Make up water	85

11.17	Combined POL and LPG facilities	86
11.18	Data monitoring system	86
12.0	Gas Monitoring System	87
12.1	Application	87
12.2	Detectors	87
12.3	Control systems	87
12.4	Annunciation systems	88
12.5	Location of detectors	88
12.6	Inspection and testing	88
13.0	Emergency Management Plan	89
13.1	Emergency plans	90
13.2	Action during emergency	92
13.3	Bomb threat	97
13.4	Industry incident	97
14.0	Safety Audit	97
15.0	Training	99
16.0	CCTV System	101
17.0	Access control system	101
18.0	References	103
19.0	Abbreviation	106
20.0	Annexure	109
Annexure-4	Statutory rules and regulations	109
Annexure-5-I	Typical design for the construction of a water seal	110
Annexure-5-II	Typical design for Closed loop degassing arrangement	111
Annexure-5-III	Typical schematic diagram of LPG bottling operation	112
Annexure-6-I	Typical pipeline connection for water drain arrangement in LPG storage vessel	113
Annexure-6-II	Typical safety fittings on an LPG storage vessel	114
Annexure-6-III	Typical sketch of a bottling pump installation	115
Annexure-6-IV	Typical sketch of compressor installation	116
Annexure-6-V	Typical sketch of a cylinder evacuation facility	117
Annexure-6-VI	Typical sketch of a new cylinder purging system	118
Annexure-6-VII	Degassing and Gas-in Procedures of Bulk LPG Storage Vessels	119
Annexure-7–I-A	Checklist for bulk LPG tank Trucks to be prepared at the unloading location	126
Annexure-7–I-B	Checklist for bulk LPG tank Trucks to be prepared at loading locations.	131
Annexure 7– II	Checklist for bulk LPG tank wagons to be prepared at unloading locations.	133
Annexure 7-III.	Liquefied gas-cargo transfer checklist	134
Annexure-7-IV	These wagons are designed by RDSO and are fitted with various devices	137
Annexure-8-I	Checklist for the safety fitting of packed trucks	138
Annexure-8-II	Typical stacking arrangement for a 14.2 kg cylinder	141
Annexure-9-I	Preventive Maintenance Schedule for Compressor	142
Annexure-9-II	Preventive Maintenance Schedule for Pump	144

Annexure-9-III	Preventive Maintenance Schedule for Filling shed equipment	145
Annexure-9-IV	Typical inspection frequency for testing of equipment	158
Annexure-9-VI	CCTV Maintenance Checklist	162
Annexure-10-I	Sequence of operations for statutory pressure testing of LPG Cylinders	163
Annexure-10-II	Criteria for Cylinder rejection	164
Annexure-10-IIII	Route Card for STP Cylinder	167
Annexure-10-IV	Colour coding scheme of LPG Cylinder	169
Annexure-10-V	Statutory Testing Certificate format	170
Annexure-10-VI	Typical format for Quality check of re-tested Cylinder	172
Annexure-11-I	Sample fire water calculation	174
Annexure-11-II	Indicative drawing of deluge layout	177
Annexure-14-I	Checklist for Daily Safety Check of LPG Installation	178
Annexure-14-II	Checklist for Electrical Audit of LPG Installation	183
Annexure-14-III	Checklist for M&I Audit of LPG Installation	198
Annexure-14-IV	Checklist for Self-audit by Location in Charge	222
Annexure-15	Major changes in the revised edition	254
Annexure-16	List of previous functional committee(s)	258

1.0 INTRODUCTION

Liquefied Petroleum Gas (LPG) is a mixture of light hydrocarbons primarily C_3 and C_4 meeting the requirement of IS4576, which is gaseous at ambient temperature and atmospheric pressure, and is liquefied at ambient temperature with the application of moderate pressure. LPG, due to its inherent properties, is susceptible to fire, explosion and other hazards. Such hazards can have an impact on the property, equipment, plant personnel and public.

Liquefied Petroleum Gas (LPG) handling, bottling and distribution pose major challenges due to its inherent properties. Major incidents, i.e. fires/explosions in LPG facilities in the world, have underlined the need for a total in-depth review of design, procedures, maintenance, firefighting and safety aspects in LPG handling.

The oil and gas industry over the years has learnt lessons from fires and explosions and has been updating and standardizing design, operations, bottling and distribution of LPG.

Some of the important properties of LPG are as below:

- LPG at atmospheric pressure and temperature is a gas that is 1.5 to 2.0 times heavier than air. It is readily liquefied under moderate pressures. The density of the liquid is approximately half that of water and ranges from 0.525 to 0.580 @ 15 ^{o}C.
- The pressure inside an LPG storage vessel/ cylinder is equal to the vapour pressure corresponding to the temperature of LPG in the storage vessel. The restriction on vapour pressure is stipulated by IS-4576.
- LPG has an explosive range of 1.8% to 9.5% volume of gas in air. This is considerably narrower than other common gaseous fuels. This gives an indication of the hazard of LPG vapour accumulated in low-lying areas in the event of a leakage or spillage.
- Water, being insoluble in LPG, if present in LPG, can cause rust and corrosion and freeze-up of valves, drain valves, excess-flow check valves, cylinder valves and regulators, thereby affecting their functioning and at times can lead to leakage of LPG.
- LPG liquid has a low viscosity of around 0.3 CS @ 45 ^{o}C and can leak when other petroleum products normally cannot, thus demanding a very high integrity in the pressurised system to avoid leakage.
- LPG, even though slightly toxic, is not poisonous in the vapour phase, but can, however, suffocate when in large concentrations due to the fact that it displaces oxygen. In view of this, the vapour possesses mild anaesthetic properties.
- LPG is colourless both in the liquid and vapour phase. During leakage, the vapourisation of liquid cools the atmosphere and condenses the water vapour contained in it to form a whitish fog, which may make it possible to see an escape of LPG.
- The coefficient of expansion is around 0.00237 per ^{o}C. At a liquid full condition, any further expansion of the liquid, the container pressure will rise by approximately 14 to 15 kg./sq.cm. for each degree centigrade.
- LPG has only a very faint smell; therefore, ethyl mercaptan (Ethanethiol, an organosulfur compound with the formula CH3CH2SH) is normally used as a stenching agent for identifying the leakage as per IS:4576.

DRAFT

- The auto-ignition temperature of LPG ranges from 410 to 580 ^{O}C.

2.0 SCOPE

This standard lays down the minimum safety requirements on design, layout, storage, loading/ unloading, bottling operation, inspection and maintenance, fire protection, emergency planning and safety audit systems for LPG Installations.

The requirement of green belt/ buffer zone beyond the plant boundary is outside the scope of this standard. Such provisions should be considered based on local environmental/ security requirements/ requirements of applicable statutory bodies.

This standard does not cover refineries and LPG Bottling Plants inside refineries, the distribution of LPG to domestic and non-domestic consumers, LPG pipeline operations, Auto LPG Dispensing stations, refrigerated and mounded storage facilities for LPG and small LPG Bottling Plants.

- Standard for Design and Safety Requirements for Liquefied Petroleum Gas Mounded Storage Facility is covered in OISD-STD-150.
- Standard for LPG Tank Trucks - Requirements of Safety on Design/ Fabrication and Fittings is covered in OISD-STD-159
- Guidelines on Small LPG Bottling Plants (Design and Fire Protection facilities) are covered in OISD-GDN-169
- Standard for Storage, Handling and Refuelling of LPG for automotive use is covered in OISD-STD-210
- Standard for Cross Country LPG Pipelines is covered in OISD-STD-214
- Standard for Design, Layout, Operation and Maintenance of Refrigerated LPG Storage is covered in OISD-STD-236.

3.0 DEFINITIONS

3.1 APPROVED TYPE

Any equipment that has specific approval for use under specified conditions by a competent authority or authorised person, as the case may be.

3.2 AREA CLASSIFICATION

It is a method of classifying an area zone-wise/group-wise based on the presence of explosive gas/ vapour-air mixture vis-a-vis the requirement of precautions for the construction and use of electrical apparatus.

3.3 AUTHORISED PERSON

An individual whose job description includes the concerned jobs.

3.4 BONDING

Bonding is the process by which two electrically conducting bodies are connected using a conductor to maintain electrical continuity to prevent sparking between two conducting bodies.

3.5 BULLET

A horizontal cylindrical pressure vessel used for the storage or transportation of LPG by rail/ road.

3.6 COMPETENT PERSON

A person recognised by the concerned Statutory Authority for the purpose in respect of which the competency is required.

DRAFT 2

3.7 COMPRESSED GAS

"Compressed Gas" means any permanent gas, liquefiable gas, or cryogenic liquid under pressure or gas mixture which in a closed pressure vessel exercises a pressure exceeding one atmosphere (gauge) at the maximum working temperature.

3.8 CYLINDER

A portable LPG container less than 1000 litres of water capacity is used for both domestic and industrial purposes.

3.9 DEADMAN VALVE

A Deadman valve (spring return valve or Deadman handle) is a type of valve designed to automatically close when the handle/ actuating mechanism is released. This is achieved through a mechanism that returns the valve to its original closed position, without manual intervention.

3.10 EARTHING

"Earthing" means a connection of the exposed conductive and extraneous parts of an installation to the main earthing grid of that installation or connection of neutral of transformer or generator, or equipment to the general mass of earth or earth grid of that installation.

3.11 EMERGENCY

Emergency means a situation or scenario that has the potential to cause serious danger to persons, the environment, or damage to property and which tends to cause disruption inside or outside the premises and may require the help of outside resources.

3.12 EMERGENCY SHUT-OFF VALVE (ROV)

A shut-off valve, which, in an emergency, operates automatically or can be operated remotely.

3.13 EXCESS FLOW CHECK VALVE (EFCV)

A valve designed to close when the liquid or vapour passing through it exceeds a prescribed flow rate.

3.14 EXPLOSIVE MIXTURE

It is a mixture of combustion agent (oxidising product-gas, vapour, liquid or solid) and a fuel (oxidisable product-gas, liquid or solid) in such proportions that it could give rise to a very rapid and lively oxidation reaction, liberating more energy than is dissipated through conduction and convection.

3.15 FILLING RATIO

"Filling ratio" means the ratio of the weight of a liquefiable gas introduced in the cylinder to the weight of the water that the cylinders will hold at 15°C.

3.16 FIRE SAFE

As applied to valves, it is the concept of controlling the leakage to an acceptable level after damage encountered during and after the fire, achieved by dual seating.

3.17 FIREPROOFING

It is an insulation that provides a degree of fire resistance to protect substrates like vessels, piping and structures for a predetermined period against fire.

3.18 FLAMEPROOF ENCLOSURE

"Flameproof enclosure" means an enclosure in which the parts that can ignite an explosive atmosphere are placed and which can withstand the pressure developed during an internal explosion

DRAFT 2

of an explosive mixture and which prevents the spread of the explosion to the explosive atmosphere surrounding the enclosure.

3.19 FLAMMABILITY

It is the percentage of volume of any flammable vapour in air-vapour mixtures capable of forming an explosive mixture.

3.20 EARTHING FLAMMABLE (OR INFLAMMABLE)

Any substance that, when tested in a specified manner, will ignite when mixed with air on contact with a flame and will support combustion.

3.21 GAS-FREE

It means the concentration of flammable or toxic gases or both in a pressure vessel or pipeline is within the safe limits specified for persons to enter and carry out hot work in such vessels/ pipelines.

3.22 HAZARDOUS AREA

It is a location where there is a potential for fire or explosion due to the presence of flammable gases, vapours, combustible dust, or ignitable fibres.

3.23 HORTON SPHERE

A spherical pressure vessel used for the storage of LPG.

3.24 HOT WORK:

An activity that can produce a spark or flame, or other sources of ignition having sufficient energy to cause ignition, where the potential for flammable vapour, gas, or dust exists. Hot work is defined as any work which involves welding, burning, soldering, brazing, sand blasting, chipping by spark-producing tools, use of certain power-driven tools, non-flame proof electrical equipment or equipment with internal combustion engines and including any other work which is likely to produce sufficient heat, capable of igniting inflammable gases.

3.25 KERB WALL

A wall of appropriate height and size constructed of suitable material and designed to contain the LPG spillage and to direct it to a safe location around the storage vessel.

3.26 LIQUEFIED PETROLEUM GAS (LPG)

The term applies to a mixture of certain light hydrocarbons, predominately C3 and C4, derived from petroleum and natural gas, which are gaseous at ambient temperature and pressure, may be condensed to a liquid state at normal ambient temperature by the application of moderate pressure and conforming to IS: 4576 or IS: 14861.

3.27 LPG Installation

The facilities where LPG is stored or loading/ unloading operations or bottling operations are carried out.

3.28 MAXIMUM WORKING PRESSURE

It is the saturated vapour pressure of LPG at 65 ^{O}C for LPG cylinders and 55 ^{O}C for bulk vessels.

3.29 PERSONAL PROTECTIVE EQUIPMENT

Devices or clothing used to insulate an individual from the chemical, thermal, explosive, electric shock, or other hazards presented by the environment in which one works. Examples of PPEs are

DRAFT 2

safety spectacles/ goggles, safety shoes, safety helmets, cotton coveralls, hand gloves, earmuffs/ plugs, etc.

3.30 PRESSURE VESSELS

"Pressure vessel" means any closed metal container of whatever shape, intended for the storage and transport of any compressed gas which is subjected to internal pressure and whose water capacity exceeds one thousand litres and includes interconnecting parts and components thereof up to the first point of connection to the connected piping and fittings.

3.31 PURGING

It is the act of replacing the atmosphere within a container with LPG vapour, in such a manner as to prevent the formation of an explosive mixture.

3.32 PURGING INTO SERVICE

It is the replacement of air in a closed system by combustible gas, vapour, or liquid.

3.33 PURGING OUT OF SERVICE

It is the replacement of the normal combustible content of a closed system by an inert substance, and the replacement of the latter by air.

3.34 SAFETY RELIEF VALVE

A pressure relief device designed to automatically open during emergency or abnormal conditions to prevent a rise of internal fluid pressure in excess of the specified design value and automatically reset/ close and prevent the flow of fluid.

3.35 TARE WEIGHT

The weight of the cylinder, together with any fitting permanently attached to it, including the weight of the valve.

3.36 THERMAL SAFETY VALVE

Also known as a thermal relief valve or temperature relief valve is a safety device used in liquid piping and pipeline systems. Its primary function is to protect the system from excessive pressure caused by the thermal expansion of the liquid. These valves are typically small in size, often with an inlet diameter no larger than 1 inch and are designed to handle relatively small volume increases.

3.37 WATER CAPACITY OF VESSEL

"Water capacity" means capacity in litres of the pressure vessel when completely filled with water at 15 °C**.**

3.38 MAY

"May" are to be considered desirable or good practices which are encouraged to implement, but not necessary.

3.39 SHALL

"Shall" indicates a mandatory requirement.

3.40 SHOULD

"Should" indicates that the provision is recommended as a good practice.

DRAFT 2

4.0 STATUTORY RULES/ REGULATIONS

The LPG installation and various facilities are covered under many regulations and require specific approval/ license from the concerned statutory authorities. The various regulations applicable to LPG installation are detailed in Annexure-4.

Provisions of all the statutory Acts and Rules, and Regulations framed thereon shall apply as per the extant version, notwithstanding the year mentioned against them anywhere in this standard.

5.0 INSTALLATION LAYOUT

5.1 GENERAL

The layout of the LPG facilities, including the arrangement and location of Plant roads, walkways, doors and operating equipment, shall be designed in line with OISD-STD-118. The layout shall permit access to the LPG facilities from at least two directions.

The general principles of layout of LPG storage, bottling and bulk handling facilities have been detailed. The various facilities within the LPG installation shall be located based on Table-I and Table-II.

5.2 LOCATION & SEPARATION DISTANCES

5.2.1 LOCATION

While assessing the suitability of any site for the location of an LPG installation, the following aspects shall be considered:

a) The location of residential quarters, other industries, railways, roads, waterways, overhead power lines, places of public assemblies, etc. This shall be covered in the risk analysis study of the proposed site. The study shall also be used to plan for emergency measures.

b) Adequate availability of water from a reliable source or alternate arrangements proposed.

c) The topographical nature of the site with specific reference to its effect on the accidental release of LPG.

d) The availability of space for future extension of LPG facilities, if any.

e) The meteorological data of the location, including predominant wind direction and velocity, high flood level, temperatures, etc.

5.2.2 SEPARATION DISTANCES

The separation distances as given in Table- I and II shall be followed.

5.3 LAYOUT

The following aspects shall be considered while establishing the layout of LPG storage vessels. Bullets or spheres are used for above ground storage of LPG.

a) The access of mobile firefighting equipment to the storage vessels shall be at least from two sides.

b) Longitudinal axis of static storage vessels (above ground bullets) shall not be pointed towards other vessels, vital equipment e.g. LPG cylinder sheds, Tank Truck/ Tank Wagon Gantries, LPG Pump House, first ROV of mounded vessels and buildings wherein control panels for fire and gas detection/ actuation panels are situated.

DRAFT 2

c) Storage vessels should be located in pre-dominant downwind of potential ignition sources at a lower elevation.

d) No foreign material/ combustible substances shall be stored in the LPG bulk storage area.

e) Storage vessels shall be laid out in a single row in each group, both in the case of the spheres and bullets. However, OISD-STD-118 and SMPV Rules shall be referred to for the same. If an entity has laid, built, constructed, expanded the Installation based on some other standard or is not meeting the requirements specified in this standard, the entity shall carry out a detailed Quantitative Risk Analysis (QRA) of its infrastructure and the entity shall thereafter take approval from its highest decision-making body or its board for non-conformities and mitigation measures. The entity's Board approval along with the compliance report, gap analysis, mitigation measures and implementation schedule shall be submitted to the OISD.

f) Storage vessels shall not be located one above the other.

g) The mounded LPG storage should be preferred as it is said to be safer compared to above ground storage vessels since it provides intrinsically passive and safe environment and eliminates the possibility of Boiling Liquid Expanding Vapour Explosion (BLEVE).

5.4 LPG STORAGE FACILITIES

There shall be a minimum of 3 no's vessels for storage of LPG in the installation.

(a) CONFINEMENT/ GRADING -

Kerb wall shall be provided around all sides of the storage vessel with concrete flooring of the ground under vessel. The minimum distance between vessel and kerb wall shall be the diameter of the vessel or five meters, whichever is less. The concrete floor shall be graded to form a slope of minimum 1:100.

Grading of the ground underneath should be levelled and directed to an area connected with a water seal away from the storage vessel. Accumulation of LPG should not be allowed underneath the storage vessel.

Kerb wall height shall be a minimum of 30 cm but shall not exceed 60 cm; otherwise, evaporation of spilled LPG may get affected.

The spillage diversion area shall be located at a distance where the flames from the fire will not impinge on the vessel. This distance shall be equal to the diameter of the nearest vessel or 15 m, whichever is more.

There shall not be any opening in the kerb wall due to the passing of pipes or cables, etc., for confinement of LPG within the kerb wall. Also, there shall not be any pits or earth pits inside the kerb wall area to prevent accumulation of LPG.

(b) PIPING –

Only piping associated with the storage vessels shall be located within the storage areas or between the storage area and the manifold system.

(c) SURFACE DRAINAGE –

Adequate water drainage system shall be provided in the installation for the various facilities such as Tank Truck gantry, Tank Wagon gantry, Bulk storage area, LPG sheds etc. In order to prevent the escape of spillage into the main drainage system, surface water from the storage area and from the manifold area shall be directed to the main drainage through a water seal to avoid the spread of LPG. The water seal of the bulk storage area shall be attached to the kerb wall. The height of the wall of

DRAFT 2

water seal shall be the same as the height of the kerb wall. Water seal shall be provided for Bulk Storage area, Tank Truck Gantry, Tank Wagon Gantry, LPG Cylinder Filling shed to prevent the LPG spillage into the main drainage system of plant. In case the Plant drain is discharging into the storm drain going outside the Plant meeting the statutory requirement as applicable, a water seal shall also be provided at the interconnecting sump inside the Plant.

Water seals shall have adequate capacity to drain firefighting sprinkler water without overflowing. Water seals shall be provided with a permanent water connection with a metallic float for auto replenishment of water. Typical design for the construction of water seal is as per Annexure-5-I.

The Plant shall prepare the drawing of the complete Plant drainage with water seal locations.

(d) GROUPING –

Vessels shall be arranged into groups, each having a maximum of six vessels. There shall be a minimum spacing as specified in Table-I between adjacent vessels. Each group shall be separated by roads having a minimum width of 3.5 m on all four sides for easy access and emergency handling, such that the access of mobile firefighting equipment to the storage vessels shall be at least from two sides.

Where LPG Pump House and Storage Vessel are situated as an integrated block (where piping manifold of the LPG Pump House and the vessel is between the vessel and the LPG Pump House), the roads from all four sides shall be provided for the combined facility of LPG Pump House and storage vessel. However, adequacy of firefighting to be ensured.

Spheres, bullets and mounded vessels shall be treated as separate groups. This inter-distance shall be as per Table-I and Table-II of this standard..

The top surfaces of all the vessels installed in a group shall be on the same elevation. Separate manifolds with independent pumping facilities should be provided for groups with dissimilar elevations. In case manifold from two groups with dissimilar elevation are interconnected, they shall be connected through fail safe device like 3-way valves or equivalent system e.g. logic controlled/ interlocked valves to prevent migration of LPG from vessels with higher elevation to the vessels at lower elevation due to gravitation so as to avoid any chances of inadvertent overfilling.

Documentary evidence to authenticate the vessel in different groups are in the same elevation shall be available in the Plant if 3-way valves or an equivalent system as mentioned above is not provided.

5.5 LPG BULK LOADING / UNLOADING FACILITIES

A. BULK LPG LOADING/ UNLOADING FACILITIES – TANK TRUCK

5.5.1 LPG tank truck loading/ unloading gantry shall be covered and located in a separate block and shall not be grouped with other petroleum products.

5.5.2 Space for turning with a minimum radius of 20 m for tank trucks shall be provided commensurate with the capacities of the tank trucks.

5.5.3 Maximum number of LPG tank truck bays shall be restricted to 8 in one group. Separation distance between the two groups shall not be less than 30 m. and number of such gantries in a premise shall not exceed two.

5.5.4 For adequate permanent protection for tank truck discharge, Pipeline Island shall be provided. The minimum width of such a pipeline island shall be 1 m.

DRAFT 2

5.5.5 The layout of the loading/ unloading location shall be such that the tank truck being unloaded shall be in drive-out position. Drive out position indicates the presence of a road in front of the parked tank truck. This road shall be directly routed to the main/ emergency exit gate.

5.5.6 The weigh bridge of adequate capacity shall be provided with proper manoeuvrability for vehicles.

5.5.7 The tank truck shall be loaded/ unloaded with a suitable arrangement for venting of loading/ unloading arm content, if used at the end of the operations.

5.5.8 A suitable earthing interlock arrangement shall be provided to ensure that the unloading/ loading operation does not commence without earthing. The system should not work when the earthing connection is made to the TLD structure or ground. The system shall work only if the earthing connection clip is connected to the Tank Truck bullet and chassis.

5.5.9 LPG tank truck parking area (Bulk/ Packed) shall be located in a secured area with entry/ exit gates. Parking area shall be provided with an adequate no. of hydrants/ monitors to provide firefighting coverage for the entire parking area from all sides. The fire hydrant network in the parking area shall be in closed loops to ensure multidirectional flow in the system. The parking area's entry/ exit gates and the parking area shall be under real-time visual supervision (Security Guard/ CCTV Cameras). Proper slotting/ marking shall be done for the safe parking of bulk LPG tank trucks and packed trucks in the parking area.

5.5.10 Suitable protection shall be provided to avoid accidental hitting of bulk LPG tank trucks/ packed trucks to hydrant lines. Slotting and marking shall be provided such that the water monitor throw is not obstructed by parked Bulk LPG tank trucks/ packed trucks. Mechanical stoppers are to be provided to restrict the throw of monitors onto the electricity overhead lines passing outside the parking area.

5.5.11 Parking area shall be adequate to accommodate (minimum) following requirements of the Plant:

(a) Bulk LPG tank trucks required for 8 hours of maximum rated capacity of the Plant bottling.

(b) In case of bulk loading location, bulk LPG tank trucks required for 8 hours of maximum rated bulk loading capacity of the Plant.

(c) Packed trucks required for 4 hours of maximum rated capacity of the Plant bottling.

(d) In addition to above, it is to be ensured that bulk LPG tank trucks and packed trucks are not parked on the Plant approach road.

(e) Capacity of the parking area with respect to the number of packed trucks and bulk LPG tank trucks to be displayed at the entrance of the parking area.

5.5.12 Movement of trucks inside the Plant and at the entrance of the Plant to be monitored to ensure smooth and safe movement of trucks to avoid congestion and for safe evacuation in case of emergency. The maximum number of packed trucks and bulk LPG tank trucks inside the Plant during operating hours shall be decided and displayed.

5.5.13 No packed truck or bulk LPG tank truck shall be kept inside the Plant during off-shift.

5.5.14 Dedicated pathway to be earmarked for the movement of pedestrians and cyclists at all places inside the Plant.

DRAFT 2

5.5.15 Convex mirrors shall be provided at blind corners at the parking area, green belt area and operating area inside the Plant.

5.5.16 Two wheelers/ vehicles other than bulk LPG tank trucks/ packed trucks shall not be parked/ driven in the truck parking area.

B. LPG BULK LOADING/ UNLOADING FACILITIES - TANK WAGON

a) LPG loading/ unloading rail gantries shall have a separate rail spur and be grouped separately at least 50 m from other rail shunting facilities.

b) Rail loading/ unloading of LPG should be restricted to a maximum of half rake as per SMPV Rules 2016. Full rake loading/ unloading shall be done on two separate rail gantries having a minimum distance of 50 m.

c) Where placement of the tank wagon is by electrical LOCO, the traction line must terminate 15 m short of the first loading /unloading point of the Tank wagon Gantry.

d) For placement, brake van/ dummy wagons shall be used between the loco and LPG wagons.

e) Main railway track shall be isolated from wagon gantry siding at least 15 m from the 1st loading/ unloading point by providing an insulation joint at the terminating point, and the loco shall stop before the insulation joint. Integrity tests of insulation joints shall be carried out quarterly as per SOP.

f) The roads in the tank wagon railway siding shall be provided on both sides of the rail tracks and connected at both ends.

5.6 LPG BOTTLING FACILITIES

5.6.1 LPG bottling facilities shall be located at a safe distance from other facilities with minimum ingress of traffic and upwind direction with respect to bulk storage**.**

5.6.2 There shall not be any deep ditches in the surrounding areas (except drainage) at least within 15 m from LPG sheds to avoid the settling of LPG vapour.

5.6.3 The LPG Bottling section shall be of a single storey. Antistatic mastic flooring conforming to IS:8374 shall be provided in the LPG filling shed/ cylinder storage, including the valve changing shed and degassing shed, to avoid frictional sparks. The shed shall be supported by RCC columns alternatively, structural steel columns shall be covered with concrete or fire-proofing material up to a full height of the columns. Anti-static mastic coating up to 1.5 m (min.) shall be applied to the supporting columns of the shed.

5.6.4 Requirement of shed for carrying out bottling operation and other allied activities is as below:

a) The bottling operation shall be carried out in the filling shed.

b) Separate sheds for Empty cum filling shed, filled cylinders' storage, cold repair shall be provided.

c) A separate shed may be provided for storing empty cylinders. The empty cylinders storage area shall be properly segregated from filling machines by 5 m (min.). Segregated storage of empty cylinders shall be marked for Good Empty, Spurious, Repairable, Due for statutory testing, and other defects. Cylinders shall always be stacked vertically with a maximum stack height of 1.5 m. Stacking zones shall be painted on floor for empty cum filling shed as well as filled cylinder storage shed for all capacity of cylinders being handled.

d) In-house statutory testing of cylinders shall be carried out in a separate shed.

e) A Degassing unit if provided in the cold repair shed, a closed-loop vent pipe arrangement shall be provided for venting out the gas released during degassing at a height not less than 1.5 m above the eaves of the shed. A vapour extraction system and a GMS sensor

DRAFT 2

shall be provided near the degassing unit. For a typical closed loop degassing vent pipe arrangement, refer to Annexure-5-II.

f) A valve changing unit without evacuation can be provided in the filling shed itself.

5.6.5 LPG cylinder filling machines and other related testing facilities shall be provided in sequential order. A typical schematic diagram of filing shed facilities is enclosed as per Annexure-5-III.

5.6.6 Cylinder storage shall be kept on or above grade and never below grade in a cellar or basement.

5.6.7 Filled cylinders shall not be stored in the vicinity of cylinders containing other gases or hazardous substances.

5.6.8 Escape routes shall be specified and marked in LPG sheds for the evacuation of personnel in case of an emergency. Further, a sufficient no. of escape routes shall be provided. The number of escape routes shall not be less than four for the filling shed and two for other sheds, i.e. filled cylinder storage shed, degassing shed, etc. All steps forming part of the escape routes shall be a minimum of 1.2 m with treads 30 cms (minimum) and a maximum rise of 15 cms.

5.6.9 There shall be sufficient no. of crossovers to avoid trapping of personnel in LPG sheds by conveyors, cylinders and other facilities. All steps forming part of the conveyor crossover shall have treads 30 cms (minimum) and a maximum rise of 15 cms. Handrails shall be provided with a minimum height of 100 cm and shall be firmly supported.

5.6.10 A suitable earthing interlock arrangement with carousel rail shall be provided so as to ensure that filling operation does not commence without earthing. Whenever earthing connections are removed for maintenance of filling guns/ filling machines, the electrical continuity of such filling guns/ filling machines should be checked, ensured, and recorded appropriately before putting into use.

5.7 PROTECTION OF FACILITIES

5.7.1 Properly laid out roads all around various facilities shall be provided within the installation area for smooth access of fire tenders, etc., in such a way that all facilities are accessed at least from two sides in case of emergency.

5.7.2 A proper industry type boundary wall at least 3 m high with 0.6 m concertina coil on top shall be provided all around the installation.

5.7.3 In case of combined POL/ LPG Installations of the same entity, a common proper industry type boundary wall at least 1 m high with a 0.6 m concertina coil on top shall be provided between the two installations. Same shall also be applicable in case bottling plant is a part of a refinery complex.

5.7.4 There shall be a pedestrian patrolling track along the inside perimeter of the boundary wall for security patrolling. Alternatively, the boundary wall should be visible from a motorable road if provided alongside of boundary wall. The Security watchmen tower (if provided) shall have clear access.

5.7.5 A separate Emergency exit with a proper gate shall be provided in the earmarked zone. The emergency gate shall be away from the main gate for evacuation of vehicles and personnel in an emergency and shall always be kept available and free from obstruction.

DRAFT 2

5.7.6 In case provision for buffer area (green belt) along the boundary wall is made, the same shall be segregated by a 1 m high wall/ chain link fencing. Access to this buffer area should be controlled through a gate. This gate shall be in a closed condition so that only authorised personnel are allowed/ entered.

5.7.7 The possibility of ingress from underneath through drains, ducts, culverts, and tunnels, etc. should be eliminated by providing suitable grating.

5.7.8 Branches of trees should never extend over the peripheral boundary wall of the Plant to facilitate scaling of the wall by any intruder.

5.8 UTILITIES

Utilities consisting of fire water pumps, Admin building, canteen, motor control center, DG room, air compressors, dryers, etc., shall be separated from other LPG facilities and located as per the separation as specified in Table-I.

DRAFT 2

TABLE – I - INTERDISTANCE FOR LPG FACILITIES

	Facility	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Above-ground pressurised LPG Storage vessel	Note-1	15	-	T-II	30	30	50	30	15	60	30	T-II	Note-3	15	60 with Note 5
2	LPG MSV	15	-	-	15	15	15	15	15	Note-2	30	30	15	15	15	30 with Note 5
3	Edge of MSV	-	-	5	5	-	-	-	-	-	-	-	-	-	-	-
4	Boundary/property line/ group of Bldgs. Not associated with LPG storage and operation facilities such as Admin building, stores, Amenities, Effluent treatment plant, Sewage treatment plant, etc.	T-II	15	5	-	30	30	50	30	30	Note-2	Note-2	Note-2	60	15	Note-2
5	Shed-LPG	30	15	-	30	15	30	50	30	15	60	30	30	30	15	30
6	LPG Tank Truck Gantry	30	15	-	30	30	30	50	50	30	60	30	30	30	30	30
7	LPG Tank Wagon Gantry	50	15	-	50	50	50	50	50	30	60	50	50	50	30	50
8	other rail spurs	30	15	-	30	30	50	50	50	30	60	30	30	50	30	30
9	LPG Pump / Compressor House	15	Note-2	-	30	15	30	30	30	Note-2	60	30	30	30	15	30
10	Fire Water Pump House/ Fire Water Storage	60	30	-	Note-2	60	60	60	60	60	Note-2	Note-2	Note-2	60	60	Note-2
11	MCC / DG / CP-TRU Room / Air Compressor house	T-II	30	-	Note-2	30	30	50	30	30	Note-2	Note-2	Note-2	60	30	Note-2
12	Non-LPG Shed – DPT shed	T-II	15	-	Note-2	30	30	50	30	30	Note-2	Note-2	Note-2	30	30	Note-2
13	Refrigerated LPG storage	Note-3	15	-	60	30	30	50	50	30	60	60	30	Note-4	15	30 With Note-5

DRAFT 2

14	Cross-country Pipeline receiving station (Note-6)	15	15	-	15	15	30	30	30	15	60	30	30	15	-	16
15	Control room	60 with Note 5	30 with Note 5	-	Note-2	30	30	50	30	30	Note-2	Note-2	Note-2	30 With Note-5	16	Note-2

SPECIFIC NOTES TO TABLE-I

a) Note-1: ¼ sum of diameters of adjacent vessels or half the diameter of the larger of the two adjacent vessels, whichever is greater.

b) Note-2: Any distance of operational convenience.

c) Note-3: The minimum distance between a refrigerated LPG tank and pressurised storage shall be three-quarters of the larger tank diameter or 30 m, whichever is more, subject to a maximum requirement of 60 m.

d) Note-4: The minimum distance between above-ground refrigerated LPG storage tanks shall be one-half the diameter of the larger tank.

e) Note-5: The type of construction (Type A / Type B) of the control room shall be as per OISD STD-163, depending on the inter-distance between the Control room and LPG storage vessels.

f) Note-6: For the measurement, the distance from the battery limit is to be considered for the measurement. Battery limit is defined as “the point (flange/valve) at which custody transfer of LPG takes place.

g) T-II: Refer to Table-II

GENERAL NOTES TO TABLE-I

a) All distances are in meters. All distances shall be measured between the nearest points on the perimeter of each facility, except in the case of LPG bulk tank truck/ tank wagon loading/unloading area and storage vessels.

b) In case of above-ground storage vessels, the distance shall be measured from the nearest point on the vertical shadow of the vessel. In case of Mounded storage vessel, all the separation distances shall be measured from the nearest point of the periphery of the vessel and also from the first exposed flange on the vessel i.e. ROV.

DRAFT 2

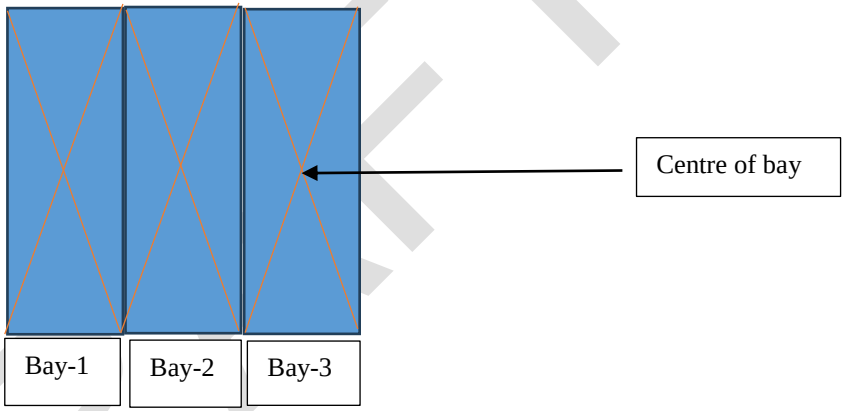
c) Shed-LPG: - Minimum 20 m inter distance shall be maintained between the cylinders loading/unloading fingers and boundary wall/property line/ group of buildings in a collinear direction of the LPG Shed fingers.

d) LPG Shed to LPG Shed distance is 15 m.
e) LPG Shed to Non-LPG Shed is 30 m.
f) LPG Shed to Fire Water pump storage is 60 m.

g) Degassing shed is considered as an LPG shed.

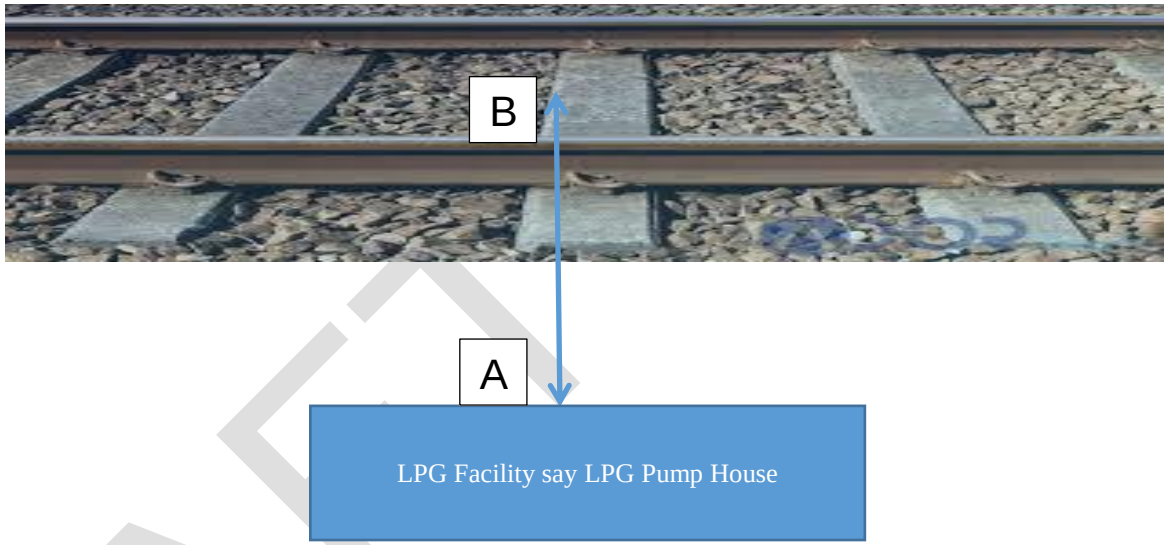
h) Distance of stabling line shall be as per Railway Standards.

i) In case of LPG bulk tank truck loading/unloading area, the distance shall be measured from the centre of each bay, where centre of the bay means inter section of two diagonals of the individual bay.



DRAFT 2

j) In case of LPG bulk tank wagon loading/unloading area, the distance shall be measured from the centre of the railway track closest to the facility from which the distance is being measured.



A is the nearest point on the perimeter of each facility, and B is the center of the railway track.

TABLE - II

DISTANCE BETWEEN LPG STORAGE VESSELS AND PLANT BOUNDARY/ GROUP OF BUILDINGS NOT ASSOCIATED WITH LPG INSTALLATION

Capacity of each vessel (Cu. Mt. of water)	Distance (in meters)
10-20	15
> 20-40	20
>40-350	30
>350-450	40
>450-750	60
>750-3800	90
> 3800	120

DRAFT

6.0 DESIGN CONSIDERATION

6.1 GENERAL

This section describes the design and safety features required in a typical LPG bottling installation.

6.2 LPG STORAGE VESSELS

The minimum requirements w.r.t. design considerations and various fittings to be provided for LPG storage vessels shall be as under:

6.2.1 MECHANICAL DESIGN

i. In line with SMPV Rules 2016, Vessels shall be designed, constructed and tested in accordance with IS:2825; ASME Section VIII Division 1 or Division 2, PD5500, EN 13458, EN 13530, AD:2000 code as amended from time to time, or such other standard or code accepted by the Chief Controller of Explosives.

   A single code shall be adopted for the design, fabrication, inspection and testing of the same storage vessel.

ii. The material of construction shall conform to the design code. ASTM A516 Gr. 60 shall be used for Refinery service (BS-5500 shall not be used for the same) and for Marketing Installation where H2S is not present, ASTM A516 Gr. 70 or A537 CLASS 1 can be used. Micro-alloyed steel containing Ni, Mo, and Va shall not be considered. Maximum specified tensile stress shall not be more than 80,000 psi.

iii. Design temperature: (-) 27°C to (+) 55°C.

iv. Design Pressure: 14.5 kg/cm^{2} (1.42 Mpa) vapour pressure of LPG at top of the vessel (as per latest IS:4576 or latest IS:14861) at 55°C.

   The recommended design pressure and temperature shall be treated as a minimum requirement, and other design considerations and statutory requirements shall also be considered.

v. Other Design Considerations

   Corrosion Allowance: minimum 1.5 mm

   Radiography: Full

   Stress Relieving: 100% irrespective of thickness.

   Wind pressure: as per IS:875

   Earthquake pressure: as per IS:1893

   Hydro test pressure: As per fabrication drawing approved by PESO/ Design Code, as mentioned in Certificate of Control (COC)/ name plate.

vi. Wherever extreme climatic conditions or security reasons warrant, suitable alterations in design can be made with approval from statutory bodies.

6.2.2 FITTINGS

- Spheres/ bullets other than underground shall have a single nozzle at the bottom for liquid inlet as well as outlet. No drainage pipe shall be provided directly from the LPG vessel. The nozzle shall be a full-welded pipe, stress relieved along with the vessel and shall extend a minimum of 3 m from the shadow of the sphere/ bullet. A remote operated shutdown valve (ROV) shall be

DRAFT 2

provided on this bottom nozzle at a distance of at least 3 m from the shadow of the sphere/ bullet. The nozzle pipe shall have a minimum slope of 1.5° to horizontal.

There shall not be any other flanges, manholes, and instrument tapping on this nozzle up to the ROV or on sphere/ bullet bottom. In order to avoid stress on the nozzle due to relative settling of the support and storage vessel, suitable supports for the bottom nozzle shall be provided.

- The top vapour zone of the vessel shall be provided with nozzles for vapour outlet, recirculation, wherever applicable. These lines shall be provided with a fire-safe ROV (Valve). ROVs for vapour and/or recirculation lines should be provided at the ground level with an isolation valve at the top. In case ROV is provided at the top of the vessel, there is no need to provide an isolation valve.
- Fire safe ROVs (Valve) shall also be provided on the liquid line and vapour line at the entry of Tank Truck Gantry, and the entry of the Tank Wagon Gantry and on the entry of the LPG filling shed. ROVs shall be provided with QB detectors and sprinkler nozzles to cool the complete ROV uniformly in case the ROV is provided within 15 m of such facilities.
- ROVs provided on the vessel inlet/ outlet liquid line and vapour line, on the liquid line and vapour line at the entry of Tank truck Gantry and Tank Wagon Gantry and on the entry of the liquid line at LPG filling shed shall be treated as Safety ROV. Location shall define safety and operational ROVs. All Safety ROVs shall have provision for electro-pneumatic actuation and shall open through manual intervention from the field only. Provision for shutdown operation in local mode shall be made for Safety ROVs from the operating station located at a minimum distance of 15 m from the facility to be protected. Safety ROV open/ close indication, along with closure facility, shall be available at the remote panel.
- Nozzles for two independent level indicators of different types, a high-level switch, at least two safety relief valves, a pressure gauge/ indicator and a manhole shall be provided on top.
- A facility for water draining to be provided through a water draw offline after the liquid ROV of the vessel. Minimum of two valves, with a suitable distance piece between them, shall be provided on the water drain line. The drain outlet points, however, must be observable by the operator from the last valve operation point. The first drain valve from the vessel should be of quick shut-off type valve, while the second valve should be a throttling type antifreeze Globe Valve. For refinery and pipeline receipt locations, there shall be a spring-loaded dead man valve in the drain line that will automatically close if the operator leaves the area. Water drain lines of storage vessels in a cluster shall not be combined.
- Provision should be made in the bulk LPG receipt header or in the water drain line for filling of water in the vessel in case of any exigencies. Water filling line to have an isolation valve with NRV and male coupling for attaching the hydrant hose, so that in case of emergency, only the hose has to be connected. Refer to Annexure-6-I for the Water drain provision and water filling arrangement.
- All the fittings shall be suitable for use at the design parameters of the storage vessels and for the temperatures appropriate to the worst operating conditions. The remote operated valves on the storage vessel connecting pipelines shall be fire-safe type conforming to API 607 or equivalent.
- The flange joints of these valves shall either have spiral wound metallic gaskets or ring joints. Plain asbestos sheet / reinforced gaskets shall not be used. The gasket used shall conform to ASME B16.20 or equivalent. The studs used shall conform to ASTM A194 Gr 2 H/ ASTM A193 Gr B7 or equivalent.

  Flange connections shall be of a minimum of 300 class rating conforming to ASME/ ANSI B16.5 class 300 or equivalent. All tapings or openings shall be of a minimum 20 mm thickness.

DRAFT 2

6.2.3 INSTRUMENTS

i) Level Instrument:

a) Each storage vessel shall have at least two independent level instruments along with an isolation arrangement, each working on different principles. However, both independent level instruments can also be radar gauge.

b) Reading of both the level gauges/ indicators to be recorded in the logbook/ dip book.

ii) High Level Switch:

a) In addition to the level instrument, one independent high level switch shall also be provided.

b) High level alarms (HLA) shall be set at not more than 85% level of the volumetric capacity of the vessel. On the actuation of high level alarm, the ROVs of the affected vessel shall close.

c) Audio-visual indication of high level alarm shall be available at the local and remote panels.

a) Each storage vessel shall have at least two safety relief valves with isolation arrangement set at different values and at not more than 110% of design pressure or maximum working pressure of the vessel and each having 100 % relieving capacity adequate for limiting the pressure build up in the vessel not more than 120% of design pressure or maximum working pressure.

i) The relieving load for the safety valves shall be based on fire condition, and no credit shall be taken for fireproofing on the vessel, if provided.

ii) The discharge of safety valves shall be connected to the flare system, wherever available. In this case, safety valves shall have lock open (or car seal open) type isolation valves on both sides of the safety valves.

iii) In case of non-availability of flare system, the discharge from safety valve shall be vented vertically upwards to atmosphere at an elevation of 3 m (minimum) for effective dispersion of hydrocarbons. In this case, an isolation valve downstream of the safety valve is not required. Isolation valve provided on upstream of the safety valve shall be lock open.

iv) A weep hole with a nipple at the low point shall be provided on the vent pipe to drain the rainwater that may accumulate otherwise. Weep hole nipples shall be so oriented that in case of safety valve lifting and consequent fire, the flame resulting from LPG coming out from the weep hole does not impinge on the storage Vessel or structure.

v) A loose-fitting non-sparking rain cap with a chain fitted to the vent pipe shall be provided on top of the safety valve.

b) Each vessel shall also be provided with one pressure and temperature measuring instrument. The pressure gauge/ indicator shall be provided with two isolation valves and an excess flow check valve. In case, temperature gauge/ indicator provided on the vessel is in a thermowell, it shall be welded to the vessel.

Typical safety fittings on a LPG storage vessel are indicated in Annexure-6-II

6.3 LPG LOADING/ UNLOADING FACILITIES

Each loading and unloading station shall consist of the following:

DRAFT 2

i. Excess Flow Check Valve (to close for excess flow of liquid LPG towards the atmosphere) and a non-return valve shall be provided in LPG loading/ unloading liquid lines along with isolation valves. Excess Flow Check Valve (EFCV) activation shall stop the liquid LPG flow from the pipeline towards the atmosphere, irrespective of loading/ unloading operation.

ii. A vapour return line with an isolation valve connected back to the storage vessel/ suction line. Additionally, a loading bay shall be provided with NRV in the vapour return line, provided either at each bay or at the common vapour return line header at the entry of the TT or Tank Wagon Gantry.

iii. Loading/ unloading arm or LPG hoses with QCDC (Quick Connect Dry Coupling) shall be provided at the end of liquid and vapour lines for connecting to the tank truck vessel or Tank Wagon. For the tank truck gantry, the loading/ unloading arm shall be provided with dry breakaway couplings with the facility to operate in both reverse and forward movement of the vehicle. Dry breakaway coupling to be provided either in the QCDC or elsewhere in the loading/ unloading arm. An isolation valve shall be provided before the breakaway coupling. Provision shall be made for venting of this section before disconnecting the arm from the tank truck. The loading arm shall be tested as per OEM recommendations. Alternatively, Hoses (hoses having an integrated valve in each end fitting, which, if the hose assembly experience a catastrophic hose failure, shall instantaneously shut down the flow in both directions) shall be used. In case LPG hoses are used, it shall conform to OISD-STD-135 or equivalent as approved by CCE.

iv. The hose-coupling/ flange joint shall be of Class 300 class rating with a metallic gasket. The hose coupling shall be provided with a cap or blind flange by which the nozzle to be closed when not in use.

v. Weigh bridges of suitable capacity or mass flow meter (MFM) shall be provided. The mass flow meters shall be certified for its use with LPG

vi. Unloading of LPG from a tank truck is carried out with vapour compressors using the pressure differential method of liquid transfer.

6.4 CYLINDER FILLING FACILITIES

i. The cylinder filling area shall be completely open type and covered from the top with a roof designed to ensure good natural ventilation. RCC roofing shall not be used.

ii. The floor area shall not have any channels or pits within 5 m of the LPG filling machine. Where these are necessary for conveyors or other equipment like a weigh machine, etc., a suitable gas detection system shall be provided. Additionally, vapour extraction system shall be provided to release LPG outside of the shed at a safe location. The whole of the LPG filling shed flooring shall be provided with mastic flooring.

iii. All carousels, including electronic ones and leak detectors, shall be of type approved by the Chief Controller of Explosives.

iv. Adequate lighting shall be provided in the cylinder filling area.

v. Proper access shall be made available for other firefighting equipment, i.e. fire tender, foam trolley, etc.

vi. Cylinder storage area requirement shall be worked out based on the stacking pattern of filled and empty cylinders as described in Annexure-8-II.

vii. A washing unit and drying unit shall be provided in the LPG filling conveyor in the filling shed, and only capped cylinders shall pass through the washing machine and drying unit. Decapping of the

DRAFT 2

cylinder should be carried out only after the cylinders are passed through the washing and drying unit. The filling machines shall be provided with an auto cut-off system so that the LPG supply is cut off when the desired quantity of product has been filled in the cylinders. The filling pressure shall not be more than the maximum working pressure of the cylinders, i.e., 16.87 **kg/cm²**.

viii. Filling machines in a carousel or stationary filling machines shall meet the requirements as specified in the Weights & Measures Rules. Maximum permissible errors on net quantity declared by weight shall be in accordance with the Legal Metrology (Packaged Commodities) Rules, 2023, as amended from time to time.

ix. Online check scales should have a minimum graduation in line with Legal Metrology (General) Rules, 2011, for example, 10 gms in case of a 14.2 kg cylinder.

x. A weight correction unit shall be provided for correction of the weight of filled cylinders, i.e. addition to or removal of LPG from filled cylinders to correct the weight.

xi. Online electronic leak detectors to check the valve and "O"-ring leak shall be provided for the cylinders filled through the carousel.

Where Valve leak cylinders are corrected using a Valve Change Without Evacuation machine, the cylinder shall be checked again through a suitable check weight scale and leak detector mechanism, like an electronic leak detector or equivalent mechanism, to ensure 100% reverification of corrected cylinders.

A new "O"-ring shall be replaced for the "O"-ring leak cylinder. These "O"-ring leak corrected cylinders shall be checked again through the main online electronic leak detector.

Compact Valve tester (CVT) to check the valve and "O"-ring shall be installed for other capacity cylinders, which are filled through a manual filling scale.

xii. Water test bath or electronic leak detectors shall be provided to detect body and bung leaks in cylinders. Facility shall be provided for all capacities of cylinders being filled in the Plant. To detect body and bung leak of 425 kg cylinder, the entity shall prepare SOP in line with PESO approval.

xiii. Vapour extraction system at strategic locations near carousel, cylinder evacuation unit, valve changing unit, weight correction unit, stationery filling machine, degassing unit at cold repair shed and at locations where leaking LPG is expected to accumulate shall be provided.

xiv. Filling machine of all types of cylinders or ROV on the liquid line at the filling shed entry shall be interlocked with the vapour extraction unit to ensure that filling does not start until or unless the vapour extraction unit is operating.

xv. Evacuation unit and valve changing without evacuation shall also be interlocked with the vapour extraction unit to ensure that this equipment does not operate until or unless the vapour extraction unit is operating.

6.5 UTILITIES

(A) Compressed air

The quality of instrument air shall conform to the requirements as recommended by the manufacturers of instruments/ equipment:

If one compressor is envisaged to run normally, another standby compressor of 100% capacity shall be provided. When more than one compressor running is envisaged, 50% standby capacity shall be provided.

DRAFT 2

(B) Service Water

Service water is required for the cylinder washing equipment and leak check bath, compressor cooling and in hose stations for washing, etc.

Water may be provided at a pressure as recommended by the OEM of the equipment for which water is being used. If one pump is envisaged to run normally, another pump with 100% capacity shall be provided as a standby. Where more than one pump running is envisaged, 50% capacity as a standby shall be provided.

Water Quality

Water shall be of potable quality conforming to IS:10500 - Specification for Drinking Water.

(C) Emergency Power

a) The installation shall be provided with a battery/ diesel generating set for operating the essential systems, such as the instrumentation and safety systems and minimum lighting during the grid power failure.

b) Emergency Feeder shall host the following equipment:

Jockey Pump, Critical lighting, Emergency Siren, Bore well, Gate Barrier, selected Yard lighting, automatic fire sprinkler system, CCTV, Gas monitoring system.

c) During a power failure, the battery bank shall be used to provide power to critical emergency lighting. Emergency lighting shall be provided at critical places such as Tank Truck Gantry, Tank Wagon Gantry, LPG shed, LPG Pump/ Compressor House, Bulk Storage Area, Fire Water Pump House, PMCC room, DG room, and security cabin.

d) For CCTV, a UPS backup for 30 min. operation shall be provided.

6.6 LPG PUMPS

i. LPG Pumps shall be designed for handling of LPG and safely withstand the maximum pressure that could be developed by the product and/ or transfer equipment. Pumps shall conform to API 610 or equivalent.

ii. Check valves shall be installed on the delivery side of all centrifugal pumps.

iii. LPG Pumps shall be provided with suction and discharge pressure gauges/ indicators, a high point vent to safe height or flare, and a suction strainer. Casing vents and drains shall be routed to a safe location, and double block valves shall be provided.

iv. Double Mechanical seal with seal failure alarms shall be provided.

v. Anti-cavitation protection through a pressure switch/ pressure transmitter, taken from the discharge end, actuating a low-pressure alarm both in the remote panel (wherever provided) as well as at the local annunciator panel in the LPG Pump House shall be provided. Actuation of the low discharge pressure alarm shall trip the LPG Pump automatically.

vi. Pumps shall be designed to build a discharge pressure such that the pressure at the carousel filling machine is at least 5.0 kg/cm² above the vapour pressure at the operating temperature.

vii. Pumps shall have a by-pass valve and other suitable protection (like pressure switch/ pressure transmitter/ auto product recirculation) against high discharge pressure on the delivery side.

viii. The electrical motor drive and switchgear shall conform to area classification as per OISD Standard-113 / IS:5572. Belt drives shall be of the fire-retardant, anti-static type.

DRAFT 2

ix. Provision shall be made for Audio Visual indications of various safety trips in the remote panel (wherever provided) as well as local annunciator panel in the LPG Pump House.

x. Carousel pump shall not be used for Tank Truck Loading without providing adequate safeguards to modify the performance parameters (head, pressure, etc.) as required.

A typical sketch of a bottling pump installation is enclosed as Annexure-6-III.

6.7 LPG VAPOUR COMPRESSOR FOR TANK TRUCK AND TANK WAGON

i. Compressors shall be suitable for handling LPG. Compressors shall conform to API 618 or API 619, or equivalent.

ii. The belts used shall be of antistatic type and fire-resistant.

iii. Compressor shall be provided with the following features as a minimum:

a) Pressure gauges or indicators in suction and discharge.
b) Temperature gauge or indicator in discharge
c) Discharge safety valve and a vent valve, their outlets leading to flare/ safe height outside the shed. Vents and drains shall be routed to a safe location, and double block valves shall be provided.
d) Low Suction Pressure Trip
e) Suction strainer
f) High Discharge Pressure Trip
g) High Temperature Trip
h) Check valve in the discharge
i) A discharge to suction recycle valve for achieving capacity turndown during startup.
j) Low cooling water pressure or flow trip

iv. Audio-visual indications of various safety trips in the local panel provided in the LPG compressor house.

v. Provision for unloading shall be provided through a local control panel or suitable mechanism without manually operated loading/ unloading of valves. Compressor shall be started in unloading condition only.

vi. A suitable size scrubber or liquid knockout drum shall be installed upstream of the vapour compressor. It shall be equipped with a safety relief valve, a drain and a high liquid level shutdown device.

A typical sketch of compressor installation is enclosed as Annexure-6-IV.

6.8 PIPING

Piping shall be designed for handling LPG. Piping shall have a minimum design pressure of 24 kg/cm² or the maximum pressure which could be developed by the transfer equipment or any other source, etc. whichever is higher and comply with the following:

i. Piping shall conform to the provisions of ANSI/ ASME B 31.3 in accordance with SMPV Rule,2016.

ii. The material shall conform to API 5L Gr. B / ASTM A106 Gr B or equivalent.

iii. Seamless pipes shall be used. Furnace butt welded or spiral welded pipes shall not be used.

iv. Pipe joints should be welded as far as practicable with a full penetration weld. Number of flanged or threaded joints should be kept to a minimum.

DRAFT 2

v. Low point drains and high point vents installed on the pipeline shall be provided with full-bore isolation valve. End points shall be blinded or plugged or capped suitably for the integrity of isolation.

vi. The vents of the LPG compressor, LPG pumps, LPG loading/ unloading arms and LPG loading/ unloading hoses shall be provided with full-bore double isolation valve.

vii. Buried piping shall be protected against physical damage and corrosion with suitable sleeves, properly sealed at both the ends, at road crossings.

viii. Same Material of Construction (MOC) shall be used for LPG liquid, vapour and venting pipeline.

ix. Pressure relief and thermal relief valves shall be constructed of steel.

6.9 THERMAL PRESSURE RELIEF SYSTEM

a) Any equipment or section of pipeline in which liquid LPG may be trapped, e.g., between shut-off valves, shall be protected against excessive pressure developed by thermal expansion of the LPG by providing suitable thermal pressure relief valve(s).

For this purpose, those sections of the pipeline which are part of larger pipeline where TSV is already provided, and intermediate valves are provided for use during shut down/ maintenance purpose shall be considered as single length of pipeline provided the entity has SOP to keep those valves in open condition during normal operating conditions including on shift and off shift time.

b) If pressure relieving devices discharge to the atmosphere, the discharge shall be vented vertically upwards at an elevation of 2 m (minimum) from the point of TSV discharge for effective dispersion. If pressure relieving device is installed on the pipe inside the shed, the discharge shall be vented at a height not less than 1.5 m above the eaves of the structure.

c) Set pressure of the Thermal Safety Valve shall not exceed the design pressure of the pipelines on which they are installed. The pressure relief devices shall be capable of preventing the pressure from rising more than 20% above the design pressure of the pipeline. ASME Section VIII Div. 1 and ASME B31.3 shall be referred to.

d) TSV shall be provided with full-bore isolation valve and shall be kept in an open lock condition. Test date and due date for TSV/ SRV shall be displayed in the field.

6.10 VALVES

Steel valves conforming to relevant API/ BIS recognized standards shall be used. Cast iron valves should not be used.

6.11 FITTINGS

i. Steel flanges and flanged fittings conforming to ASTM A105 or equivalent, ASME B16.5 shall be used. Slip-on or weld neck flanges should be used. Screwed flanges for sizes 50 mm or smaller may be used.

ii. Steel screwed fittings and couplings shall conform to ASMI B 16.11 or equivalent. Steel unions shall have ground metal-to-metal seats. Gasket-type unions shall not be used.

iii. Plugs shall be of steel. Cast iron or brass plugs shall not be used.

iv. All flanges shall be connected for bonding for electrical continuity.

DRAFT 2

6.12 EVACUATION FACILITY FOR DEFECTIVE CYLINDERS

(A) The cylinder evacuation facility shall consist of:

- Cylinder emptying vessel (s)
- Compressor
- Four-way valves or equivalent

Each of the cylinder emptying vessels mentioned above shall be equipped with the following:

- Pressure gauge/ indicator
- Level gauge/ indicator
- High level alarm switch set at 85 % with trip/ switchover arrangement.
- Pressure relief valves with isolation arrangement and set at values not exceeding 110% of the design pressure of the vessel. In case one SRV is considered for a vessel less than 1000 litres capacity, such a unit shall be kept idle till the time a calibrated SRV is replaced.
- Vent valves discharging to 1.5 m above the shed.
- Other trims like the drain valve, utility connection.

A typical sketch of cylinder evacuation facility is enclosed as Annexure-6-V

OR

(B) Any other suitable facility, like a pneumatically operated hydraulic cylinder, for evacuating LPG from the cylinder and pushing it into the main LPG return header. It is a positive displacement, double-acting reciprocating type pump directly coupled to an air motor/ cylinder specially designed for pumping LPG. The equipment consists of a suitably sized reciprocating Pump, coupled to a pneumatic cylinder which acts as the prime mover. The Pneumatic cylinder and LPG Pump are connected by a common piston rod but physically separated, thereby eliminating the chances of LPG and Air getting mixed. The pneumatic cylinder drives the LPG Pump and creates the required differential pressure, thereby sucking product from the connected LPG cylinders. The evacuated product is delivered into the liquid return line in the Filling Shed.

For both the provisions as mentioned in (A) and (B) above, the following shall be provided:

i) Cylinder rack, header to be provided with pressure gauge/indicator and a strainer with isolation valves.

ii) The LPG line exiting from the cylinder evacuation area is to be provided with a non-return valve before joining the main LPG return header.

iii) Independent earthing connections.

6.13 PURGING OF NEW CYLINDERS/ TANKERS

a) The new LPG cylinders containing air shall be evacuated with a vacuum pump. The evacuation facility shall consist of:

i) Vacuum pump (water/ oil/ air-cooled type)

ii) Purging manifold

DRAFT 2

iii) Vacuum receiver fitted with vacuum gauge, vent and drain.

iv) LPG vapour header

v) Purging adapters

vi) Vacuum pump with suction strainer

vii) Pressure regulator, relief valves

A typical sketch of the new cylinder purging system is enclosed as Annexure-6-VI.

b) Purging of tankers/ vessels shall be done using either Nitrogen or by filling water and displacing with LPG vapours.

c) Degassing and Gas-in Procedures of Bulk LPG Storage Vessels are as per Annexure 6-VII.

6.14 EQUIPMENT FOR ETHYL MERCAPTAN SERVICE

6.14.1 Material of Construction of Ethyl Mercaptan containers

Stainless steel and copper-free steel alloys are the preferred materials of construction of equipment for mercaptan service. Aluminium of the desired pressure rating can also be used. Use of Iron or carbon steel shall be avoided as there is the hazard of formation of Iron-Sulphur complexes, which are pyrophoric. Copper or copper-bearing alloys shall not be used for mercaptan service, as mercaptan readily attacks and contaminates them.

6.14.2 Pumps for Mercaptan service

Only hermetically sealed pumps shall be used.

6.14.3 Piping and Fittings

i. Seamless stainless-steel piping shall be used. Threaded fittings can be used provided they meet the service requirement ratings. A fluoroplastic tape sealant is recommended.

ii. Flexible connections shall be seamless stainless steel with pipe nipples welded to each end.

iii. Relief valves shall be full nozzle, disc type, closed bonnet, carbon steel body with SS304 stainless steel trim and stainless or aluminized steel spring.

iv. For transfer valves, stainless steel ball valves with fluoroplastic seats and stainless steel (SS316) trim or their equivalent are recommended.

v. Fluoroplastics shall be used for gasketing.

vi. For pressure gauges/indicators, a stainless-steel diaphragm type is recommended with stainless steel socket and tip.

vii. Provision for analysis in line with IS:4576 or IS:14861, as applicable, shall be made.

6.14.4 Handling of Ethyl Mercaptan

Leaks and spills of mercaptan can be treated with commercially available masking agents (e.g., Aldor and Neutroleum Alpha, etc.) to make them odourless. Large mercaptan spills shall be covered with sand or activated carbon or any other absorbing material, which are then buried or incinerated. The area over the spill should then be washed with ordinary household bleach solution and then thoroughly with water. Dry bleaching powder should never be used in treating mercaptan spills. A violent reaction could occur. MSDS shall be referred to for the safe handling of ethyl mercaptan.

The empty containers of ethyl mercaptan shall be disposed of as per the MSIHC Rules, 1989

DRAFT 2

6.15 SELECTION OF ELECTRICAL EQUIPMENT

i. Electrical equipment shall be selected, sized and installed so as to ensure adequacy of performance, safety and reliability. The equipment in general shall conform to relevant Indian Standards and shall be suitable for installation and satisfactory operation in the service conditions envisaged.

ii. All electrical equipment shall be selected considering the system neutral earthing.

iii. Electrical equipment and apparatus shall conform to hazardous area classification and be selected in accordance with IS:16724. These shall be tested by agencies such as CIMFR, ERTL, CPRI or an independent test laboratory of the country of origin for such equipment. Indigenous flameproof equipment shall comply with the relevant BIS standard as per the requirements of the statutory authorities. All electrical equipment used in a hazardous area shall be approved by the Chief Controller of Explosives.

iv. For details on hazardous area classification, enclosure protection, etc., OISD- STD–113, OISD-RP-149, NEC–1985, IS/IEC:60079, IS:5572, IS:16724 Petroleum Rules, 2002 shall be referred.

v. Electrical equipment installed inside the LPG sheds with water sprinkler provision, suitable Ingress Protection in line with IS:60529 shall be provided.

vi. Fail-safe (Mechanical/ Electrical) Interlock/ change-over switch shall be provided between the Grid Power and the DG power to ensure that the equipment gets supply from one source only.

6.16 INSTALLATION LIGHTING

i. Sufficient lighting shall be provided so as to enable Plant operators to move safely within the accessible areas of the installation and to perform routine operations. In the event of a normal power failure, emergency lighting shall be provided in critical areas.

ii. Normal lighting system shall be on a 415/ 240V AC supply, whereas emergency lighting will be DC supply based.

iii. Lighting requirements provided during the failure of the power supply for Normal lighting are intended broadly,

   a) To facilitate the carrying out of specified operations, for the safe shutdown of the installation.

   b) To gain access and permit ready identification of firefighting facilities such as fire water pumps, fire alarm stations, etc.

   c) For the visibility of the escape route for the safe evacuation of operating personnel.

iv. Under normal operation, both emergency and normal lighting shall be fed by a normal power source. On failure of normal supply, critical lighting shall be transferred immediately to the emergency source. Critical lighting (D.C. supply based) shall be normally kept ‘ON’ so as to ensure that in case of power failure, the critical lighting gets instantly ON. During a power failure, the battery bank shall be used to provide power.

v. On failure of normal supply, the DG set for lighting purposes should start through the AMF panel. (Auto mains fail). In case of LPG leakage, the SOP for disabling of auto start of the DG is to be displayed.

vi. Low pressure sodium vapour lamps shall not be installed in hazardous areas.

DRAFT 2

vii. The illumination levels in different areas shall be as per good engineering practice. The Illumination in the operational areas, including inside the kerb and manifold, shall be such that adequate visibility is there at all times for emergency and normal operations.

viii. Min. one number calibrated lux meter shall be kept in the location.

ix. The following correlation shall be referred, with the areas/ sheds, etc., as normally defined in LPG Installations, along with the lux levels required.

Area or Facility in the Plant	Average Maintained Illumination Level, lux
All roads in the LPG Installation, including the parking area for packed trucks and Bulk LPG tank trucks.	20
Patrolling the pathway alongside the boundary wall	10
LPG Storage vessels within the kerb wall	20
LPG Pump and compressor shed	100
Service water pump shed, Air compressor shed, Connecting platforms between sheds	100
Filling shed, Cylinder storage shed, Valve change shed, DPT shed, Degassing shed, Loading/ unloading sheds, Tank truck gantry, Tank wagon gantry	100
Manifold with operating valves, ROV, drain valves, etc, in the LPG storage vessel area, Coastal terminal, LPG Jetty	60
Crossovers outside sheds, Deluge Valve operating areas	20
Pig receiving/ launching area, Pipeline transfer receiving section, LPG pipeline sections with valves, strainers, ROVs requiring operation during normal functioning	60
Non-operational areas outside the sheds	10
MCC room, DG Room, VFD room	150
Inside the MCC Room, behind the panels	150
Cable cellar	70
Transformer yard/ room, Double pole structure with switch	100
Switchyard (i) operating area	100

DRAFT 2

Switchyard (ii) other areas	50
Battery room, UPS room	100
Monitoring room, Control room, QC lab	400
All stores	100
Office building, Security cabin	300
Toilets	150
Stairs, Corridors, Lifts, Entrance, other areas in the office building, canteen, washrooms,	70
POTA cabin, Main gates, Emergency gate, wicket gates, Licensed area entry gate barriers, Access control devices and security checking areas, Door frame metal detector areas	100
Security watch towers, including stairs	50
Near and outside the main gate and licensed area entry	150
Bore wells, Open wells	100
Fire Water Pump House, Parking area for cars and two-wheelers	100

x. The lighting fixtures on various circuits shall be suitably interlaced so that failures of any one circuit do not result in complete darkness.

xi. Switches controlling the lighting fixtures and exhaust fan shall be installed outside the battery room.

xii. Switches of lighting panels installed in hazardous areas shall have a pole to break the neutral in addition to the poles for phases.

xiii. When the means of isolation is located in a non-hazardous area, the switch shall break all poles, including neutral, or alternatively may break only live poles, the neutral being isolated by a removable link.

xiv. A sufficient number of welding sockets shall be provided in LPG Pump House, LPG shed, Tank truck gantry, Tank wagon gantry, Storage area, etc., for carrying out any welding job inside the Plant. There shall not be any temporary/ loose electrical connections. Earthing of the welding machine shall be ensured. Provision to be made for tripping of power to the welding machine in case of operation of ESD/ DV.

xv. For details on inspection practices relevant OISD standards to be referred to.

6.17 INSTALLATION EARTHING

i. All electrical equipment shall be suitably earthed. The earthing system shall cover the following:

- Equipment earthing for personnel safety.

DRAFT 2

- System neutral earthing, and
- Protection against Static and Lightning discharges.

The earthing system shall have an earthing network with the required number of earth electrodes connected to it. The following shall be earthed:

- System neutral
- Metallic non-current carrying parts of all electrical apparatus, such as transformers, switchgears, motors, lighting/ power panels, terminal boxes, control stations, lighting fixtures, receptacles, etc.
- Steel structures, loading platform, etc.
- Cable trays and racks, lighting mast and poles.
- Spheres, Bullets, etc.
- Electrical equipment fencing (e.g., transformer, yard, etc.)
- Cable shields and armour.
- Flexible earth provision for wagon, truck.
- Conveyer in LPG shed (bonding and earthing)
- LPG pipeline
- Carousel including filling guns.
- Data Processing system/ Automation

ii. Installation earthing design shall be carried out in line with IS:3043, relevant OISD standards, CEA Regulations 2023. The installation earthing design document shall be available at the installation. The earthing system shall have an earthing network with a required number of earth electrodes connected to it. All earth connections should be visible for inspection to the extent possible.

iii. The lightning protection system shall be provided as per the requirements of OISD-GDN-180, IS/IEC 62305-1, IS/IEC 62305-2 and IS/IEC 62305-3.

iv. The resistance values of an earthing system to the general mass of earth should be as per requirement of relevant OISD standard & IS:3043.

v. The earth conductor shall be adequately sized to carry the applicable maximum earth fault current without undue temperature rise. All joints shall be protected against corrosion.

vi. All the electrical equipment operating above 250 volts shall have two separate and distinct connections to the earth grid.

vii. The main earthing network shall be used for the earthing of equipment to protect it against static electricity. The main earth loop shall be laid at a depth of 500 mm below ground level as per IS3043.

viii. Every Sphere/ bullet shall be electrically connected with the earth in an efficient manner by not less than two separate and distinct connections. The connections and the contacts required shall have as few joints as possible. All joints shall be riveted, welded or bolted and also soldered to ensure both mechanical and electrical soundness.

DRAFT 2

ix. All joints in pipelines, valves in installations, and associated facilities and equipment for LPG shall be made electrically continuous by bonding or otherwise. End flanges of product pipelines shall be earthed through a grid/ running earthing strip.

x. All Steel structures of sheds, loading/ unloading platforms/ gantries, watchtowers, etc. shall have two separate and distinct connections to earth.

xi. Bonding shall be provided in the joints of vapour extraction ducts.

xii. The electrical earthing layout diagram with earth pit numbers needs to be prepared, approved, and should be kept available.

6.18 EARTH PITS

i. Earth pit design shall be as per IS:3043, CEA Regulations 2023, relevant OISD standards.

ii. The jointing of the earth pit connector shall be as per IS:3043, relevant OISD standards. In case joints are through connectors/ lugs, no gap shall be ensured between the surface of stripes by providing a minimum of two or more bolts.

iii. Earth pit testing & its maintenance shall be done as per requirement of relevant OISD standard & IS:3043. The inspection and testing of the earth pits shall be done six monthly, once in dry and once in wet weather, and records shall be maintained. The electrical continuity between equipment and the earth pit shall be tested twice a year.

6.19 ELECTRICALLY INDEPENDENT EARTH ELECTRODES

i. The Neutral of the Transformer and Diesel Generator shall be earthed with two distinct earth pits separately using insulated cable/ strip.

ii. All Metallic non-current carrying parts of all electrical apparatus, like cable trays, fuel tanks of DG sets and fire engines, etc., shall be earthed to ensure that the exposed metallic parts do not become dangerous by attaining high voltages in case of faults.

6.20 PROTECTION

i. The protective system shall be designed to ensure Protection of Personnel and Plant equipment against damage which can occur due to internal or external short circuits, overloading, abnormal operating conditions, switching, lightning surges, etc. Accordingly, relays and protective devices shall be suitably selected and installed.

ii. All the protective relays for the Generator, Transformer, Motors, and Switchgears shall be tested at least once a year in line with relevant OISD standard and test records maintained.

iii. Surge protection device to be provided for critical electronics equipment, including automation. ELCB/ RCCB of rating 30 mA shall be provided.

6.21 CABLE

i. Cable size, voltage grade and type should comply with Approved for Construction (AFC) drawings.

ii. Cable selection and laying shall be done as per relevant OISD standard and Indian Standard.

iii. Instrument and signal communication cables should not be laid in the same trench/ tray along with electrical cables.

DRAFT 2

iv. Cable route markers are provided in a permanent way at a location of changes in the direction of cables and at intervals of not more than 30 m in line with relevant OISD standard and at cable joint locations.

v. Identification number tags of the cable, for the equipment to which the supply is fed, shall be provided at both ends of the cable.

vi. Double compression cable glands shall be used for zone-1 and zone-2 areas as per the area classification drawing.

vii. Cable insulation resistance test shall be done.

6.22 PMCC PANEL

i. The identification number of the panel shall be displayed on the front and the rear of the panel. Fuse/ MCCB and relay rating shall be displayed.

ii. The door of the MCC panel shall not be openable when the panel is in the switched-on condition

iii. Live phases of the busbar inside the MCC panel shall be protected with a Bakelite sheet or equivalent protection.

iv. Vermin-proofness of the panel shall be ensured.

v. Lock Out Tag Out (LOTO) provision for panel isolation shall be provided.

vi. Thermography of panels shall be carried out every six months, and the record to be maintained

vii. For further details, relevant OISD standard, CEA Regulation 2023 and applicable Indian standard shall be referred to.

6.23 OTHERS

i. Electrical shock restoration chart (in English or Hindi and the local language of the district, and where Hindi is the local language, in English and Hindi) shall be displayed in Motor Control Centre (MCC) as per CEA Regulations 2023.

ii. IS:13774 marked electrical hand gloves for LT/ HT voltage shall be available with test reports.

iii. Insulating floors or mats conforming to IS:15652 of the appropriate voltage level shall be provided in front of the panels for the safety of operating personnel.

iv. Relays/ Cables shall be tested once a year, and records to be maintained.

v. Transformer oil shall be tested once a year, and records to be maintained.

vi. Thermography of panels shall be carried out every six months, and the record to be maintained.

7.0 STORAGE AND HANDLING OF BULK LPG

7.1 GENERAL:

This section deals with the safe practices and provisions applying to the loading and unloading of bulk LPG and the storage of bulk LPG at installations. There should be strict compliance with respect to the selection, deployment of proper skilled manpower for effective operation and maintenance.

7.2 BULK HANDLING FOR MOVEMENT BY BULK LPG TANK TRUCK

The tank truck for road movement shall be designed, constructed and tested in accordance with the Static and Mobile Pressure Vessels (Unfired) Rules, 2016, latest and OISD-STD-159.

DRAFT 2

7.2.1 SAFETY PRECAUTIONS

The following precautions should be taken due to associated hazards during the transfer of LPG to or from a tank truck.

1. No source of ignition must be allowed in the area where product transfer operations are carried out.
2. An authorized person of the entity shall supervise the complete loading/ unloading operation and respond immediately in the event of an emergency.
3. Fire extinguishers shall be placed near the tank trucks during transfer operations.
4. After positioning the tank truck, place wheel chokes and provide proper earthing, i.e., by connecting the first clip to the designated earthing point on the tank body and the second clip to the chassis. A suitable earthing interlock arrangement shall be provided so as to ensure that unloading/ loading operation does not commence without earthing. Earthing shall be disconnected only just before the release of the tank truck. Wheel chokes should be removed after the disconnection of the earthing.
5. Before removing plugs or caps from the loading/ unloading arm, always crack open the vent valve to allow for the release of trapped LPG and further ensure that the valves are effectively sealing. Loading/ unloading hose/ arms shall be plugged when not in use.
6. LPG Hoses shall be provided and tested as per OISD-STD-135.
7. A metallic gasket shall be used for flange joints, and ensure all studs are tightened. Bonding of flanges through copper jumpers shall be ensured.
8. For connecting and disconnecting arms or hoses, only non-sparking tools shall be used.
9. While disconnecting hoses/ arms, connections shall be loosened only slightly at first to allow release of trapped pressure, if any.
10. Do not expose hands, face or clothing to liquid LPG. Always use personal protective equipment such as Safety Shoes, Safety Helmets, Cotton Coverall etc., as defined in the respective entity's PPE policy/ guidelines. Usage of low temperature gloves, Safety Goggles to be ensured while making or breaking the connections to avoid cold burns.
11. The master switch shall be put off immediately after parking the tank truck in position. No electrical switch on the tank truck shall be turned "on" or "off" during the transfer operation.
12. No repairs shall be made on the tank truck while it is in the loading/ unloading area.
13. New tank trucks, vessels, or vessels received after repair shall be purged as per the laid-down procedure.
14. In case the receiving vessel has a pressure of less than 1 kg/cm², it shall not be filled. Such a vessel shall be taken for filling only after detailed safety checks.
15. Venting and purging of LPG during the transfer operation shall not be carried out in an open atmosphere. It shall be piped to a suitable flare system. Where such facility does not exist, vent pipes shall be provided to carry the vented product to a safe distance and released at a height not less than 1.5 m above the eaves of the structure.
16. The Tank Truck crew and person responsible for the loading/ unloading operation should be present near the tank truck throughout the operation.

DRAFT 2

17. Filling/transfer operations should be stopped immediately in the event of -

i. Any leakage occurring.

ii. A fire occurring in the vicinity

iii. Lightning and thunderstorm

iv. Any emergency.

7.2.2 PROCEDURES FOR OPERATION

(a) TANK TRUCK LOADING OPERATIONS

1. Check for the following in a tank truck as per statutory regulations before accepting it for filling:

- Provision of two safety valves, a level gauge/ indicator, Internal Excess flow check and control valve on the liquid and vapour lines, pressure gauge/ indicator, temperature gauge/ indicator on the vessel/ bullet. Protection to the gauges/ indicators and safety relief valves as required in SMPV Rules,2016 (Latest) and OISD-STD-159 to be provided.
- A fire screen between the cabin and the vessel is provided. For this purpose, cabins with a metallic back cover without any opening will be considered as a fire screen.
- Two portable IS marked 9 kg. DCP fire extinguishers shall be provided on each vehicle, one on each side and should be accessible from outside the cabin and another DCP extinguisher of 1 kg capacity should be kept in the driver's cabin, and all extinguishers (minimum 3 nos.) should have the marking indicating the next due date for pressure testing. TT number to be painted on each fire extinguisher.
- Spark arrester welded to the exhaust pipe of the prime mover/horse, and the spark arrester shall be of a type approved by the CCE.
- Any alternate spark elimination system duly approved by CCE shall also be acceptable, e.g. in case the exhaust of a diesel engine is based on a design having electronic fuel management with unit injectors and electronic control unit coupled with turbocharger and intercooler arrangement, no separate spark arrester shall be provided.
- No leakage in the exhaust silencer pipe exists.
- The manufacturer's name plate with the date of testing is fitted on the vessel.
- Yearly testing of SRV & IEFCV as per statutory requirement.
- A valid Explosive License and RTO certificate is available.
- Approved drawings of a vessel are available.
- Blind flanges/caps are provided on the vessel.
- Bonding between the vessel and chassis and between flanges is satisfactory.
- An earthing/ bonding point is available.
- Third-party inspection/ test certificates for the vessel/ fittings are available.
- Liquid/ vapour line valves are in good condition.
- Air brakes are not bypassed. ABS shall be available in the vehicle.
- Master switch (300-amp rating) with double pole wiring

DRAFT 2

- Endorsement in the driving licence by licensing authority to the effect that he is authorised to drive a goods carriage carrying goods of dangerous or hazardous nature to human life.
- No Voltage step up device is available in the cabin.
- VTS shall be available in the vehicle as per AIS140 and any other statutory regulations as applicable.
- A rear and lateral under-run protection shall be available in the vehicle.
- Dedicated second crew in a tank truck.
- SRV shall be replaced by a new SRV every ten years and if found defective during any of the earlier yearly inspections, it shall be replaced earlier.
- Each IEFCV shall be replaced by a new IEFCV every ten years and if found defective during operation or any of the earlier inspection, it shall be replaced at that time itself i.e., earlier than ten years.

2. Move the tank truck to the weight bridge and record the initial weight. Place the tank truck on the loading bay in drive-out position and adjust the position of the tank truck to facilitate easy connection of loading arms/ hoses, and place choke blocks at the front and rear wheels. Keep the truck in neutral mode with hand brakes "ON".
3. Stop the engine and switch off all electrical equipment. The master switch shall be put off immediately after parking the tank truck in position. No electrical switch on the tank truck shall be turned “on" or “off" during the loading operation.
4. All persons shall leave the driver's cabin. The crew shall be available near tank truck till the completion of loading/ unloading.
5. Make earthing connections of the vehicle at the specified point to the fixed grounding system.
6. Connect the liquid and vapour lines suitably with the loading point. Crack open valves on loading and vapour return lines and inspect Arms/hoses and connections for leakage. Loading shall start only when the system is leak free.
7. Start the loading pump.
8. Crack open the valve on the liquid loading arm and gradually open the valve.
9. Slowly open the valve on the vapour loading arm/ hoses and gradually open the valve fully.
10. The quantity loaded into the tank truck can be determined by -
    i. Liquid level
    ii. Weighment
    iii. Positive displacement meter
    iv. Mass flow meter

    The quantity of LPG filled shall not exceed the RLW of the tank truck.
11. Liquid level may be determined by roto-gauge or fixed liquid level gauge, and the percentage volume to be filled shall not exceed 85 % at ambient temperature or the allowable filling limit as per PESO, whichever is lower.

DRAFT 2

12. Where a weigh bridge is used, it is necessary to determine the density of the product being loaded to avoid excess filling in terms of volume. The weigh bridge shall be periodically calibrated and stamped by the Weights & Measures Authorities.

13. When the filling operation is in progress, the pressure within the tank truck vessel shall be observed to ensure that it does not approach the start-to-discharge pressure of the relief valve. Filling rate may be regulated as required.

14. The couplings may then be removed, and plugs/ caps replaced on the tank truck valves. Re-check the tightness with soap solution.

15. After loading is completed, close the liquid and vapour return line valves. Close all the valves of the tank truck (including the EFCV lever) and loading arms.

16. Roto gauge reading, pressure, and temperature shall be noted.

17. Vent out the product between the tank truck valves and the valves of loading arms through the vent line. After completion of venting, the loading arm shall be disconnected.

18. Ensure vapour and liquid loading arms are properly secured to the post and plugged/ capped when not in use.

19. Disconnect the earthing connection.

20. Remove wheel chokes.

21. After completion of the loading operation, move the tank truck to the weigh bridge and record the weight of the loaded tank truck.

(b) Tank Truck Unloading Operations

1. Operations described under Items 1 through 5 in para 7.2.2(a) should be carried out.
2. The liquid line and vapour line of the tank truck shall be connected to the respective unloading arms or hoses fixed to the designated unloading point.
3. Crack open the valves on the LPG liquid and vapour line and inspect arms/ hoses and connections for leakage. Unloading shall start only when the system is leak free. When satisfied, valves on the tank truck and the receiving vessel shall be opened. Make sure that the receiving vessel shall have enough ullage.
4. Start the LPG compressor.
5. Slowly pressurize the tank truck by opening the vapour line valves on the Tank truck, vapour unloading arm and pipeline.
6. Start unloading/ decanting the tank truck by opening the liquid line valves (including EFCV lever) on the Tank Truck, the liquid unloading arm and the pipeline.
7. Vapours will be sucked from the receiving vessel and will be discharged into the vapour space of the tank truck vessel creating pressure differential thereby pushing the liquid from the tank truck vessel to the receiving vessel.
8. Observe the roto-gauge reading and pressure reading of the tank truck at regular intervals to ensure that the product is being transferred from the Tank truck to the storage vessel.
9. Care should be exercised to see that the pressure created within the delivering tank truck vessel does not reach or exceed the set pressure of the relief valve.

DRAFT 2

10. An authorised person of the entity shall supervise the transfer operation and respond immediately in the event of an emergency.
11. When liquid transfer is completed from any tank truck of the batch, close the liquid and vapour valves of that tank truck and its unloading arm and wait till the liquid decantation from the whole batch is completed.
12. After the liquid decantation is completed from all the tank trucks of the batch, the vapour recovery operation may be started by opening the vapour line valves in the manifold and the tank truck. Care should be taken not to bring down the pressure of the tank truck vessel below 1.5 kg/cm².
13. On completion of vapour recovery, stop the LPG Compressor, close all the valves of the Tank truck, unloading arms and the pipeline.
14. Roto gauge reading, pressure, and temperature shall be noted.
15. Vent out the unloading arm content and disconnect the unloading arm.
16. Ensure vapour and liquid unloading arms are properly secured to the post and plugged/ capped when not in use.
17. Disconnect the earthing connection.
18. Remove wheel chokes.
19. After completion of the unloading operation, move the tank truck to the weigh bridge and record the weight of the unloaded tank truck.

A checklist for bulk LPG tank Trucks to be prepared at unloading and loading locations as per Annexure-7-I-A and Annexure-7-I-B, and a record for deviation and corrective action shall be maintained.

7.3 BULK HANDLING FOR MOVEMENT BY RAIL

LPG is moved in Tank Wagons by the Railways. These wagons are designed by RDSO and are fitted with various devices as shown in Annexure-7-IV.

The loading facilities are similar to those for tank truck loading, while the unloading of LPG from tank wagons shall be done with the help of a compressor. The compressor is used to create a differential pressure between the receiving and discharging vessels by withdrawing vapours from the receiving vessel and forcing them at high pressure into the discharging vessel, thereby establishing a smooth flow. The content of tank wagons to be ascertained by weighment on a weigh bridge before and after emptying or by using mass flow meter.

A checklist for bulk LPG tank wagons to be prepared at unloading locations as per Annexure 7-II, and a record for deviation and corrective action shall be maintained.

7.3.1 SAFETY PRECAUTIONS

1. Suitable SOP in line with the weigh bridge OEM recommendations to be framed and adhered to.
2. Sufficient number of dummy wagons shall be used to avoid hauling locos/ electric coming within 15 m from the first fill point during placement or withdrawal of rake. Main railway track shall be isolated from wagon gantry siding at least 15 m from 1st loading/ unloading point by providing insulation joint and loco shall stop before the insulation joint. The dummy wagons shall not be placed on the track covering the insulation joint during the loading/ unloading operation so that the insulation joint is not short-circuited.

DRAFT 2

3. The first operation after positioning the wagon shall be to provide for proper earthing. Earthing shall be disconnected just before the release of the wagon.
4. For connecting and disconnecting arms or hoses, only non-sparking tools shall be used.
5. Always use personal protective equipment such as Safety Shoes, Safety Helmets, Cotton Coverall etc., as defined in the respective entity's PPE policy/ guidelines. Usage of low temperature gloves, Safety Goggles to be ensured while making or breaking the connections to avoid cold burns.
6. Ensure that the lower portion of the Flapper Bridge at the wagon side is fitted with rubber or wooden padding.
7. Ensure that the electrical continuity of the system is intact.
8. Ensure that all fittings on the wagons are checked physically.
9. Hoses, if available, shall be provided and tested as per OISD-STD-135, and records shall be maintained.
10. The loading/unloading operation shall be carried out under the close supervision of an authorised person.
11. During the unloading operation, after the liquid transfer is over, the wagon pressure shall not be reduced below 1.5 kg/cm^2.

The maximum quantity of LPG filled in any tank wagon shall be limited so that the tank wagon shall not be liquid full due to expansion of the contents with rise of the temperature to 55 °C. or limited to specified gross weight on the wagon, whichever is less.

7.3.2 OPERATING PROCEDURES (TANK WAGON)

(a) LOADING OPERATION

1. Ensure that flapper bridges are in the lifted position and secured with the help of the lock provided.
2. Ensure that the Loading Arms are not swinging and are secured inside the gantry with the help of the arm lock provided.
3. Place the wagon on the loading bay to facilitate easy connection of loading arms/ hoses after noting the tare weight through a weigh bridge/ online in motion weigh bridge, wherever provided.
4. Ask the loco to move away and exhibit a caution sign at a minimum 15 m away from the wagons on both ends.
5. Switch off the loco engine if parked nearby.
6. Apply hand brakes on all wagons.
7. Record T/Wagon Nos, POH and ROH due dates, for barrel and for under frame.
8. Check the TWs visually for the following, to determine their fitness prior to loading:

   a) Whether Periodic Over Hauling (POH) is due for the barrel / under frame.

   b) Whether Regular Over Hauling (ROH) is due for the barrel / under frame.

   c) For any visible damage to the tank barrel.

   d) For any damage to the tank fittings.

DRAFT 2

e) For any leakage from the dome fittings or barrel

f) Whether all valves and fittings are in position and are in working condition.

g) Whether the surface of the barrel is free from rust and the paint on the barrel is in good condition.

h) Whether the tank wagon details like wagon serial number, its tare weight and carrying capacity, POH and ROH due dates are legible.

9. Ensure the fire system, safety interlocks, and communication system are functional.
10. Take loading advice.
11. Make the earthing and bonding connection for each wagon. Each Wagon shall be bonded at each loading bay at 2 places: one at the dome of the wagon and one at the body of the wagon.
12. Lower the flapper bridge slowly on the wagon.
13. Rail loading/unloading of LPG should be restricted to a maximum of half rake as per SMPV Rules 2016. Full rake loading/ unloading shall be done on two separate rail gantries having a minimum distance of 50m.
14. Open the lid of the wagon.
15. Connect the filling hose or loading arm and vapour return line hose or arm to the wagon. Ensure that the vent connection valves are closed.
16. Ensure that the header is charged with LPG and the bulk loading pump is running.
17. Open the tank wagon filling line valve and vapour return line valve.
18. Check the system for leaks. Loading shall start only when the system is leak-free.
19. Open the valve on the vapour return line. Now, slowly open the valve on the filling line. Increase the valve opening and gradually open the valve fully. Ensure filling is up to a safe filling level, keeping in view temperature factors. Take mass flow meter readings if used.
20. Once the filling is over, close the wagon filling and vapour return lines valves and also valves on filling and vapour return lines at the loading point.
21. Open the valve on the vent line connection to both the feed line and vapour return line slowly while depressurizing to avoid ice formation. Then, close the vent line connection valves.
22. Disconnect the filling and vapour return line arm or hose connections from the wagon. Replace and tighten the plugs on the filling and vapour return lines.
23. Close the top cover of the wagon and seal it properly. Remove earthing and bonding connections.
24. Release the hand brake of the wagon.
25. Release all the wagons on the loading points in the lot.
26. Check the gross and net weight at the weigh bridge.

(b) UNLOADING OPERATIONS

1. Operations described under Item 1 through 12 in para 7.3.2(a) should be carried out.
2. Check the seals, open dome covers, and check fittings.

DRAFT 2

3. Rail loading/unloading of LPG should be restricted to a maximum of half rake as per SMPV Rules 2016. Full rake loading/ unloading shall be done on two separate rail gantries having a minimum distance of 50m
4. Record the dips.
5. Ascertain that the liquid discharge valve and the vapour valve within the tank wagon cover are in the closed position.
6. Open the port covers on the side of the dome shell, if they exist. Unscrew the plugs in the outlets of the vapour valve and the liquid valves using a box wrench. This must be done slowly.
7. If there is any sound of escaping vapour or if there seems to be pressure behind the plugs, the pressure must be allowed to relieve itself past the threads before the plugs are entirely disengaged.
8. If the vapour discharge continues or if there is evidence of a liquid discharge, the valves should be re-tightened.
9. Apply small quantity of sealant to the connecting pipe male threads, keeping the sealant away from the end of the thread. Tighten pipe with a pipe wrench.
10. Connect the two liquid transfer unloading swing arms or hoses to the nipples attached to the liquid discharge valves. Connect the vapour or equalizing swing arm or hose to the nipple attached to the vapour valve. In most of the cases, these connections to be made by means of either a ground joint union or a hose coupling.
11. In the event, a ground joint union is used, no gasket will be required. If a hose coupling is employed, ensure that the appropriate gasket is in place. Make sure that they are secured tightly by appropriate means.
12. After the vapour and liquid arms or hoses have been connected and before any valve is opened, the valves on the tank wagon are crack opened in order to apply pressure to the hoses / loading arms as a test for leaks.
13. If any leak appears, the valve should be immediately closed and corrective measures applied.
14. Recheck the lines and connections to make sure that they are connected correctly. Unloading shall start only when the system is leak-free.
15. After the liquid and vapour lines have been secured and tested, both liquid isolation valves should be opened slowly and completely. Then, open all other valves in the liquid line, working from the tank wagon to the storage tank.
16. If the tank wagon pressure is higher than that in the storage tank, do not open the valves in the vapour line or operate the compressor. When the rate of liquid flow drops to an unsatisfactory level with the storage tank filling valve wide open, open the vapour valves between the tank wagon and the storage tank.
17. At this point, make sure that the control valves at the compressor are in a position that allows the compressor to draw vapours from the storage tank and force it into the tank wagon, then start the compressor.
18. When the tank wagon is held at a pressure of 2.0 to 2.5 kg/cm². above the storage tank pressure, the tank wagon should be emptied into the storage vessel.

DRAFT 2

19. When the tank wagon is emptied of all liquid, stop the compressor and close the liquid valves beginning at the storage tank and progressing to the tank wagon.
20. If the facilities are so arranged that vapours may be removed from the tank wagon, the pipeline at the compressor should be arranged so that the compressor will draw vapour from the tank wagon and force it into the storage tank.
21. In this operation, the vapour should be discharged below the surface of the liquid in the storage tank to hasten the liquefaction and, in turn, help prevent excessive pressure in the storage tank.
22. Restart the compressor and when the tank wagon pressure is reduced to about 1.5 - 2.0 kg/cm². stop the compressor and close all the valves in the vapour line.
23. After bleeding off the pressure in the arms or hoses, disconnect both the liquid and vapour lines. Replace all the plugs in the tank wagon valves and the unloading fittings.
24. Recheck the sample valve, gauging device and thermometer well to determine that they have been returned to their original condition and are closed tight. Lower the dome cover carefully and lock it in place with the locking pin or secure it by appropriate means.
25. Remove earthing and bonding connections.
26. Any defect observed in the tank wagon should be noted on the appropriate forms and routed in accordance with acceptable procedure.
27. Notify the railways in writing about the release of the wagon and ensure that it is removed from the siding promptly.
28. Gauge the storage tanks within the installation, which have received the LPG, to determine that the liquid level is appropriate.
29. In the event of LPG received on a weight basis, the tank wagon may require weighing following the completion of the unloading operation.

7.3.3 DEGASSING OF SICK/ LEAKY TANK WAGONS

In the degassing system recommended for LPG wagons, the vacuum cycle purging technique or water displacement technique should be followed as outlined below.

a) Vacuum cycle purging technique:

i) The vessel should be evacuated repeatedly by a vacuum pump.
ii) Vapour should be discharged through high high-rise vent, keeping the steam on.
iii) Maximum possible vapours should be sucked out in first step depending on the capacity of vacuum pump/ compressor. As per RDSO, the vessel can also sustain a full vacuum also.
iv) Between the two vacuum cycles, nitrogen/inert gas should be used for breaking the vacuum.
v) The cycle should be repeated till the LPG concentration is found below the end point of LPG (for butane, it is 4% by volume when purging is done by nitrogen). Periodically, samples should be drawn from a suitable location and analysed for LPG concentration.
vi) Finally, the vessel should be flushed with air to displace the nitrogen. The barrel should be tested finally with an explosive meter to ensure that the hydrocarbon concentration is below 10% of LEL.

b) Water displacement Technique:

DRAFT 2

i) Personal protective equipment, e.g., Hand Gloves, Safety Goggles, Safety Shoes, Safety Helmet, Cotton Coverall etc., as defined in the respective entity's PPE policy, shall be worn by all the operating crew members.
ii) Check the Wagon for liquid LPG. In case Wagon has liquid LPG, unload the same by starting the LPG compressor.
iii) During the degassing operation on a spur, all loading or unloading operations on that spur shall be stopped.
iv) Earthing to be ensured on the degassing bay before commencing the degassing.
v) Connect the Vapour line or sampling valve of the tank wagon with the vent header and depressurize the wagon in a controlled manner under constant supervision by venting out LPG vapour.
vi) Confirm the vessel is depressurized with the help of the pressure gauge on the tank wagon.
vii) The plugs of the liquid and vapour valves should be unscrewed, and then the valves to be opened.
viii) The method adopted to make the wagon free from LPG:

a) Fill the wagon completely with water.
b) Allow to overflow.
c) Keep the tank wagon for one hour in a water-filled condition to remove the LPG presence in the vessel.
d) Alternatively, steam or inert gas can be used for degassing.

ix) Remove the manhole cover. The heating of studs/bolts shall not be adopted as a method for opening studs/ bolts. If required, rust remover should be used.
x) Check thoroughly for the presence of LPG vapour inside with an explosive meter.
xi) Repeat the procedure in point 8 above if the meter shows the presence of vapour inside.
xii) Ensure that the wagon vessel is dry after degassing in case water or steam purging is adopted.
xiii) Now the vessel is degassed completely and ready for any testing/ repair.
xiv) A degassing certificate is to be issued.
xv) A safety instruction checklist should be exhibited at the degassing station, containing all relevant points in the degassing procedure.
xvi) Degassing operation should be discontinued during a thunderstorm or lightning.
xvii) All valves, manhole covers and hoses shall be disconnected.
xviii) Sludge from the LPG wagon should be shifted from the siding in a wet condition and shall be disposed of in a safe manner.
xix) Suitable systems should be used for the reuse of water used for degassing. Like connecting a flexible hose from a wagon into which vapour is being filled to the TW liquid line of the next wagon, which is to be degassed. This enables water from the purged wagon to be utilized for the purging of the next wagon.

7.3.4 PURGING OF TANK WAGON

In the Purging system recommended for LPG Tank wagon, below technique should be followed as outlined below:

a) Using water and LPG Vapour

Procedure:

1. Take the tare weight of each wagon on the In-motion weighbridge and record the same.

DRAFT 2

2. Loco engine is to be moved out and exhibit a caution sign at a suitable distance away from the wagons on both ends.
3. Switch off the loco engine if parked nearby.
4. Connect two earthing clips to each tank wagon.
5. Lower the flapper bridge (swing platform) slowly on the wagon.
6. Open the cover of the wagon valve platform.
7. Connect the TW liquid line with a 2'' flexible hose to the service water line and fill water up to 100%. Let the water overflow from the vapour valve.
8. Once water is filled, connect the TW vapour line with a 2'' vapour line to give vapour up to 50 PSI.
9. Purging to be started by opening the valve on the 2'' vapour line. Water transfer is to be monitored continuously.
10. Once water is drained completely, disconnect the flexible hose. Close 2'' vapour line, TW liquid line and TW vapour line.
11. After purging, the TW vessel vapour pressure must not be less than 1.5 kg/cm².
12. Close the top cover of the wagon and seal it properly. Remove earthing connections. Lift up the swing platform.
13. After completion of the process, TW is to be pulled out from the TW gantry, and the gross weight of each wagon is to be done on the In-motion weighbridge.
14. Purging certificate to be issued.
15. Suitable systems should be used for the reuse of water used for purging, like connecting a flexible hose from a wagon into which vapour is being filled to the TW liquid line of the next wagon, which is to be purged. This enables water from the purged wagon to be utilized for the purging of the next wagon.

b) Using a Vacuum Pump and LPG Vapour

1. Procedure as outlined in clause 7.3.4 (a) shall be followed**.**
2. Connect the vapour line of the Tank wagon with the vacuum pump
3. Start the vacuum pump up to 0.35 kg/cm² absolute to remove the air from the tank wagon
4. Connect the LPG Line to introduce LPG vapour into the tank wagon.
5. Ensure the LPG Vapour pressure in the tank wagon to 1.5 to 2 kg/cm².
6. Issue the certificate for having purged the tank wagon.

7.4 BULK HANDLING FOR MOVEMENT BY PIPELINE TRANSFER

For the interface between the pipeline operator and consumer terminal, compliance with all the relevant provisions of OISD-STD-214 shall be ensured.

The battery limit of the pipeline receipt line between the supplier and receiver shall be identified and documented, and to be shown in SCADA.

7.5 BULK HANDLING FOR MOVEMENT BY SEA

Transportation of LPG in bulk by ocean tankers may be carried out under fully pressurised, semi-pressurised (semi-refrigerated) or fully refrigerated at atmospheric pressure conditions.

7.5.1 CARGO CARRIER DESIGN & CONSTRUCTION

The design and construction of the cargo carrier shall be in line with the International Code of the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code).

DRAFT 2

7.5.2 TANKER FACILITIES

The cargo handling equipment in an ocean tanker usually comprises of Cargo pumps (submersible, deep well and booster), Compressors, Condensers, Heat Exchangers, Vapourisers, Cargo heaters, etc.

The deep well pump supplies liquid to the booster pump to send the product ashore. In a pressurised ship, liquid is withdrawn by pressurising the tank through vapour compression from other tanks. All semi and fully refrigerated tankers are provided with cargo heaters to enable the vessels to discharge into pressure storage ashore and a booster pump if the discharge pressure is significantly above 9 bars. Each cargo tank is provided with the following equipment:

a) Two cargo pumps, one each on either side of the longitudinal bulkhead.
b) Liquid discharge line from the tank dome, connected to the main liquid line.
c) All emergency pump trunk way.
d) A liquid loading line connected to the main liquid line.
e) Two liquid level indicating devices, one on each side. These usually consist of a float attached to a self-winding tape that moves up and down, either on guide wires or inside a guide tube. The liquid level is read off the tape through a gas-tight window at the top of and outside the tank.
f) Two sets of purge lines at the top and bottom of the tank. These are used to distribute inert gas or vapour for gas-freeing or gassing-up of the cargo tanks.
g) A vapour line for withdrawal of vapour to the compressor.
h) Sample tubes.

Safety Devices

a) At least two safety valves in each cargo tank
b) High and Low level alarms.
c) Overfill alarm. When actuated, this will shuttle the main loading valve and sound an alarm.

However, the design of the tankers with regard to provision of facilities, equipment, accessories and safety features must be in accordance with the requirements of the International Code of the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk

An ESD pendant is to be provided to the terminal from the ship for emergency activation. It is recommended to have a system where the Terminal presents the ships with a pendant by means of which the ship may actuate the terminal's ESD, or the discharging ship may provide the jetty with a pendant by means of which the ship's ESD may be actuated from the shore.

7.5.3 TRANSFER PIPING

The following facilities for transfer piping between the Terminal and Shore tanks should be provided:

a) ROVs at both ends of the transfer line. (Additional ROV at critical locations in the pipeline route.)
b) Relief valves on the liquid line to the surge vessel with an audible alarm system connected to a high safe venting system.
c) Physical protection against impact to vent/ drainpipes
d) Protection of the pipeline against corrosion, particularly when the pipeline passes under a public roadway, which is likely to be waterlogged.

Pipe work passing a public roadway should be designed to the acceptable public highway authority standard in terms of roadway axle weights.

DRAFT 2

7.5.4 TERMINAL FACILITIES

Transfer of LPG from the ship to the terminal and vice versa is accomplished using loading arms of approved type as per OCIMF guidelines. Provision of vapour return facility shall be made.

In case of pressurised gas transfer, it is observed that the pumping rate falls off gradually due to the back pressure of the shore tank. A suitable vapour recovery system or re-liquefaction of the gas from the receiving tank shall be provided.

7.5.5 UNLOADING OPERATIONS

It is essential that the ship and terminal operators are familiar with the basic characteristics of each other's facilities, are aware of the precise division of responsibilities and are able to communicate effectively during the time they are together involved in the joint operation of cargo handling.

7.5.6 COMMUNICATION

Reliable and effective communications, irrespective of whether these are directly between the tanker and the terminal or indirectly via a third party, are prerequisites of safe and efficient cargo operations before the tanker comes alongside and during the period of cargo operations and until the tanker departs. Terminal communication shall be compatible with the tanker's system.

7.5.7 PRE-CARGO TRANSFER DISCUSSIONS

Before any cargo transfer operation, it is imperative that the intended procedures are thoroughly discussed between the responsible personnel from the tanker and the terminal with a view to make both sides fully conversant with the characteristics of the tanker and shore cargo handling systems, the envisaged operational and safety procedures and requirements and the parameters to be adhered to during the transfer.

The broad outlines for the discussions are as under:

a) The names and roles of terminal and ship personnel who will be responsible for cargo transfer operations may be noted.
b) The terminal representatives shall check that pre-arrival instructions to the ship on cargo, cargo disposition, and cargo conditioning have been carried out. They shall also check that all necessary tanker equipment inspections and tests have been performed.
c) Terminal representatives and customs and/or independent surveyors, where necessary, will be informed of the cargo tank data, e.g. temperature, pressure, whether free of cargo, liquid heel or arrival dip, composition of tank vapour and quantity of cargo on board

7.5.8 SHIP / SHORE SAFETY CHECKLIST

Checklist for Ship/Shore safety is given in Annexure-7-III.

7.5.9 DISCONNECTION OF HOSE/ UNLOADING ARM

On completion of the unloading operation, the tanker discharge pipeline needs to be purged to push liquid LPG to the receiving tank. Purging may be done by LPG vapour, inert gas or nitrogen as per the design of the system. However, before disconnecting hoses or unloading arm, it is to be ensured that there is no liquid LPG left between the tanker main valve and the shore isolation valve. Product (LPG vapour, inert gas or nitrogen) from this length of pipeline is to be safely vented.

Unloading Arms shall have with Quick Release Coupling with suitable design. In the event of any emergency Quick Release Coupling shall dis-engage from Cargo Tanker. Yearly Inspection/Testing of Quick Release Coupling shall be ensured and records to be maintained.

DRAFT 2

7.6 STORAGE OF BULK LPG

Precautions for storage of LPG in bulk:

a) The liquid level in the vessel shall not exceed the permissible limits. Filling shall not exceed the safe permissible filling ratio.
b) The vessel shall be inspected daily for any LPG leakage and corrective action taken.
c) All gauges/indicators, viz. High-level alarm, measuring gauges/ indicators, pressure gauges/ indicators, temperature gauges/ indicators shall be kept in operating condition at all times.
d) High level alarms of storage vessels shall be physically checked by raising the bulk level up to 85% capacity of vessels once a year under the supervision of an authorised person of the entity.
e) Daily recording of both primary and secondary level gauges/indicators shall be done and ensure that readings of both the gauges/ indicators should match at all times.

7.7 SAMPLING AND DRAINING

7.7.1 SAMPLING

Sampling is primarily required at supply locations, i.e. Refineries, Gas Processing Plants, Import Terminals. Sampling of LPG for different analytical tests shall be carried out in accordance with IS:1448. During the handling of LPG samples, the following safety precautions shall be followed:

i) Special sampling bombs shall be used for LPG sampling.
ii) Sampling bombs shall be inspected periodically and tested hydrostatically yearly. A record of such inspection/test shall be maintained.
iii) Sample bombs shall be earthed effectively before and during the discharge of the sample.
iv) LPG samples shall be collected only in open/ventilated areas, and the person taking the sample shall stand on the windward side.
v) The sampler shall wear personal protective equipment, e.g. Safety goggles, low temperature hand gloves, safety shoes, safety helmets, cotton coverall, etc. as defined in the respective entity's PPE policy while taking samples.
vi) Samples, after collection, shall be properly labelled and kept at the designated place.
vii) Sampling of LPG should preferably be done in bombs with provision of pre-charge at one end. This will not only eliminate the possibility of undesirable presence of air in the bomb, which may become a safety hazard, but also minimise flashing of the sample during collection, thereby providing a truly representative sample.

7.7.2 DRAINING

The storage vessel should be drained regularly for its water content, sediments, caustic contents, etc. Draining operation should be carried out under the supervision of an authorised person. Safety instructions for the water draining operation shall be displayed prominently.

For draining of water from LPG vessels, the following procedure shall be followed:

i) The valve nearer to the vessel shall be opened first and closed. Then, draining shall be carried out by gradually opening the valve farther from the vessel. The operation shall be repeated till LPG release starts.
ii) The point at which LPG is exposed in the open for the first time during draining shall be at least 3 m away from the shadow of the LPG storage vessels. They shall also be outside the shadow of the associated pipeline manifold. The individual discharge end of the water draw off piping shall be visible and shall not directly discharge into a closed drainage system.

DRAFT 2

iii) After the completion of the draining operation, there shall be no LPG left in the pipeline after the first valve (nearest to the vessel) in the drain line.

It is important not to attempt the removal of the entire quantity of water, as considerable liquid LPG may escape in the process. The escape of LPG may lead to the formation of an ice block at the valve seat, preventing closure of the valve. This will result in the escape of uncontrolled vapour to the atmosphere, causing a very serious hazard.

The valve shall be operated slowly to avoid ice formation.

8.0 BOTTLING OPERATIONS

8.1 GENERAL

This section covers the operations in the LPG Bottling Plants.

a) Standard operating procedure (SOP) shall be established to ensure safe operation, maintenance, and modification activities while carrying out various jobs. These practices shall be documented and approved. The SOP shall be updated once in three years in line with OISD-GDN-206.
b) Activity on the bottling line, including valve change, shall be carried out by trained personnel.
c) Entry of daily activities shall be recorded in the logbook of the respective area where work is being carried out. Logbooks shall be reviewed and countersigned by the role holder officer on a daily basis. Standard formats for operation/ maintenance/ security activities shall be issued by the Head Office. Formats used and maintained by the location for the above activities shall be in line with standard formats issued by the Head Office.

8.2 BOTTLING OPERATIONS

The Bottling Operations broadly classified under the following heads:

i) Receipt of LPG cylinders
ii) Filling and dispatch of LPG Cylinders
iii) Handling and storage of LPG cylinders
iv) Auxiliary Operations

8.2.1 RECEIPT OF LPG CYLINDERS

a) The plant shall receive and dispatch cylinders of various capacities through packed trucks. Facilities provided for the positioning of packed trucks shall be 15 m away from the LPG shed for the receipt/ loading of cylinders.
b) Cylinders shall be received in capped condition. Cylinder shall conform to IS:3196.
c) Check for the following in a packed truck as per statutory regulations before accepting it for unloading/ loading of the cylinder:
    i) A properly fitted spark arrestor of the make and design approved by CCE or any spark elimination system duly approved by CCE.
    ii) 2 nos. 9 kg DCP fire extinguishers are in an easily accessible and removable position with Truck No., the last date of checking and charging painted on it.
    iii) Air brakes are not bypassed.
    iv) Anti-Lock braking system (ABS) in working condition
    v) The fuel tank is protected by means of a stout guard, and the fuel tank cap is locked.
d) In addition to the above, Plant shall carry out checking of the safety fitting of the packed truck on a quarterly basis as per Annexure-8-I. The plant shall maintain a record of deviation and subsequent corrective action.
e) VTS shall be available in the vehicle as per AIS140 and any other statutory regulations as applicable

DRAFT 2

f) Handling of 425 kg cylinders inside the LPG sheds shall be carried out using suitable equipment like hand-operated hydraulic pallet trolleys, EOT, etc.
g) Loading and unloading operations of 425 kg cylinders shall be done using suitable equipment like manual hydraulic scissor lifts, etc.
h) SOP for soap solution topping up/ replacement in the chain conveyor soap tray work to be displayed in each shed, and adherence to established procedures to be ensured.

8.2.2 FILLING OF LPG CYLINDERS

8.2.2.1 PRE-FILLING OPERATION

Cylinder before filling shall be checked for the following, and if found faulty, it shall be segregated:

i) Visually inspected as per IS:15966
ii) Spurious cylinders -Identification of spurious cylinders:
  a) Cylinders having less or more number of stay plates.
  b) The width of the stay plates are not as per OMC's specification.
  c) Cylinder having abnormal height.
  d) New cylinders fitted with old valves.
  e) Cylinders with improper colour codes.
  f) Cylinders having more than one circumferential weld.
  g) Cylinders having vertical welds.
  h) Cylinders with no stampings on the stay plate/bung.
  i) Cylinders with non-standard foot ring.
iii) Cylinders "Due for Pressure Testing" (DPT)
iv) Hot Repairable cylinders (HR)
v) Other Marketing Company (OMC) Cylinders
vi) Cylinder having burn marks.
vii) Bulged cylinders
viii) Cylinder having a dent / sharp cut.
ix) Cylinders having pitting / corrosion / crack / lamination / dig.
x) "O"-ring (joint packing) is present, and there is no apparent damage to the valve.
xi) Tare weight is legible.

8.2.2.2 FILLING OPERATION

a) Purged capped cylinder shall pass through the washing machine and drying unit for external cleaning.

b) De-capping of the cylinder should be carried out only after the cylinders have been passed through the washing and drying unit.

c) Purged Cylinders of all capacities shall be filled on a gross weight basis on a filling machine (electronic carousel or unit filling machine).

d) Negative and positive tare weight settings shall be maintained in electronic carousels to avoid filling of cylinders having variation in actual tare weight beyond the set limit. Ensure positive and negative tare weight settings and SOP for its handling are displayed at the carousel. Test cylinders for checking the setting of positive and negative tare weight at the carousel shall be available.

e) Maximum permissible errors on net quantity declared by weight shall be in accordance with the Legal Metrology (Packaged Commodities) Rules 2011.

DRAFT 2

f) Suitable provision by the means of a differential pressure valve or any other equivalent arrangement shall be provided to ensure filling pressure not to exceed the maximum working pressure of the cylinder, i.e., 16.87 kg/cm² at Carousel/ Unit Filling Machine (UFM) pipeline manifold.

8.2.2.3 POST FILLING OPERATION

After LPG filling, the cylinder of all capacities shall undergo below operation:

8.2.2.3.1 COUNTER CHECK OF WEIGHT

Cylinders of all capacities shall be counterweighed with a check scale. Variation of net weight shall not exceed statutory limits.

8.2.2.3.2 WEIGHT CORRECTION OF CYLINDER

A weight correction facility shall be available for cylinders of all capacities. All under-filled/ over-filled cylinders shall be corrected before dispatch.

8.2.2.3.3 CHECKING OF VALVE LEAK OF CYLINDER

a) Cylinders of all capacities shall be checked for valve leaks by means of an electronic leak detector or Compact Valve Tester (CVT).

b) Online electronic leak detectors to check the valve leak shall be provided for the cylinders filled through the carousel. The valve of the cylinder shall be replaced for a valve leak cylinder.

c) Where Valve leak cylinders are corrected using the Valve Change Without Evacuation machine, the cylinder shall be checked again through a suitable check weight scale and leak detector mechanism, like an electronic leak detector or equivalent mechanism, to ensure 100% reverification of corrected cylinders. The weight of these cylinders shall also be corrected to ensure the correctness of the quantity of LPG in the cylinder.

d) Valve Change Without Evacuation machine, shall be checked for containment integrity with regard to proper sealing. The machine shall be further checked every day with respect to torque settings before being put in use. The torque shall not exceed 20 +/-2 kg.m. The maintenance of the Valve Change Without Evacuation machine shall be carried out as per OEM recommendations.

e) A Compact Valve Tester (CVT) to check valve leaks should be installed for other capacity cylinders that are filled through the stationary scale.

f) CVT or electronic leak detector (valve leak) shall be calibrated to detect valve leakage beyond 0.5 gm/hr.

8.2.2.3.4 CHECKING OF "O"-RING LEAK OF CYLINDER

a) Cylinders of all capacities shall be checked for "O"-ring leak by means of an electronic leak detector or Compact Valve Tester (CVT).

b) Online electronic leak detectors to check "O"-ring leaks shall be provided for the cylinders filled through the carousel.

c) A new "O"-ring shall be replaced for the "O"-ring leak cylinder. These "O"-ring leak corrected cylinders shall be checked again through the main online electronic leak detector.

d) Compact Valve tester (CVT) to check the "O"-ring is installed for other capacity cylinders, which are filled through a stationary scale.

DRAFT 2

e) CVT or electronic leak detector ("O"-ring leak) shall be calibrated to detect leakage.

8.2.2.3.5 VERIFICATION & CALIBRATION OF VALVE & "O"-RING DETECTOR

a) Verification of "O"-ring and Valve leak detectors (Manual or Electronic) shall be done a minimum of twice in each shift with a test cylinder.

b) Before each verification, the test cylinder shall be checked and verified with an approved manometer in line with the procedure given by the OEM.

c) Calibration of "O"-ring and Valve leak detector (Manual or Electronic) shall be done minimum once in a week or whenever detectors fail during the above verification in line with the procedure given by OEM.

d) Repeatability of the electronic "O"-ring and Valve leak detector shall be carried out in line with the procedure given by the OEM.

8.2.2.3.6 CAPPING OF CYLINDER

Cylinders of all capacities shall be capped.

8.2.2.3.7 BODY & BUNG LEAK CHECK OF CYLINDER

a) Cylinders of all capacities shall be checked for body leak and bung leak by dipping in water test bath or by an established electronic leak detection system.

b) In case a water test bath is used, it is to be ensured that the water in the test Bath is clean and free of dirt and mud, and the water level in the test bath is above the cylinder bung and below the safety cap of the valve. All the cylinders passing through the test bath shall be visually inspected as per SOP for any leak, and the test bath shall never be unattended. The cylinder shall be capped prior to dipping in the test bath water.

c) If the cylinder has a body leak, then, cylinder shall be sent for evacuation and degassing. If the cylinder has a bung leak, then the same shall be sent for evacuation and necessary further action, including checking the condition of the Bung.

8.2.2.3.8 SEALING OF CYLINDER

Only those cylinders found passing all the above checks shall be processed further for sealing. Seal shall have an identification mark of the oil marketing company. The cylinder shall be sealed using an online or portable hot air sealing machine.

8.2.2.3.9 EVACUATION OF CYLINDER

a) Any defective cylinder, such as body leak, bung leak, valve leak (which is not being handled by Valve changing without evacuation machine), etc., shall be send for evacuation before further handling.

b) An evacuation facility shall be provided to handle all types and capacities of cylinders being filled at the location.

8.2.2.3.10 No Bypassing of interlocks, auto fire water protection system, gas detection system, vapour extraction system shall be permitted. In case same is unavoidable, bypass management in line with OISD-GDN-206 shall be followed for interlock/ safety critical equipment.

DRAFT 2

8.2.3 HANDLING & STORAGE OF LPG CYLINDERS

Handling and storage of LPG cylinders shall be governed by the Gas Cylinders Rules, 2016. The following safe practices shall be observed:

8.2.3.1 HANDLING OF LPG CYLINDERS

a) LPG cylinders shall not be dropped, rolled on the body and shall not be subjected to any violent contact with any other cylinder or object.

b) The cylinder shall be moved by rolling on its foot ring or on conveyors or hand trolleys.

c) Telescopic conveyors should be provided for the unloading/ loading of cylinders from/to the packed trucks at the unloading/ loading fingers of the shed. The telescopic boom shall have a provision to automatically stop the boom upon impact.

d) Rubber mats shall be provided at points where cylinders are being unloaded from the conveyor.

8.2.3.2 DEFECTIVE FILLED CYLINDERS

a) Any cylinder having a body leak, bung leak, bulge, fire ravaged, or spurious shall be evacuated immediately.

b) Any cylinder with a valve leak shall be immediately capped and thereafter evacuated or rectified immediately.

c) Cylinders requiring repairs other than valve leaks shall be degassed for repairs.

d) All above cylinders shall be clearly identified, with markers, according to their nature of defects and shall be kept capped during storage.

8.2.3.3 LPG CYLINDER STORAGE

a) All capacity cylinders shall be stored in the LPG shed and stacked vertically (with valve in upright position) in not more than 1.5 m height. The cylinders of capacity 14.2/19 kg shall be stacked in small lots of 4 rows and 25 cylinders in length or as approved by PESO in the layout drawing. A minimum access path of 1 m must be maintained on both sides of 4 rows and 2 m after every 5 such lots. After each length of 25 cylinders, there must be a passage of 2 m. The cylinder storage space shall be properly marked/ painted on the flooring. Typical stacking arrangement for 14.2 kg cylinder is given in Annexure-8-II. For the stacking of cylinder of other capacity such as 2/ 5/ 10/ 35/ 47.5/ 425/ composite type etc., marking size of 14.2/19 kg cylinder shall be used.

b) Cylinders of different capacities and types shall be stacked separately.

c) All cylinders shall be capped during storage.

d) Empty and filled cylinders shall be demarcated clearly.

e) Minimum distance of 15 m from the cylinder loading area and minimum distance of 10 m from the filling point shall be observed for all filled cylinder stacks.

f) Empty cylinders storage area shall be properly segregated from filling machines by 5 m (min.).

g) No purged LPG cylinder containing gas (liquid/ vapour) to be kept on the ground/ outside the shed.

DRAFT 2

8.2.4 AUXILIARY OPERATIONS

8.2.4.1 CYLINDER REQUIRING VALVE REMOVAL AND VALVE FIXING

A cylinder having a defective valve and with a bung leak shall have to undergo valve removal and valve refixing operations.

The following safe procedures shall be observed for the operation:

i. Cylinder shall be evacuated of its product up to a pressure, 1.5 kg/cm² or less, thereafter depressurised to near atmospheric pressure through a vent before opening the defective valve.

ii. The valve of cylinders with a bung/ valve leak shall be replaced.

iii. Cylinder bung shall be cleaned with a 3/4" NGT tap, and cylinder bung threads checked with L1 and L9 gauge.

iv. Only cylinders passing above tests shall be taken up for valve change operation.

v. The good valve shall be fixed using a torque wrench. The torque shall not exceed 20 +/- 2 kg · m. Before fixing the valve suitable jointing compound shall be used. Jointing compound shall be compatible with LPG, water and the metallurgy of cylinders and valves. Teflon tape/ paste of suitable valve joining material is recommended as a jointing material.

vi. For composite cylinder, refer to clause 8.2.6 of this standard. The valve tightening torque depends on the specific PESO approval given to the manufacturer.

8.2.4.2 PURGING OF CYLINDERS

a) New cylinders and cylinders degassed for repairs shall be purged of air prior to being taken up for refilling with LPG.

b) Following safe procedure shall be observed for purging of LPG cylinders:

i. Connect the cylinder, except composite cylinders, valve to a vacuum pump and evacuate air so that the pressure in the cylinder is brought down to a maximum of 0.35 kg/cm² absolute using vacuum based cut-off switch.

ii. Inject LPG vapour in the cylinder so that the pressure is a minimum of 1 kg/cm² and not above working pressure of the cylinder, i.e., 16.87 kgf/sq.cm at 65 °C. Timer based vapour insertion may be used.

c) A purging facility shall be provided to handle all types and capacities of cylinders being filled at the location. For purging of 2/ 5/ 35/ 47.5/ 425 kg capacity cylinders, a common manual vacuum connection may be provided along with a vapour insertion facility. The same facility may be used for different valve types (VOT/ LOT, etc) with the provision of a suitable filling adaptor.

d) For purging of composite cylinders, clause 8.2.6.4 of this standard shall be referred.

8.2.4.3 DEGASSING OF CYLINDERS

a) A degassing facility shall be provided to handle all types and capacities of cylinders being filled at the location.

b) Cylinders shall be evacuated and depressurised to near atmospheric pressure. The residual LPG in the cylinder should be vented in the atmosphere with vent pipe outside the shed at 1.5 m height above the eaves of the shed.

DRAFT 2

c) The valves shall be removed with a suitable torque wrench or by using online valve changing machines.

d) Cylinders shall be filled with water fully, and water should be allowed to overflow. The LPG displaced by the water shall be vented in the atmosphere with pipe outside the shed at 1.5 m height above the eaves of the shed or a height of 6 m from floor level, whichever is more.

e) Thereafter, cylinders shall be emptied of water and checked with an explosimeter for confirmation of degassing. In case the gas detector indicates the presence of LPG above 20 % of the Lower explosive limit, the process shall be repeated. Not more than ~~6~~ 5 cylinders shall be degassed at a time.

8.2.5 OPERATION AND HANDLING OF 425 kg CYLINDERS

8.2.5.1 Facility for 425 kg cylinders filling, prefilling and post filling checks and processes shall be provided. Filling and allied facilities shall be set up as per approved by PESO.

The following process shall be followed for pre-filling, filling, and post-filling operations of 425 kg cylinders:

a) All the cylinders shall be counterweighed using post filling check scales.
b) Leaks through the liquid valves and bung shall be checked using CVT or soap solution.
c) Vapour valves shall always be kept closed and provided with a blind plug. Leaks from the blind plug shall be checked using a soap solution.
d) The cylinder surface shall be visually checked for body leaks after filling.
e) Evacuation of defective cylinders shall be carried out using suitable equipment.
f) Handling of 425 kg cylinders inside the LPG sheds shall be carried out using suitable equipment like hand operated hydraulic pallet trolleys.
g) Loading and unloading operations shall be done using suitable equipment, like manual hydraulic scissor lifts.
h) Purging of 425 kg cylinders shall be carried out using suitable equipment.
i) Degassing of cylinders should be carried out by filling water through the liquid port and venting the vapour at a height through the vapour port. After the retention water shall be drained from the drain plug provided at the bottom of the cylinder.
j) Replacement of defective valves shall be carried out after the defective cylinders are evacuated and depressurized.

8.2.5.2 SRV Testing of 425 kg CYLINDERS

a) SRV installed on 425 kg cylinders shall be inspected on a yearly basis or as per respective state factory rules at prescribed pressure by a competent person.
b) The cylinder with the test due SRV will be depressurised.
c) The vapour will be vented (above roof level) safely using a vent arrangement with vent pipe outside the shed at 1.5 m height above the eaves of the shed.
d) The SRV on 425 kg cylinders is mounted on an adaptor that is fixed on the cylinder.
e) The adaptor has a bottom spring-loaded plate that will close the opening once the SRV is removed from the adaptor. The adaptor should not be removed, and only the SRV should be removed for testing.
f) After testing the SRV number, the test date and due date should be painted on each cylinder.
g) 425 kg Cylinder Design Pressure: 16.9 kg/cm² at 65 °C.
h) SRV Test Pressure shall be within 110% of the design pressure.

8.2.6 OPERATION AND HANDLING OF COMPOSITE CYLINDERS

8.2.6.1 RECEIPT OF NEW COMPOSITE CYLINDERS FROM MANUFACTURERS/REPAIRERS

DRAFT 2

a) Cylinders shall be transported by using pallets or with protective body cover, i.e. nylon mesh, etc., to protect cylinders from any damage and scratches to the outer body.
b) Name/logo of manufacturer stencilling/ embossing of OMC, test dates, due dates, tare weight, cylinder capacity, serial number, etc., to be checked and verified with respect to Purchase Order, approved PESO drawings, BIS test reports, etc., before unloading the lot. The existing quality check for receipt of new cylinders is to be followed as per the format given below.

TARE WEIGHT INSPECTION AND REPORTING											
Sr. No.	Delivery No. and Date	Category (New/ST / HR)	Manufacturer	Cylinder Sr. No.	Observed Tare Weight Data			Difference with Actual		Cylinders were found not to be meeting quality	Details of defects
					Actual	Punched	stencilled	Punched	Stencilled		

c) Stack the cylinders separately as per capacity and as per manufacturer wise at a designated place.

8.2.6.2 HANDLING & STACKING OF COMPOSITE CYLINDER

a) All cylinders shall be stacked vertically (with valve in upright position) as defined in clause 8.2.3.3
b) Composite Cylinders of different capacities shall not be stacked together.
c) Cylinders without cages shall not be stacked one above another.
d) Composite cylinders shall not be stacked along with steel cylinders. The composite cylinder shall be stacked in a separate, earmarked place.
e) New unpurged composite cylinders of the same capacity but of different manufacturers, without a cage, shall not be stacked together. However, cylinders of different manufacturers of the same capacity with a cage may be stacked together.
f) Cylinders of different capacities shall not be stacked together in the same delivery cage if the capacity of the cage is more than one cylinder.
g) Composite cylinders shall not be scratched, scraped, rolled, dropped, dragged, knocked, thrown or burnt during any stage of handling.
h) Movement of individual composite cylinders in the shed, if required, shall be done with a cage. It shall be done manually or by using a trolley made for cylinder movement.
i) Individual composite cylinder shall be provided with an SS cage for safe handling, transportation and carrying in the conveyor system.

8.2.6.3 WASHING UNIT FOR COMPOSITE CYLINDER

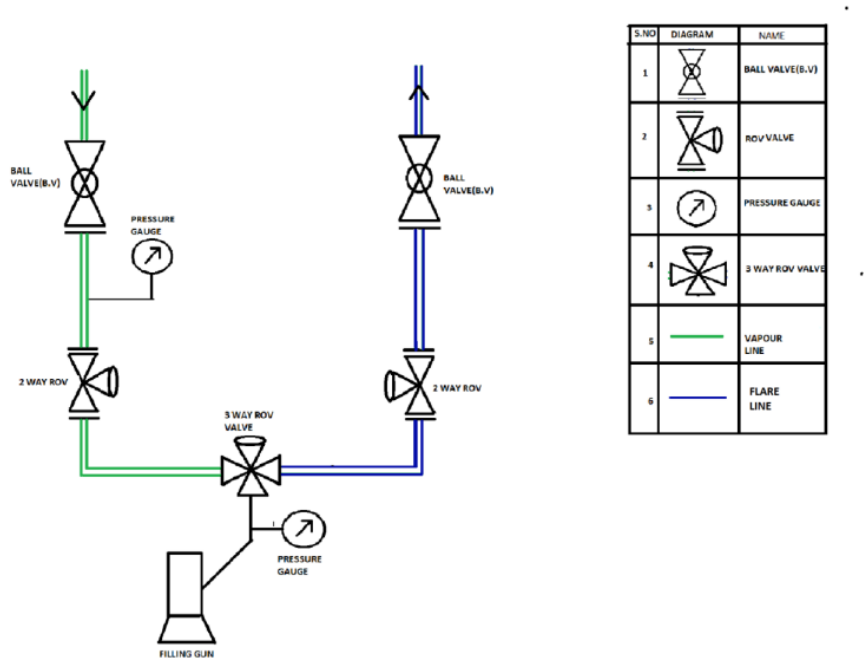
a) The composite cylinder surface shall be cleaned with water in the cylinder washing unit, both for manual and conveyorized filling.
b) When empty cylinders with individual cages are received from the distributor, the same shall be unloaded in the conveyor line and sent through the washing unit, before filling or stacking.
c) Ensure cylinders are capped condition while entering the washing unit.

8.2.6.4 PURGING OF COMPOSITE CYLINDER

a) The composite cylinders shall not be subjected to vacuum.

DRAFT 2

b) Depressurize the new cylinders using suitable adaptors and hoses with suitable vent arrangement in the atmosphere with vent pipe outside the shed at 1.5 m height above the eaves of the shed.
c) A purging facility shall be provided with one flare line and one vapour line.
d) Ensure that the pressure gauge reading shows zero-gauge pressure before disengaging the adaptors.
e) Ensure that the surface of the cylinder is made wet with a water spray. Water mist, consisting of water and pressurized air, can also be used for saving water.
f) Fill LPG vapour so that the pressure inside the cylinder is around 2.5 kg/cm^2.
g) Depressurize the cylinder using a suitable adaptor till the pressure gauge reading shows zero-gauge pressure.
h) Repeat the steps as per Sr. no. d), e), f) and g) continuously for five cycles. The cylinder will be purged with LPG vapour.
i) A typical drawing of the composite cylinder purging unit with connections is shown below:



8.2.6.5 FILLING OF COMPOSITE CYLINDER

a) Ensure that the surface of the cylinder is completely wet with water spray/water mist before the commencement of filling operations. This is to ensure proper dissipation of static charge generated or deposited before or during the filling of cylinders with LPG.
b) Ensure proper bonding and grounding of filling gun, filling station, conveyors and other connected equipment for filling of composite cylinders.
c) Ensure the vapour extraction unit and LPG pumps are ON. Also, ensure vapour extraction unit and filling unit shall be interlocked.
d) Position the cylinder in the filling unit and lift the platform of the machine. Movement of filling guns is to be controlled so as not to hit the composite cylinder while connecting for filling.
e) Set the tare weight of the composite cylinder and commence filling by pressing the start button on the filling unit panel (which will automatically start ROVs/ Gas stop valves).
f) Filling will stop automatically once the cylinder is filled completely as per the set value.

g) Avoid any kind of rubbing, striking, etc., while handling the composite cylinders to avoid the accumulation of static charges and damage to the cylinders.

8.2.6.6 POST FILLING CHECKS

a) Existing SOPs for all the post-filling checks of the unit filling machine, viz., manual Check weighing, Valve leak check through CVT, "O"-ring Leak Checking through CVT shall be followed.
b) Ensure that all the CVT guns, frames, conveyors, etc., are properly grounded for dissipation of static charges.
c) Cylinder shall be accepted if the net weight is within +/- 100 gms and +/-50 gms for 10 kg and 5 kg cylinders as per Legal Metrology (Packed Commodity) Rules, respectively, then sent for further quality check.
d) If the cylinder is underfilled then, then send it for correction in the filling machine and repeat the processes as mentioned in "Filling Operation of composite cylinders".
e) If the cylinder is overfilled, then, cylinder shall be sent for partial evacuation then repeat the process of underfilled cylinder.
f) Valve leak (VL) and "O"-ring leak (OL) shall be checked with the manual CVT. If it is VL, then the defective cylinder shall be sent for valve replacement. If it is OL, then the defective "O"-ring shall be replaced with a new "O"-ring and rechecked.
g) For Body and bung leak tests, the cylinder(s) shall be immersed in a water test bath as per the system available in the Plant, duly following all SOPs of handling of composite cylinders. Check for a bung and body leak of the cylinder.
h) If, cylinder has a body leak, then, cylinder shall be sent for evacuation and degassing. If the cylinder has a bung leak, then the same shall be sent for evacuation and necessary further action, including checking the condition of the Bung.

8.2.6.7 SEALING FACILITY OF COMPOSITE CYLINDER

All composite cylinders shall be sealed before dispatch.

8.2.6.8 EVACUATION OF COMPOSITE CYLINDER

a) The composite cylinders shall be clamped only with proper cushioning so that no undue /stress is exerted on the cylinder body. The surface area of cushioned clamps shall be twice that of the normal clamping, so that the stress is reduced considerably, and pressure exerted on the cylinder body shall not be more than 4-5 kg/cm^2. The outer sheath can be removed after discussion with the Manufacturer during the valve removal process to ensure proper grip of the cylinders.
b) Adopter/ facility for evacuation of pin-broken cylinder and single cylinder mechanical tilting mechanism to be used for evacuation of composite cylinders.
c) Composite cylinders shall be placed manually in a single cylinder tilting mechanism in an upright position and secured properly after ensuring proper earthing of the tilting mechanism. Tilting mechanism shall have a proper cushion for holding and resting of cylinder body so that any scratch on the body is avoided.
d) The adopter available for evacuating pin broken cylinders shall be connected to the valve of the composite cylinder and ensure no leaks after opening the valve of the adopter.
e) Tilt the tilting mechanism along with composite cylinders using appropriate levers/ manual mechanisms. In case a manual mechanism is used to tilt the cylinder, there shall be a provision for locking it in the desired position.
f) All the approved SOPs for the evacuation of steel cylinders shall be followed for evacuating composite cylinders.
g) The cylinder is made upright, and valves are closed after ensuring that all the liquid is evacuated.

DRAFT 2

h) The adopter with a hose is removed from the composite cylinder, and the same is transferred to the depressurizing manifold for bringing down the cylinder pressure to atmospheric pressure. The cylinder is now ready for further operations of replacing the SC valve/ degassing.

8.2.6.9 VALVE CHANGING OPERATION

a) Torque for tightening SC Valve to composite cylinders is different from regular/ normal LPG cylinders and is NOT 145+/- 15 lb ft.
b) Valve tightening torque of the composite cylinders shall be as approved by PESO for the individual manufacturer. The valve tightening torque shall be clearly marked on the cylinder/metal boss of the cylinder. Valve screwing and unscrewing shall be done manually. It is to be ensured to avoid rotation of cylinder, metal boss to be hold with special spanner as recommended by OEM before screwing/unscrewing of SC valve.
c) The valve tightening operation shall be carried out, and supervision of the floor officer and a record of each cylinder is maintained in a separate register as per the following format:

S. No.	Date	Serial Number of Cylinder	Name of Manufacturer	Capacity of cylinder (in Kgs)	Torque is mentioned on the Composite Cylinder	Torque used for tightening the SC Valve	Remarks

d) Hold the composite cylinder firmly in position using the special tool/ fixture on its metal boss, to avoid free rotation of HDPE liner inside the composite reinforcement during valve screwing/ unscrewing operations.
e) Clamping of the composite cylinders should be done using cushioned clamping pads having a clamping area twice that of the clamps used for normal cylinders and with a clamping force not more than 4-5 kg/cm², to avoid any damage to cylinders/ outer casing due to undue clamping load.

8.2.6.10 DEPRESSURIZING & DEGASSING OF COMPOSITE CYLINDERS

a) After evacuation, cylinders shall be depressurized using the existing depressurizing arrangement.
b) Valve removal shall be done as per the above-cited procedure.
~~c)~~ After valve removal, the cylinder shall be sent for degassing.

8.2.6.11 TRANSPORTATION OF COMPOSITE CYLINDER

a) Composite cylinders shall be handled and transported only in cages of appropriate size and dimensions, preferably in a single cylinder cage, so that unloading becomes easy.
b) Stacking of individual composite cylinders with cage shall be stacked up to 3 layers in the packed truck.
c) Outer casing of cylinders and delivery cages shall be checked thoroughly for breakage or damage before dispatch and on receipt.
~~d)~~ Cages and Cylinders found damaged shall be segregated and kept separately for further inspection and should not be dispatched.

8.2.6.12 RECEIPT OF COMPOSITE CYLINDERS FROM DISTRIBUTORS

a) Composite cylinders to be dispatched and received only with the delivery cages.
b) Delivery cages to be placed and moved to the designated place and stacked manually or by using a trolley.
c) Physically check the condition of delivery cages, composite cylinders for any damage.

DRAFT 2

d) Physically check each cylinder for its genuineness.
e) Physically check the composite cylinders for the following defects:
   i) Abrasion damage
   ii) Damage due to cuts on the surface of cylinders
   iii) Impact damage
   iv) Delamination
   v) Heat or fire damage
   vi) Structural damage
   vii) Chemical attack in terms of discoloration, change in surface structure, change in hardness, etc.
   viii) Identification markings other than those approved by PESO.
   ix) Plug or neck inserts.
   x) Permanent attachments
   xi) Neck (metal boss) threads
f) Any cylinders/ cages found damaged and unfit for further use shall be segregated separately.
g) Composite cylinders shall not be moved along with other normal cylinders on the conveyor.
h) Composite cylinders shall be stacked separately, capacity-wise wise at a designated place in delivery cages.

8.2.7 MERCAPTEN RECEIPT, STORAGE, DOSING & TESTING

8.2.7.1 SAFETY PRECAUTIONS

a) During loading/ unloading of mercaptan a full-face cartridge gas mask and Hazmat suit, safety shoes, and nitrile hand gloves shall be used.
b) Job to be done by standing in the upwind direction.
c) Ensure proper ventilation in the area to avoid inhalation of mercaptan fumes.
d) Be aware of emergency procedures in case of a spill or exposure.

8.2.7.2 UNLOADING OF MERCAPTAN FROM THE BULK TANKER

a) Unloading of bulk Mercaptan is to be done by the qualified technicians of the supplier.
b) Bulk tanker carrying mercaptan to be decanted using nitrogen.
c) The tanker is to be positioned in the designated area in the drive-out position.
d) Ensure placement of wheel chocks.
e) Connect the earthing clips to the Mercaptan storage vessel inside the tanker and ensure continuity to the Earth pit.
f) Position fire extinguishers near the tanker.
g) Ensure all the operators are equipped with appropriate PPE.
h) Ensure Mercaptan suppressing agents are kept handy.
i) Connect the liquid line of the tanker to the main receipt line of the bulk Mercaptan storage tanks through a Quick release coupling (QRC)
j) Connect the return line of the Mercaptan receiving tank to the tanker scrubber.
k) Connect the Nitrogen cylinder to the mercaptan tanker.
l) Crack open the Nitrogen line connected to the tanker and check for leakage.
m) Open the nitrogen valve to pressurize the tanker to the desired extent and open the main receipt and return line valves of the bulk Mercaptan storage tanks.

DRAFT 2

n) Note the initial level of the vessel under receipt and continuously monitor the levels till the transfer is completed.
o) Continuously keep spraying the suppressing agent near the joints and monitor the wind direction throughout the transfer process.
p) After completion of loading, safely disconnect the hoses by operating the QRCs, remove the earthing, take the final weighment and release the tanker.

8.2.7.3 UNLOADING OF MERCAPTAN FROM BARREL TO VESSEL

a) Activity to be done in the presence of the area in-charge/ shift in-charge.
b) Ethyl Mercaptan (C2H5SH) transferred from the Mercaptan drum to the Mercaptan storage vessel using a pneumatic/ hand pump. Ensure that the pneumatic pump is in working condition one day before unloading.
c) Bulk vessel pressure to be continuously monitored, and if required, flaring to be done.
d) Keep all the barrels for unloading near the unloading point area (Specified area for EM Barrels) with the help of a forklift. Open slowly the cap of the barrel.
e) Care should be taken because of the pressure inside the barrel.

8.2.7.4 MERCAPTAN DOSING

a) Ethyl mercaptan is C2H5SH or C2H6S
b) SYSTEM COMPONENTS:

- Storage Tank: Contains the mercaptan compound.
- Metering Pump: Controls the flow rate of mercaptan.
- Injection Points: Locations where mercaptan is injected into the gas stream.
- Monitoring Equipment: Ensures the correct dosage is being injected.

c) STARTUP PROCEDURE

- Inspection: Check all system components for any signs of damage or leaks.
- Calibration: Ensure the metering pump is calibrated correctly.
- System Check: Verify that all valves and connections are secure.
- Startup: Gradually start the system and monitor for any irregularities
- The mercaptan dosing pot has a calibrated level gauge for determining the quantity delivered for dosing.
- The dosing pot is connected to 2 nos. reciprocating dosing pumps, each with a variable flow rate up to a maximum of 10.5 litres/hour at full stroke length. The stroke length is adjustable to achieve the desired flow rate.
- For dosing operations, one pump is used while the other is kept as a standby.

d) ONLINE DOSING:

- Mercaptan dosing is done directly into the designated LPG pipeline.
- The mercaptan dosing pump is started, ensuring the dosing line is through up to the dosing point.
- Flow rate is controlled by adjusting the stroke length of the pump, depending upon the receipt flow rate and density of the product.
- Dosing of 20 ppm of Mercaptan is ascertained by checking the Mercaptan consumption against the LPG quantity transferred through the pipeline.
- On completion of LPG transfer through the line, the dosing pump is stopped.
- Storage Tank: Continuously monitor the storage tank level, and in case of the tank level goes down below the safe operating level, the tank is to be topped up in a closed environment.

DRAFT 2

- Monitoring: Continuously monitor the system to ensure proper operation.
- Adjustments: Make necessary adjustments to the flow rate based on monitoring data.
- Venting: Venting of Ethyl Mercaptan either through a pressure relieving device or through atmospheric vents must be resorted to through a suitable scrubber system before releasing to the atmosphere. The venting height shall be maintained for effective dispersion.
- Record Keeping: Maintain a log of all operations, including any adjustments made.

e) SHUTDOWN PROCEDURE

- Gradual Shutdown: Slowly reduce the flow rate and turn off the system.
- Inspection: Check for any leaks or issues that need to be addressed.
- Cleaning: Clean the system components as per the manufacturer's recommendations.

f) MAINTENANCE

- Regular Inspections: Conduct regular inspections to ensure the system is in good working condition.
- Component Replacement: Replace any worn or damaged components as needed.
- Calibration: Periodically recalibrate the metering pump to ensure accuracy.

8.2.7.5 EMERGENCY PROCEDURES:

a) Spill Response: Follow the spill response plan to contain and clean up any mercaptan spills.
b) Exposure Response: Provide immediate medical attention to anyone exposed to mercaptan.
c) Overfilling Response: Do transfer excess material into the intermediate and drain vessel if required. Ensure enough space is left for thermal expansion.
d) Fire Response: Follow the fire emergency response as per ERDMP.

8.2.8 QUALITY CONTROL

a) Daily Calibration of all the filling scales.
b) Calibration of Electronic OLD and VLD shall be checked every 2 hours in a shift by a test cylinder.
c) Calibration of the post-filling check scale using dead weight shall be checked twice in a shift.
d) Calibration of CVT shall be checked twice in a shift by a test cylinder.
e) SQC of a minimum of 2% of the total bottling shall be carried out.
f) SQC of Cylinder before despatch, SQC of STP/ New/ HR cylinder shall be ensured in line with entity's policy.

8.2.9 PERSONAL PROTECTIVE EQUIPMENT

Personal protective equipment shall be used as per OISD-STD-155. The entity shall create a PPE policy/ guideline in line with OISD-STD-155**.** Identification of Personal Protective Equipment (PPE) shall be based on a detailed Hazard Identification and Risk Assessment. The entity shall carry out hazard and risk assessment and develop the PPE policy/ guideline applicable to various types of installations/ facilities under its domain of operations. The requirement of the type of such PPE shall clearly be identified, documented, and implemented. Safety Shoes, Safety Helmet and Cotton Coverall shall be deemed as basic mandatory PPE for all personnel entering into the operational area of any installation in line with OISD-STD-155.

9.0 MAINTENANCE & INSPECTION OF EQUIPMENT

9.1 GENERAL

This section covers the maintenance and inspection practices to be followed to ensure safe and trouble-free operation of various equipment.

DRAFT 2

9.2

a) To facilitate the maintenance service to be rendered in a planned manner, a preventive maintenance schedule covering the necessary work to be done, mentioning the periodicity, i.e., daily, weekly, monthly, half yearly and yearly schedules, shall be worked out.

b) While preparing the maintenance schedule, basic recommendations given by the manufacturers shall be considered and modified bearing in mind the criticality of use/ local conditions.

c) Records about the preventative schedule vis-a–vis actual work done shall be maintained either in soft or hard form. The record formats as attached at Annexure-9-I to Annexure-9-III are suggested to ensure proper documentation.

d) History cards for critical equipment, i.e., pressure vessels/ Fire Pumps/ LPG Pumps/ LPG Compressors/ Air Compressors/ Diesel Generator/ Carousel/ Electronic Leak detectors/ VCWE/ Vision Unit, etc., shall be maintained. The history card should contain at least Equipment Tag No., Equipment Description, location of equipment, Capacity, Make and Model, Manufacturing year, date of Commissioning, Cost of equipment, breakdown period, reasons for breakdown, description of major repair/replacement/modification, cost of repair, date of decommissioning. The history card being used for different types of equipment shall be standardised across similar installations.

e) Plant to also maintain breakdown analysis of major equipment such as LPG Pump, LPG Compressor, Carousel, Diesel Generator, Fire Pump, Jockey Pump, etc.

f) All AMC reports/ test certificates issued in-house/ by third party shall be reviewed and signed by the Location-in-Charge or his representative, and corrective action on any recommendation shall be ensured.

Inspection frequency shall be followed as per Annexure-9-IV.

9.3 SAFETY CRITICAL EQUIPMENT

Safety Critical Equipment are the equipment, instrumentation, controls, or systems (including computer software) etc. whose malfunction or failure would cause or contribute substantially to a major incident or whose proper operation is required to mitigate the consequences of a major incident. Entity to identify list of safety critical equipment. The status of safety critical equipment is to be regularly monitored to ensure their availability all the time. Exception reports for non-availability of safety critical equipment should be available to designated persons and alternate risk mitigation plans should be in place. OISD-GDN-206 shall be referred.

9.4 WORK PERMIT SYSTEM

An essential part of controlling workplace hazards is an effective implementation of Work permit System to prevent incidents, provide safe working conditions and execute the work safely.

The primary purpose of the work permit system is to ensure that work is carried out in the safest possible manner to prevent injuries to personnel, protect property/ environment from damage and avoid fire/ explosion.

The entity shall define the list of jobs exempted from the work permit system. Such activities involving routine production and process operations. The entity shall develop and implement procedures for executing such jobs safely.

The work permit is required for jobs like various construction activities, inspection, repair and maintenance work, entry into confined spaces, work at height, excavation, pressure testing of

DRAFT 2

pipelines, use of electrical equipment that is not recommended for hazardous areas, and entry/ operation of vehicles that is not recommended for hazardous areas.

For the work permit system, OISD-STD-105 shall be referred.

9.5 INSPECTION & MAINTENANCE OF VARIOUS FACILITIES IN LPG INSTALLATIONS

9.5.1 STATIC EQUIPMENT

9.5.1.1 PRESSURE VESSELS & STORAGE FACILITIES

OISD Standard OISD-STD-128 on "Inspection of Unfired Pressure Vessels" shall be followed.

9.5.1.2 SHUT-OFF VALVES, CONTROL VALVES, NON-RETURN VALVES, EXCESS CHECK VALVES, PIPELINES, VALVES AND FITTINGS:

OISD Standard OISD-STD-130 on "Inspection of Piping Systems" shall be followed.

9.5.2 FLANGES, GASKETS & BOLTS

OISD Standard OISD-STD-130 on "Inspection of piping systems" shall be followed.

9.5.3 PIPELINES

i) Inspection data as well as thickness data of newly constructed pipelines shall be collected at the earliest and within two years of their commissioning to function as a base for the establishment of corrosion rates.

ii) External on-stream inspection of the piping system shall be carried out during an in-service condition once in three years. External On-stream inspection to monitor deterioration shall be visual and instrument-aided (ultrasonic, radiographic, thermographic), subject to feasibility as permitted by process parameters (fluid, pressure and temperature). Any abnormalities noticed during on-stream inspection shall be investigated and corrective steps initiated at the earliest. External on-stream inspection shall comprise:

   a) Visual Inspection
   b) Ultrasonic thickness survey

For carrying out the above inspections, the respective clauses of OISD-STD-130 shall be referred to.

The thickness survey shall be maintained on an isometric sketch, and a report shall be prepared in line with annexure-V of OISD-STD-130.

The results of the thickness test are to be compared with the original thickness and the last tested thickness. Retiring thickness shall be documented.

iii) Comprehensive inspection of LPG pipelines shall be carried out in line with OISD-STD-130 and other statutory regulations during shutdown/ breakdown/ idle time or as per schedule.

Comprehensive inspection of the piping system comprises external on-stream inspections as covered above. In addition, hydro testing shall be carried out during a comprehensive inspection under the following conditions, as applicable:

a) After any major alteration/ re-rating/ replacement
b) Inaccessible piping and complicated manifold
c) System after chemical cleaning
d) When the piping is out of service for more than six months.

DRAFT 2

For the hydro testing procedure respective clause of OISD-STD-130 shall be referred to.

iv) Piping in coastal installations and in corrosive environments shall be visually inspected once a year and records maintained thereof.

v) For inspection of all underground piping with coating and wrapping with/ without cathodic protection, respective clause of OISD-STD-130 shall be referred to. Long-range ultrasonic guided wave technology should be deployed to assess metal loss in inaccessible locations (e.g., pipeline under culvert) not covered under conventional ultrasonic thickness measurement.

vi) The loading/ unloading arms shall be tested along with the LPG pipelines at the same frequency.

vii) The breakaway coupling (BAC) of anyone (selected at random) loading/unloading arms shall be tested during annual internal safety audits. If the BAC fails during the test, all the BACs of all loading and unloading arms shall be tested. The BAC shall be tested for decoupling for movement of the unloading/ loading arm in forward and reverse directions during the above tests as per OEM recommendation. In the absence of OEM recommendation, testing has to be done once in six months and records to be maintained.

viii) The piping shall be supported at regular intervals not exceeding 6 m or as per design approved. For the indicative drawing of pipe support, Annexure-XV of OISD-STD-130 shall be referred to.

Provisions of OISD Standard, OISD-STD-130 on "Inspection of Piping Systems" shall be followed.

9.5.4 HOSE AND FLEXIBLE CONNECTION

LPG hoses shall be inspected and tested at a maximum interval of 4 months as detailed in OISD Standard, OISD-STD-135 on "Inspection of Loading and Unloading Hoses for Petroleum Products" and records maintained thereof.

LPG hoses shall be discarded after 3 years of service or earlier in case the permanent elongation exceeds 5% or in case of any defect.

9.5.5 STRAINERS AND FILTERS:

Strainers and filters shall be inspected and cleaned as per frequency mentioned in Annexure-9-IV, unless sluggish operation warrants earlier inspection.

9.5.6 SAFETY RELIEF VALVES

Safety Relief valves shall be tested once a year. Furthermore, an on-stream visual inspection should be carried out at least once every 6 months to check the following:

a) Gags, blinds do not exist.

b) Upstream and downstream isolation valves, if any, are opened and sealed.

c) Seals protecting the spring setting have not been broken.

d) The relief device is not leaking. This may be checked visually and by thermography or contact thermometers or by hand touch at the outlet nozzle, wherever practicable.

e) The continuous operation of heat tracing provision, if any, provided for low temperature application on valve and discharge piping.

f) Condition of insulation and cladding on the heat-traced piping and valves.

Provisions of OISD Standard, OISD-STD-132 on "Inspection of Pressure Relieving Devices" shall be followed.

DRAFT 2

9.5.7 ROTARY EQUIPMENT

9.5.7.1 COMPRESSORS

OEM recommendations for the maintenance of compressors shall be followed. Alternatively, OISD-STD-240 on "Asset Integrity Management of rotary equipment" shall be followed. In addition, periodic maintenance checks, as detailed in Annexure-9-I to be referred.

9.5.7.2 PUMPS

OEM recommendations for the maintenance of pumps shall be followed. Alternatively, OISD-STD-240 on "Asset Integrity Management of rotary equipment" shall be followed. In addition, periodic checks as detailed in Annexure-9-II are to be referred.

9.5.7.3 DIESEL ENGINES

OEM recommendations for the maintenance of diesel engines shall be followed. Alternatively, OISD-STD-240 on "Asset Integrity Management of rotary equipment" shall be followed.

9.5.7.4 Periodic vibration analysis shall be conducted for critical equipment like Fire engine, Air Compressor, LPG Compressor, LPG Pump, Jockey Pump and DG set, etc. Frequency of periodic vibration analysis shall be decided basis recommendation of OEM.

9.5.8 FILLING EQUIPMENT

It shall be ensured that the equipment is in good working condition to ensure quality output and also safe operation. Inspection checks, as advised by the OEM, shall be followed for the following equipment. Inspection checks are given at Annexure-9-III for the following equipment:

a) **CAROUSEL**
   i) Drive units
   ii) Centre column
   iii) Running Rail
   iv) Introduction and Ejection devices
   v) Filling machines

b) Check Weigh Scales

c) Chain/ Roller conveyor

d) Drive Units for Conveyors

e) CVT-leak detector

f) Cylinder Test Bath

g) Vision unit

h) Purging unit

9.5.9 INSTRUMENTATION

9.5.9.1 TEMPERATURE INDICATING DEVICES

Temperature indicating devices shall be checked daily for its proper functioning and shall be calibrated once in 6 months.

DRAFT 2

9.5.9.2 LEVEL GAUGING DEVICES

The gauging devices shall be checked daily for proper functioning.

9.5.9.3 PRESSURE GAUGES/ INDICATOR

a) Pressure gauges/ indicator shall be checked daily for its proper functioning and shall be calibrated once in 6 months.

b) Pressure Gauges/indicators installed in the cylinder hydro testing facility shall be calibrated at least once a week, and the Pressure Gauges/ indicators used for verification shall be tested once every three months.

c) All the Pressure Gauges/ indicators used for the pneumatic test of the cylinder shall be calibrated at least once in 3 weeks, and the Pressure Gauge/ indicator used for verification shall be tested once in every three months.

9.5.9.4 FLOW MEASURING DEVICES

All flow measuring devices shall be checked daily for proper functioning. Calibration shall be carried out as per OEM recommendation/ requirement of the Department of Legal Metrology.

9.5.9.5 Record of all such calibrations mentioned above, along with their master calibrator, shall be available.

9.5.10 FIRE FIGHTING EQUIPMENT

Firefighting equipment shall be inspected and tested as per. OISD-STD-142 - "Inspection of Fire Fighting Equipments and Systems".

Refer to Annexure-9-IV for inspection frequency.

9.5.11 ELECTRICAL EQUIPMENT

Proper functioning of electrical equipment can only be ensured by means of periodic preventive and predictive maintenance of the equipment. This enhances equipment life and also ensures the safety of the equipment, installation and operating personnel.

Maintenance shall be daily, weekly, quarterly or annual, depending upon the type of equipment.

Adequate logs shall be maintained to ensure that maintenance is carried out as per approved checklists. Preventive maintenance should be carried out as per the schedules laid down in OISD-STD-137.

Special emphasis shall be laid on the maintenance of equipment installed in hazardous areas.

All electric apparatus and wiring in a hazardous area shall at all times be so maintained as to retain the characteristic on which their approval has been granted.

Precautions to be taken for repairs and testing of flameproof equipment shall be as below:

i. No flame-proof or intrinsically safe apparatus shall be opened, and no work likely to impair the safety characteristics of such apparatus or electric wiring connected thereto shall be carried out until all electric supply has been cut off from said apparatus or wiring. The electric supply shall not be restored thereto until the work has been completed and the safety characteristics provided in connection with the apparatus and wiring have been fully restored.

ii. Use of soldering apparatus or other means involving flame, fire or heat or use of industrial type of apparatus in a zone “1” area shall be permitted for the purposes of effecting repairs and testing and alterations, provided that the area in which such apparatus or wiring has been installed, has first been

DRAFT 2

made safe and certified by a authorised/ agency after verifying that area is safe and free from inflammable vapours, gases or liquids and is maintained in such conditions, so long as the work is in progress.

iii. No alteration shall be made to any apparatus that might invalidate the certificate, or any other document relating to the safety of the apparatus.

Replacement fasteners, nuts, studs and bolts shall be of the type specified by the manufacturer for the particular apparatus. No attempt shall be made to replace or repair a glass in a flameproof enclosure, e.g. in a luminaire or other enclosures, except by replacement with the complete assembly or part obtainable from the manufacturer, complying with the approval certificate.

If replacement components such as cable glands, conduit or conduit accessories are available only with thread forms that differ from those provided on the apparatus, suitable adaptors having the necessary certification and approval shall be employed.

iv. Equipment enclosures and fittings shall be examined to see that all stopping plugs and bolts are in position and properly tightened. Locking and sealing devices shall be checked periodically to ensure that they are secured in a prescribed manner.

v. If at any time, there is a change in the area classification or the characteristics of the flammable material handled in the area or if the equipment is relocated in the area, the area classification drawing should be correspondingly revised, and a check shall be made to ensure that the equipment selection corresponds to the revised area classification.

vi. A system shall be established to record the results of all inspections and the action taken to correct defects.

9.5.12 LIST OF CRITICAL SPARES OF EQUIPMENT

Entity needs to identify a list of critical spares of all the equipment. The status of critical spares is to be regularly monitored to ensure their availability and usability at all times.

9.5.13 PAINTING

Painting is essential for installation for protection against environmental factors, i.e., moisture, preventing corrosion and extending the lifespan of metal. Colour coding through paint helps in the identification of different equipment, storage areas, and safety zones. The entity shall have a colour coding scheme for various pipelines, product storage tanks, water tanks, structures, etc and same shall be displayed. A rolling plan for painting of various Plant facilities shall be available, including the frequency of painting of various structures/ pipelines along with actual implementation of the plan.

Coastal areas experience high levels of humidity and salt-laden air which significantly accelerates the corrosion of metals, especially steel and iron. In such places, additional measures to be taken. Durable, multi-layered painting systems with high-quality surface preparation and regular maintenance are essential to prevent structural degradation. Considering the corrosive ambience, Rolling plan should be suitable amended and followed.

9.5.14 DRAWINGS

a) The plant shall maintain an updated drawing of

1. PESO approved Plant layout showing all facilities.
2. PESO approved chain conveyor layout.
3. Factory Department approved the Plant layout.

DRAFT 2

4. Fire zone diagram detailing all the zones and identifying their Deluge Valve system.
5. Fixed firefighting system showing position of water monitor, double hydrant and isolation valve along with their identification number (Hydrant Layout).
6. Diagram showing the positioning of all types of fire extinguishers.
7. Plant drainage system with location of water seals available.
8. Hazardous Area Classification diagram in line with OISD-STD-113.
9. Single-line diagram of power/ lighting distribution.
10. Earth pit diagram showing earthing grid and connection including earth pit number, size of strip, type of earth pit indicated by symbol clearly differentiating between various types of earth pit (viz. Pipe, Plate, Chemical, etc.).
11. Lightning Protection System Diagram.
12. Electric Cable route diagram with cable route marker.
13. Piping and Instrumentation diagram.
14. Emergency evacuation routes, along with the emergency exits for personnel and vehicles.

b) The latest updated as-built diagram shall be used for the preparation of various Plant diagrams. All the above drawings shall be displayed as per the requirement at the appropriate place.

c) Plant shall also ensure Control certificate of the bulk storage vessel, fabrication drawing of the bulk storage vessel and fabrication drawing of the fire water tank are available.

d) The plant should prepare a summary document wherein details of the total number of Plant equipment, like LPG hose, Fire hose, earth pits, pressure gauge/indicator, temperature gauge/indicator, level gauge/indicator, TSV, SRV, are shown.

e) Plant shall prepare a master list along with serial number for Plant equipment like LPG hose, Fire hose, earth pits, pressure gauge/ indicator, temperature gauge/ indicator, level gauge/indicator, TSV, SRV etc.

9.5.16 DOCUMENTATION

Standard formats for operation/ maintenance/ security activities shall be issued by the Head Office. Formats used and maintained by the location shall be in line with standard formats issued by Head Office. Entry of daily activities shall be recorded in the logbook of the respective area where work is being carried out. Logbooks shall be reviewed and countersigned by the role holder officer on a daily basis.

9.5.17 Below is the list of applicable OISD standards used in LPG installation for inspection and maintenance of equipment. However, this list is not exhaustive. Provision of all the OISD standard shall apply as per the extant version.

Sr.no.	Inspection and Maintenance of Equipment/ Facility	OISD standard
1	Pumps, Compressors, Rotary Equipment, Predictive Maintenance Practices, Mechanical Seals, Rotating Equipment, Diesel Engines	OISD-STD-240
2	Inspection of Unfired Pressure Vessels	OISD-STD-128

DRAFT 2

3	Storage tanks	OISD-STD-129
4	Piping systems	OISD-STD-130
5	Pressure reliving devices	OISD-STD-132
6	Loading and unloading hoses for petroleum products	OISD-STD-135
7	Electrical equipment	OISD-STD-137 OISD-RP-146 OISD-RP-147 OISD-RP-148
8	Firefighting equipment and systems	OISD-STD-142
9	Mounded Storage Vessel	OISD-STD-150
10	Maintenance & inspection of safety instrumentation in hydrocarbon industry	OISD-STD-153
11	Personal protective equipment	OISD-STD-155
12	Cross-country LPG pipelines	OISD-STD-214

10.0 STATUTORY TESTING OF CYLINDERS

All types and capacities of cylinders used for filling LPG are required to be tested periodically in line with the applicable statutory requirements to ensure the fitness of the cylinders for further service. Such tests shall be conducted at the PESO recognised cylinder testing facilities only. Person meeting the requirements of the manager, supervisor and operators shall be deployed for inspection and approvals meeting qualification criteria in line with Schedule IV of the Gas Cylinder Rule, 2016.

The sequence of operations for statutory pressure testing of LPG Cylinders is given at Annexure-10-**I.**

10.1 TESTING PROCEDURE

10.1.1 DEGASSING OF LPG CYLINDERS

All the cylinders identified for periodical testing shall be depressurized to atmospheric pressure at the approved designated location/ degassing station. Residual LPG shall be vented using a suitable adaptor through a pipe 1.5 m above the eaves of the shed to ensure that the internal cylinder pressure is reduced to atmospheric pressure.

Only after depressurising the cylinder, the cylinder valve shall be removed, and the LPG cylinder is taken up for water filling, and water is overflown and retained in the cylinder for not less than one hour to remove any traces of LPG. Degassing can also be achieved by stripping, Nitrogen purging, etc. After degassing, the inside of the LPG Cylinder shall be checked by an explosimeter to ensure that the cylinder is gas-free. The cylinder shall be degassed at a site approved by the statutory authority.

Refer to clause 8.2.4.3 for degassing of the cylinder.

DRAFT 2

10.1.2 SURFACE CLEANING

a) After depressurising the cylinder and emptying the water, the same is kept in an inverted position and stroked gently with a non-sparking hammer, such as a wooden mallet, to remove any dust, rust or any loose remnants in the cylinders.

b) The depressurized and degassed cylinder bung is plugged and is taken up for surface cleaning by immersing the cylinder in the caustic soda bath and soap water, and the same is then rinsed with water. Sand blasting/ shot blasting can also be used for surface cleaning of the cylinder.

10.1.3 CHECKING TARE WEIGHT

All the LPG cylinders shall be weighed, and if the loss in tare weight is more than 5% of the original weight, such cylinders shall be segregated and scrapped. At the time of weighing, the weight of the valve is also accounted for.

10.1.4 VISUAL INSPECTION

a) The detailed visual inspection of the cylinder body shall be carried out as per IS:16054/ or any other code or standard accepted by the Chief Controller of Explosives.

b) Depending upon the visual inspection, cylinders are segregated for scrapping, for cold repairs and for hot repairs, in case required.

c) Internal cleaning of the cylinder shall be done by a vacuum pump. After internal cleaning, a magnetic bar shall be used to remove any ferrous particles. The interior of the cylinder shall be visually examined with low low-voltage FLP lamp. The internal examination shall be carried out as per IS:16054.

d) Dents, cuts, gouges, bulges, cracks, laminations or punctures, bottom shell thickness, foot ring bottom clearance, etc., applying the criteria for rejection given in Annexure-10-II. In case of dented/ buckled cylinders, visual inspection of welds needs to be done more carefully since the welds are more susceptible to cracking. Denting or buckling by more than the nominal plate thickness may be attended by repairs/ replacement.

e) Cylinders shall be rejected in case the observed thickness of the undamaged remaining wall in the affected portion is less than the minimum calculated wall thickness. Rejection criteria with respect to thickness shall be formulated by Entities and displayed at the location.

10.1.5 HYDROSTATIC TESTING

a) All the new LPG cylinders shall be taken up for hydrostatic test within an interval of 10 years from the date of manufacturing for first testing and subsequent testing after every 5 years in accordance with IS:16054. Only water should be used as a medium for hydrostatic testing, and the pressure to be applied should be the test pressure marked on the cylinder body (25.35 kg/cm^2), and the same should be retained for a minimum period of 30 seconds. During the test, the cylinder shall be gently tapped with a wooden/ nylon hammer near the weld joints. Any reduction in pressure noticed during the test, any leakage, visible bulge or deformation, or sweating of the cylinder shall be treated as a case of failure. Such cylinders shall be properly marked, observation recorded and segregated for scrapping and disposal.

b) Pressure Gauges/ indicators shall be calibrated at least once a week, and the Pressure Gauges/ indicators used for verification shall be tested once in every three months.

c) Test equipment shall be in line with IS16054. An auto cut-off arrangement shall be provided at the required hydro test pressure of the cylinder.

DRAFT 2

d) A suitable device (SRV, additional pressure switch, etc.) shall be fitted in the test equipment to ensure that no cylinder is subjected to pressure in excess of its test pressure by more than 10%.

e) Dedicated pressure gauges/ indicators for each cylinder to read the cylinder test pressure in accordance with the relevant Indian Standard shall be provided. In addition to this, two numbers of main gauges of 150 mm dial size shall also be provided on the inlet line of the equipment after the isolation valve.

10.1.6 VALVE FIXING

a) Before commencing the fitting of valve operation, all the water inside the cylinder should be drained and the cylinder should be made dry with the use of dry compressed air and a vacuum pump. Before refitting the valve, internal threads of the cylinder bung should be cleaned with a 3/4" NGT thread tap and checked against thread damage using L1 and L9 gauges. Cylinders in which the valve pad thread has worn out or does not meet the requirements of IS:15894 shall be rejected.

b) Teflon paste/ suitable sealing compound should be used as a sealing material for tightening the valve, and the torque of 20+/-2 kg.m. should be applied. Spanners and ordinary wrenches shall not be used for tightening the valves.

10.1.7 PNEUMATIC TEST

a) The Pneumatic test shall be applied to the cylinder at a pressure of 12 kg/cm² for a minimum of 60 seconds as per IS:16054. This is basically to check the tightness of the valve cylinder joints. A pressure gauge of suitable capacity shall be provided at each air filling point to check the pressure of air in the cylinder prior to starting the test. Test equipment shall be in line with IS:16054.

b) With the cylinder under pressure, a soap solution should be applied to the cylinder valve bung joint to ensure that there is no leak. Alternatively, leak to be checked by dipping the cylinders in the water bath.

c) After pneumatic testing, air shall be removed from the cylinder.

d) All the Pressure Gauges/ indicators used for Pneumatic tests shall be calibrated at least once in 3 weeks, and the Pressure Gauge/ indicator used for verification shall be tested once in every three months.

10.1.8 PAINTING OPERATIONS:

a) To enhance the life of the cylinder and to project a better image to the customers, maintaining a well painted surface of the cylinder is most desirable.

b) All LPG cylinders are painted with one coat of zinc chromate primer conforming to IS:2074 and a coat of super synthetic enamel paint conforming to IS:2932 as per the colour scheme as per PESO approved drawing. Care should be taken to paint the cylinders in vulnerable areas like underneath the foot ring. Both primer coating and enamel paint application shall be done by using a spray gun.

c) The valve should be protected while painting the cylinder.

d) The combined minimum coat thickness for layers of primer and paint shall not be less than 30 µ in line with IS:16054.

DRAFT 2

10.1.9 CHECKING AND MARKING THE TARE WEIGHTS

a) The Cylinder fitted with the valve shall be weighed for tare.

b) Weighing machines used for checking the tare weight shall be duly certified by the Weights & Measures Authority. The machine shall be tested daily with the standard weights to verify accuracy. The test used for checking should be tested against standard weights, and the same must be stamped periodically by the Weights & Measures Authority. The minimum graduation of 10 gms. must be available.

c) Weighing scale error should not be greater than 1.0%.

d) The tare weight should be recorded with an accuracy of +/-50 gms. and rounded off to the nearest 100 gms. as per IS:2- Rules for rounding off numerical value.

e) Observed tare weight of the cylinder should be stencilled with black paint within the V.P. Ring circle diameter at 3 places for easy identification by filling operators. The recommended lettering size is 25 mm, and the thickness of the numerals is 6-7 mm.

f) Next periodic inspection date shall also be stencilled on the cylinder.

g) It is also recommended that the net weight of LPG to be filled should be stencilled on the body of the cylinders outside the V.P. Ring diameter. The size of the lettering recommended is 18 mm, and the thickness of the numerals is 4-5 mm.

10.1.10 PUNCHING OF MARKS:

After satisfactory completion of periodical inspection and testing, the following shall be marked with following on the vertical stays of the cylinder:

a) New test date while retaining the previous test dates.

b) Identification mark of the testing unit as registered with the statutory body on the stay plate near the retest date punching.

c) New tare weight and gross weight (in case of any change in tare weight) while retaining the old tare and gross weight.

d) Loss in tare weight shall be punched on the foot ring. For example, if the loss in tare weight is 200 gms, it shall be punched as “LTW200”.

10.1.11 ROUTE CARD

Route card process and batch formation for statutory pressure testing activities shall be followed as per IS 16054. Refer Annexure-10-III for Route Card.

10.1.12 COLOUR CODING OF CYLINDER

This is done primarily to facilitate easy segregation of cylinders for repairs at installations without going through the test date of manufacturing dates punched on each and every cylinder. Currently, the following colour codes are used for the determination of the hydro-testing cycle of cylinders, and the same colour coding shall be repeated after every ten years cycle. Revised colour coding schemes as per the latest guidelines are attached as Annexure-10-IV

10.1.13 CERTIFICATION

After completion of the testing, the re-testing agency shall record the details for cylinders accepted and cylinders rejected/ recommended for repair. The re-testing agency shall issue a certificate as per Annexure-10-V towards periodic testing of cylinders. This certificate, along with the list of cylinders

DRAFT 2

accepted and the list of cylinders rejected/ recommended for repair, shall be available at LPG installation.

10.1.14 OTHERS

a) For the cylinder testing facility, adequate CCTV coverage, along with a recording facility to be provided and monitored to ensure sample checks of the procedure being followed.

b) The responsibility of carrying out RQC of ST Cylinder shall be clearly defined in the SOP. Before releasing the batch for production, RQC shall be carried out by an officer as per Annexure-10-VI

c) Quality checks of pressure tested cylinders shall be included in the checklist of internal audit/ multi-disciplinary safety audit.

d) Training shall be imparted to officers and workers handling cylinder testing activity, covering all the aspects of the statutory testing of cylinders.

11.0 FIRE PROTECTION FACILITIES

11.1 GENERAL

LPG Installations shall have well defined in-built fire prevention and protection system to mitigate any exigency. The requirements of the fire prevention and protection system shall be as under.

11.2 DESIGN CRITERIA

i) The single largest fire risk shall be considered.

ii) All LPG storage Vessels, Cylinder Storage/ Filling/ Repair Sheds, LPG Pump/ Compressor Houses, remote operating valves/ motor operated valves on LPG lines, Bulk Lorry and Tank Wagon Gantries, pipeline receiving station shall be fully covered by a medium velocity water spray system.

iii) Fire Protection Facilities shall have firefighting access, means of escape in case of fire and also segregation of facilities so that the adjacent facilities are not endangered during the fire.

iv) Heat Detectors for the detection of fire for automatic actuation of medium velocity water sprinkler system shall be provided. The QB/EP detectors shall be placed directly overhead or inside the hazard. In areas without specific hazards, detectors shall be placed evenly across the ceiling or with a maximum spacing of 3 m inside the shed. Any other detectors, if provided, shall comply with the design requirements.

11.3 FIRE WATER SYSTEM

The main components of the fire water system are:

- Fire Water Storage
- Fire Water Pumps
- Fire Hydrant/ Monitor distribution piping network.
- Water Sprinkler/ Deluge system.

11.4 FLOW RATE DESIGN

The Fire Water pumping requirement for medium velocity spray system shall be calculated based on minimum spray density as 10.2 lpm/m^2 of the exposed surface or area to be cooled. In case of LPG Pump/ compressor, it shall be 20.4 lpm/m^2.

DRAFT 2

a) STORAGE VESSELS

A MV sprinkler system with automatic heat detection, having a remote/ local operated Deluge Valve with spray density of Minimum 10.2 lpm/m^2 shall be provided on all above-ground storage vessels.

For water flow calculations aggregate surface area of all vessels within a distance of 30 m from the periphery of the affected LPG vessel shall also be considered as a single risk.

b) SHEDS

A MV sprinkler system with automatic heat detection, having a remote/ local operated Deluge Valve with spray density of minimum 10.2 lpm/m^2 shall be provided in the entire shed, including the loading/ unloading fingers.

For spray water calculations, the shed can be divided into a suitable number of zones, each served by an independent deluge system. The adjacent zones shall be operative around the zone under fire, and the same shall be considered as a single risk. The width of a zone shall not be less than 10 m except for the zone on either end of the shed.

As there is no storage of cylinders in the loading/ unloading fingers, the spray system can be taken as a separate zone with an independent Deluge Valve, which need not be in logic operation with the zones in the empty / filled cylinder shed to optimize the fire water requirement.

c) LPG PUMP / COMPRESSOR HOUSE

A MV sprinkler system with automatic heat detection, having a remote/ local operated Deluge Valve with spray density of Minimum 20.4 lpm/m^2 shall be provided.

The entire LPG pump /compressor house, including the valves of the pipeline manifold, shall be considered as a single risk area; alternatively, it can be divided into a suitable number of zones, each of which shall have a width not less than 10 m. Each zone shall be served by an independent deluge system.

d) TANK LORRY GANTRY

In case of Tank Lorry Gantry, automatic detection of heat for automatic actuation of MV sprinkler system having remote and local operated Deluge Valve with spray density 10.2 lpm/m^2 of surface area shall be provided. A maximum of 8 bays shall be considered as a single risk area. In addition to the tank lorry surface, the MV spray system shall effectively cover the tyres, manifold and dish ends of various sizes of tank trucks. There shall be at least 3 tiers of MV sprinkler systems to uniformly cool the top, middle and bottom of the tank lorry. Each zone shall be served by an independent deluge system.

e) TANK WAGON GANTRY

A MV sprinkler system with automatic heat detection, having a remote and local operated Deluge Valve with effective spray density of 10.2 lpm/m^2 shall be provided. The gantry can be divided into a suitable number of zones with a minimum zone length of 30 m, and adjacent zones shall operate around the zone under fire, and the same shall be considered as a single risk. Each zone shall be served by an independent deluge system. There shall be at least 3 tiers of MV sprinkler systems to uniformly cool the top, middle and bottom of the tank wagons. The spray system shall also cover the associated LPG pipelines in the gantry.

The fire water system in the Plant shall be designed to meet the highest fire water flow requirement of a single largest risk of any of the above cases at a time, plus 288 m^3/hr for operating 2 nos. Fire water Monitors/ Supplementary Hose requirements.

DRAFT 2

The MV Sprinkler Zone drawing shall be prepared detailing all the zones and identifying their independent Deluge Valve systems. The drawing shall be kept readily available in the Plant and displayed prominently in the field.

11.5 FIRE WATER SYSTEM DESIGN

i) The Fire Water pressure system shall be designed for a minimum residual pressure of 7.0 kg/cm² at the remotest place of application in the Plant.

ii) A fire water ring main shall be provided all around the perimeter of the LPG Installation facilities, e.g., LPG storage vessel, Tank truck gantry, Tank wagon gantry, LPG sheds, LPG Pump/ Compressor House, etc. Mounded Storage Vessel and its associated LPG Pump/ Compressor House should be considered as a single facility for the provision of a fire water main ring.

iii) A fire water ring main shall have hydrants/ monitors spaced at intervals not exceeding 30 m when measured aerially. Fire hydrants and monitors shall not be installed within 15 m of the facilities/ equipment to be protected.

iv) The fire hydrant network shall be in closed loops to ensure multidirectional flow in the system. Isolation valves shall be provided to enable isolation of any section of the network without affecting the flow in the rest. The isolation valves shall be located normally near the loop junction. Additional isolation valves shall be provided in the segments where the length of the segment exceeds 300 m. Isolation valves shall have a visible indication for open and closed conditions.

v) A suitable over pressure relief mechanism of appropriate size shall be provided in the hydrant line against thermal expansion, sudden closure of valves, low water requirement in the field with respect to the rated capacity of the Fire engine. This is to ensure the pressure in the line never exceeds the shut-off pressure of the fire water pump, even with the bypass valve in a closed condition.

11.6 FIRE WATER STORAGE

i) Water for the hydrant service shall be stored in any easily accessible surface or underground lined reservoir or above-ground tanks of steel, concrete or masonry. The effective capacity of the water tank above the level of the suction point shall be a minimum of 4 hours' aggregate working capacity of pumps. For a sample fire water calculation, refer to Annexure-11-I. Where the make-up water supply system is 50% or more, this storage capacity may be reduced to the 3 hours aggregate rated capacity of pumps.

   Large natural reservoirs (within installation premises) having water capacity exceeding 10 times the aggregate water requirement of fire pumps may be left unlined.

ii) Storage tank/ reservoir shall be in two compartments to facilitate cleaning and repairs. In case of steel tanks there shall be a minimum of two tanks.

11.7 FIRE WATER PUMPS

i) Centrifugal pumps having flooded suction shall be installed to meet the design fire water flow rate and head. If fire water is stored in underground tanks, an overhead water tank of sufficient capacity shall be provided for flooded suction after accounting for leakage in the network, if any. Pumps shall be provided with suitably sized strainers on the suction line and NRVs on ~~and~~ discharge lines.

ii) The system pressure has to be maintained at 7 kg/cm² at the farthest end of the hydrant system with the help of a Jockey Pump operating automatically. The jockey pump shall be sized adequately to compensate for the pressure drop in the fire water header arising from system

DRAFT 2

leakage and normal pressure losses and shall maintain the fire water header pressure at 7.0 kg/cm² at the farthest point. The Jockey Pump shall start-stop automatically, actuated by pressure switches set appropriately, and the start/ stop pressure shall be displayed. A standby jockey pump shall be provided.

iii) The fire water pump(s), including the standby pump(s) shall be diesel engine driven. The pumps shall be capable of discharging 150% of its rated discharge at a minimum of 65% of the rated head. The shut-off head shall not exceed 120% of the rated head for horizontal centrifugal pumps. Each engine shall have an independent fuel tank of suitable size for 6 hrs. continuous running at the desired head and discharge of the fire pump concerned. Fuel tanks should be installed outside the Fire Water Pump's House and shall have provision for venting. If tanks are located inside the Fire Water Pump House, the vent shall have provision for venting outside the Pump House.

iv) At least one standby fire water pump shall be provided when the no. of main pumps does not exceed 2 nos. When the said requirement exceeds 2 pumps, at least 2 standby pumps of the same capacity, head and characteristics shall be provided. All the fire water pumps shall be of the same rated capacity, head and similar characteristics.

v) Fire water pumps shall be used exclusively for Fire protection and prevention purposes only.

vi) Suction and discharge valves of the fire water pump shall be kept open all the time.

vii) Fire engines shall have provision for high coolant temperature and low lube oil pressure alarms.

viii) All the Fire Engines shall be set at the rated RPM of the fire pump. As all the fire engines are to start in auto mode, throttling of the fire engines' governors for setting the engine speed below the rated RPM shall not be permitted.

ix) The OEM drawing of the cooling system of the fire engines, clearly marking the minimum and maximum inlet and outlet cooling water pressures permitted in the line, location of the pressure regulator and its settings, etc., shall be available. Valves of cooling water lines shall be in an open condition for the engine kept in auto/standby mode. The pressure of the cooling water shall be regulated as per OEM recommendation.

x) The fire water pumps shall be provided with auto starting facility. The auto starting logic shall ensure that, if the pump - designated to start first - fails to build up the required pressure of 7 kg/cm² at the farthest end of the hydrant system in the fire water ring main system within 20 seconds (max.) or fails to start at all, the next designated pump shall start and continue the same cycle. Auto starting logic and set pressure shall be displayed for fire engines.

xi) Provision shall be available to check the fire water pump without compromising the auto mode of the system.

11.8 FIRE HYDRANT NETWORK

i) Fire hydrant ring main shall be laid above ground, ensuring that: -

a) The pipeline shall be laid at a height of 300 mm to 400 mm above finished ground level. For existing piping, wherever the ground clearance is less than 300 mm, location specific protections, e.g., wrapping/ coating/ PCC/ paver blocks, should be provided.

b) The mains shall be supported at regular intervals not exceeding 6 m or as per design approved. For the indicative drawing of pipe support, Annexure-XV of OISD-STD-130 shall be referred.

DRAFT 2

c) The system for above ground portion shall be analysed for flexibility against thermal expansion, and necessary expansion loops, where called for, shall be provided.

ii) However, the ring main may be laid underground at the following places:

a) Road crossings.

b) Places where above-ground piping is likely to cause obstruction to operation and vehicle movement.

c) Places where above-ground piping is likely to get damaged mechanically, particularly in the LPG storage area, where water supply lines are laid for feeding the sprinkler deluge system.

d) Where Frost conditions warrant, and ambient temperature is likely to fall below zero deg. centigrade underground piping at least 1 m below the ground level should be provided. Alternatively, for above-ground pipelines, water circulation to be carried out.

iii) For a fire water ring main laid underground, the following shall be ensured:

a) The ring main shall have at least 1 m earth cushion in open ground, 1.2 m cushion under the road crossings and in case of rail crossing, provisions stipulated by Indian Railways shall be complied with.

b) The ring main shall be suitably protected against soil corrosion.

c) In case of poor soil conditions, it may be necessary to provide concrete/ masonry supports under the pipeline.

iv) Fire water ring main shall be sized for 120% of the design water flow rate. The velocity of water shall not exceed 5 m per second in the fire water ring main. Design flow rates shall be distributed at nodal points to give the most realistic way of water requirements in an emergency.

The hydraulic analysis of the network shall be done at the design time. Also, whenever fire water demand increases due to the addition of facilities or extensive extension of the network, fresh hydraulic analysis shall be carried out. In case of sea water service, the fire water main pipes shall be concrete mortar lined internally, or other suitable coating material shall be used.

Design details of the fire hydrant system, Deluge Valve and MV Sprinkler system, including design pressure, operating pressure, test pressure and hydraulic calculations, shall be kept readily available at the Plant.

v) A fire water ring main shall be provided all around the perimeter of the LPG Installation facilities. Hydrants/ monitors shall be located considering various fire scenarios at different sections of the premises to be protected and to give the most effective service. Fire hydrants and monitors shall not be installed within 15 m of the edge of the facilities/ equipment to be protected. At least one hydrant post/ monitor shall be provided spaced at intervals not exceeding 30 m when measured aerially in case of hazardous areas. For a non-hazardous area, they shall be spaced at 45 m intervals.

vi) Hydrants/ Monitors shall be located with branch connections and not directly over main header for easy accessibility and to avoid corrosion of main header due to dripping from hydrants/accumulation of water. Hydrant/ Monitors shall be located along roadside berms for easy accessibility as far as possible.

vii) The horizontal range and coverage of hydrants with hose connections shall not be considered more than 45 M. In case of buildings, distance from the location of hydrant shall not be less than

DRAFT 2

2 m and not more than 15 m from the face of the building. Provision of hydrants within the building shall be provided in accordance with IS:3844.

viii) In case of Monitor, the distance of monitors shall not exceed the performance range of the monitor from the hazard to be protected.

ix) Double-headed hydrants with two separate landing valves on a 3"/ 4" stand post shall be used. All hydrant outlets shall be 1.2 m above ground level.

x) Monitors shall be located to direct water on the object as well as to provide a water shield to firemen approaching a fire. The requirement of monitors shall be established based on hazard identification, risk assessment involved and layout considerations.

xi) Hydrants and monitors shall not be installed inside the kerb areas. However, in exceptional circumstances where coverage of the SRV (installed on the top of the vessel) from two sides, by the monitor (s) installed outside the kerb, is inadequate, remote operated monitors can be provided inside the kerb area to cover the inaccessible portion of the SRV through external monitors. The monitor shall be provided with an isolation valve or ROV outside the kerb area. The monitor shall be placed on an independent elevated structure. The supporting structure shall be fireproofed. The riser pipe to monitor shall also be fireproofed. The Hydrant pipeline inside the kerb area shall be underground. In case, above ground hydrant pipeline is provided inside the kerb area, the same shall be fireproofed. The remote-control panel for HVLR monitor shall be at a distance of minimum of 60 m away from the vessels to be protected.

xii) LPG Tank wagon gantry and Tank truck gantry area should be provided with hydrants having multipurpose combination nozzles for jet spray and fog arrangement. Fire hydrants shall be provided on both sides of the Tank Wagon gantry. The hydrants and monitors shall be located at a minimum distance of 15 m from the Tank Wagon/ Tank Trucks, measured from the edge of the facilities.

xiii) Fire water monitors shall be provided with independent isolation valves.

xiv) A hose box with 2 nos. hoses and a jet nozzle shall be provided at each hydrant point.

xv) The fire water Deluge Valve shall be located at a minimum of 15 m away from the shadow of the storage vessel and from the edge of other LPG sheds and bulk LPG loading/ unloading gantry.

xvi) A firewall shall be provided for the protection of the Deluge Valve and personal protection of the operator.

xvii) Pipes made of composite material shall be laid underground.

xviii) In case of underground mains, the isolation valves shall be located in an RCC/ brick masonry chamber.

11.9 MATERIAL SPECIFICATIONS

All the materials used in the fire water system using fresh water shall be of the type as indicated below.

i) **Pipes**: Carbon Steel as per IS:3589/ IS:1239/ IS:1978 or Composite Material or its equivalent for freshwater service. In case saline, brackish or treated effluent water is used, the fire water ring main of steel pipes, internally cement mortar lines or glass reinforced epoxy coated or pipes made of material suitable for the quality of water, able to withstand the temperature and pressure, shall be used. Alternately, pipes made of composite materials shall be used. The composite material to be used may be as per API 15LR/API 15HR / IS:12709.

DRAFT 2

ii) **Isolation Valves**: Gate valve or quick shut-off type isolation valves made of Carbon Steel, having open/ close indication, shall be used. Other materials, such as cupro-nickel for saline/ brackish water, can be used. The material of the valve shall be suitable for the service.

iii) **Deluge Valve:** Carbon Steel

iv) Hydrant

Refer OISD-GDN-115 for specification

v) **Monitors**:

Refer OISD-GDN-115 for specification.

vi) The above-ground fire water main and the fire hydrant stand post shall be painted with corrosion resistant “Fire Red" paint as per IS:5. Water monitor, hydrant point, and hose box shall be painted " Fire Red/ Luminous Yellow' as per IS:5- colours for ready mixed paints and enamels.

11.10 MEDIUM VELOCITY SPRINKLER SYSTEM

i) The medium velocity spray system shall be provided in all critical areas like LPG Storage Area**,** LPG Sheds having cylinder filling, cylinder storage, cylinder loading/ unloading fingers (including packed trucks covering at least the cylinder thereon), LPG Pump/ Compressor House, Bulk Tank truck/ tank wagon loading/ unloading gantries, valves of piping manifold, cold repair sheds, painting shed used exclusively for painting of purged ND cylinder etc.

ii) Spray nozzles shall be directed radially to the exposed surface of above above-ground storage vessel at a distance not exceeding 0.6 m from the vessel.

iii) Only one type and size of spray nozzle shall be used in a particular facility. The horizontal extremities of water flow from spray nozzles shall at least meet to ensure no gap between the water spray from the adjacent nozzles.

iv) Where projections (manhole flanges, pipe flanges, staircase, and support brackets) obstruct the water spray coverage, including rundown or slippages on vertical surfaces, additional nozzles shall be installed around the projections to maintain the wetting pattern. The first valve of the vessel shall be adequately covered with sprinklers.

v) Piping downstream of the block valve and after the Deluge Valve shall have adequate drain facilities at selected locations.

11.11 AUTOMATIC FIRE PROTECTION SYSTEM

i) Automatic fire protection (Fixed) system based on heat detection through thermal fuses/ quartz bulbs/ EP detectors shall be employed. Sensors shall be installed at all critical places wherever a medium velocity spray system has been installed as described below.

   a. For LPG storage vessels, such detectors shall be provided in multiple rings (at least two rings for bullets and three rings for spheres) encircling each vessel, detectors equi-spaced with a maximum horizontal spacing of 1 m, the lowest ring starting at an elevation of within 1.5 m from the bottom of the vessel. Also, a minimum of 2 detectors shall be provided at the top of the vessel at each flange or flange cluster, as the case may be and at least one near the liquid line ROV to take care of the failure of flanges. In case of an automatic thermal fuse-based fire protection system, the instrument air supply pressure to thermal fuses shall be maintained through a pressure control valve and a restriction orifice and/or solenoid valves.

DRAFT 2

b. Heat Detectors shall be placed at critical locations in LPG sheds (cylinder filling, cylinder storage, testing, evacuation, cylinder loading/unloading fingers including packed trucks covering at least the cylinders thereon, etc.), bulk Tank truck and Tank wagon loading/ unloading gantries, LPG Pump and Compressor House, piping manifold, cold repair sheds etc. At bulk Tank truck and Tank wagon loading and unloading gantries, heat detectors shall be provided at least in 2 tiers.

ii) In case Quartzoid Bulbs are used for detection, the same shall be designed to blow at 79 °C (max.) and Quartzoid Bulb network shall be maintained with Plant air at a pressure not more than 3.5 kg/cm^2 and shall be such that the discharge of air through one Quartzoid Bulb will depressurize the system to actuate the Deluge Valves. Alternatively, water can also be used in place of air.

iii) In case of Electro pneumatic (E.P.) heat detectors, it shall actuate the Deluge Valve in any of the following conditions:

a) Rate of rise: 10 °C minimum of temp.

b) At 79 °C (max.)

The EP detectors shall be divided into groups, and alternate detectors shall be connected in one circuit. Two detectors from two different groups shall function/ operate for the actuation of the sprinkler system.

iv) Water spray nozzles and heat detection system shall be of approved type and duly certified for performance.

v) The actuation of detectors shall initiate the following:

a. Opening of the Deluge Valve of the affected zone as well as adjacent zones within 20 seconds.

b. Audio-visual alarm indicating the affected zone as well as adjacent zones (feedback from air/ electrical switch for DV activation) and status indications for Deluge Valves (feedback from water pressure switch/ transmitter installed at discharge of Deluge Valve) with facility for DV actuation at the Fire Water Pump House and additionally, at the control panel which is manned 24X7.

c. Electrically operated Emergency Siren, audible up to 1 km from the periphery of the Plant.

d. Closure of all Safety Remote Operated Valves in the affected facility within 30 seconds.

e. Tripping of power supply, barring the emergency power.

f. The water spray from all nozzles within 30 seconds.

g. The fire water pump(s) shall start based on their set pressure to supplement/ to maintain the fire water pressure in the ring main.

vi) Additionally, system like push buttons and/or air release valves in QB circuit (local provision for DV actuation) etc. shall also be provided for each Deluge Valve in the field at easily accessible locations placed at a minimum distance of 15 m from the protected facility for enabling actuation by an operator for initiating all the actions as per above.

vii) In case the zoning concept is used for the MV sprinkler system, the concerned Zone shall be clearly marked on each of the push buttons.

DRAFT 2

viii) In the field, manual bypass valves of fire water Deluge Valves shall also be provided. A strainer shall be provided before the Deluge Valve. It shall be ensured that in case of cleaning of the strainer and maintenance of DV, the sprinkler system is available for operation using the bypass line. Indicative deluge layout is as per Annexure-11-II

ix) DV activation logic with cause and effect for each DV to be displayed on the remote operating panel and in the field.

x) In addition to the above three provisions (QB, local provision, remote DV panel) for DV activation, if the Deluge Valve is having local provision (air release/ water release/ manual bypass valve) for opening for maintenance and testing purpose, adjacent DV need not be actuated.

xi) Care shall be taken in positioning nozzles so that water spray does not miss the targeted surface and reduce the efficiency or calculated discharge rate.

xii) PLC based logging (with printing option) of events of the firefighting system (Deluge Valve operation, ESD operation, MCP operation, Fire Pump operation, etc.) shall be ensured.

11.12 FIRST AID FIRE FIGHTING EQUIPMENT

11.12.1 PORTABLE FIRE EXTINGUISHERS

i) All fire extinguishers shall conform to respective IS/UL or equivalent codes, viz. 9 kg DCP Type (IS:15683/ UL299), 4.5/ 6/ 8 kg CO2 Type (IS:16018/ UL154) and 25/ 50/ 75 kg DCP Type (IS:16018/ UL299) and bear IS/ UL mark. BIS/ UL or Equivalent certificates of all extinguishers shall be maintained at the location. Refer to OISD-GDN-115 for the guidelines.

ii) Portable fire extinguishers shall be located at convenient locations and shall at all times be readily accessible and clearly visible.

iii) The maximum running distance to locate an extinguisher in working areas shall not exceed 15 m.

iv) The top surface of the extinguisher shall not be more than 1.5 m high.

v) The fire extinguishers shall be provided at various locations as under:

AREA	Number of Portable Fire Extinguishers
LPG Storage Vessel	2 nos. 9kg DCP fire extinguishers for each Vessel
LPG cylinder Sheds	2 nos. 9kg DCP fire extinguishers per 200 m^2 area.
LPG Pump House	2 nos. 9 kg DCP fire extinguishers per 50 m^2 area
Tank Truck loading/ unloading Gantry	1 no. 9 kg DCP fire extinguisher in each bay.
Tank Wagon loading/ unloading Gantry	1 no. 9 kg DCP fire extinguisher for every 15/20 m of gantry on the gantry platform
Other Pump Houses	2 nos. 9 kg DCP fire extinguishers
Office/Canteen/ Stores	2 nos. 9 kg DCP fire extinguishers in each floor of the building

DRAFT 2

MCC / DG Room / HT Room	2 x 4.5 kg CO2 fire extinguishers in each room or for a floor Area of about 100 m^2, and 4 sand buckets and a stand shall be provided in the MCC/ DG Room

vi) 100% spare CO2 cartridges and 50% (min) spare DCP (as per Fire Extinguisher Capacities) shall be stored in the LPG installation.

vii) For pressurised type fire extinguishers: 10% spare DCP fire extinguishers shall be stored in the LPG installation.

viii) The sand buckets shall have a round bottom with a bottom handle having 9 litre water capacity conforming to IS:2546. The sand stored in the bucket shall be fine and free from oil, water or rubbish.

ix) Rain protection of suitable design should be provided for all extinguishers and sand buckets.

11.12.2 WHEELED/ MOBILE FIRE FIGHTING EQUIPMENT

i) One No. Mobile 75 kg DCP fire extinguisher shall be provided in filling shed, LPG storage vessels, LPG Pump/ compressor House, Tank truck loading/ unloading gantry and tank wagon loading/ unloading gantry. This shall be provided in each group of the above-mentioned facilities.

ii) The Dry chemical powder used in the extinguishers shall be Potassium/ Urea based or Sodium Bicarbonate as per OISD-GDN-115. Nitrogen/ Carbon Dioxide shall be used as expelling gas.

11.12.3 HOSES, NOZZLES AND ACCESSORIES

i) Reinforced rubber lined hoses (63mm) conforming to IS:636 (type A or B) shall be provided.

ii) The hoses shall be of 15 m standard length and shall be provided with gunmetal/ aluminium alloy

iii) 30% spare Hoses shall be stored in the LPG Installation.

iv) In addition to the nozzles provided in the hose boxes, there shall be at least 2 nos. spare nozzles in each category, viz. Jet Nozzles with branch pipes, Fog Nozzles, universal Nozzles, water curtain Nozzles, Spray Nozzles and Triple purpose nozzles in the Plant.

v) An Emergency trolley containing Fire Proximity Suit, low temperature suit, Breathing Apparatus Set, Water gel Blanket, Resuscitator, First Aid Box, Stretcher with blanket, Spare fire hoses, Special purpose nozzles, Explosimeter, P. A. System such as Megaphone etc. shall be readily available at the Installation and positioned to have easy access to it during emergency.

vi) The following accessories/ first aid items shall be provided in the Plant (including those provided in the trolley):

1. Fire hoses: Two nos. for each Hose Box
2. Safety helmet: min. 10 nos.
3. Hose Box: At each hydrant point.
4. Stretcher with min. 2 nos. blankets.
5. First Aid Box: min. 2 nos.
6. Rubber hand gloves for electrical (BIS approved) purpose: - min. 2 pairs each for HT and LT
7. Low temperature rubber hand gloves for LPG emergency as per OISD-GDN-115: 4 pairs

DRAFT 2

8. Low temp. Protective clothing for LPG emergency: min. 2 sets
9. Fire proximity suit: min. 1 No.
10. Resuscitator: min. 2 No.
11. Red/Green Flags: As reqd.
12. Self-contained breathing apparatus with one spare cyl (Cap. 30 minutes): min. 2 no.
13. Water gel blankets: min. 2 nos.
14. Portable Gas detectors / Explosimeter: min. 2 plus one spare.
15. Flameproof Torch: min. 2 nos.
16. Non-Sparking Tools: min. 1 set
17. Megaphone Ex-proof: min 1 no.
18. Multipurpose nozzle: min 2 nos. (in store)
19. Safety harness: min 2 nos.
20. Lifeline: min 2 nos.

vii) Content of First Aid box should be as per OISD-STD-227 and as per statutory requirement. First Aid boxes shall be checked periodically, and records to be maintained. Trained first aid personnel shall be available in the installation and a list of such personnel shall be displayed at a prominent place. Minimum two first aid trained personnel shall be available in each working shift and minimum one in nonworking shift. First aid chart for treating electrical shock shall be displayed prominently in MCC room.

viii) The above are guidelines and minimum requirements of each item and can be increased depending upon the scale of operations, statutory/ mandatory requirements of local bodies/ State Governments or any other expert body.

11.13 EMERGENCY CONTROL / ALARM SYSTEM

11.13.1 FIRE ALARM SYSTEM

i) Manual call points (MCP) near to Filling shed, Bulk storage area, Tank truck gantry, Tank wagon gantry, LPG Pump/ compressor House, bulk LPG tank truck and packed truck parking area etc. shall be provided. Operation of these MCP shall raise the audio-visual alarm in a 24 x 7 manned control panel with identification of the alarm location.

ii) Emergency Shut Down (ESD) provision near to Filling shed, Bulk storage area, Tank truck gantry, Tank wagon gantry, LPG Pump/ compressor House, Pipeline transfer battery limit, etc. shall be provided. Operation of these ESD points shall trip the power supply, barring the emergency power and raise the audio-visual alarm in the control panel with identification of the alarm location.

iii) The cause and effect of activation of MCP and ESD shall be displayed in the field for each MCP and ESD. MCP and ESD shall be tested once in month.

iv) The electricity-operated Emergency Siren shall be audible up to 1 km from the periphery of the Plant.

DRAFT 2

v) Manually operated emergency sirens shall be provided at strategic places such as Bulk storage area, Filling shed, Valve change shed, Degassing area, Tank truck gantry, Tank wagon gantry, LPG Pump/ compressor House, Bulk LPG tank truck and packed truck parking area, Admin bldg, PMCC, Pipeline transfer battery limit, etc.

The following emergency siren codes shall be followed for different emergency situations:

a) **Level - I**: The siren code for Level - I shall be decided by an entity depending upon the size and complexity of the installation. Entity can keep same siren code for Level - I and Level - II. However, no siren code for Level - I shall not be accepted.

b) **Level - II**: A wailing siren for two minutes.

c) **Level - III**: Same type of siren as in case of Level - II but the same will be sounded for three times at the interval of one minute i.e. (wailing siren 2 min + gap 1 min + wailing siren 2 min + gap 1 min + wailing siren 2 min) total duration of disaster siren to be eight minutes.

d) **ALL CLEAR**: Straight run siren for two minutes.

e) **TEST:** Straight run the siren for two minutes at the frequency at least once a week.

The emergency siren code shall be displayed at a prominent location.

Electricity operated and manually operated emergency siren shall be tested at least once a week**.**

11.13.2 COMMUNICATION SYSTEM

i) A communications system, like a telephone, in the administrative/ security office shall be provided.

ii) PA paging or walkie-talkie or intrinsically safe mobile phone shall be provided at critical facilities such as Bulk Storage Area, LPG Pump/ compressor House, Cylinder Sheds Area, Tank truck gantry, Tank Wagon gantry, bulk LPG tank truck and packed truck Parking Area, Fire Water Pump House, Location-in-Charge, Safety Officer, Security, Motor Control Center (MCC) etc.

iii) Communication system provided in the hazardous areas shall be flame-proof/ intrinsically safe and approved by PESO.

iv) Wherever possible, a hotline connection between the City Fire Brigade / nearby major Industries should be provided.

v) The Plant shall have a 'Mutual Aid' arrangement with nearby industries to pool in their resources during an emergency. OISD-STD-227 shall be referred for preparing mutual aid agreements. Mutual aid agreement to include contact details of the officials of the mutual aid members, details of a minimum number of firefighting equipment, manpower and quantity of consumables to be shared among the mutual aid members, planned frequency of meeting with mutual aid members and validity of the Agreement. Records of meetings shall be available. Contact details of the mutual aid members shall be updated quarterly.

11.14 SPECIAL FIRE PROTECTION

Fireproofing shall be provided on the structural supporting legs of the vessel. The minimum rating for fireproofing shall be 2 hrs.

Fireproofing is not required for diagonal bracing, including tie rods, or redundant members that are not necessary for supporting the static loads.

DRAFT 2

Before fire-proofing application, protected metal surfaces shall be prepared through such means as sandblasting and corrosion-protective primers. Particular attention shall be given to the top junction of the fireproofing with the protected metal to prevent water ingress. OISD-STD-164 shall be referred to for details on fireproofing. Fire water line to each sphere should be so routed that it is not exposed to direct fire. This is to protect it from failure in the initial period when water flow has not commenced. It is recommended that the riser should be located away from the bottom ROV. The horizontal run of the Fire Water line may be buried if fire engulfment cannot be avoided otherwise.

11.15 INSPECTION AND TESTING

The fire protection equipment shall be kept in good operating condition at all times.

The firefighting system shall be periodically tested for proper functioning and logged for record and corrective actions.

In addition to the following routine checks/ maintenance, the requirements of OISD-STD-142 in respect of periodic inspection, maintenance and testing of firefighting equipment shall be complied with.

11.15.1 FIRE WATER PUMPS

i) Every Pump shall be in a test run at least once a week, for half an hour at the rated conditions. Suitable provision should be made in the Fire Water Pump House to ensure that during the test run of any individual fire pump, the firefighting system of the entire Plant continues to be in auto mode.

ii) Once a month, each pump shall be checked and tested, and the shut-off pressure observed and logged.

iii) Each pump shall be run continuously for 4 hours at its rated head and flow using the circulation line of fire water storage tanks at least once a year. Various critical parameters of the fire engine, e.g., discharge pressure, RPM, lube oil pressure, temperature, etc., shall be recorded minimum of every hour, and observations pertaining to the health of the fire engine shall be logged.

iv) Fire engine and Fire Pump RPM should match to get rated performance. Fire pumps should be operated at rated RPMs to get the desired/ rated head and flow.

v) Each pump shall be checked and tested for its performance once in six months by opening the required nos. of hydrants/monitors, depending on the capacity of the pump, to verify that the discharge pressure, flow and motor load are in conformity with the design parameters.

vi) Start/ stop set pressure of the Jockey pump operation shall be checked and recorded in the Fire Water Pump House logbook at least once a week.

vii) Number of start/ stop operations to be recorded on a daily basis in the Fire Water Pump House logbook. Frequent start/ stop condition of the Jockey pump indicates that there are water leaks in the system, which should be attended to promptly.

11.15.2 FIRE WATER RING MAIN

i) The ring main shall be kept pressurized at 7 kg/cm² with the help of jockey pumps.

ii) The ring mains, hydrant, and monitor shall be visually inspected for any missing accessories, defects, damage and corrosion every month, and records maintained.

iii) All valves on the ring main/ hydrant/ monitor valves shall be checked for leaks/ operation and lubricated once a month.

DRAFT 2

iv) The ring main shall be checked for leaks once a year by operating one or more pumps and keeping the hydrant points closed to get the maximum pressure.

v) For inspection, testing and maintenance of the fire water network, OISD-STD-142 shall be referred.

11.15.3 FIRE WATER SPRINKLER SYSTEM

i) All Deluge Valves and sprinkler systems shall be operated and checked once in a quarter by actuating DV through remote or field operation for proper positioning, clogging of each nozzle and working of the system in total.

ii) The performance test certificates after installation in respect of spray density, flow rate, and response time for each facility to be protected, provided by the manufacturer, shall be maintained at the LPG Installations and verified once in 6 months by the installation and records maintained thereof. This can be carried out as part of the testing carried out as mentioned at point i) above.

iii) Testing of all Deluge Valves with the outlet valve of DV closed shall be carried out once in a quarter by Field actuation (Emergency Push button or Break Glass or air release provision), remote actuation from the panel and by breaking QB.

iv) The strainers provided in the Fire water sprinkler system shall be cleaned once in a quarter, and records maintained. Cleaning of the strainer and or maintenance of DV shall be done when the protected facility is not in operation.

11.15.4 FIRE WATER TANK/ RESERVOIR

i) The water reservoir shall be emptied out and cleaned once in 5 years.

ii) In case of a steel tank, Routine in-service inspection, external inspection and internal inspection shall be conducted in line with OISD-STD-129.

iii) In case of water reservoir and steel tank of open roof type, floating leaves, material, or algae, if any, shall be removed once in 6 months or as and when required and record to be maintained.

11.15.5 PORTABLE & WHEELED FIRE EXTINGUISHERS

For inspection, testing and maintenance of portable and wheeled fire extinguishers, OISD-STD-142 shall be referred to.

11.15.6 FIRE HYDRANT, WATER MONITOR, ISOLATION VALVE, HOSES, NOZZLES AND ACCESSORIES

i) For inspection, testing and maintenance of Hoses, Nozzles and other accessories, OISD-STD-142 shall be referred.

ii) Pressure testing of hydrant hoses in service, refer Annex G (IS636:2018). Once in service, a hydrant hose should be hydrostatically tested at 12 month intervals and after each occasion of operational use. The hose should be tested at a pressure 50% in excess of the intended working pressure or at 10 kg/cm^2, whichever is the greater, and maintained for 1 min. If the hose shows any sign of leakage, it should be repaired and retested. If it cannot be satisfactorily repaired, it should be discarded.

11.16 MAKE-UP WATER

Facilities to make up water during firefighting shall be provided.

DRAFT 2

If bore wells are available within the Plant, yields should be recorded on a quarterly basis for each bore well.

11.17 COMBINED POL AND LPG FACILITIES:

It is not preferred to have common water storage facility and fire pump house between the POL and LPG Installation. No new POL and LPG installation shall be made with common water storage facility and pump house from the date of publication of 3rd revision of this standard.

However, in existing location, where the common water storage facility for firefighting purposes is shared between the POL Installation and the LPG Installation, following conditions shall be ensured:

a) Each POL/ LPG facility shall independently meet the design, layout and fire protection system requirements of the corresponding OISD standards and have a common boundary wall and ownership of both facilities under the same entity. Common boundary wall between POL and LPG installation shall be a 1 m high wall with 0.6 m barbed wire on top.

b) The fire water requirement shall be based on two fire contingencies simultaneously in the combined facility, and the fire water storage capacity shall be fixed accordingly.

c) The fire water system shall ensure availability of pressure of 7 kg/cm² at the farthest point.

d) The entire fire water system shall remain pressurized and kept in auto mode.

e) The responsibility of maintenance of these water tanks and maintaining the water level in the storage tank at all the times shall be clearly defined and documented.

f) The pump house may be common/ separate. In case a common pump house is provided, the control of the pump house shall be clearly defined and documented.

g) SOP on operating, maintenance and access of such facility should be prepared mutually, signed by both the LPG and POL Location-in-Charge. It shall be ensured that there is uniformity in such documents across various locations of an entity. A copy of the same should be available with both POL and LPG installations.

h) Other than common fire water facility and common fire water pump house, all other firefighting facilities such as hydrant network, monitor/ hydrant etc. at the respective area of POL/LPG shall be maintained by respective group.

i) In case of audit of LPG Installation, firefighting facility (Fire water Tank and Fire Water Pump House) shall also be audited and vice versa.

11.18 DATA MONITORING SYSTEM

Data monitoring system shall be provided for various operational technology (OT) equipment, including control systems, automation, Gas monitoring systems, LPG filling and quality checks control systems, CCTV, access control systems, Fire-fighting automation/ PLC systems with SCADA for the operation of the LPG installations.

i) All such control equipment should have data logging facilities to provide reports of the history of the occurrence of events with date and time.

ii) Such reports should be customized to ensure identification of each event and shall be monitored.

iii) The timestamp of all OT systems, including SCADA, PLC, fire-fighting systems, and gas monitoring systems, etc., must be synchronized to ensure uniformity across all systems within the LPG installation.

iv) Entity should have appropriate safety measures and policy to safeguard against cyber-attack.

DRAFT 2

12.0 GAS MONITORING SYSTEM

12.0 GENERAL

The best method of prevention of an explosion is to avoid the basic buildup of gas concentration immediately on the occurrence of a leakage. This would require basically a reliable and continuous detection system with warning annunciation to alert the Plant personnel to take corrective action.

The Gas Monitoring system (GMS) shall provide early warning on the buildup of dispersed gas concentration below the Lower Explosive Limit (LEL) limits.

The system shall be designed considering small leaks (leaks which have secondary closures) such as mechanical seals failure, sampling point left open, gasket leaks, hose pinholes, valves gland leakage, drain point left partially open, TSV's relief discharge in the manifold area, filling/ evacuation hoses leakage.

12.1 APPLICATION

i) The detectors for the gas monitoring system shall be strategically located in the LPG Installations at all facilities close to the potential source of leakage.

ii) The monitoring display with audio visual alarm to show Real Time status of all sensors and field transmission units (FTU) in a Plant Layout form shall be provided in the 24 hrs. manned location for continuous monitoring even during power failure.

iii) In case of failure of power at FTU or failure of communication with FTU, monitoring display shall show the status of that particular Gas monitoring sensor as faulty.

12.2 DETECTORS

Amongst the various methods of detection available, the following proven systems for LPG can be considered.:

i) Catalytic detectors

ii) Infra-red detectors

12.3 CONTROL SYSTEMS

i) This is critical equipment for the Plant's safety. Hence, the system should have control equipment, power supply, UPS to ensure that the system remains in operation at all times.

ii) In case of failure or during maintenance when the entire GMS system is not in operation, the Plant Safety Officer should keep the Plant personnel on alert to be more vigilant. In such case, bypass management in line with OISD-GDN-206 shall be followed.

iii) The control equipment should have data logging facilities to provide print outs of the history of the events with date and time of leakages. Suitable graphical representation for activation of each sensor shall be generated for analysis of leakages on daily basis. A summary report of all sensors shall be generated for monitoring GMS alarms by the Plant Officer on a daily basis, and the action taken shall be recorded. Ensure the report should contain:

   a) Historical records of all Events (Warn, Alarm, Fault, etc) for a user-selected period.

   b) Historical Peak and average % LEL readings of all sensors along with events count (Warn, Alarm, Over range, and Fault) for a user selected period.

   c) Historical Peak % LEL readings of a user selected sensor with event counts for a user selected period.

iv) The control equipment shall be able to generate at least two alarms at different levels of LEL concentration of gas. Audio-visual alarms shall be provided with the first alarm set at 20% of the Lower Explosive Limit (LEL) and the second level alarm at 60% of Lower Explosive Limit (LEL).

DRAFT 2

v) The control equipment is not required to have automatic corrective action capabilities on sensing leakages and alarm shall not get reset automatically, as this is basically a warning device.

12.4 ANNUNCIATION SYSTEMS

An appropriate field and remote annunciation system shall be available to ensure that all the alarms generated, both audio and visual, are available to the Plant personnel, who are authorized to take corrective action. Depending on the deployment pattern of the Plant, the alarms, both audio and visual, to be repeated at additional locations to ensure fast corrective action.

12.5 LOCATION OF DETECTORS

i) The behaviour of the gas leakage governs the positioning of the gas detector. As LPG is heavier than air, the height of the detector should not be more than 0.3 m from the mounting level.

ii) The predominant wind direction should be considered with respect to the potential source of leakage to ensure positioning of the detector on the downwind side.

iii) The detectors, especially the catalytic type, should not be positioned very close to the potential source of leakage to avoid poisoning of the detectors (temporary malfunctioning). The detectors should be located at least 0.3 m away from the potential source of leakage. However, the location of the detector shall not be more than 1.5 m from the potential source of leakage. The location can be decided by assessing the behaviour of gas drift and consequential safety risk.

iv) In case of infra-red detectors, the same shall be installed on the downwind side, ensuring the path is free from obstructions. In case any additional expansion/ construction is undertaken, the detectors will have to be pre-positioned.

v) The minimum detector facilities-wise are as given below:

a) Storage vessels (above ground): - 1 No. on top, near the SRVs
- 1 No. near bottom ROV.
- One each near water draining/sampling points.

b) LPG Pump House: 1 No. in Pump House and 1 No. in manifold.

c) Filled cylinder shed: - 2 nos.

d) Valve change shed: - 1 No. near valve change unit, 1 no near degassing unit.

e) Empty-cum-filling shed: - 2 nos. near carousel, 1 No. near evacuation unit, 1 No. weight correction unit, 1 no. near the Valve Changing Without Evacuation, near the manual filling unit.

f) TLD: - one at each manifold and at the ends.

g) Tank wagon gantry: - 1 no. sensor to be provided at ground level for every two bays or at every 30 m whichever is less. Additionally, some detectors at selected locations shall be provided at the top of the platform.

12.6 INSPECTION AND TESTING

i) The gas monitoring system shall be kept in good operating condition at all times. The inspection of the system shall be done at an interval of 1 month during which the gas shall be released at all the detectors, and the performance of the system shall be established.

ii) The calibration of the equipment shall coincide with the monthly inspection schedule every three months for calibration of the gas detectors. Calibrated gas with a known and certified level of concentration shall be used.

DRAFT 2

iii) The drift in the sensitivity of the individual detectors shall be recorded in the maintenance history logbook during calibration, and the detectors with abnormal or wide drift in sensitivity.

iv) The Plants should have at least 2 nos. spare detectors for each system maintained at all times to facilitate immediate replacement of defective detectors.

13.0 EMERGENCY RESPONSE AND DISASTER MANAGEMENT PLAN

13.0 GENERAL

Several major accidents leading to grave disasters have occurred across the world in the past. The lessons learnt from the disasters made it essential to draw an Emergency Response and Disaster Management Plan to handle such an eventuality. An Emergency Management Plan is essential to obviate such an eventuality by providing the measures to contain the incident and minimise the after affects.

The best way to manage any emergency is to prevent it. The following guidelines for emergency prevention shall be followed:

a) Sound engineering practice in the design, fabrication, installation and maintenance of facilities.

b) Careful selection and correct use of equipment.

c) Observance of safety and security regulations.

d) Deployment of trained manpower.

e) Development and adherence to site specific Standard Operating Procedure (SOP).

f) Periodic training and guidance to all personnel working in the Plant, with particular reference to product knowledge (MSDS) and maintenance practices.

g) Following the Management of Change (MOC) procedure.

h) To maintain good housekeeping.

i) Constant supervision and alertness.

j) Evaluation of the emergency plan through regular tabletop exercises and mock drills.

CLASSIFICATION OF EMERGENCIES: -

Emergencies can be categorized into three broad levels based on seriousness and response requirements, namely: -

a) **Level-I**: This is an emergency or an incident which

i) can be effectively and safely managed and contained within the site, location, or installation by the available resources.

ii) has no impact outside the site, location, or installation.

b) **Level-II:** This is an emergency or an incident which

i) cannot be effectively and safely managed or contained at the location or installation by available resources, and additional support is alerted or required.

ii) is having or has the potential to have an effect beyond the site, location, or installation and where external support of a mutual aid partner may be involved.

iii) is likely to be a danger to life, the environment, industrial assets, or reputation.

c) **Level-III**: This is an emergency or an incident with off-site impact which could be catastrophic and is likely to affect the population, property, and environment inside and outside the installation, and management and control are done by district administration. Although the Level-III emergency falls under the purview of the District Authority but till they step in, it should be the responsibility of the unit to manage the emergency.

DRAFT 2

Note: Level-I and Level-II shall normally be grouped as an Onsite emergency, and Level-III as an Off-site emergency.

13.1 EMERGENCY PLAN

The objective of Emergency response management is to mitigate adverse effects of incidents through effective risk assessment, emergency preparedness, mitigation and response measures.

a) Each organization should develop, implement and establish Emergency Response and Disaster Management Plan (ERDMP) in accordance with all applicable statutory/ regulatory/ OISD guidelines such as OISD-STD-227/ MOP&NG DMP/ PNGRB ERDMP Regulation-2010 and amendments thereof.

b) Hazard Identification and Risk Analysis (HIRA) of the installation shall be carried out to identify, evaluate, and reduce the likelihood and/ or minimize the consequences of uncontrolled releases and other safety or environmental incidents within the entity's risk tolerance criterion by recommending appropriate prevention and/ or mitigation measures. Also, the HIRA process is a worthwhile continuous improvement tool, identifying operability concerns and opportunities for improvement.

c) The entity should develop a written approach document clearly defining applicability and methodology/ techniques of HIRA (viz. qualitative, semi-quantitative and/ or quantitative).

d) HIRA shall be carried out at the layout stage with the objective of arriving at any specific mitigation measures required for Hazards identified. Risk reduction/ mitigation measures shall be given due credit. Risk assessment shall include unconfined Vapour cloud explosion (UVCE) scenarios, etc. In case the Plant receives bulk LPG via pipeline transfer, various hazards associated with the adjoining pipeline receiving station must be included in these studies. The risk analysis of the location shall estimate the risk of all the potential on-site and off-site hazards (most credible and worst-case scenarios) using consequence analysis methodology to reduce the risk and prioritize the emergency response plans.

   Thereafter, Quantitative Risk Assessment (QRA) HIRA shall be done whenever there is a major addition(s) in facilities or major changes in the surrounding areas, changes in the operating parameters, change in the product grade stored in the tanks takes place, recommended by Incident Investigation report, or once in every five years whichever is earlier.

   For detailed guidelines on HIRA, OISD-GDN-206 "Guidelines on Safety Management System in Oil & Gas Industry" shall be referred.

e) The Occupier of the Installation shall thoroughly review the HIRA document and sign the document endorsing the outcome and obligations of the HIRA report.

f) The outcome of the HIRA study shall guide in preparation of the On-site and Off-site Emergency Response and Disaster Management Plan of the installation. ERDMP shall be prepared as per the PNGRB ERDMP Regulation-2010 and amendments thereof ERDMP amended regulations, 2020 and certified by the PNGRB accredited party.

g) The location-specific fire organisation chart shall be available in the ERDMP document enumerating the roles and responsibilities of the personnel during emergencies round the clock.

h) The ERDMP shall contain the location-specific security threat plan, emergency evacuation routes/ plan, Mutual Aid Arrangements, interface issues/ procedures with refinery/ cross-country pipeline to take care of normal and emergency situations as per the guidelines prescribed in ERDMP amendment regulations, 2020.

i) The list of emergency contact numbers shall be correctly captured in the ERDMP document and updated Quarterly.

j) ERDMP should be revised/ vetted once in 5 years or with any major change/ modification/ recommendation of a major incident.

DRAFT 2

k) ERDMP Plan of the Plant should cover various hazards associated with adjoining pipeline receiving station as well.

l) Location-in-Charge shall ensure that suitable instructions are issued to both the Corporation's entity's and contractor's personnel identifying the action to be taken by each one in an emergency. This should be achieved by the display of a firefighting organisation chart/ posters, not only for firefighting but also for other likely emergencies.

m) As a part of the Emergency Response and Disaster Management Plan, a communication organogram shall be drawn up giving the flow of communication from the originating location to various local agencies and also to Statutory Authorities and upwards within the entity to mobilize support and to consider alternatives for maintaining essential supplies.

13.1.1 LIKELY LOCATION OF LPG LEAKAGE IN THE PLANT

Check for LPG leakage at following locations:

a) Through water drain valve while draining

b) Leaking sample valve.

c) Breakage of glass on the Sight Flow Indicator on the pipeline.

d) Opening of pop-off valve, gland/ packing, etc.

e) Leakage through LPG compressor/ pump.

f) Leakage due to bursting/ rupture of the hose or through the swivel joints of the body arms.

g) Leakage from the filling guns.

h) Escape of liquid/ vapour LPG during maintenance/ repairs

13.1.2 SOME OF THE COMMON CAUSES OF FIRE

i) **ELECTRICAL**:

ii) Short circuiting due to loose wiring, including that of an automobile moving in the area.

iii) Use of non-flame proof fittings such as lights, torch, motors, fan, telephone, temporary lighting etc. (Note: Flameproof fittings become non-flame proof due to poor maintenance e.g. increase in gap in junction boxes, due to missing nuts and bolts or improper glands used for connection to junction box, or sealing compound not used in junction boxes etc.)

iv) Static electricity as a result of improper bonding/ grounding to tanks, conveyor systems, tank trucks, tank wagons, etc.

v) **VEHICLE:**

a) Exhaust of the vehicle, if effective spark arrestors are not provided.

b) Operation of non-FLP electrical switches, such as vehicle ignition.

c) Spark from the loose contact on vehicle batteries.

d) Spark from the exhaust of a railway engine.

e) Cinders from the steam locomotives.

vi) OTHERS

a) Landing of crackers (flying type) in hazardous areas.

b) Chiselling/ hammering.

DRAFT 2

c) Spark from studded shoes, with a steel button, by friction against ground/ metal.

d) Lighted agarbatti in a truck.

e) Excessive release of vapour beyond the safe limit and fire due to an outside source of ignition.

f) Lighting

g) Smoking.

13.2 ACTION DURING EMERGENCY

13.2.1 GENERAL

Immediate action is the most important factor in emergency control because the first few seconds count, as LPG fires develop and spread very quickly unless prompt and efficient action is taken. In the event of a fire/ gas leak within the LPG Installation, the following action shall be taken as quickly as possible:

a) Take immediate steps to stop LPG leakage/ fire and raise an alarm simultaneously.

a) Initiate action as per the Fire Organisation Chart (FOC) or Emergency response and Disaster Management Plan, based on the gravity of the emergency.

b) Stop all operations and ensure closure of all ROVs and isolation valves.

c) All-out effort should be made to contain the spread of leakage/ fire.

d) Saving of human life shall get priority in comparison to stocks/ assets.

e) Plant personnel without specific duties should assemble at the assembly point.

f) All vehicles except those required for emergency use should be moved away from the operating area, in an orderly manner along a pre-nominated route.

g) The electrical system, except for control supplies, utilities, lighting and firefighting system, should be isolated.

h) If the feed to the fire cannot be cut off, the fire must be controlled and not extinguished.

i) Start the water spray system at areas involved in or exposed to fire risks.

j) In case of leakage of LPG without fire and inability to stop the flow, take all precautions to avoid the source of ignition.

k) Block all roads in the adjacent area with Police support for the purpose.

13.2.2 FIRE FIGHTING OPERATION

a) Enlist the support of the local fire brigade and neighbouring industries. A valid mutual aid scheme and records of meetings or discussions held shall be available.

b) If escaping vapour cannot be stopped, jets of water should be directed at the point of leakage to assist controlled release of vapour, and in between, water fog should be used for dilution and rapid dispersion of the vapour cloud.

c) Firefighting personnel working in or close to unignited vapour clouds or close to fire must wear protective clothing and equipment, including a safety harness and a manned lifeline. They must be protected continuously by water sprays. Water protection for firefighters should never be shut

DRAFT 2

off, even though the flames appear to have been extinguished, until all personnel are safely out of the danger area.

d) Exercise care to ensure that static charge is not generated in the LPG vapour cloud. For this purpose, solid jets of water must be avoided, instead, fog nozzles should be used.

e) Firefighters should advance towards a fire from upwind direction.

f) If the only valve that is require to be closed/ attended to stop the leakage is surrounded by fire, the attempt should be directed by trained persons only. The person attempting the closure should be continuously protected by means of water spraying (through fog nozzles), fire entry suit, water gel blanket or any other approved equipment. The person must be equipped with a safety harness and a manned lifeline.

g) Any rapid increase in pressure or noise level of product discharged through the safety relief valve of the vessel/ pipeline should be treated as a warning of over pressurisation. In such cases, all personnel should be evacuated immediately.

h) In case of any emergency situation, it is of paramount importance to avoid endangering human life in the event of fire involving or seriously exposing LPG equipment or serious leakage of LPG without the fire.

13.2.3 ACTION IN THE EVENT OF LPG LEAKAGE WITHOUT FIRE

a) Take basic action as detailed in 13.2.1 above.

b) If escaping LPG is not on fire, close any valve that will stop the flow.

c) Refer to OISD-GDN-161 for the LPG Tank truck incident and relief operations.

d) In case of a leaking Bulk LPG tank truck in the Plant or in case of leakage from a Bulk LPG tank truck parked in Truck Parking Area.

i) Carry out steps as listed in OISD-GDN-161 for arresting the leakage.

ii) Tank truck to be decanted in TLD inside the Plant. Care to be taken, during the decantation of such a tanker, other bays of TLD shall be vacant. Pressurising of the tanker, if required, should be in a controlled manner under the supervision of an officer.

iii) Concentration of vapour shall be checked intermittently using an explosimeter.

e) Take necessary actions as detailed in 13.2.2.

13.2.4 ACTION IN THE EVENT OF LPG FIRE

i) Take basic action as detailed in 13.2.1 above.

ii) Extinguishing Fires - A small fire at the point of leakage should be extinguished by enveloping with a water spray or a suitable smoothening agent such as carbon dioxide or dry chemical powder. LPG fire should not be extinguished until the escape of product has been stopped.

iii) Firefighting procedure - Firefighting procedures would vary depending upon various factors such as the nature, sources, size, location, etc. of the fire. Basic firefighting techniques have been explained earlier in section 13.2.2. However, for the purpose of guidelines, firefighting techniques for a few common cases are as follows:

iv) **Cylinder Fire:** If a filled cylinder is involved in fire, internal pressure may start rising, and if not relieved, the built-up pressure could rise above 70 kg/cm^2 and ultimately rupture the cylinder.

DRAFT 2

Cylinders in the vicinity should be kept cool by water sprays until the contents of the cylinder have burned away. If the gas leakage does not ignite, the cylinder should be approached from upwind (if in the open air) and be removed to a safe place away from sources of ignition. Cylinders not directly involved in the fire should be moved away from heat exposure, while applying cooling water sprays on cylinders directly involved.

v) **Fire on storage vessel/ exposed to radiant heat:** If a pressure vessel is exposed to radiant heat from external fire, it shall be kept cool by water sprays to prevent excessive pressure rise in the vessel. Cooling water sprays must be applied without delay using a fixed water sprinkler system or equivalent spray water coverage, through fixed monitors or other equipment. Cooling the vessel with water sprays reduces the heat input to the vessel and thereby reduces the pressure, thus reducing the rate of discharge from the relief valves.

Monitor pressure in LPG vessels.

If LPG vessel's SRVs have activated, start/ continue cooling.

Do not try to empty LPG vessels exposed to fire. They become more vulnerable to overheating as their liquid content decreases.

vi) **Leakage of LPG burning at the outlet of a cylinder valve:**

Extinguish the fire by means of a portable fire extinguisher, water spray or smothering with a thick, non-synthetic wet cloth etc.

Fix the safety cap on the valve to prevent further leakage.

After the fire is extinguished, if the leakage cannot be stopped, the cylinder should be moved carefully to a safe location to vent out the contents under controlled conditions.

vii) **Fire of LPG vapour escaping from a damaged storage vessel fitting (flame not impinging on the vessel):**

Allow the fire to continue till exhaustion or stoppage of LPG leakage.

If vessel or Plant equipment is being heated by radiation from the flame, spray water to cool.

Cool adjacent vessels and structures.

viii) **Fire at Safety Relief Valves of vessels exposed to adjacent fire.**

Cool LPG vessels exposed to fire/ heat.

13.2.5 FIRE FIGHTING ORGANISATION CHART

DRAFT 2

a) A plan of action for use in the event of a major LPG product leakage with a fire or risk of fire is essential. Such a plan must be carefully prepared for each Plant. It should be fully understood by all LPG Plant supervisory personnel and other personnel responsible for action as per the plan. The plan shall identify an emergency response with appropriate lines of authority, with succession planning and activating the response management. Responsibilities for decision making shall be clearly shown in a fire organisation chart. The plan shall identify each responder's position, mission, duties and reporting relationship. Petroleum and Natural Gas Regulatory Board (Codes of Practices for Emergency Response and Disaster Management Plan (ERDMP)) Regulations, 2010 and amendment thereof ERDMP amended regulations, 2020, shall be referred to for the preparation of the firefighting organisation chart. It should cover all the aspects for ON-Shift, General Shift and Off-Shift and be displayed at a prominent location. Roles during an emergency shall be displayed at the workstation. It shall be based on the following:

i) Plant personnel shall be fully trained for the specialized techniques necessary for combating LPG leakages and LPG fires.

ii) If leakage and/ or fire occur, all personnel should use the equipment provided to carry out their allotted tasks as detailed in the firefighting organisation chart.

iii) All personnel should be conversant with fire control equipment and also its location.

iv) LPG Plant personnel should know the position and method of operation of all Remote Operated valves (Safety ROVs) in the Plant.

v) Plant personnel should be familiar with the safety signage, first aid and all safety equipment, location of emergency exits, and the location of water hydrant points/ monitors, location of assembly points and the sound of the emergency siren.

b) The Fire Fighting Organisation Chart, together with the layout of firefighting and safety devices, shall be displayed at prominent places and explained to all personnel. It shall include the following functions, expanded to suit the location facilities/ requirement: -

i) Initiating the emergency siren.

ii) Shutting off the LPG supply.

iii) Calling the fire brigade/ police.

iv) Fire control, with first aid, firefighting equipment.

v) Dispersal of LPG leakage.

vi) Operation of fixed water sprinkler system, starting of fire pump (s) and application of water hose jets/ sprays for containing/ extinguishing fires.

vii) Cooling of vessels/ cylinders and other equipment/ facilities.

viii) Closing down all operations in the Plant, including closure of valves and stopping of pumps.

ix) Preventing all sources of ignition in case of LPG leakage.

x) Evacuation of vehicles if needed and if condition permits.

xi) Assembly and mustering of personnel (head count).

xii) Establishing an emergency fire-control center.

xiii) Traffic control.

DRAFT 2

xiv) Stations and duties of all personnel.

xv) Policing of affected areas.

xvi) Any other specialized duties.

xvii) Display of fire brigade, ambulance, Police telephone numbers, etc.

xviii) All clear signal by a Site Incident Controller.

c) **Liaison with local fire brigade**:

Close co-operation with the local fire authorities is essential and shall take the following form:

i) The fire brigade should be made familiar with the layout of the Plant and the location of important equipment/ facilities provided, and their method of use. Mock fire drills/exercises jointly by Plant personnel and local fire brigades shall be planned.

ii) Firefighting equipment at the Plant shall be compatible with the fire brigade equipment, otherwise, adapters shall be kept ready for hoses, hydrants, etc.

iii) The fire brigade shall be aware of the Plant's firefighting organisation plan.

iv) In the event of an emergency/ fire, the Location-in-Charge and/ or his representative shall advise the Fire Officer about particular or potential hazards that may be present, e.g. stocks of filled cylinders at that particular point of time.

d) Fire Drills & Training:

i) Mock drills for all Plant personnel, making use of the "Firefighting Organisation Chart" and practicing the specialized techniques required for fighting LPG fires or dispensing/ diluting LPG vapour clouds, shall be held minimum of once a month.

ii) The drills should cover various types of incidents, e.g. major spillage, LPG bulk lorry fire, cylinder fire, etc. It shall be ensured that all scenarios as defined in ERDMP are covered.

iii) Mock drills shall also be conducted at least once a year at night and off-shifts to check the effectiveness of response in case of emergency during non-operation of the Plant.

iv) Extinguishers used in fire drills shall be replenished subsequently.

v) During practice drills, the following should be tested:

a) Emergency siren

b) Fire Extinguisher operation

c) Fire Pump

d) Sprinkler system

e) Monitor/ hydrant operation

vi) Fire alarms shall be sounded/ tested (neighbouring Plants/ factories and the fire brigade shall be warned in advance of this test).

vii) Protective clothing, breathing apparatus and any other specialized safety equipment available shall be tried out during drills to train all concerned in their application.

viii) The local fire brigade and mutual aid members should be encouraged to participate in fire drills periodically.

DRAFT 2

ix) Observer should be appointed to witness the mock drill.

x) Area of improvement noticed during the drill shall be rectified.

13.3 BOMB THREAT:

The entity shall prepare a Crisis Management Scheme for bomb threats. A bomb threat drill shall be conducted as defined in the scheme. An adequate number of sandbags shall be kept in the Plant to be used in case of a bomb scare.

13.4 INDUSTRY INCIDENT

Installation shall maintain the following:

1) List of major On-Site accidents in the LPG industry that have taken place in the last 5 years
2) The OISD case study and safety alert related to the above are to be available at the location.
3) List of any minor accidents that have taken place in the last 2 years at the LPG installation.
4) Compliance with recommendations of accidents as listed in points 1), 2) and 3).

Learning from the above incident should be communicated to all relevant stakeholders by means of safety committee meetings, training, etc., to prevent the occurrence of similar incidents at the installation.

14.0 SAFETY AUDIT

14.1 GENERAL

Safety audit is a well-recognized tool to improve the effectiveness of safety programs and loss prevention measures by carrying out a systematic and critical appraisal of potential safety hazards involving personnel, Plant and equipment.

For a structured and systematic safety audit of any Plant/ facility, checklists are the most important prerequisite. Checklists should cover all the operational areas, major equipment, operating/ maintenance procedures, safety practices, fire prevention/ protection systems, etc. Observations/ recommendations of the safety audit team should be presented in the form of a formal report, and an action plan for corrective measures should be prepared and subsequently monitored for implementation. The entity shall define a matrix for closure of audit recommendations, and the Audit team shall review the compliance with the recommendations of the previous audit.

OISD-STD-145 on Internal Safety Audits shall be referred.

14.2 TYPE OF SAFETY AUDIT AND PERIODICITY

Two types of Safety Audits are proposed to be carried out as below:

I) Internal Safety Audit

II) External Safety Audit

Audit conducted by the Internal Audit teams of the entity is categorized as an Internal Safety Audit. Internal Safety Audits will be coordinated by local management under the overall direction of the respective Corporate Offices. OISD-STD-145 shall be referred for an internal safety audit.

External safety audits will be carried out by outside teams consisting of technical experts from other entity in Oil Industry, professional bodies and statutory authorities etc.

In view of the high hazard potential of LPG, safety audits at frequent intervals by the teams/ personnel of various levels are advocated. Broad checklists for different recommended safety audits are given in

DRAFT 2

Annexure. These checklists are indicative and can be further supplemented and modified suitably to meet the specific requirements of the individual LPG installation. Recommended minimum safety audits at different levels, their frequencies and broad checklists to be used are indicated below. However, depending on the nature of the audit and the type/ complexity of the installations, the duration of the audit shall be fixed by the entity.

AUDIT FREQUENCY TABLE

S.No.	Type of Safety Audit	By whom	Frequency	Checklist Annexure
1	Safety Check of LPG Installation	Designated Safety Officer of LPG Installation	Daily	Annexure-14-I
2	Self-audit	Location-in-Charge	Quarterly	Annexure-14-IV
3	Electrical audit of LPG Installation	Electrical Safety Officer as per C regulation, 2023	Annual	Annexure-14-II
4	Internal safety audit	Multi-disciplinary team of Region/ HQ	Annual	Checklists will be made available by OISD to the respective Principal Panellists. Only the latest OISD checklists are to be referred to for preparing internal audit checklist.
5	Inspection and maintenance audit of LPG Installation (M&I)	Personnel from Inspection/ Maintenance	Annual	Annexure-14-III
6	External safety audit of LPG Installation	OISD	As per OISD	OISD ESA checklist as provided by OISD. Only the latest OISD checklists are to be referred to.
7	Pre-commissioning inspection	OISD	Before the commissioning of a new LPG installation and any addition of Bulk LPG Storage	OISD Pre-Commissioning checklist provided by OISD

The daily safety check of LPG installation essentially shall cover the compliance of recommended operating procedures/ safety practices. Reports of audit shall be reviewed by the Location-in-Charge on daily basis and corrective action shall be undertaken, along with the target dates, shall be done immediately. A compliance monitoring mechanism for the observation of the daily safety check shall be prepared. Summary of recommendations and compliance to be maintained monthly basis.

DRAFT 2

The compliance to other audit recommendations, along with target schedules, shall be monitored on a monthly basis. Alternately, safety measures shall be taken till the recommendations are compiled in totality.

In case of an external safety audit, the following areas, besides the points given in the checklist of OISD-STD-145**,** shall be covered.

1. Safety Organisation
2. Security System
3. Fire prevention/ protection system
4. Preparedness to handle any emergency- Emergency Response and Disaster Control Plan and mock drills.
5. Skill level and training needs for HRD.
6. Operating manual/ instructions, display of Do's/ Don'ts, etc

For the duration of the safety audit and formation of a multi-disciplinary safety audit team, OISD-STD-145 shall be referred. The entity shall nominate internal auditors from the approved panel. These internal auditors shall be imparted training/ refresher courses on auditing.

Electrical Safety Officer shall be designated for all installations with an electrical connected load of more than 250 KW in line with Chapter 3 Clause 5-(3) of CEA regulation 2023.

15.0 TRAINING

15.1 GENERAL

Products handled at the LPG installations are hazardous in nature. Therefore, safety education and training require great attention. Training courses, including the refresher courses, shall be conducted to develop the skills and safety awareness of employees, contractor workers, security staff and crew members.

a) Training on safety, firefighting and rescue operations shall be compulsory for all officers, clericals, operators, security, T/T drivers and contract workmen who are likely to be present/ working in the installation and record maintained. The above training shall be imparted as per OISD-STD-154 guidelines.

b) A refresher course shall be ensured in line with a frequency set in OISD-STD-154 after initial training.

c) For new employees and contractor personnel, there should be an induction program addressing the relevant competence requirement, job coverage and possible HSE risks. Training for new employees and contract personnel shall be done as per OISD-STD-178.

d) All operating personnel (covered in the Fire organisation chart), including Security staff, should be trained as first responders for firefighting and rescue operations (including evacuation).

e) All employees, contract workmen (covered in fire organisation chart) and security staff (covered in fire organisation chart) at operating location shall be given hands on training on Live Firefighting at any of the recognised/ reputed training institutes/ by expert agencies having facilities for simulations, representing various fire scenarios likely to occur at LPG installations.

f) Every employee or authorized person of a contractor working in the installation shall be familiarized with emergency siren codes and the location of the ESD push button nearest to their place of work.

DRAFT 2

g) Maintenance officers and Safety Officers to be trained on electrical safety from any of the recognised/ reputed training institutes/ by expert agencies.

h) Whenever any officer is transferred to a new function, adequate training/ work knowledge should be imparted within the specified time period, and a record to be maintained. For new officers and officers joining on lateral transfers from other functions, structured induction training of a minimum of one week, at the new location, is to be considered.

15.2 DESIGNATED SAFETY OFFICER (DSO)

A qualified and experienced officer shall be designated as the 'Safety Officer' of the installation after training. He should be given exposure to Hazop, Risk Assessment, Safety Audit and upkeep of firefighting facilities and conducting Safety Meetings.

Duties and responsibilities of the designated Safety Officer shall be to build and ensure a safe working environment at the location, and to advise and assist the location in-charge in complying with statutory requirements or otherwise concerning health, safety, security and environment of the location. These duties shall include the following:

a) Duties and responsibilities of designated safety officer shall be in line with the state factory rules.

b) Advise various functions within the location in planning and organizing measures necessary for a safe and secure work environment, including effective prevention and control of personal injuries.

c) Identify, coordinate, organize and conduct health, safety, security and environment related training for all sections of personnel connected in terminal activities.

d) Conduct routine safety inspection rounds, safety checks on LPG installation facilities and equipment to ensure conformity with prevalent norms/ guidelines.

e) Updation and facilitate implementation of the fire organisation chart (FOC), Emergency Response and Disaster Management Plan (ERDMP), Risk Assessment and other documents related to fire and safety, training, etc., required by statutes.

f) Co-ordinate and conduct regular mock emergency drill (Level I/ II/ III), DCMP drill, and Mutual aid meeting.

g) Conduct safety committee meeting and monitor compliance.

h) Implementation of sound safety culture to promote safe practices amongst all sections of personnel at the location - employees, contract labour, TT crew, Security staff and Visitors.

i) Conduct safety audit and coordinate/ facilitate for conducting other internal and external audits and monitor compliance.

j) Prepare and monitor periodical compliance status of various audit and major accident investigation report recommendations (action taken report-ATR).

k) Prepare and submit to appropriate authorities', after due approval, various periodical reports on safety, security, health, environment including training. Maintain all related records for the purpose of reference and audits.

l) Monitor the appropriate administration of security measures such as gate passes, CCTVs, area surveillance, antecedent verification, access control procedures as per entity's policy

m) Ensure and monitor implementation of effective work permit system and maintain record maintained including JSA and TBT.

DRAFT 2

n) To promote identification and reporting of unsafe acts, unsafe conditions and near misses by all sections of location personnel. An appropriate reward and recognition scheme may be introduced, encouraging employees to identify and report unsafe acts, conditions and near miss events.

o) Near miss analysis with specific focus on the root causes, reporting, dissemination of findings with all concerned across the location and corrective action required for preventive measures.

p) Accident reduction measures.

q) To keep focus on rainwater harvesting, renewable energy (e.g. solar) and tree plantation, energy efficient lights.

r) Periodical health assessment in line with statutory requirement and entity's policy.

16.0 CCTV SYSTEM

i) CCTV shall be installed in the LPG installation covering entry/ exit gates, and all critical areas of the location (viz. Bulk storage area, LPG Pump House, Tank truck Gantry, Tank wagon Gantry, LPG Sheds, cylinder loading/ unloading area, cylinder testing shed, Pipeline receipt manifold area, Fire Water Pump House, TT parking area, electrical substation, etc, and any other place as per the policy issued by entity.

ii) High risk areas such as bulk loading/ unloading operations at Tank truck gantry and Tank wagon gantry, carousel, and cylinder testing shed should be covered by fixed cameras for continuous monitoring. However, if such areas are covered by pan-tilt-zoom (PTZ) cameras, the camera movement should be adjusted for proper monitoring of high-risk areas.

iii) The CCTV monitoring station shall be provided in 24 hrs. manned station. The size of the monitor shall be adequate to accommodate all the cameras simultaneously, or an auto changeover of CCTV monitoring screen for continuous surveillance shall be provided. The screen size of the monitor shall be such as to ensure clear visibility of the area being monitored.

iv) Location to monitor the area surveillance through CCTV system, and a record of any abnormality, unsafe act, or unsafe condition observed shall be maintained. A record of a random review of the CCTV recording by a Plant Officer shall be maintained.

v) Power to the UPS shall be routed through the emergency lighting panel. UPS shall be provided for the CCTV system, including cameras, network switches, monitor and shall have battery back-up of a minimum 30 min.

vi) The CCTV data shall be stored/ preserved for a minimum period of 90 days or in line with the Industrial Security manual of OMC or prevailing IB norms, whichever is higher.

vii) Suitable technologies like CCTV should be utilized to monitor site conditions continuously when high risk non routine work is being carried out. In case fixed CCTV does not have coverage of such areas, the use of a portable CCTV camera (appropriate for the area classification) to be explored. Details of CCTV cameras shall be recorded in the Work permit for all such jobs.

viii) CCTV should have a provision for recording with "date and time" data on screen for ease of analysis/ review. Time of all independent monitoring systems, e.g., GMS, CCTV, etc., to be synchronized for correct recording of time and date for analysis purposes.

17.0 ACCESS CONTROL SYSTEM

The LPG Installation shall have a minimum of two control points for access.

1) Main Gate

DRAFT 2

2) Licensed area entry gate

a) Main Gate

i) Persons shall be allowed into the Plant through the main gate after due verification of ID card for the person issued by the concerned contractor/ transporter and certified by the Location-in-Charge, and approval. Gate/ Visitor Pass for entry to the de-licensed area shall be prepared for the other visitor as per the security guidelines for installation.

ii) Security guard deployed at Main Gate shall monitor the movement of traffic of packed trucks and bulk LPG tank trucks entering the Plant.

iii) Under vehicle screening of all the vehicles entering the Plant premises shall be ensured by using trolley mirror or any other device.

iv) Prohibited items such as cigarettes, matches, lighter or explosive items, stored in the truck, truck cabin and any other vehicle entering the Plant premises.

b) Licensed/ Hazardous Area Entry Gate

i) Entry to licensed area to the visitor shall be permitted only after due authorization from concerned officer.

ii) Entry to hazardous areas shall be permitted only through an access control system as per security guidelines for installations. However, the access control system shall uniquely identify the movement of personnel and vehicles on real time basis entering and exiting the hazardous areas and record shall be maintained.

iii) Door frame metal detector and handheld metal detector shall be available at entry to licensed area gate.

iv) All the persons shall be checked for their authorisation (ID card/ Gate pass) to enter the licensed gate and shall be checked using handheld metal detector.

v) Verification of character and antecedents of all security employees, contract/casual labourers, truck crews of regular nature should be completed through District Police Administration.

vi) Records maintained by the security staff for personal, material, vehicle including Visitor register, Material in/out register, Gate pass register etc. should be checked by the concerned officer at least once in a fortnight to ensure correctness and no entry of any unauthorized person in the Plant. It shall be ensured that all details in the registers are filled as per the format for registers given in Security Manual. It shall be ensured that all registers are serially numbered, and list of registers maintained in the location is available with the Location-in-Charge.

DRAFT 2

18.0 ABBREVIATION

ABS	Anti-Lock Braking System
AFC	Approved for Construction
ATR	Action Taken Report
BAC	Break Away Coupling
BLEVE	Boiling Liquid Expanding Vapour Explosion
CCE	Chief Controller of Explosives
CCTV	Closed Circuit Television
CEA Regulations	Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations,
CIMFR	Central Institute of Mining and Fuel Research
CPRI	Central Power Research Institute
CVT	Compact Valve Tester
DCP	Dry Chemical Powder
DPT	Due for Pressure Testing
DV	Deluge Valve
EFCV	Excess Flow Check Valve
ELCB	Earth Leakage Circuit Breaker.
ELD	Electronic Leak Detector
EOT	Electric Overhead Traveling
ERDMP	Emergency Response and Disaster Management Plan.
ERTL	Electronics Regional Test Laboratory
ESD	Emergency Shut Down
FLP	Flameproof
FOC	Fire Organisation Chart
GCR Rules	Gas Cylinders Rules
GMS	Gas Monitoring System
HAZCHEM	Hazardous Chemicals
HAZOP	Hazard and Operability Study
HEL	Higher Explosive Limit
HIRA	Hazard Identification and Risk Assessment
HLA	High Level alarm
HR	Hot Repairable
HT	High Tension
IEFCV	Internal Excess Flow Check Valve
JSA	Job Safety Analysis
LA	Lightning Arrestor
LEL	Lower Explosive Limit
LIC	Location-in-Charge
LOT	Liquid Off Take
LOTO	Lock Out Tag Out
LPG	Liquefied Petroleum Gas

DRAFT 2

LT	Low Tension
MAWP	Maximum allowable working pressure
MCP	Manual Call Point
MFM	Mass Flow Meter
MOC	Management Of Change
MSDS	Material Safety Data Sheet
NDT	Non-destructive testing
NPT	National Pipe Thread
NRV	Non-return valve
OEM	Original Equipment Manufacturer
OMC	Oil Marketing Company
OTI	Oil Temperature Indicator
P&ID	Piping & Instrumentation Diagram
PAV	Pressure Actuating Valve
PCC	Plain Cement Concrete
PESO	Petroleum and Explosives Safety Organisation.
PLT	Pipeline Transfer
PMCC	Power cum Motor Control Centre
PNGRB	Petroleum & Natural Gas Regulatory Board.
POH	Periodic Over Hauling
PPE	Personal Protective Equipment
PVC	Poly Vinyl Chloride
QAP	Quality Assurance Plan
QCDC	Quick Connect Dry Coupling
QRA	Quantitative Risk Assessment
RCC	Reinforced Cement Concrete
RCCB	Residual Current Circuit Breaker
RDSO	Research Design and Standards Organisation
ROH	Regular Over Hauling
ROV	Remote Operated Valve
RQC	Random Quality Check
SCABA	Self-Contained Breathing Apparatus
SCADA	Supervisory Control and Data Acquisition
SMPV Rule	Static and Mobile Pressure Vessels (Unfired) Rules
SOP	Standard Operating Procedure
SRV	Safety Relief Valve
TBT	Toolbox Talk
TLD	Tank Lorry Decanting
TLF	Tank Lorry filling
TPIA	Third Party Inspection Agency
TREM	Transport emergency
TRV	Thermal Relief Valve
TSV	Thermal Safety Valve

DRAFT 2

TWG	Tank Wagon Gantry
UPS	Uninterruptible Power Supply
UVCE	Unconfined Vapour cloud explosion
VCB	Vacuum Circuit Breaker
VCWE	Valve Change Without Evacuation
VOT	Vapour Off Take
VTS	Vehicle Tracking System
WTI	Winding Temperature Indicator
XLPE	Cross-linked polyethylene

DRAFT 2

19.0 REFERENCES

The following codes, standards and publications have either been referred to or used in the preparation of this standard, and the same shall be read in conjunction with this standard.

a) **BIS CODES PERTAINING TO LPG OPERATIONS.**

IS 2:2022	Rules for rounding off numerical values
IS 5:2007	Colours for ready mixed paints & enamels
IS 1196:2023	Laying Bitumen Mastic Flooring - Code of Practice
IS 16514 (Part-1 & 2)	Platform weighing machine - Non-Automatic Weighing Instruments
IS: 1448(P: 70-2004)	Residue in LPG
IS: 1448(P: 71- 2018)	Vapour pressure of LPG
IS: 1448(P: 72- 2023)	Volatility of LPG
IS:1448(P:73- 2004)	Hydrogen sulphide in LPG
IS:1448(P:74-2022)	Dryness of LPG
IS: 1448(P: 76-2019)	Density of LPG
IS: 1448(P: 111-2025)	Analysis of LPG and propylene concentrates by gas chromatography
IS/IEC 60079-1:200	Explosive atmospheres part 1 equipment protection by flameproof enclosures “d”
IS: 2808-1964	Method for computation of capacity tables for pressurized storage tanks
IS: 2825-1969	Code for unfired pressure vessels
IS: 2827-1976	Methods for calibration of pressurized storage tanks
IS: 3196 (PART 1)-2013	Welded low carbon steel gas cylinder exceeding 5 litre water capacity for low pressure liquefiable gases cylinders for lpg.
IS: 3710-1978	Filling ratios for low pressure liquefiable gas.
IS: 4246-2002	Specification for domestic gas stoves for use with liquefied petroleum gases
IS: 4379-2021	Identification of contents of industrial gas cylinders
IS: 4576-2021	Specification for liquefied petroleum gases
IS:16280-2021	High Pressure Regulators and Adaptors for Use with Liquefied Petroleum Gases (LPG) — Specification
IS: 4991-1968	Criteria for blast resistant design of structures for explosions above ground
IS: 5116-2020.	General requirement for domestic and commercial equipment for use with LPG
IS 16724: 2018 IEC 60079-14: 2013	Explosive atmospheres - Electrical installations design, selection and erection
IS: 5572-2009 IEC 60079-10:2002	Classification of hazardous areas (Other Than Mines) having flammable gases and vapours for electrical installation
IS: 5572-2009	Classification of hazardous areas (Other Than Mines) having flammable gases and vapours for electrical installation
IS: 5844-2014	Recommendations for hydrostatic stretch testing of compressed gas cylinders

DRAFT 2

IS: 5903-2025	Types of Safety Device for Gas Cylinders and Gas Cylinder Valves
IS: 6044(PART 1)-2018	Liquefied petroleum gas storage installations - Code of practice: Part 1 residential, commercial and industrial cylinder installations
IS: 6044(PART 2)-2021	Industrial, commercial and domestic bulk storage installations
IS: 6665-1972	Code of practice for industrial lighting
IS: 7142-2025	Welded low carbon steel gas cylinder for low pressure liquefiable gases, not exceeding 5 ltr water capacity
IS/ISO 6789:2017	Assembly tools for screws and nuts - hand torque tools - requirements and test methods for design conformance testing, quality conformance testing and recalibration procedure
IS: 7241-2024	Glossary of terms used in gas cylinder technology
IS: 8198(PART 5)-2022	Code of practice for steel cylinders for compressed gases; liquefied petroleum gas (LPG)
IS: 8374-1977	Specification for bitumen mastic, anti-static electrically conducting grade
IS: 8451-2018	Periodic inspection and testing of seamless steel gas cylinders - Code of practice
IS: 8737-2017	Valve Fittings for Use with Liquefied Petroleum Gas (LPG) of up to 250 litre Water Capacity - Specification
IS: 8776-2024	Valve Fittings for Use with Liquefied Petroleum Gas (LPG) Cylinders up to and Including 13 Litre Water Capacity - Specification
IS: 8790(PART 1)-1978	General requirements of powered industrial trucks working in hazardous areas: Part 1 internal combustion engine powered trucks
IS: 8790(PART 2)-1979	General requirements of powered industrial trucks working in hazardous areas: Part 2 electric - Battery - Powered industrial trucks
IS: 9122-2023	Inspection Gauges for Checking Type 2 Taper Thread of Gas Cylinder Valves, Taper 3 in 25 — Specification
IS: 9200-2021	Method of disposals of unserviceable compressed gas cylinders
IS: 9281	Electronic weighing systems
IS: 9281(PART 1)-1979	Terms and definitions
IS:9281(PART 2)-1979	Methods of measurements
IS: 9281(PART 3)-1981	Requirements
IS: 9281(PART 4)-1983	Code of practice for use & installation
IS: 9639-2017	Code of practice for visual inspection of low pressure welded steel gas cylinders in use
IS: 9777-2023	Data sheet for selection of weighbridges for bulk handling equipment
IS: 9798-2013	Low pressure regulators for use with lpg
IS: 9902-2004	Recommended practice for leak testing
IS:17891-2023	O-rings
IS: 10733-2023	Electrically bonded road and rail tanker hose of rubber, resistant to petroleum product
IS: 10908-2023	Flexible rubber tubing for LPG
IS: 11498-2023	Geometrical Product Specifications GPS Dimensional Measuring

DRAFT 2

	Equipment: Dial Test Indicators Lever Type Design and Metrological Characteristic
IS: 12936-2017	Code for basic requirements for delivery persons engaged in the delivery of lpg cylinders
IS: 13258-2014	Welded low carbon steel cylinders exceeding 5 lt water capacity for low pressure liquefied petroleum gas code of practice for inspection and reconditioning of used lpg cylinders
IS: 13432-2003	Mechanical type gas leak detector for use with low pressure liquefied petroleum gas burning appliances specification
IS:15325-2020	Design and installation of fixed automatic high and medium velocity water spray system — code of practice
IS: 16054:2013	Periodic inspections and testing: welded low carbon steel cylinders exceeding 5 litres water capacity for liquefied petroleum gas (lpg) – code of practice
API Standard 2510	Design and Construction of LP Gas Installations
API PUBLICATION 2510A	Fire-Protection Considerations for the Design and Operation of Liquefied Petroleum Gas (LPG) Storage Facilities
ASME Boiler and Pressure Vessel Code Sect. VIII Division 1 American Society of Mech. Engrs., New York	
NFPA 58	Storage and Handling of Liquefied Petroleum Gases
NEPA 59	LP Gases at Utility Gas Plants
OISD-STD-105	Work Permit System.
OISD-STD-113	Classification of Areas for Electrical Installations at Hydrocarbon processing and handling facilities
OISD-STD-135	Inspection of Loading & Unloading hoses for Petroleum Products
OISD-STD-150	Design and Safety Requirements for Liquefied Petroleum Gas Mounded Storage Facility
OISD-RP-158	Recommended Practices on Storage and Handling of Bulk Liquefied Petroleum Gas (LPG)
OISD-STD-159	LPG Tank Trucks - Requirements of Safety on Design/ Fabrication & Fittings
ASME B - 31.3	Chemical Plant and Petroleum Refinery Piping
ASME B - 31.4	Liquid Transportation Systems for Hydrocarbons, Liquid Petroleum Gas, Anhydrous Ammonia and Alcohols
PD-5500	Specification for unfired fusion welded pressure vessels

DRAFT 2

Annexure-4

STATUTORY RULES AND REGULATIONS

Below is the list of various regulation applicable to LPG installation. However, this list is not exhaustive. Provision of all the statutory acts and rules and regulation frames thereon shall apply as per the extant version, notwithstanding the year mentioned against them anywhere in the standard.

1) The Gas Cylinder Rules, 2016 (Under Explosives Act, 1884)

2) The Static & Mobile Pressure Vessels (Unfired) Rules, 2016 (under the Explosives Act 1884).

3) The Factories Act, 1948 (with latest amendments)

4) Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023

5) The Legal Metrology (General) Rules, 2011 under the Legal Metrology Act, 2009. (With latest revisions and amendments)

6) Legal Metrology (Packaged Commodities) Rules, 2011 (With latest revisions and amendments) - Legal Metrology (Packaged Commodities) (Amendment) Rules, 2023

7) Liquid Petroleum Gas (Regulation of Supply & Distribution) Amendment Order (With latest revisions and amendments)

8) The Motor Vehicles Act,1988 with Central Motor Vehicle Rules (With latest revisions and amendments)

9) Environmental Regulations:

- Water (Prevention & Control of Pollution) Act 1974 - with Water (Prevention and Control of Pollution) Rules (With latest revisions & amendments)
- Air (Prevention & Control of Pollution) Act, 1981 - with Air (Prevention and Control of Pollution) Rules, 1982. (With latest revisions & amendments)
- The Environment (Protection) Act, 1986 – with Environment (Protection) Rules, 1986 - (With latest revisions & amendments)
- Manufacture, Storage and Import of Hazardous Chemicals Rules, 1989.

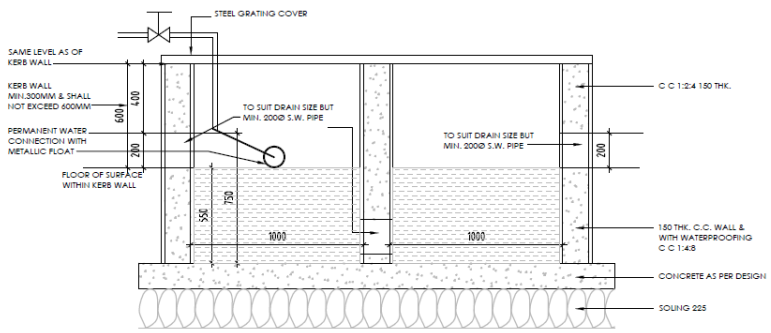
10) The Hazardous and Other Wastes (Management and Transboundary Movement) Rules,2016 Amendment Rules, 2022. (With latest revisions & amendments)Petroleum and Natural Gas Regulatory Board (Codes of Practice for Emergency Response and Disaster Management Plan (ERDMP)) Regulations, 2010 with 2nd Amendment on 17.09.2020

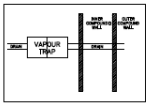
11) Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for LPG Storage, Handling and Bottling Facilities) Regulations, 2019. (T4S Regulations)

DRAFT 2

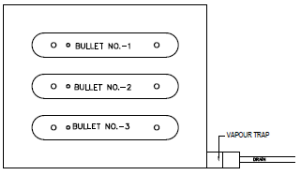
Annexure-5- I

TYPICAL DESIGN FOR CONSTRUCTION OF WATER SEAL

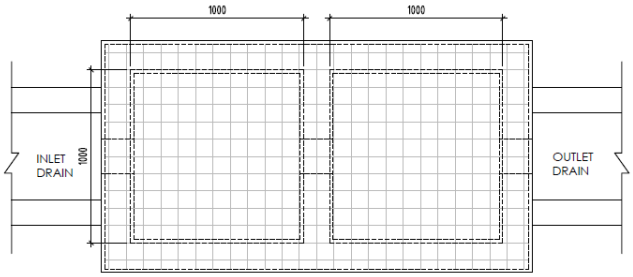


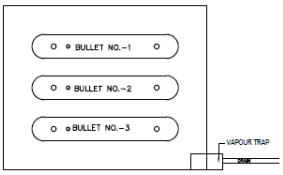


LOCATION OF VAPOUR TRAP AT DRAIN

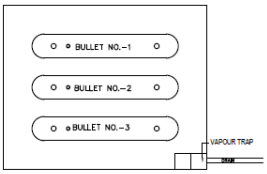


LOCATION OF VAPOUR TRAP





LOCATION OF VAPOUR TRAP - ALTERNATE-1



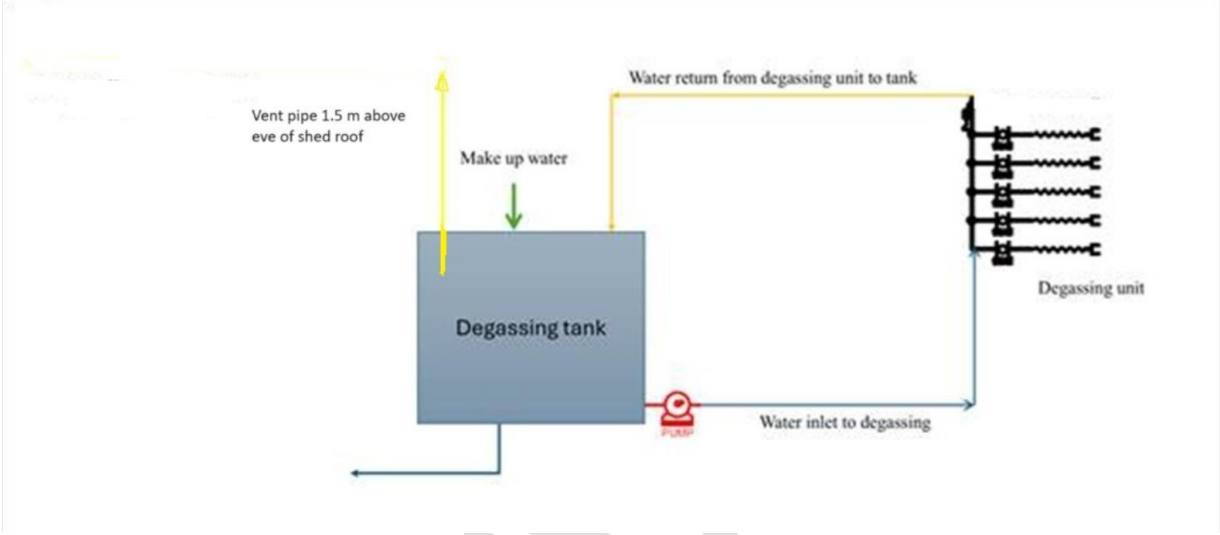
LOCATION OF VAPOUR TRAP - ALTERNATE-2

NOTES:

* ALL DIMENSIONS ARE IN MM
* PROPER WATER SUPPLY ARRANGEMENT TO BE PROVIDED NEAR VAPOUR TRAP TO MAINTAIN WATER SEAL
* OPENING OF 200MM DIA. CAN BE INCREASED TO SUIT THE SIZE OF DRAIN
* DRAIN TO BE ADEQUATELY SIZED TO PREVENT WATER FROM OVER FLOWING THE KERB WALL DURING MOCK DRILLS OR FIRE FIGHTING ACTIVITIES.

Annexure- 5-II

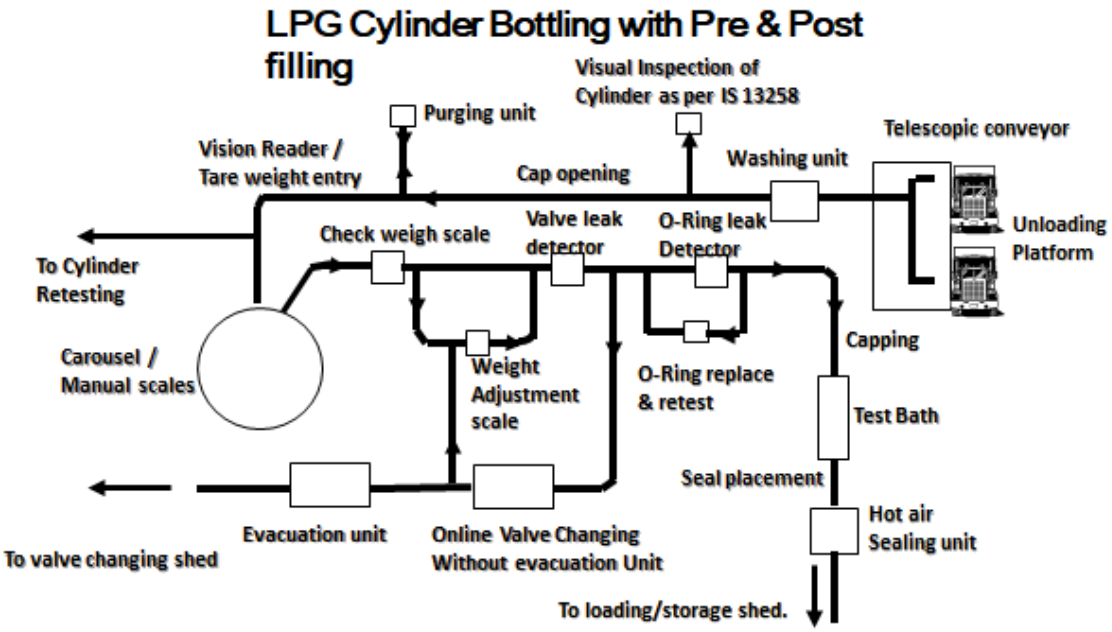
TYPICAL DESIGN FOR CLOSED LOOP DEGASSING ARRANGEMENT



DRAFT

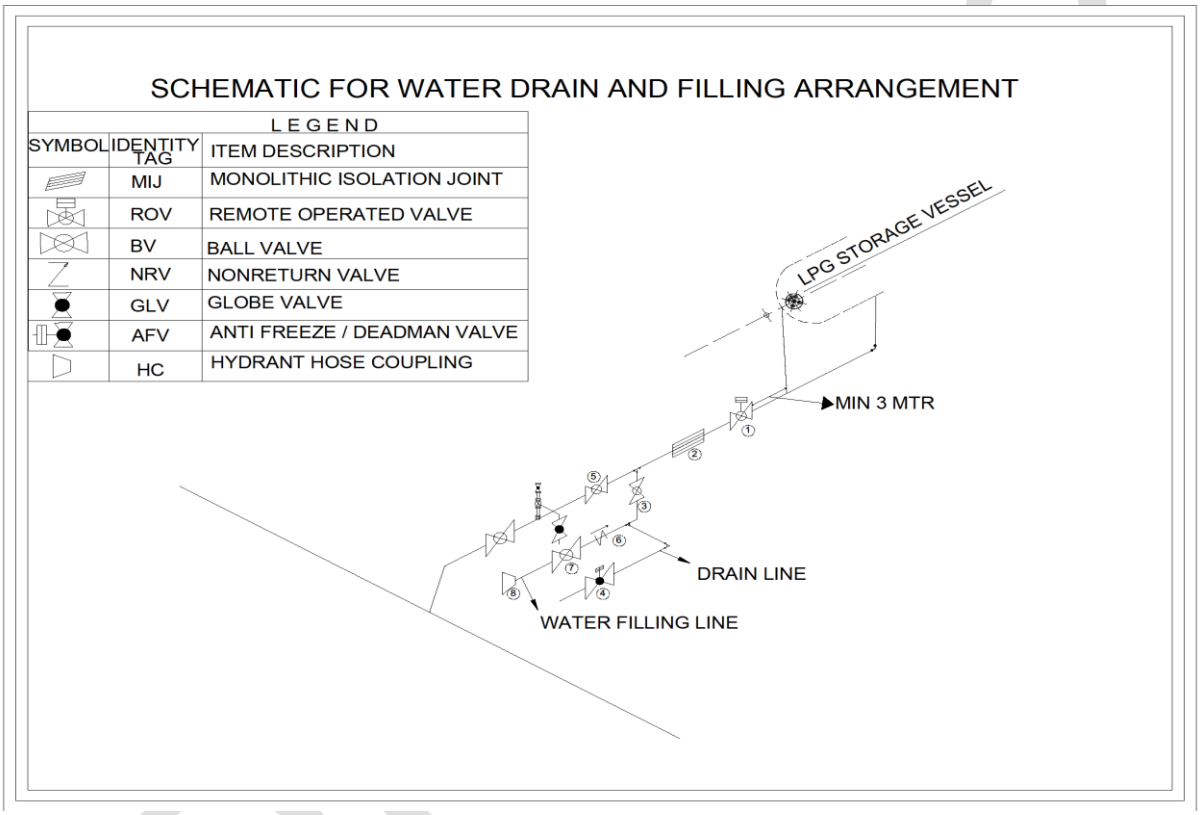
Annexure-5- III

SCHEMATIC DIAGRAM FOR LPG BOTTLING OPERATION



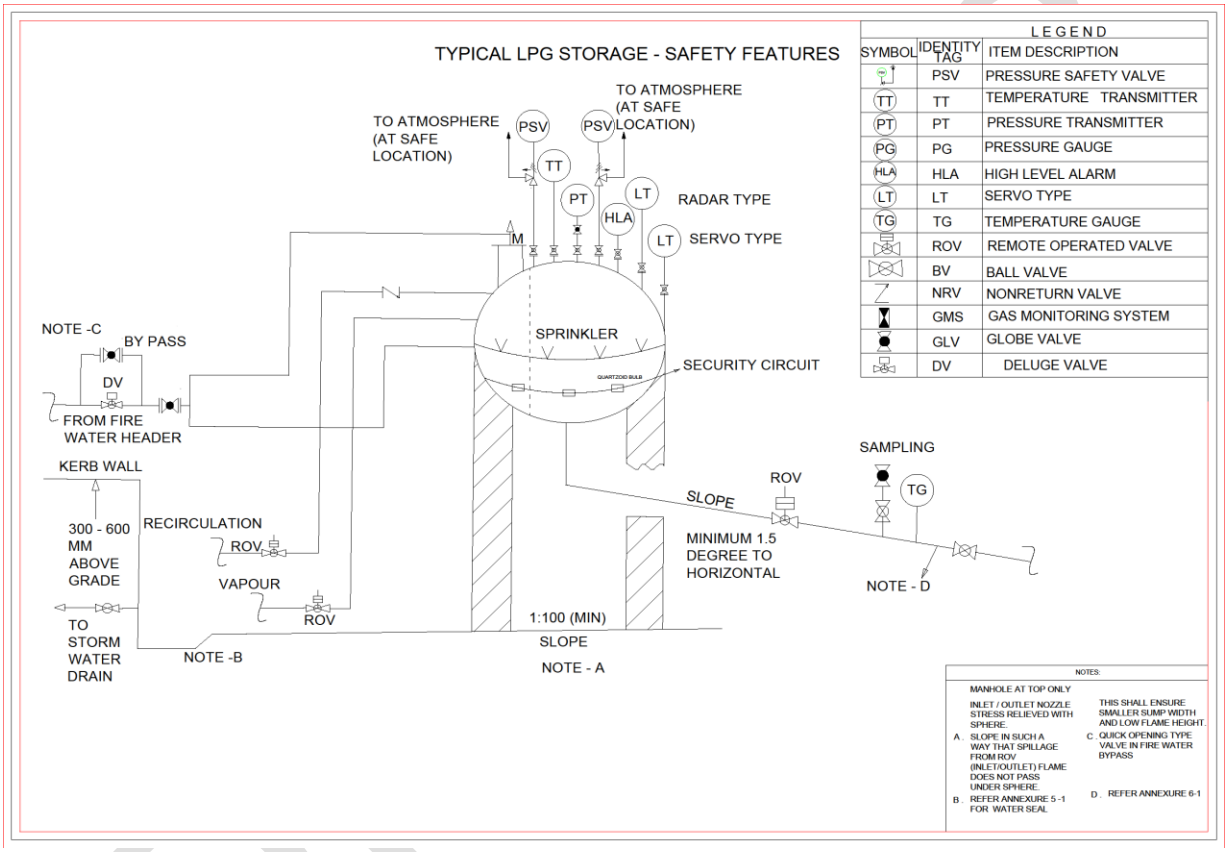
Annexure- 6-I

TYPICAL PIPELINE CONNECTION FOR WATER DRAIN ARRANGEMENT IN LPG STORAGE VESSEL



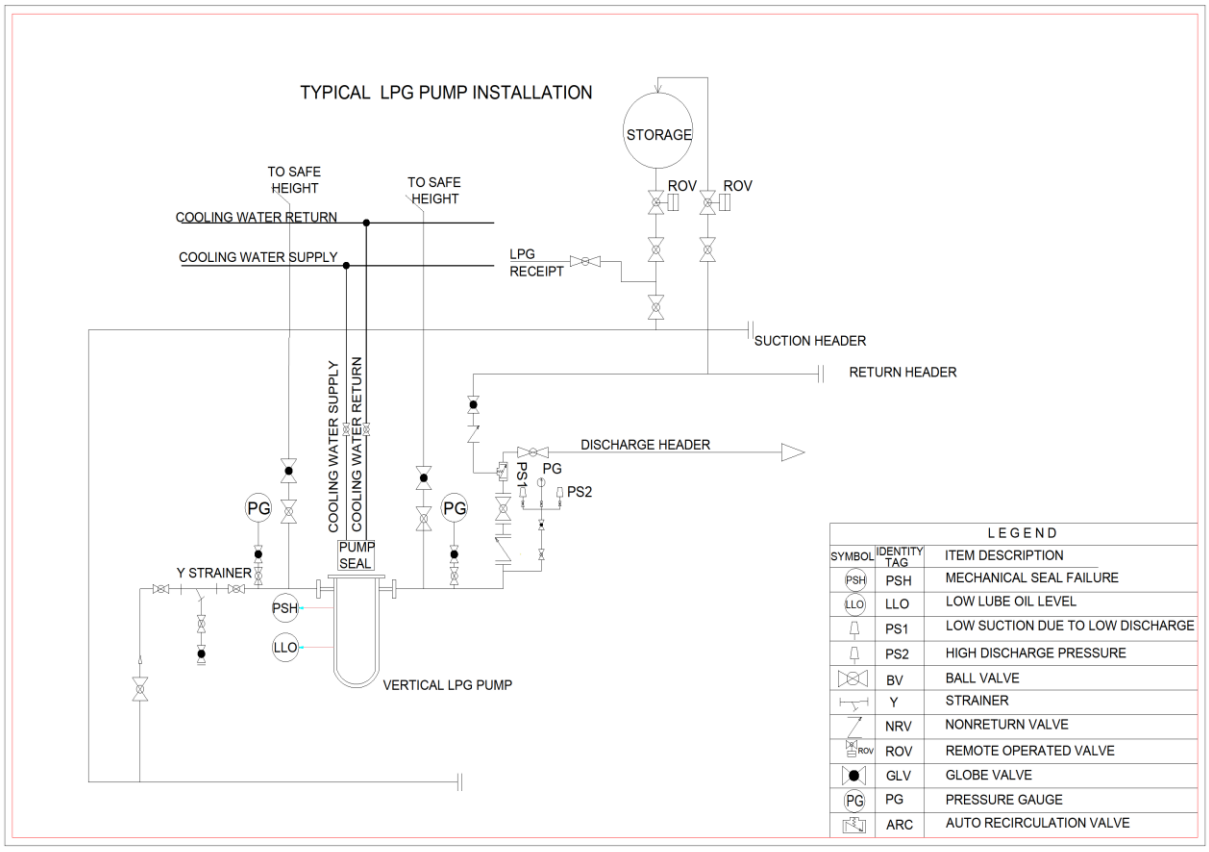
Annexure- 6-II

TYPICAL SAFETY FITTINGS ON LPG STORAGE VESSEL



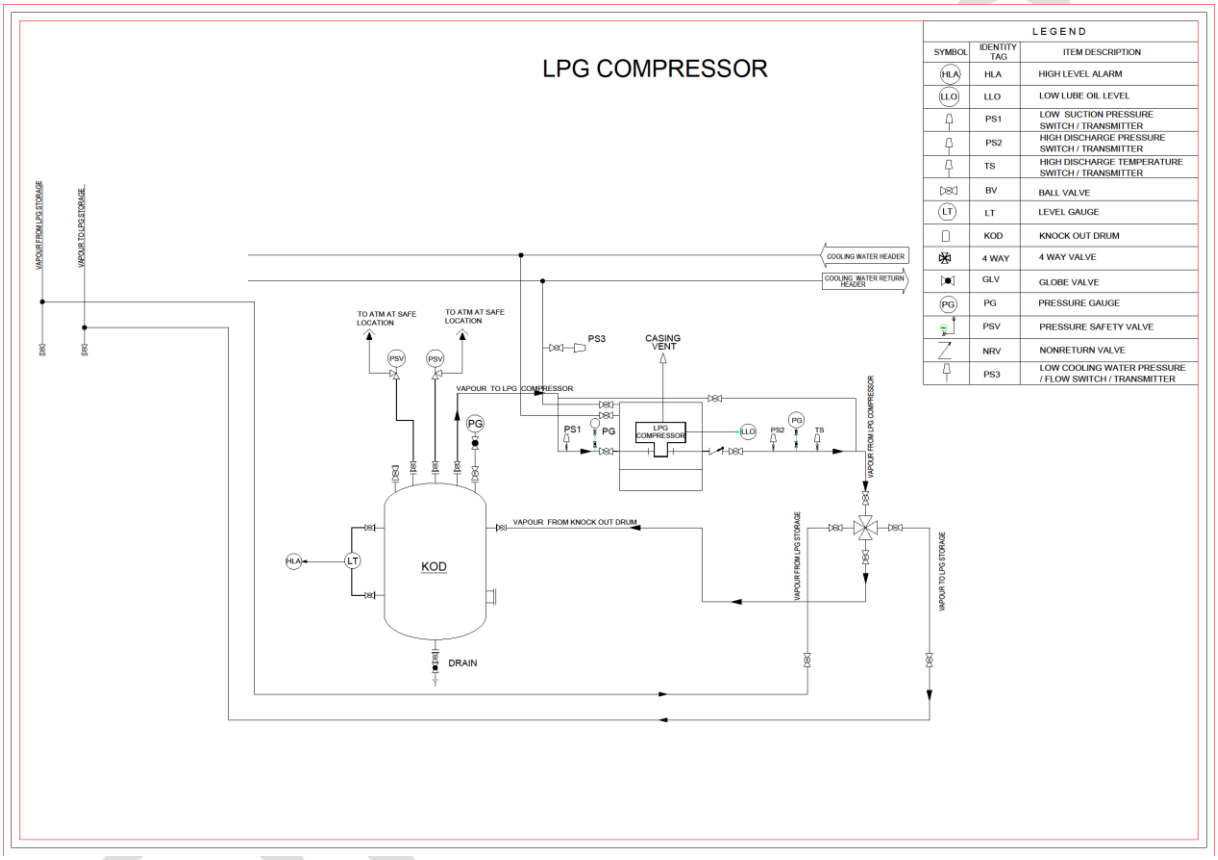
Annexure- 6-III

TYPICAL BOTTLING PUMP INSTALLATION



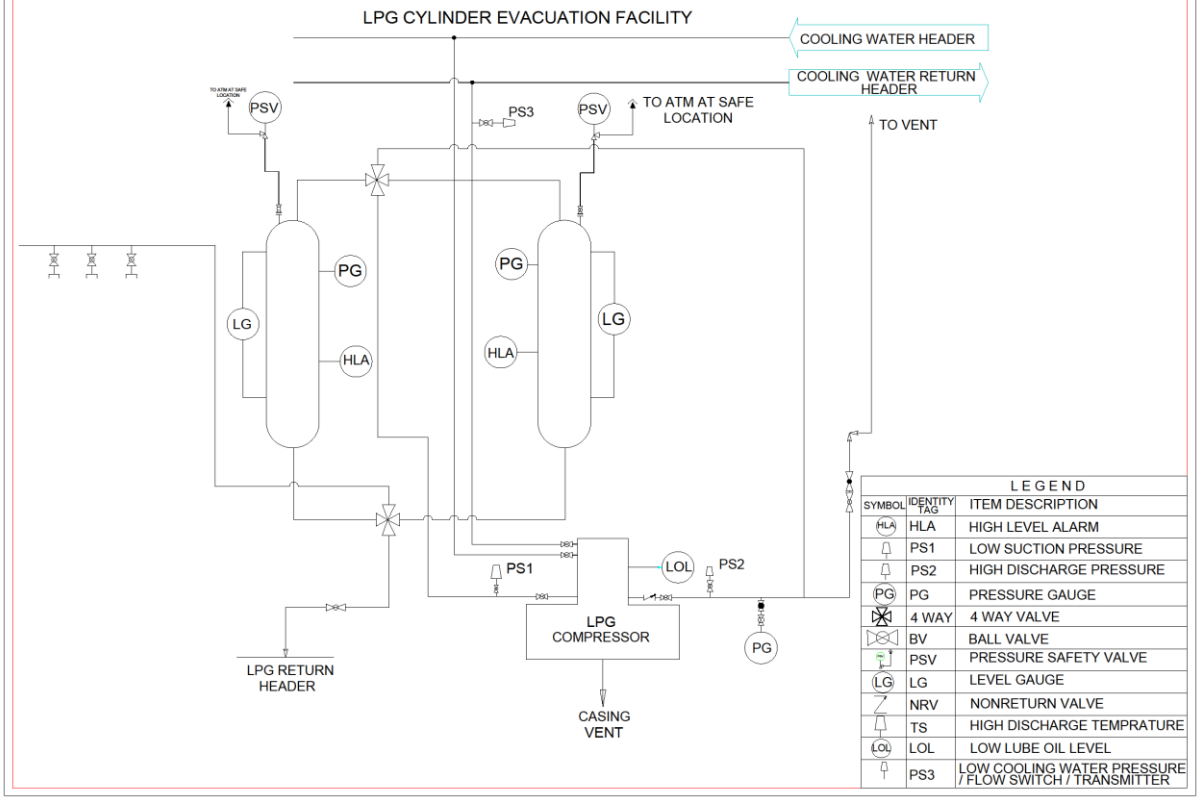
Annexure- 6-IV

TYPICAL COMPRESSOR INSTALLATION



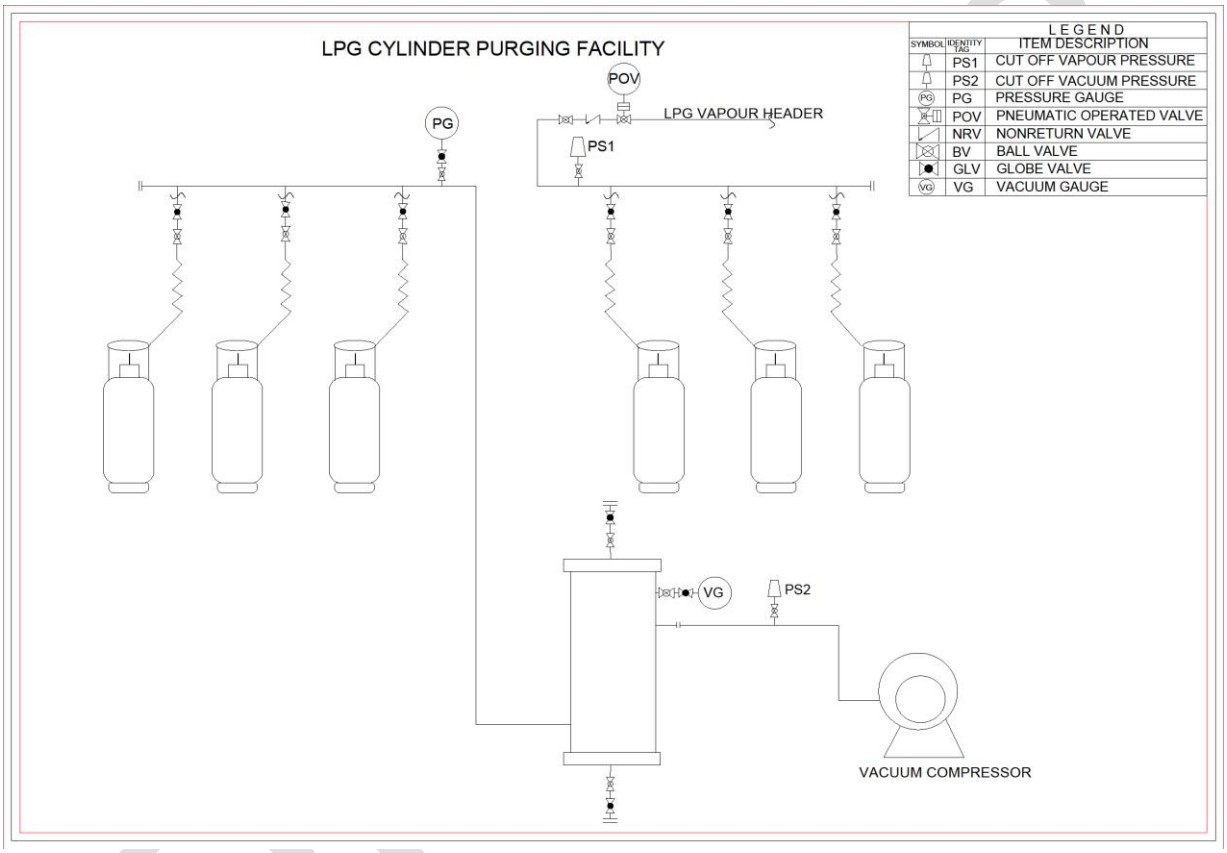
Annexure- 6-V

TYPICAL CYLINDER EVACUATION FACILITY



Annexure- 6-VI

LPG CYLINDER PURGING FACILITY



Annexure- 6-VII

DEGASSING & GAS-IN PROCEDURES OF BULK LPG STORAGE VESSELS

The following steps are to be followed before handing over of Storage Vessels for Hydro testing:

1. Hot work permits and JSA will be separate for depressurisation, degassing, vessel entry, NDT hydrotesting, boxing up, charging of the vessel, and water draining till such a time, normal operation of the vessel is taken up.
2. No work shall be carried out in the storage vessel in the absence of the Permit receiver and authorized representative of the contractor.
3. All the works pertaining to a storage vessel should be carried out under the strict supervision of a competent person who should be present during the entire operation.

Preliminary Procedure and Emptying out Liquid LPG:

1. PESO Construction Approval and final as-built approval before recommissioning the vessels shall be obtained for any hot repair on storage vessels or replacement of fittings in line with SMPV Rules.
2. Rule 33 certificate to be obtained from PESO recognised competent person after completion of the works, in case of hot works/ vessel modifications are carried out.
3. PESO approved procedure and TPIA approved QAP should be available and followed for vessel modifications that involve hot works.
4. PESO approved procedure and approved QAP to be available for replacing/ changing the existing fittings (which require degassing of the vessel) on vessels.
5. MOC approval to be obtained before carrying out repairs to storage vessels, replacement/ change of fittings/ instruments, including ROVs, gauges, etc.
6. A joint inspection and detailed discussions to be carried out for finalizing dos and don'ts, procedure to be followed, site conditions, operation of various fittings, instruments, etc. This has to be ensured at the site by the Location in-charge, Safety Officer, Maintenance Officer, Permit receiver, Contractor, Contractor's authorized representative and workers to be deployed by the contractor.
7. Contractor's authorized representatives, workmen who would carry out the work, employees associated with work, officers who are responsible for the subject work, shall be given training on important do's and don'ts, procedure, safety precautions, P&ID, work permit system, JSA, guidelines as applicable and same to be documented. Signed copies of documents are to be kept as training records. It must be ensured that the workmen being deployed for undertaking critical activities are competent and trained to undertake the task. A test check w.r.t. competency level should be carried out before the commencement of the job.
8. A site register with serially numbered pages shall be maintained at the site for storage vessels testing and repair works for recording daily activities.
9. All activities, including Depressurization, Degassing, Vessel entry, NDT, and Hydro Testing, to be carried out only during daylight hours.
10. Ensure that PPEs, including Personal Life Saving equipment such as safety harness, safety lifeline SCABA, rescue tripod, etc., are available and used as required, and all personnel are well acquainted with the usage of safety equipment.

DRAFT 2

11. Pressure gauges used for Hydrotesting shall have a valid calibration certificate, and the same shall be ensured in advance.
12. Caution boards and barricades shall be ensured right from day one.
13. Ensure the usage of only non-sparking tools for all work connected with the statutory testing of storage vessels.
14. Toolbox Talk to be imparted to all the maintenance/ contractor personnel at the beginning of each shift before giving clearance for maintenance activity and as and when felt necessary by the safety/ maintenance officer.
15. Only approved reduced voltage lights, not exceeding 24 volts, are to be allowed inside the vessel for consideration of personal safety.
16. In case of statutory testing, the vessels to be recommissioned after issuance of the Rule 19 certificate by a PESO competent person.

Process:

17. Storage Vessel should be emptied out by transferring the Liquid LPG to another storage vessel or by giving the vessel for bottling/ TLF till suction failure of the LPG pump.
18. Both level gauges of the storage vessel shall be monitored for Liquid LPG quantity.
19. It shall be ensured that Liquid LPG is emptied out in total by the LPG Pump initially and then by transferring the Liquid LPG to another vessel by the pressure differential method.
20. It is to be ensured that the vessel is free of LPG Liquid by opening the water drain line to check the presence of liquid LPG, duly following SOPs for water draining.
21. Depressurize LPG Liquid in the spool piece between ROV and first isolation valves after ROV using the water drain line by closing both ROV and first isolation valves.
22. Ensure that LPG vapour is not available in the spool piece by opening the drain line before inserting the blind in the flange joint of the first isolation valve (manual globe/ gate valve) to ensure positive isolation of the vessel from the LPG liquid manifold.
23. Reduce the vapour pressure in the storage vessel up to 1.5 kg/cm² pressure either with the help of LPG vapour compressors or by interconnecting the vessel under repair/ testing with another vessel having a lesser vapour pressure. Vapour compressors shall be used for reducing the vapour pressure after equalization of the vapour pressure in both the vessels.
24. Connect leak-free hydrant hose, which was hydrotested at 10.5 kg/cm² with required end fitting/ couplings and NRV in the water drain line for water filling into the vessel. Ensure that the Liquid ROV of the vessel is in auto mode for immediate closure in case of any unforeseen emergency.
25. Start the pumping of water and ensure that water pressure is always 2 kg/cm² more than the LPG vapour pressure of the storage vessel into which the water is being pumped.
26. Fill water in the drain line up to ROV, open the ROV when water pressure in the drain line is 2 kg / sq cm (minimum) more than the vapour pressure of the vessel and continue with the process of water filling in the storage vessel up to 85% gradually.

Note: The above process is given, assuming that the water is pumped into the vessel using the fire hydrant pumps and the water pressure at the inlet of the vessel is more than 7 kg/cm².

27. During the process of pumping water, vapour from the vessel shall be withdrawn continuously either with the help of vapour compressors or by interconnecting the vessel under maintenance

DRAFT 2

with another vessel having a lesser/ equal vapour pressure. During the process, it is to be ensured that the vapour pressure inside the vessel (which is being filled with water) is not more than 5 kg/ sq cm at any point in time. In case the vapour pressure is more than 5 kg/cm², the water pumping is to be stopped, and the vapour pressure is to be brought down below 5 kg/cm², since the hydrant pressure is considered as 7 kg/cm².

Note: SERVO gauge in the storage vessel if present, needs to be hoisted up and close the isolation valve before starting the water pumping.

28. Continuously monitor the water level in the storage vessel with the help of other gauges and a fixed tube gauge.
29. Stop filling of water when the level in the storage vessel reaches 85% and reduce the vapour pressure to 1.5 kg/cm² with the help of vapour compressors. Keep constant vigil on the liquid level of knock-out drums/ suction bottles of vapour compressors during the process, and more so when the water level in the vessel is at 85%.
30. Close Vapour ROV and other isolation valves of vapour recovery and inlet lines of the storage vessel and insert a blind in the vapour line near the second isolation valve and isolate the vapour line of the vessel from the manifold. Remove the spool piece and EFCV if available in the vapour line.
31. Close the isolation valves of SRVs and remove one of the SRVs from the storage vessel.
32. Start venting vapour through SRV nozzle opening under controlled condition by duly ensuring that the LEL % is not more than 20% in the vicinity of storage vessel, at bottom and around the vessels in dyke area. The process of venting continues up to “0” kg/cm²/ till the storage vessel pressure is near atmospheric pressure. Venting of vapour shall be stopped, if the concentration of vapour is found to be more than 20% of LEL in the vicinity of storage vessels/ dyke.
33. Restart the pumping of water while venting is continued. However, the venting and pumping of water is to be stopped if the vapour concentration in the vicinity is found to be more than 20% of LEL. The process is to be continued till the vessel is overflowed with water through the opening of the SRV nozzle. Retain water in the vessel for at least 60 minutes after allowing overflow of water through the SRV.
34. Stop and restart water pumping intermittently 3 to 4 times to eliminate the possibility of vapour pockets.
35. Drain the water through the water drain valve for 15 minutes. Then top up water and allow water to overflow through the SRV/ brim of the vessel.
36. Give clearance for removing measuring instruments, gauges, transmitters, except the manhole cover, to avoid accidental fall of personnel inside the vessel.
37. Ensure that the vessel is filled with water up to brim level at least for 2 hours, including the time taken for removing the gauges/ instruments/ fittings as above.
38. Issue clearance for opening the Manhole Cover, duly following the hot work permit.
39. A suitable mesh shall be provided on the top of the manhole to avoid accidental man entry.
40. Start the water pumping again and ensure that the water overflows through all the nozzles for at least 10 minutes to remove vapour pockets left in other nozzles.
41. Drain the water at a controlled rate from the storage vessel by keeping the SRV nozzle open for air entry into the vessel and to avoid vacuum lock. The isolation valve of the SRV is to be kept in the open condition to enable atmospheric air to flow into the vessel. As a precaution, a vacuum

DRAFT 2

gauge should be temporarily fitted at the top of the spheres for the operation to ensure that the pressure inside the vessel is atmospheric.

Note: Necessary precautions shall be taken to let out the water directly into the vapour trap and then into the drain, without accumulating/ stagnating near the foundation/ mound of the vessels, which otherwise may affect the foundation

42. Adequate FLP JBs, connections with double compression FLP glands shall be provided for exhaust fans and FLP lighting for the vessel.

43. Ventilate the vessel either thru FLP air blower/ FLP exhaust fan fixed on one of the manholes by keeping all other piping and instrument nozzles open for two days (blowers/ exhaust fan to be operated during daytime and natural ventilation is allowed during night hours with all safety precautions to avoid accidental entry of animals/ birds/ personnel inside vessel thru any of the nozzles or manhole.

44. Oxygen, CO, and LEL levels at the bottom of the vessel shall be checked using an appropriate intrinsically safe multi-gas detector having sufficient probe length/ data logging facility. The probe or explosimeter is to be lowered into the vessel up to the bottom of the vessel without man entry.

45. The required safe limits of various gases for man entry shall be as below:

a. Oxygen content should not fall below: 19.0%

b. HC: less than 5 % of LEL

c. CO: 9 ppm

Note: The forced/ natural ventilation process as above is to be repeated till such time the gas concentrations as above are reached. Forced ventilation may be preferred by taking all precautions to reduce the time of vessel outage.

46. Take repeated readings at an interval of 2 hours for a total period of 6 hours to ensure the gas concentrations are within safe limits continuously before clearing the vessel for man entry.

47. A mechanical retrieval system, such as Tripod with Winch and Leg support/ safety harness with lifeline of sufficient length, shall be used for man entry into the vessels. It is to be ensured that there are always two personnel outside the vessel to keep watch on the personnel inside the vessel.

a. The length of the lifeline must be sufficient to allow the person inside the tank to move around.

b. Provision of an Airline breathing apparatus to be made before vessel entry for emergency rescue. It is advisable to enter in confined space/ vessel with the help of an airline breathing apparatus (Cylinders/ Supplied Airline Respirator system are placed outside of the tank, and only a mask is used by the person, and supply of air is through the tube)

Note: Designated Safety Officer, all the officers associated with the work, and the receiver of the permit (of the vendor) must be trained on checking the critical parameters.

48. Oxygen level to be measured near the manhole and inside the vessel.

49. The supervisor deputed by the Contractor must be trained and experienced enough for testing LEL, Oxygen, CO at the bottom of the tank and allied facilities using multi gas detectors without man entry into the vessel. Safety Officer and all officers associated with the subject work shall be fully conversant with using the suggested PPEs and SCABA.

a. The calibration certificate of the multi gas detector should be valid and available at the site.

DRAFT 2

b. Location-in-Charge and Safety Officer, both shall give clearance jointly for man entry after ensuring that the gas concentrations are within the safe limits and recording the same in vessel entry/ hot permit and site register. The onus of ensuring safe working conditions and necessary precautions at the worksite until permit closure rests with both the permit issuer and receiver.

c. At the time of vessel entry after degassing of vessels during periodic testing or any other maintenance work, suitable equipment such as a hose mask with an air blower/ Supplied Airline Respirator system and a full body safety harness with double lanyard shall be used.

Cleaning NDT and Hydro testing of the vessel:

50. Necessary scaffolding must be ensured before internal cleaning.
51. Carry out internal cleaning by using brass brushes/ plates for NDT, duly ensuring PPE, safety harness as required at the site.
52. No unauthorized personnel shall be permitted in the storage vessel area, which is under statutory testing.
53. Carry out servicing/ calibration/ repair of Servo Gauge, Radar Gauge, any other gauges, High level alarm, Vessel's fittings as required.
54. Carry out servicing and testing of the ROVs of the vessel.
55. Sludge to be collected, kept wet and kept in a pit in a wet condition till disposal. Sludge should be disposed of as per prevailing statutory/ PCB Guidelines.
56. Ensure all NDT, Repairs, and Servicing, etc., are complete in Toto. The storage vessel to be handed over for water filling and hydro testing as per the instructions/ clearance by a PESO competent person.
57. Ensure that no person/ unwanted materials are left inside the vessel before giving clearance for water filling.
58. Manhole cover should be fixed back before filling water in the presence of the Location-in-Charge and Designated Safety Officer. Document for boxing up the vessel to be signed by two officers as assigned by Location-in-Charge for personally inspecting and giving clearance.
59. Refix the spool piece between the liquid ROV and the first isolation valve for water filling and keeping the other end of the second valve (first isolation valve after the ROV) dummied for positive isolation.
60. All used gaskets and studs are to be replaced with new ones.
61. Blind all the nozzles of the vessel, except the SRV, without tightening for venting the air while pumping in the water.
62. Start filling water in the storage vessel through the water drain line till the vessel is filled 100%.
63. Ensure that the isolation valve of the servo gauge is kept blinded with the spectacle blind/ servo gauge removed during the process of water filling in the storage vessel.
64. Tighten the blinds provided on the nozzles/ openings. Before tightening the blinds, ensure that there are no air pockets inside the vessel by allowing the water to overflow till such time no air pockets are observed.
65. Clearance be given for Hydrotesting by a PESO Competent person who will witness the Hydro Testing and an authorised Officer in writing to this effect in the Site Register and work permit for Hydrotesting.

DRAFT 2

66. Start pumping the water inside the tank to increase the tank pressure. The rate of increase of pressure should not be more than 2 kg/cm²/ minute. Hydro Testing Pressure to be retained for 60 minutes.
67. Hydro testing shall be witnessed by a PESO authorized Competent Person along with other officer(s) as authorized by Location-in-Charge.
68. Ensure that the hydro testing pressure is as mentioned in the control certificate of the vessel, which was duly vetted by TPIA.
69. After Satisfactory completion of Hydro Testing, clearance to be given for releasing the water pressure by an authorised Officer, PESO authorized Certifying agency concerned.
70. Commissioning of storage vessel after hydrotesting. Start releasing the hydro pressure slowly by gradually opening a drain valve. Do not reduce the water pressure by more than 2 kg/cm² per minute while releasing the hydrotest pressure.
71. Remove all the blinds where the connections of vapour lines, instruments, valves, etc., are to be fixed. The blind in the liquid line is not to be removed at this stage and ensure that the vessel is full of water and is not drained at this stage.
72. Fix all the instruments, valves, vapour connections, etc.
73. Crack open the vapour line in a controlled manner so that the pressure inside the vessel is gradually increased to 2 kg/cm² and check for any hissing sounds/ leaks of vapour through the connected instruments/ valve joints after the hydrotest. Any leaks/ hissing sounds are to be immediately corrected by tightening the studs. In case the same could not be arrested, then the vessel needs to be depressurized again after blinding the vapour line. Serrations of leaking flange joints to be cleaned, clean MP grease to be applied to serrations, a new gasket to be inserted, and the studs to be tightened.
74. Repeat the process as above till such time it is ensured that there are no leaky joints in the storage vessel.
75. Keep removing/ draining the water through the water drain line in a controlled way. Maintain vapour pressure at 1.5 to 2.0 kg/cm² only during the process of replacing the water with LPG Vapour. Ensure Nil water in the storage vessel by repeatedly draining the water through the water drain line of the storage vessel, duly following the water drain procedure.
76. Close the ROV of the liquid line and all the valves of the drain line once it is ensured that there is no water in the storage vessel.
77. De-hoist the Servo Gauge by taking all precautions such as fully opening the isolation valve and following the SOP/ manualized instructions of the manufacturer.
78. Remove the blind in the liquid line near the first isolation valve after ROV of the vessel, once it is ensured that there is no water in the vessel.
79. Crack open the first isolation in the liquid line and let liquid flow in from another vessel or from the LPG receipt line. Record the reading in the liquid level gauges periodically.
80. Close the liquid line valves once the vessel gauge readings show 10% of the liquid level. Allow the product to settle for one hour and check whether there are any traces of water in the vessel. After making sure that there are no traces of water, by repeated water draining, the vessel to be put into regular use.

Note: Responsible person to be available at site during all activities of storage vessel testing including depressurization, degassing, removal of vessel fittings, removal of manhole cover, Vessel entry,

DRAFT 2

hydrotesting, fixing of storage vessel fittings, boxing up, vapour intake after Hydrotesting (purging in), liquid intake, water draining and while taking the vessel for first time operation of Bottling/ Filling/ Bulk Loading.

DRAFT 2

Annexure-7-I-A

CHECKLIST FOR BULK LPG TANK TRUCKS AT UNLOADING LOCATION

S.No. ……..

Date ……

Time ……

Bulk LPG Tank Truck No. ……...

Driver Name…….....

Transport Name……

Loading Location……

Sr. No.	When	Parameter	CHECK POINTS FOR BULK LPG TANK LORRY	Status	Remarks
1	Before Entry	Electrical	Readily accessible master switch with di-pole wiring of minimum 300 Amp rating for switching off the engine is provided inside the cabin.	Yes/ No	
2	Before Entry	Electrical	Truck is self-starting.	Yes/ No	
3	Before Entry	Electrical	a) All Junction boxes are properly sealed. b) Any loose electrical wiring terminal?	Yes/ No	
4	Before Entry	Electrical	Electrical wiring is insulated and provided with suitable overcurrent protection	Yes/ No	
5	Before Entry	Electrical	1) The battery shall be mounted in an easily accessible position. 2) Battery, if mounted outside, shall be kept in well well-ventilated box with a lock. 3) Battery terminals shall be electrically insulated, or the battery shall be covered by an insulating cover.	Yes/ No	
6	Before Entry	Tank Fitting	Leakage from any fittings or joints	Yes/ No	
7	Before Entry	Tank Truck	Properly filled spark arrestor of make and design approved by CCE, Nagpur?	Yes/ No	
8	Before Entry	Tank Truck	No leakage in exhaust silencer pipe exists.	Yes/	

DRAFT 2

				No	
9	Before Entry	Tank Truck	a) Two portable IS marked 9 Kg. DCP fire extinguishers, one on each side and accessible from outside the cabin. b) One DCP extinguisher of 1 Kg capacity in the driver's cabin c) All extinguishers marked for next due date for pressure testing. d) The TT number painted on each fire extinguisher.	Yes/ No	
10	Before Entry	Tank Truck	No Visible dent on the bullet?	Yes/ No	
11	Before Entry	Tank Truck	Provision of a quick closing manifold valve with a lever indicating closed and open status.	Yes/ No	
12	Before Entry	Tank Truck	Proper fittings, SRV, IEFCV (Internal excess flow check valve), temperature and pressure gauges, etc., as per extant SMPV (U) Rules and OISD-STD-159?	Yes/ No	
13	Before Entry	Tank Truck	Air brakes are not bypassed.	Yes/ No	
14	Before Entry	Tank Truck	A dedicated second crew is available in the tank truck.	Yes/ No	
15	During Unloading	Tank Truck	Vehicles are protected by a robust steel bumper with rear and side under run protection devices	Yes/ No	
16	During Unloading	General	a) When was the vessel last hydro tested. b) Whether the vehicle carries the half-yearly and annual test certificates for the fittings, including SRV, IEFCV? c) Certificates of bullet hydro testing, IEFCV and SRV testing from a competent agency?	Yes/ No	Mention Test Dates and valid till
17	During Unloading	General	a) Non-sparking tools for all fittings is available. b) First aid kit is available. c) Whether the Emergency Tool kit as per OISD-GDN-161 is available?	Yes/ No	
18	During	General	4 nos. caution sign boards with luminous paint for stopping traffic in case of any	Yes/ No	

DRAFT 2

	Unloading		emergency?		
19	During Unloading	General	Whether the vehicle is provided with retractive type seat belt and driver is using it?	Yes/ No	
20	During Unloading	General	Whether the vehicle tyres in good condition?	Yes/ No	
21	During Unloading	General	Whether the carrier has been provided with Anti-lock braking system (ABS)?	Yes/ No	
22	During Unloading	General	Whether VTS is available and operational	Yes/ No	
23	During Unloading	General	Whether the Carrier Legal Liability Insurance and Public Liability Insurance are available with Tank Truck?	Yes/ No	
24	During Unloading	General	Whether the driver of TT has undergone 3-day training and one day refresher training?	Yes/ No	Date of training
25	During Unloading	General	Driver's medical certificate- no night/ colour blindness	Yes/ No	
26	During Unloading	General	Carries RTO permits for hazardous goods and a fitness certificate?	Yes/ No	Valid till
27	During Unloading	General	a) Carries TREM CARD, an instruction booklet detailing instruction on handling emergencies on route? b) Whether the TREM card contains all the important phone numbers, including "112" - Emergency response system of GOI - the Single Emergency number?	Yes/ No	
28	During Unloading	General	Carries Route Map?	Yes/ No	
29	During Unloading	General	RLW-ULW > Licensed capacity?	Yes/ No	
30	During Unloading	General	Endorsement in the driving licence by licensing authority to the effect that he is authorised to drive a goods carriage carrying goods of dangerous or hazardous nature to human life.	Yes/ No	Valid till…
31	During	Tank	Liquid/Vapour pipelines are adequately anchored and well protected by means of a	Yes/ No	

DRAFT 2

	Unloading	Fitting	stout steel guard?		
32	During Unloading	Tank Fitting	a) Liquid/ vapour pipelines are in a single piece from the internal excess flow check valve (IEFCV) to the discharge valves. b) All safety fittings viz. safety valves, roto gauge, pressure gauge and temperature gauge are adequately protected.	Yes/No	
33	During Unloading	Tank Fitting	Operative fittings like roto gauge, pressure gauge and temperature gauge are operational?	Yes/ No	
34	During Unloading	Tank Fitting	a) The bullet has no sign of external corrosion b) The paint of the bullet is not peeling off?		
35	During Unloading	Tank Fitting	Flanges and bend pipes on liquid/ vapour lines are rated of 300 class.	Yes/ No	
36	During Unloading	Tank Fitting	The fuel tank is protected by means of a stout guard, and the fuel tank cap is locked.	Yes/ No	
37	During Unloading	Tank Fitting	Height barrier provided as per specifications.	Yes/ No	
38	During Unloading	Tank Fitting	SRV fitting date. (To ensure SRV is replaced in 10 years)	Date of fitting	
39	During Unloading	Tank Fitting	IEFCV fitting date. (To ensure IEFCV is replaced in 10 years)	Date of fitting	
40	During Unloading	Tank Fitting	Has a valid CCE license and an authenticated copy of the drawing?	Yes/ No	Valid till
41	During Unloading	Tank Fitting	The manufacturer's name plate with the date of testing is fitted on the vessel	Yes/ No	
42	During Unloading	Tank Fitting	HAZCHEM Sign, name of contractor/ transporter with address and telephone number is displayed prominently?	Yes/No	
43	During Unloading	Tank Fitting	A fire screen between the cabin and the vessel is provided	Yes/ No	

DRAFT 2

44	During Unloading	Tank Fitting	Bonding between the vessel and chassis and between flanges is satisfactory	Yes/ No	
45	During Unloading	Electrical	The voltage step-up device is not available in the cabin.	Yes/No	
46	During Unloading	Tank Fitting	The earthing boss/ point is available	Yes/ No	
47	During Unloading	Electrical	Head/ parking/ indicator lamps are in working condition	Yes/ No	
48	During Unloading	General	Double-sided view mirrors are available	Yes/ No	

Signature of Officer Signature of Operator Signature of Driver

Remarks:

DRAFT 2

Annexure-7-I-B

CHECKLIST FOR BULK LPG TANK TRUCKS AT LOADING LOCATION

S.No. …….

Date …… Time ……

Bulk LPG Truck No. ……..... Driver Name ……

Transport Name……

Unloading Location……

Sr. No.	When	Parameter	CHECK POINTS FOR BULK LPG TANK LORRY	Status	Remarks
1	Before Entry	TT checklist	Whether the Bulk TT checklist is filled out at the Unloading location is available	Yes/ No	
2	Before Entry	TT checklist	Whether any anomalies mentioned in the above checklist been rectified?	Yes/ No	
3	Before Entry	TT checklist	Minimum Vapour pressure of Bulk TT bullet is 1.5 kg/cm²?	Yes/ No	Pressure observed
4	Before Entry	Electrical	Readily accessible master switch with dipole wiring of a minimum 300 Amp rating for switching off the engine is provided inside the cabin.	Yes/ No	
5	Before Entry	Electrical	Truck is self-starting?	Yes/ No	
6	Before Entry	Tank Truck	Properly fitted spark arrestor of make and design approved by CCE, Nagpur?	Yes/ No	
7	Before Entry	Tank Truck	a) Two portable IS marked 9 Kg DCP fire extinguishers, one on each side and accessible from outside the cabin. b) One DCP fire extinguisher of 1 Kg capacity in the driver's cabin c) All fire extinguishers marked for next due date for pressure testing. d) The TT number painted on each fire extinguisher.	Yes/ No	
8	Before Entry	Tank Truck	Whether VTS is available and operational	Yes/ No	

DRAFT 2

9	Before Entry	Tank Truck	No leakage in the exhaust silencer pipe exists.	Yes/ No	
10	Before Entry	General	Whether the driver of TT has undergone 3-days training and one-day refresher training?	Yes/ No	
11	Before Entry	General	Endorsement in the driving licence by licensing authority to the effect that he is authorised to drive a goods carriage carrying goods of dangerous or hazardous nature to human life	Yes/ No	
12	Before Entry	General	Carries TREM CARD, an instruction booklet detailing instruction on handling emergencies on route?	Yes/ No	
13	Before Entry	General	Carries Route Map?	Yes/ No	

Note: Tanker must be loaded when all above check points are complied with.

Remarks

Signature of Officer Signature of Operator Signature of Driver

DRAFT 2

Annexure-7-II

CHECKLIST AT UNLOADING LOCATION FOR BULK LPG TANK WAGONS

VISUAL EXAMINATION OF WAGON FOR

Check Point	**Remarks**
POH/ ROH OVERDUE: FOR BARREL	Yes/ No
FOR UNDERFRAME	Yes/ No
Damage to the Tank Barrel	Yes/ No
Damage/ missing tank fittings	Yes/ No
Seal intact	Yes/ No
All valves, fittings are in working order	Yes/ No
The surface is free from rust	Yes/ No
The paint on the barrel is in good condition and has not lost its reflecting characteristics.	Yes/ No

The matter regarding the tank wagon not complying with any of the above checks shall be taken up with the loading location for corrective action.

Remarks

Signature of Officer

Signature of Operator

DRAFT 2

Annexure-7-III

LIQUEFIED GAS-CARGO TRANSFER CHECKLIST

(Items to be checked before cargo transfer begins)

SHIP: _____ DATE: _____

PORT & BERTH: _____ TIME: _____

S.NO	CHECK POINT	FOR SHIP	FOR SHORE
1	Has information on cargo and ship-shore connection been supplied?		
2	Is the agreed ship-shore communication system operative?		
3	Are fire and other emergency procedures agreed?		
4	Are local regulations being observed?		
5	Has safe access been provided, and warning notices posted?		
6	Are moorings secure and agreement reached on the use of tension winches?		
7	Are emergency towing-off wires correctly positioned?		
8	Is the ship ready to move under its own power?		
9	Are smoking restrictions in force and notices posted?		
10	Are naked light restrictions being observed?		
11	Are portable electrical equipment cables disconnected?		
12	Are all hand torches and portable R/T sets of approved type?		
13	Are the ship's main transmitting aerials and radar switched off and earthed?		
14	Some doors and ports have to be closed: are they actually shut?		
15	Are all air conditioning intakes correctly trimmed, and window type units closed?		
16	Is the water main ready for immediate use?		

DRAFT 2

17	Is the water spray system ready for immediate use?		
18	Are dry powder and all other firefighting appliances correctly positioned and ready for immediate use?		
19	Is necessary protective clothing is being used?		
20	Are void spaces properly inerted?		
21	Is the required ventilation equipment in operation?		
22	Is the cargo system set for the operation?		
23	Are all remote-control valves in working order?		
24	Are cargo tank relief valves correctly set and in good order?		
25	Are the required cargo pumps and compressors in good order?		
26	Is re-liquefaction or boil-off control equipment in good order?		
27	Is gas detection equipment set for the cargo, calibrated and in good order?		
28	Are cargo system gauges and alarms correctly set and in good order?		
29	Are scuppers plugged, and are suitable drip trays in position?		
30	Are cargo marine unloading arm and bunker hoses in good condition and properly rigged, have certificates being checked?		
31	Are unused bunker connections blanked, and bunker tank lids closed?		
32	Are unused cargo connections (including inert gas line) securely blanked?		
33	Are automatic shutdown systems working properly?		
34	Does shore know the closing rate of the ship's automatic valve at operating temperature; does the ship have similar details of the shore system?		
35	Are all personnel (including supernumeraries and new arrivals) aware that cargo transfer is to begin?		

DRAFT 2

36	Have all personnel been allocated emergency stations?		
37	Are non-essential personnel clear of the cargo area?		
38	Are those directly involved aware of the agreed cargo transfer sequence?		

REMARKS:

We have checked with each other the items on the above checklist and have satisfied ourselves that the entries we have made are correct to the best of our knowledge.

CHECKED BY

Name, designation & Signature

Name, designation & Signature

For Ship

For Terminal

DRAFT 2

Annexure- 7-IV

TYPICAL LPG TANK WAGON SAFETY FITTING

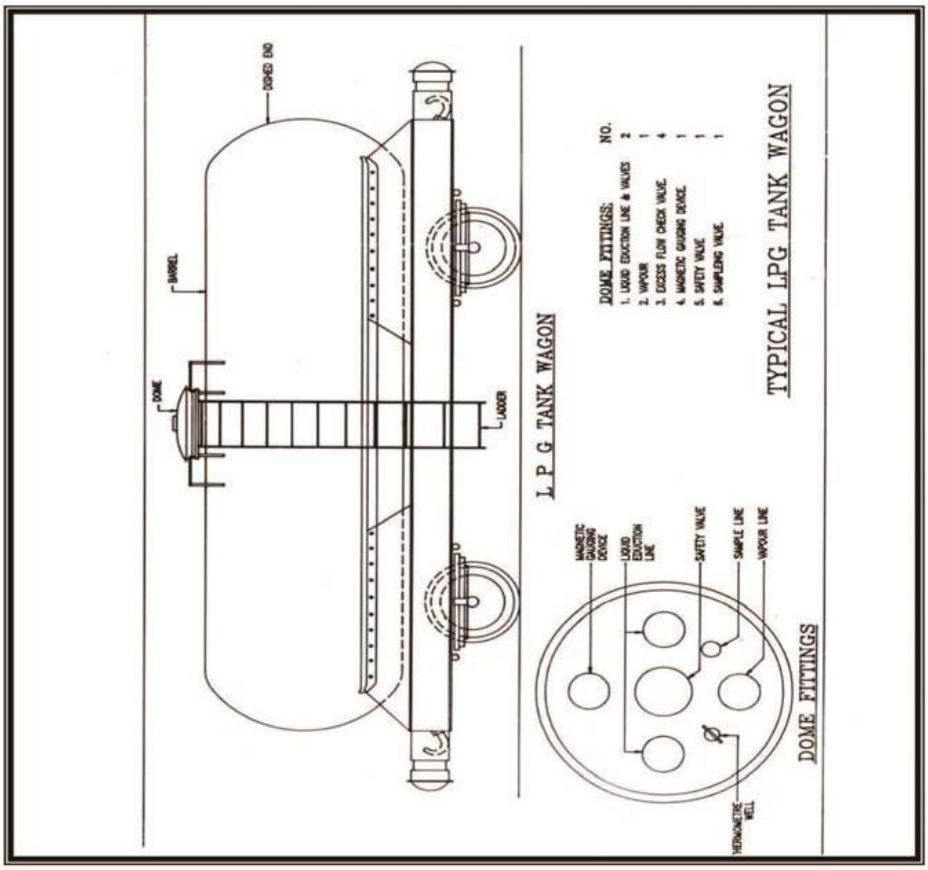


Figure - VII

Annexure-8-I

CHECKLIST FOR THE SAFETY FITTING OF PACKED CYLINDER TRUCK

S.No. …….

Date …… Time ……

Packed Truck No. ……..... Driver Name …….....

Transport Name ……

Sr. No.	When	Safety Check	Status	Remarks
1.	Before Entry	a) Validity of RC, b) Goods Permit to carry hazardous Goods (Rule 86-90 CMRV) c) Vehicle fitness certificate d) PUC available e) Comprehensive Insurance for the vehicle and product	Yes/ No	Mention Valid till
2	Before Entry	a) Validity of driving License, b) Endorsement in the driving licence by licensing authority to the effect that he is authorised to drive a goods carriage carrying goods of dangerous or hazardous nature to human life.	Yes/ No	Mention Valid till
3	Before Entry	Properly filled spark arrestor of make and design approved by CCE, Nagpur? (If CCE approval is available for the non-fitment of the spark arrestor, the exemption document from CCE is to be checked, and the document reference is to be recorded)	Yes/ No	
4	Before Entry	The hand brake is in working condition	Yes/ No	
5	Before Entry	A dedicated second crew is available in the packed truck.	Yes/ No	
6	Once in quarter	a) The battery shall be mounted in an easily accessible position. b) Battery if mounted outside, shall be kept in well-ventilated box with a lock. c) Battery terminals shall be electrically insulated, or the battery shall be covered by an insulating cover.	Yes/ No	

DRAFT 2

7	Once in quarter	a) Double pole master cut-off switch of minimum 300 amp rating for switching off the engine is provided inside the cabin in a readily accessible position. b) The cable connections on the battery master switch shall have a protection degree IP54 in accordance with IEC Standard 60529 or any other equivalent standard	Yes/ No	
8	Once in quarter	a) Are all Junction boxes properly sealed? b) Any loose electrical wiring terminal? c) Electrical wiring is insulated and provided with suitable overcurrent protection?	Yes/ No	
9	Once in quarter	The voltage step-up device is not available in the cabin.	Yes/ No	
10	Once in quarter	The vehicle is self-starting	Yes/ No	
11	Once in quarter	Head/ parking/ indicator lamps are in working condition	Yes/ No	
12	Once in quarter	a) Two portable IS marked 9 Kg. DCP fire extinguishers, one on each side and accessible from outside the cabin. b) One DCP extinguisher of 1 Kg capacity in the driver's cabin c) All extinguishers marked for next due date for pressure testing. d) Truck number painted on each fire extinguisher.	Yes/ No	
13	Once in quarter	Whether the Carrier has been provided with an Anti-lock braking system (ABS)?	Yes/ No	
14	Once in quarter	Whether VTS is available and operational.	Yes/ No	
15	Once in quarter	Air brakes are not bypassed	Yes/ No	
16	Once in quarter	Whether the Vehicle tyres in good condition?	Yes/ No	
17	Once in quarter	Condition of type-tread depth of min 2 mm, or check the depth indicator	Yes/ No	
18	Once in quarter	Engine exhaust is provided in the front below the Driver's cabin (for vehicles without CCE spark arrestor exemption)	Yes/ No	
19	Once in quarter	Vehicles are protected by a robust steel bumper	Yes/ No	

DRAFT 2

		with rear and side under run protection devices		
20	Once in quarter	The fuel tank is protected by means of a stout guard, and the fuel tank cap is locked.	Yes/ No	
21	Once in quarter	Double-sided view mirrors are available	Yes/ No	
22	Once in quarter	Truck floor and sides are provided with protection to prevent cylinder damage (e.g., Rubber/ wood)	Yes/ No	
23	Once in quarter	Whether the Vehicle is provided with a retractable type of Seat belt and the driver is using it?	Yes/ No	
24	Once in quarter	A first aid box is provided in the driver's cabin	Yes/ No	
25	Once in quarter	4 nos. caution sign boards with luminous paint for stopping traffic in case of any emergency?	Yes/ No	
26.	Once in quarter	HAZCHEM code, name of contractor with address and telephone number is displayed prominently. Flammable motor spirit to be mentioned on three sides.	Yes/ No	
27.	Once in quarter	Driver's medical certificate- no night/colour blindness	Yes/ No	
28	Once in quarter	Whether the driver of TT has undergone 3-day training & one day refresher training?	Yes/ No	
29	Once in quarter	a) Carries TREM CARD, an instruction booklet detailing instruction on handling emergencies on route? b) Whether the TREM card contains all the important phone numbers, including "112" - Emergency response system of GOI - the Single Emergency number?	Yes/ No	

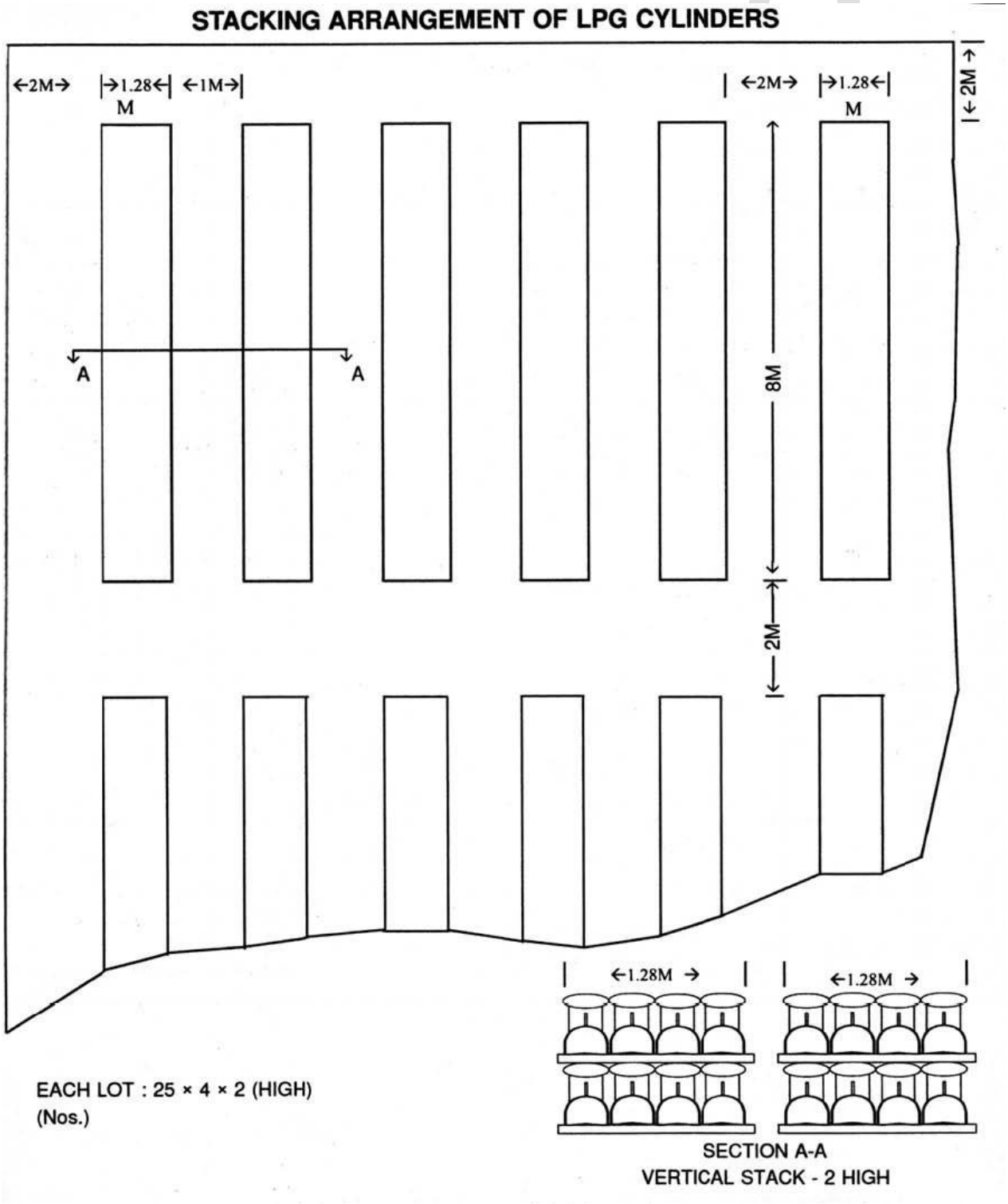
Remarks……
……
……

Driver’s Sign Security’s Sign Officer’s Sign

DRAFT 2

Annexure-8-II

TYPICAL STACKING ARRANGEMENT FOR LPG CYLINDER



Annexure-9-I

MAINTENANCE SCHEDULE FOR COMPRESSORS

This preventive maintenance schedule is indicative.

A record of daily checks may be maintained in the logbook.

Sr.	Equipment checks	Periodicity					
		Daily	Weekly	Monthly	Qtrly	Half Yearly	Yearly
1	Check the lube oil level of the plunger pump unit and top up if necessary.	Y					
2	Check the oil level in the crank case and top up if necessary.	Y					
3	Before starting, check the flow of cylinder jacket cooling water.	Y					
4	Check oil flow from the plunger pump to the gland packing assembly and compressor cylinder.	Y					
5	Check for flow rate from plunger pump	Y					
6	Drain water from receiver/ liquid trap.	Y					
7	Check belt tension and adjust if necessary.			Y			
7.1	Check Filters: Inspect and clean the air intake and oil filters. Replace any clogged or worn filters.		Y				
8	Ensure the tightness of foundation bolts			Y			
9	Flush out the lube oil system: a) Crank case b) Pump the chamber and fill with fresh oil. Replace compressor oil if necessary,				Y		

DRAFT 2

	based on operating hours or manufacturer recommendations.						
10	Flush jacket water cooling system.			Y			
11	Check the alignment of the compressor and motor pulleys.				Y		
12	Change lube oil filter and clean LPG Suction filter. Replace, if necessary, based on operating hours or manufacturer recommendations.			Y			
13	Clean oil strainer		Y				
14	Clean Breather			Y			
15	Open and inspect suction and discharge valves.				Y		
16	Check instrumentation and calibrate if necessary.			~~Y~~		Y	
17	Test safety release valve.						Y
18	Visual inspection for leakage, suction and discharge valves of compressor cylinder	Y					
19	Interlock with High level alarm				Y		
20	All Safety Trips				Y		
21	Pressure and Temperature Gauges are functioning	Y					
22	Checking of Lamp Test and visual indication of Control Panel, wherever provided.			Y			

DRAFT 2

Annexure-9-II

MAINTENANCE SCHEDULE FOR CENTRIFUGAL PUMPS

This preventive maintenance schedule is indicative.

A record of daily checks may be maintained in the logbook.

Sr.	Equipment checks	Periodicity					
		Daily	Weekly	Monthly	Qtrly	Half Yearly	Yearly
1	Check lube oil and top up level "Y" if necessary.	Y					
2	Check cooling water flow (where provided).	Y					
3	Check mechanical seal/ gland leakage	Y					
4	Change lube oil as per OEM recommendation.			Y			
5	Check integrity of coupling and coupling guard.			Y			
6	Check tightness of foundation bolts.			Y			
7	Clean suction strainer of: a) Product pumps b) Other pumps				Y		
8	Check alignment of pump and motor				Y		
9	Greasing of Bearings			Y			
10	Testing of All Safety Trips				Y		

DRAFT 2

Annexure-9-III

MAINTENANCE SCHEDULE FOR FILLING SHED EQUIPMENT

This preventive maintenance schedule is indicative.

Record of daily checks may be maintained in logbook.

A	VISION UNIT.						
Sr.	Equipment checks	Periodicity					
		Daily	Weekly	Monthly	Qtrly	Half Yearly	Yearly
1	Cleaning of Vision Unit Outside.	Y					
2	Cleaning of Vision Unit Inside.		Y				
3	Background image and Cy Ring setting on Computer.			Y			
4	Cleaning and Painting of Vision Unit Railing.			Y			
5	Camera Alignments Check on Vision.				Y		
6	Check the Camera Trigger Antenna Sensor.	Y					
7	Check the alignment of Stopper and mounting.		Y				
8	Clean the Pusher guide rod and check tightness of locking nut.		Y				
B	TANGENTIAL INLET UNIT						
1	Drain the Main air line and Filter Regulator.	Y					
2	Cleaning and check the alignment of Waiting Stopper.		Y				
3	Clean and apply slim layer of grease to Guide rods. (Bearing No = ø20MM Linear		Y				

DRAFT 2

	ball bearing)						
4	Clean the adjustable throttle of quick exhaust valve.			Y			
5	Clean the Bar Code Plates.		Y				
C	**CENTER COLUMN**						
	Center column acts as a rotating axis of Carousel and distributes LPG, Air and Power to all filling machines. Center column does not have any specific preventive maintenance since all Bearings are greased for lifetime. However, following precautions needs to be taken for long time smooth functioning. 1. LPG pipeline, which is connected to center column top, should not have any Vibration and should be perfectly in-line with center column axis. 2. Clean the LPG Manifold strainer regularly and ensure LPG to center column Does not contain any foreign particles like dust, dirt, metal particles, etc. Presence of which results in leakage/damages of Gas Stop valve and Filling Gun.						
D	**FILLING MACHINES**						
1	Drain the Main airline and Filter Regulator Inside Car.	Y					
2	Cleaning and calibration of Filling Machines.	Y					
3	Check the Rubber Bush and load cell H frame.		Y				
4	Check the adjustments leg and balancing of machine.		Y				
5	Ensure all filling machines are at same height also 5-8mm lower than Inlet plate and 5-8 mm upper w.r.t. Outlet plate.		Y				
6	Centering of filling gun			Y			

DRAFT 2

	w.r.t. centering arm.						
E	**CAROUSEL RAIL AND WHEEL AND FRAME**						
1	Cleaning of rail.		Y				
2	Cleaning of frame.	Y					
3	Greasing of wheel sliding plates.			Y			
4	Check the wheel Bearings.				Y		
F	**CAROUSEL DRIVE UNIT**						
1	Check the Hydraulic Oil Level. (Max)	**Y**					
2	Check the oil Temp. (Should be less than 60 degree) (Oil = MOBIL DTE H32 or compatible) (Pump Filter Replacement = Every 500 hours) (Pump Oil Replacement = Every 1500 hours)	**Y**					
3	Check the oil pressure on the manometer. **(**Not exceed normal working pressure by more than 2 kg/cm^2**)**	**Y**					
4	Lubricate the Duplex Chain in drive unit. **(**Pulley Bearing = Ball bearing 6206-2RS1**)**		**Y**				
5	Check the Drive belt tightness.	First 6 months =M, 6 to12 Months = Q, After 1 year = HY					

DRAFT 2

6	Greasing of drive unit rollers. (Ball bearing 6005-2RS1)			**Y**			
7	Check the Belt Joining link.				**Y**		
8	Check the Carousel Speed.		**Y**				
9	Check Pump and motor for noise, vibration, etc.	**Y**					
G	**TANGENTIAL OUTLET UNIT**						
1	Drain the Main air line and Filter Regulator.	**Y**					
2	Clean and apply slim layer of grease to Guide rods. (Bearing No = ø20MM Linear ball bearing)	**Y**					
3	Clean the adjustable throttle of quick exhaust valve.			**Y**			
4	Check the tightness of ejection arm with Cylinder.				**Y**		
5	Check the tightness of guide rod holding blocks.					**Y**	
H	**DOUBLE STATIC CHECK SCALE.**						
1	Drain the Main air line and Filter Regulator.	**Y**					
2	Cleaning and calibration of Check Scale.	**Y**					
3	Check the alignment of Stopper and mounting.		**Y**				

DRAFT 2

4	Check the alignment of CS and ensure it is parts not touching to conveyor.		**Y**				
5	Clean the Pusher guide rod and check tightness of locking nut.		**Y**				
6	Ensure height of DCS Chain top w.r.t. Conveyor chain top is between 15-20 mm.			**Y**			
7	Check the condition of rubber bush and H frame.				**Y**		
8	Servicing of check scale.			**Y**			
9	Change the rubber bush and 63 x 25 cylinder with new one.						**Y**
I	**WEIGHT ADJUSTMENT SCALE / CORRECTION UNIT.**						
1	Drain the main air line and Filter Regulator.	**Y**					
2	Cleaning and calibration of machines.	**Y**					
3	Check the rubber bush and load cell H frame.			**Y**			
4	Check the adjustments leg and balancing of machine.				**Y**		
J	**VALVE LEAK DETECTOR (VLD) CIR**						
1	Drain the main air line and Filter Regulator.	**Y**					
2	Check the centering of the machine w.r.t. test head Gun.			**Y**			

DRAFT 2

3	Cleaning and lubrication of the centering mechanism and the lifting cylinder. (Ball bearing 6301-2RS1/C3)		**Y**				
4	Check the tightness of toggle joints.		**Y**				
5	Check the alignment of the stoppers and mounting.		**Y**				
6	Clean the Pusher guide rod and check the tightness of Locking nut.		**Y**				
7	Ensure the gun's down and centering clamp movement is smooth and slow.			**Y**			
8	Cleaning the receiver and transmitter lens and test head.	**Y**					
9	Check the jet air coming from Both the nozzles.	**Y**					
10	Ensure earthing resistance is less than 1 ohm.			**Y**			
11	Cleaning of test head and antistatic hoses.			**Y**			
12	Calibration of CIR heads.			**Y**			
K	**'O'-RING LEAK DETECTOR (OLD)**						
1	Drain the main air line and Filter Regulator	**Y**					
2	Check the centering of the machine w.r.t. Test		**Y**				

DRAFT 2

	head Gun						
3	Cleaning and lubrications of centering mechanism and Lifting Cylinder. (Ball bearing 6301-2RS1/C3)		**Y**				
4	Check the tightness of toggle joints.		**Y**				
5	Check the alignment of the Stopper and mounting.		**Y**				
6	Clean the pusher guide rod and check the tightness of locking nut.		**Y**				
7	Ensure the Gun's down and centering clamp movement is smooth and slow.			**Y**			
8	Check the functioning of the Cylinder NRVs.			**Y**			
9	Check/ Change the Test Air and Transmitter Pneumatic 3 mm PU tubes.				**Y**		
10	Calibration and gun services for the machine.		**Y**				
L	**LPG MANIFOLD & DP VALVE.**						
1	Cleaning of Manifold and DP Valve.	**Y**					
2	Check the DP valve Function on the LPG Pressure Gauge. (Function tolerance: +/-	**Y**					

DRAFT 2

	0.5-0.8 kg/cm²)						
3	Clean the Dirt trap/Strainer.				**Y**		
N	**INLINE CAP PRESSING MACHINE**						
1	Drain the Main air line and Filter Regulator.	**Y**					
2	Cleaning and check the any air leakage on Machine.	**Y**					
3	Check the alignment of Stopper and mounting.		**Y**				
4	Check the Tightness of cap pressing head.		**Y**				
O	**INLINE SEALING MACHINE.**						
1	Drain the Main air line and Filter Regulator.	Y					
2	Cleaning and check the any air leakage on Machine.	Y					
3	Check the alignment of Stopper and mounting.		Y				
4	Check the Tightness of Sealing head and hot air hoses.		Y				
5	Greasing of all Moving Parts.		Y				
6	Check the Temperature of air heating (180° – 220°)			Y			
P	**SQC**						
1	Drain the Main air line and Filter Regulator.	Y					
2	Cleaning and check the any air leakage on	Y					

DRAFT 2

	Machine.						
3	Check the Calibration of Check Scale and CVT.	Y					
4	Check the kerosene/ MTO level of CVT.	Y					
5	Check the alignment of Lifting and Stopper and mounting.		Y				
6	Greasing of Stopper and Moving Parts.		Y				
Q	**CHAIN CONEYERS**						
1	Top up soap solution trays	Y					
2	Check position of tension roller of chain and adjust if required	Y					
3	Lubricate tension rollers and wheels		Y				
4	Clean soap solution tray.			Y			
5	Grease the bearing			Y			
6	Check foundation bolts of conveyor frame for tight- ness					Y	
R	**DRIVE UNIT OF CONVEYORS**						
1	Check and top up oil level.	Y					
2	Check coupling bolts and rubber bushings. Replace the cracked or worn-out bushings.			Y			
3	Check alignment of gear box and correct if necessary					Y	

DRAFT 2

4	Clean and lubricate chain.			Y			
5	Tighten foundation bolts			Y			
6	Drain and change oil.					Y	
7	Inspect for wear and tear					Y	
S	**LOADING & UNLOADING BOOM'S**						
1	Drain the Main air line and Filter Regulator.	**Y**					
2	Cleaning and check the any air leakage on booms.	**Y**					
3	Check the alignment of boom main rollers and support guide rollers and drive unit's duplex chains.		**Y**				
4	Check the Tightness of roller bearings and conveyor bearings.		**Y**				
5	Clean and check the function of dead men switch.	**Y**					
6	Greasing or oiling on duplex chain and boom cy guide pipe and 8 teeth sprockets.		**Y**				
7	Check the coupling spider rubber of boom drive unit.	**Y**					
8	Check the oil level of conveyor drive unit gear box.	**Y**					
T	**MANUAL SEALING MACHINE.**						

DRAFT 2

1	Drain the Main air line and Filter Regulator.	Y					
2	Cleaning and check the any air leakage on Machine.	Y					
3	Check the Tightness of Sealing head and hot air hoses.		Y				
4	Check the smooth rotation of traveling wheel.		Y				
5	Check the Temperature of air heating (180° – 230°)			Y			
U	**WASHING UNIT.**						
1	Clean and check the washing unit nozzle.	Y					
2	Clean the water tank and filling with service water.	Y					
3	Clean the inside of washing unit.		Y				
4	Check the water pump oil level.	Y					
5	Clean the washing unit pump and motor.	Y					
6	Check the noise, vibration, etc. on motor and pump.	Y					
V	**CYLINDER TEST BATH:**						
1	Fill with clean water.	Y					
2	Lubricate moving parts.		Y				
3	Clean tank walls from inside.		Y				

DRAFT 2

4	Check and lubricate pneumatic circuitry of all equipment's.		Y				
5	Check all fasteners and tighten.		Y				
6	Paint tank walls from inside.		~~Y~~			**Y**	
W	**MANUAL FILLING MACHINES**						
1	Clean and check the washing unit.		Y				
2	Clean and check the water pump and motor.	Y					
3	Check the noise, vibration, etc. on motor and pump.	Y					
4	Clean and check the LPG prefilling machine.	Y					
5	Clean and check the main LPG filling machine	Y					
6	Drain the Main air line and Filter Regulator.	Y					
7	Cleaning and check the any air leakage on any Machine.	Y					
8	Check the Calibration of filling machine, Check Scale and CVT.	Y					
9	Check the kerosene level of CVT.	Y					
10	Check the alignment of Lifting unit and stopper and mounting.		Y				
11	Greasing of Stopper and Moving Parts.		Y				

DRAFT 2

12	Clean and check the manual sealing unit.		Y				
X	**FILLING SHED EQUIPMENT - PNEUMATICS (ACTUATORS; FRLS; TUBING; PNEUMATIC INSTRUMENTS; ETC.)**						
1	Condition of Filter Regulator is in satisfactory condition. No water/dust in filters / condensate drained	Y					
2	No air leaks from lines, ferrules, valves and Pneumatic dressing are ok	Y					
3	Air pressure at buffer tanks in sheds is functional and condensate drained from all vessels	Y					
4	Air leakage from pipeline, pneumatic valves etc. to be checked in non-working day/ night by ensuring absolute silence (minimum once in a quarter				Y		

DRAFT 2

Annexure-9-IV

TYPICAL INSPECTION FREQUENCY

Item	Minimum Frequency
Excess Flow Check Valve of the TT loading gantry	Once in five years
Jockey pump start or stop pressures	Once a week
Main fire pump set pressure	Once a month
Earthing interlock test in the TT loading/ unloading gantry and carousel	Once a month
ESD system	Once a month
Alarms and trips in the pipeline transfer system in line with OISD-STD-214	Once a month
Testing of Deluge Valve with outlet valve of DV closed by Field actuation (Emergency Push button or Break Glass or air release provision), remote actuation from panel and by breaking QB	Once in a quarter
Sprinkler system checks for working and clogging	Once a quarter
performance test of sprinkler system	Once in six months
Testing of Gas monitoring system by releasing gas	Once a month
Calibration of Gas monitoring system	Once in quarter
shut off tests of ROV	Once a month
ERC or Breakaway Couplings of TLD	As per OEM. In the absence of OEM recommendation, at least once in six months
Manual Call Points	Once a month
Checking of High-level alarms through simulation	Once a quarter
Checking of High-level alarms by raising the level	Once in a year
Radar Gauge calibration	Once in five years
Servo Gauge calibration	Once in five years
Mass flow meter calibration	Once a year

DRAFT 2

CP system potential check by reference Electrode	Once in quarter
Interlock Carousel/ filling machine and vapour extraction Blower	Once a month
Emergency Siren test	Once a week
Cleaning of FR/ FRL	Once a week
Cleaning of all Strainers on LPG Lines	Once in three months
Cleaning of all strainers provided on deluge, fire water pump suction line, Jockey pump, etc.	Once in three months
LPG Hoses testing as per OISD-STD-135	Once in four months
Calibration of Pressure Gauge	Once in six months
Calibration of the Temperature gauge	Once in six months
Calibration of the Pressure gauge in the Cylinder hydro testing unit	Once a week
Calibration of the Pressure gauge in the Cylinder pneumatic testing unit	Once in three weeks
Checking of trips of the LPG Pump and the LPG Compressor	Once in three months
LPG Pipeline testing	
External On-stream inspection – a) Visual Inspection b) Ultrasonic thickness survey Note- Piping in coastal installations and in corrosive environments shall be visually inspected once a year	Once in three years
Comprehensive Inspection a) Visual Inspection b) Ultrasonic thickness survey c) Hydro testing – (If applicable, i.e., after major alteration, inaccessible piping, after chemical cleaning, out of service more than six months)	Once in nine years
Loading/ unloading arms – As applicable for pipelines with the same frequency	
Testing of Fire Water Tank (in line with OISD-STD-129 & statutory requirement as applicable)	
a) Visual External Inspection	Once a year

DRAFT 2

b) Detailed External Inspection along with ultrasonic thickness survey (Without Corrosion Rate Based Assessment)	Once in four years
c) Detailed External Inspection along with ultrasonic thickness survey (With Corrosion Rate Based Assessment)	Once in six years
d) Inter inspection of Fire water tank (Without Corrosion Rate Based Assessment)	Once in ten years
e) Inter inspection of Fire water tank (With Corrosion Rate Based Assessment)	Years determined by Corrosion Rate and Integrity Assessment or 15 years, whichever is lower
Cleaning of the firewater reservoir	Once in six months
NDT of LPG Vessel	Once in five years
Hydro testing of LPG Vessel (Above ground)	Once in five years
Hydro testing of LPG Vessel (MSV)	Once in Ten year
Testing of all SRV/TSV	Once in a year
Settlement of MSV	Once in six months
Testing of Insulation joint for MSV using IJ tester	Once in a year
Testing of Insulation joint for MSV using PSP method	Once in a three month
Testing of OFF-PSP of MSV	Once in three months
Testing of SSD/ kirk Cell	Once in three months
Testing of surge arrestor	Once in three months
Earth resistance on individual electrodes (Earth pit Testing)	Once in six months
Continuity of earthing conductors	Once a year
Thermography of Panel	Once in six months
Checking of VCB bottles for healthiness	Once in three years
Electromechanical relays testing	Once a year
Insulating Mats checking	Once a year

DRAFT 2

Condition of cable end sealing and terminal boxes	Once a year
Sealing of the bus duct entry on both sides of the wall	Once a year
Transformer oil testing as per IS:1866	Once a year
OLTC oil testing as per IS:1866	Once a year
DGA of Transformer oil	Once in three years
Lighting Illumination level (Lux Level) checking of various areas	Once a year
Third harmonic current testing of the lightning arrestor	Once in three years
Testing of evacuation vessel, Knock out drum, air vessel etc.	As per requirement of Factories rules of the state.
Pressure relief mechanism is provided in the hydrant line	Once a month
Auto start logic of the fire water pump	Once a month
Thickness survey of Hydrant line	Once in three years
Comprehensive testing of Hydrant line	Once in Ten years
Testing of Fire Water Pump alarm	Once a month
Inspection and maintenance of Isolation valve in hydrant network	Once in six months
Shut off the pressure testing of the Fire Water pump	Once a month
Four hour testing of the Fire Water Pump	Once a year
Hydro testing of the Fire hose	Once a year
Hydro testing of DCP Fire extinguisher	Once in three years
Hydro testing of Co2 fire extinguisher	Once in five years
Checking of Electrical Insulation Block provided for Railway siding	Once in three months

DRAFT 2

Annexure-9-VI

CCTV MAINTENANCE CHECKLIST

1. Regularly clean camera lenses

Regularly check the lenses of ~~your~~ outdoor and indoor security cameras. It is recommended to perform this cleaning ~~task~~ at least once every three months.

2. Check the camera's housing

Ensure that the enclosures of cameras are waterproof to protect them from moisture damage.

3. Ensure that all cables are fully connected

It is essential to regularly check all cable connectors to ensure they are accepting power properly. Additionally, inspect the cabling for any signs of wear and tear, and replace any uncovered wires instantly to prevent potential issues.

4. Check the power supply

Regularly inspect the power supplies of CCTV security systems to ensure there are no power losses caused by thunderstorms, tampering, brownouts, or other unexpected events. It is also advisable to check.

5. Check for corrosion

Inspect all connectors for signs of corrosion regularly.

6. Dust your Recording devices

Regularly clean the recording devices to prevent the accumulation of dust.

7. Trim the landscaping

Trim bushes, vines, trees, and other Plants near the cameras to prevent them from blocking the camera's field of vision.

8. Check the lighting

Make sure that critical areas have adequate lighting for effective illumination. Proper lighting is essential for capturing clear and usable footage during nighttime surveillance.

9. Periodically review the camera's placement

Regularly review the placement of your cameras to ensure they are positioned optimally and have the correct focus.

10. Ensure system recording functionality

Verify that CCTV system is recording properly on a regular basis.

Annexure-10-I

SEQUENCE OF OPERATIONS FOR STATUTORY PRESSURE TESTING OF LPG CYLINDERS

1. Receipt of Cylinders
2. Depressurising
3. Recording details (like S. No., Test Dates, Tare weight, Manufacturer's name)
4. Valve Removal
5. Degassing
6. External surface cleaning
7. Internal cleaning and internal inspections
8. Visual Inspections of the external surface and bung threads by L-1 and L-9 plug gauges.
9. Tare weight checking
10. Measurement of Bottom Shell Thickness
11. Hydrostatic testing
12. Internal cleaning and checking
13. Primer coating
14. Signal red painting.
15. Punching of new test date and repairer's identification mark
16. Tare weighing
17. Colour code painting to indicate ownership of cylinders, stencilling new tare weight, new weight and next test date.
18. Tapping the bung thread, cleaning the same with a brass wire brush and checking the bung threads by L-1 and L-9 plug gauges.
19. Valve fixing
20. Pneumatic test
21. Certification and dispatches.

DRAFT 2

Annexure-10-II

REJECTION CRITERIA OF CYLINDER DURING STATUTORY TESTING

Physical Defects in the Cylinder Wall

S No.	Defects	Description	Rejection Criteria
1	Bulge	Visible swelling of the cylinder	Rejection in all cases
2	Dent	Depression in the cylinder that has neither penetrated nor been removed from the metal	a) Depth of dent is greater than 1/10th of the mean diameter of the dent **or** b) The mean diameter of the dent is greater than one-quarter of the diameter of the cylinder **or** c) Dent on or adjacent to the weld
3	Cut or gouge	Sharp impression where metal has been removed or redistributed	75 mm or more in length, or when the depth of cut is such that the undamaged (remaining) wall thickness is less than the minimum calculated wall thickness
4	Intersecting cut or gouge	Point of intersection of two or more cuts or gouges	Rejection in all cases
5	Dent containing cut or gouge	Depression in the cylinder, within which there is a cut or gouge	When the size of a dent or cut, or gouge exceeds the dimension for rejection as an individual defect
6	Crack	A split or rift in the cylinder shell	Rejection in all cases
7	Lamination	Layering of the material within the cylinder wall appears as a discontinuity, a crack. Lap or bulge at the surface	Rejection in all cases
8	Dig	A sharp depression where the surface material has been penetrated	75 mm or more in length and 4 mm or more in width
9	Bottom shell Thickness	Thickness of the bottom of the cylinder surface	Less than the minimum calculated wall thickness

DRAFT 2

10	Footring bottom Clearance	Clearance of the body bottom from the ground	a) Less than 8 mm for cylinders of water capacity up to 34 litres b) Less than 15 mm for cylinders of water capacity exceeding 34 litres and up to 50 litres c) Less than 25 mm for cylinders of water capacity exceeding 50 litres.

Corrosion Defects in the Cylinder Wall

S. No.	Defects	Description	Rejection Criteria
1	Isolated corrosion Pits	Pitting of metal occurring in isolated areas at a concentration not greater than 1 pit/ 500 mm^2 of surface area	Depth of penetration of any pit exceeds 0.4 mm. Note (1)
2	Area corrosion	Reduction in wall thickness over an area not exceeding 20 percent of the cylinder surface, including the ends (top and bottom)	Depth of penetration of any pit exceeds 0.4 mm. Note (1)
3	General corrosion	Reduction in wall thickness over an area exceeding 20 percent of the cylinder surface	Rejection in all the cases
4	Line corrosion	A series of pits or corroded cavities of limited width along the length or around the corrosion circumference	a) Total length of corrosion in any direction exceeds 50 percent of the circumference of the cylinder b) Depth of penetration exceeds 0.4 mm, note (1) c) Depth of corrosion cannot be measured
5	Cervice corrosion	Cervice corrosion occurs in the area of the intersection of the footring or stay plate with the cylinder	When the depth of penetration exceeds 0.4 mm or when the depth of corrosion cannot be measured, note (1)
note (1)	Cylinders shall be rejected in case the observed thickness of the undamaged remaining wall in the affected portion is less than the minimum calculated wall thickness.		

Other Defects

S. No.	Defects	Description	Rejection Criteria

DRAFT 2

1	Depressed Bung	Damage to the bung, which alters the profile of the cylinder	Rejection in all cases
2	Arc or torch Burns	Burning of the cylinder base metal, a hardened heat affected zone, the addition of extraneous weld metal, or the removal of metal by scarfing or cratering	Rejection in all cases
3	Fire Damage	Excessive general or localized heating of the cylinder is usually indicated by: a) Charring or burning of paint (note) b) Fire damage to the metal c) Distortion of the cylinder d) Melting of metallic valve parts e) Melting of the safety cap Note: In case of superficial charring of the paint cylinder may be accepted.	Rejection in all cases

DRAFT 2

Annexure-10-III

ROUTE CARD OF CYLINDER

Name of LPG Cylinder Testing Station: ……..…Owner of LPG Cylinders: .....

Batch No.: ..... Batch Size: ..... Date: .....

Control No.	**1**	**2**	**3**	**……**
Sl. No. of LPG cylinders				
Name of manufacturer				
Date of manufacture (that is, first test date)				
Previous test dates				
Internal inspection - Ok/ Rejected				
Bottom shell thickness in mm				
Foot ring bottom clearance in mm				
External inspection RT/R/Rej* (In case of rejection, write code for rejection)				
Tare weight check Ok/ Rejected				
Bung thread inspection Ok/ Rejected				
Hydrostatic test at 2.45 MPa (25.35 kg/cm^2) Ok/ Rejected				
Old Tare weight				
New tare weight				
Final inspection for marking, painting, stencilling Accepted/ Not accepted				
Any other defects Accepted/ Not accepted				
Pneumatic testing Ok/ Rejected				
Remarks, if Any				

DRAFT 2

Name and signature of inspector: _____

Date of completion: _____

* RT — Accepted for re-testing, R — Repair, Rej. — Rejected.

The code used shall be as given below:

Error Description	**Error Code**	**Error Description**	**Error Code**
Bulge	E1	General corrosion	E8
Burn	E2	Bung thread damaged	E9
Dent	E3	Loss in tare weight	E10
Dig	E4	Identity lost	E11
Cut	E5	Bottom clearance is less than the limits	E12
Pit	E6	Wall thickness is less than the limits	E13
Line corrosion	E7	Internal defect cannot be assessed	I1

DRAFT 2

Annexure-10-IV

COLOUR CODING SCHEME OF LPG CYLINDER

Due date of Testing		Background		Colour of Lette
Quarter	**Year**	**Colour**	**Shed as per IS:5**	
A/B/C/D	2025	Aircraft Blue	108	White
A/B/C/D	2026	Salmon Pink	443	Black
A/B/C/D	2027	Indian Brown	415	White
A/B/C/D	2028	Satin Blue	177	Black
A/B/C/D	2029	Bus Green	299	White
A/B/C/D	2030	Black	521	White
A/B/C/D	2031	Yellow	355	Black
A/B/C/D	2032	White	127	Black
A/B/C/D	2033	Sea Green	217	Black
A/B/C/D	2034	Dove Grey	694	Black

DRAFT 2

Annexure-10-V

CERTIFICATE ON PERIODIC TESTING OF LPG CYLINDERS

Certificate No: Date:

This is to certify that LPG cylinders having nominal water capacity of _____liters as per enclosed list 1 have been found acceptable in accordance with IS _____ and hydrostatic test at 2.45 Mpa (25.35 kg/cm^2) pressure in accordance with IS 3196 (Part 1): 2006. These cylinders are fitted with self-closing (SC) type LPG valves in accordance with IS 8737:1995 of Type 1 threads (3/4"-14 NGT OF ANSI/CGA V-1-1987).

Batch details:

Batch No.: _____ Batch Size: _____

No. of cylinders accepted: _____

No. of cylinders rejected: _____

Duration of testing: From _____ to _____

The cylinders accepted and tested are as per attached **List 1**.

The cylinders rejected/recommended for repair are as per attached **List 2.**

Re-testing agency's signature and seal

LIST 1: LPG CYLINDERS ACCEPTED

Name of Testing Station: _____

Owner of cylinders: _____

Certificate No. :_____

Date: _____

S. No.	Cylinder No.	Name of Manufacturer	Date of Manufacture	Old Tare weight	New Tare weight
1					
2					
3					
…					

Re-testing agency's signature and seal

DRAFT 2

LIST 2: LPG CYLINDERS REJECTED/ RECOMMENDED FOR REPAIR

Name of Testing Station: _____

Owner of cylinders: _____

Certificate No.: _____

Date: _____

S. No.	Cylinder No.	Name of Manufacturer	Date of Manufacture	Rejected/ For Repair	Reason
1					
2					
3					
…					

Re-testing agency's signature and seal

DRAFT 2

Annexure-10-VI

QUALITY CONTROL CHECKS ON RE-TESTED CYLINDERS

(RECEIVED FROM REPAIRER/ IN-HOUSE TESTING)

Name of Repairer: _____ **Test Certificate Number:** _____

Batch size: _____

Truck Number: _____ **DC Number:** _____

1) Tare weight and Quality of Markings checks

(Sample to be drawn 5% of the consignment with a minimum sample size of 25 cyls)

(Allowed Deviation: for 14.2 Kg and 19 Kg cylinders: +/- 50 grams, for 35 Kg and 47.5 Kg cylinders: +/-100 grams)

Sr. No.	Serial No. of the cylinder	Punched Tare wt. on on stay plate	Stencilled Tare wt. , Shroud	Observed Tare wt.	Variation in grams	Remarks

2) **Visual Inspection**

(In case of any critical observation total lot to be checked)

Sr. No.	Serial No. of the cylinder	Quality of Painting / Stencilling and Screen Printing	Quality of OMC on the body and Stay Plate	Air inside the cylinder	Test date and marking of the repairer	Overall Workmanship	Loss in Tare Wt.	Remarks

3) **Cylinder Bottom Inspection:**

(Sample to be drawn 5% of the consignment with a minimum sample size of 25 cyls)

Sr. No.	Serial No. of the cylinder	Bottom Shell Thickness	Foot ring bottom clearance	Quality of Painting on the bottom surface of the cylinder	Remarks

DRAFT 2

Foot ring bottom clearance (Clearance of the body bottom from the ground): Should be more than

a) 8 mm for the cylinder with water capacity up to 34 litres

b) 15 mm for cylinders of water capacity more than 34 and up to 50 litres

c) 25 mm for water capacity more than 50 litres

Bottom shell thickness (Thickness of the bottom of the cylinder surface): Should be more than the minimum calculated wall thickness.

4) **Valve Tightening**:
(Quantity to be checked 1% of the consignment, subject to a minimum of 3 nos. Torque to be within the 145 +/- 15 lbft)

Sr. No.	Serial No. of the cylinder	Any damage to Valve	Observed Torque	Remarks

All above noted serial nos. to be cross-checked with the available certificate, and any deviation, if found, to be noted.

Inspected by: Name and designation of Officer

Sign:

DRAFT 2

Annexure-11-I

SAMPLE FIRE WATER CALCULATION

Fire water calculation of LPG Installation

Sizes of various sheds are considered as below for the indicative purpose only.
A) Unloading Platform including cylinders on truck: 25 x 23 m

B) Filling cum Empty shed Size : 40 x 92 m
C) Filled Shed : 40 x 50 m
D) Loading Platform including cylinders on truck: 25 x 23 m
E) Horton Sphere (18 m dia), 4 HS in one group
F) TLD Shed (8 bay) : 50 x 17 m
G) LPG Pump House : 45 x 10 m

AREA	ZONE Number	Length	Width	AREA (Sq. M) (Length x Width)	SPRAY DENSITY LPM per Sq.M	WATER REQUIRED (KL/Hr)	Zones under operation in case of emergency, including adjacent zones	Add 288 KL/ Hr for hydrant/monitor operation	TOTAL Water Required (KL/ Hr)
A) Unloading Platform, including cylinders on the truck	Zone No:01	23	25	575	10.2	351.9	1	288	639.9
B) Filling cum Empty shed									
	Zone No:02	8	40	320	10.2	195.84	2 & 3	288	851.04
	Zone No:03	15	40	600	10.2	367.2	2, 3 & 4	288	1218.24
	Zone No:04	15	40	600	10.2	367.2	3, 4 & 5	288	1389.6
	Zone No:05	15	40	600	10.2	367.2	4, 5 & 6	288	1389.6
	Zone No:06	15	40	600	10.2	367.2	5, 6 & 7	288	1389.6

DRAFT 2

	Zone No:07	15	40	600	10.2	367.2	6, 7 & 8	288	1242.72
	Zone No:08	9	40	360	10.2	220.32	7 & 8	288	875.52
C) Filled Shed									
	Zone No:09	25	40	1000	10.2	612	9 &10	288	1512
	Zone No:10	25	40	1000	10.2	612	9 &10	288	1512
D) Loading Platform, including cylinders on the truck	Zone No:11	23	25	575	10.2	351.9	11	288	639.9
E) Horton Sphere (4 Horton Sphere of 18 m dia in a group) **The area of the Horton Sphere is $4\pi r^2$**									
HS1	Zone No:12	Radius is 9		1018.3	10.2	623.2	12 &13	288	1534.4
HS2	Zone No:13	Radius is 9		1018.3	10.2	623.2	12, 13 & 14	288	2157.6
HS3	Zone No:14	Radius is 9		1018.3	10.2	623.2	13,14 & 15	288	2157.6
HS4	Zone No:15	Radius is 9		1018.3	10.2	623.2	14 & 15	288	1534.4
F) TLD Shed									
TLD Shed	Zone No:16	17	25	425	10.2	260.1	16 &17	288	808.2
TLD Shed	Zone No:17	17	25	425	10.2	260.1	16 &17	288	808.2
G) LPG Pump House									
	Zone No:18	45	10	450	20.4	550.8	18	288	838.8
	Single Largest Risk Area						**Detail**	**Unit**	

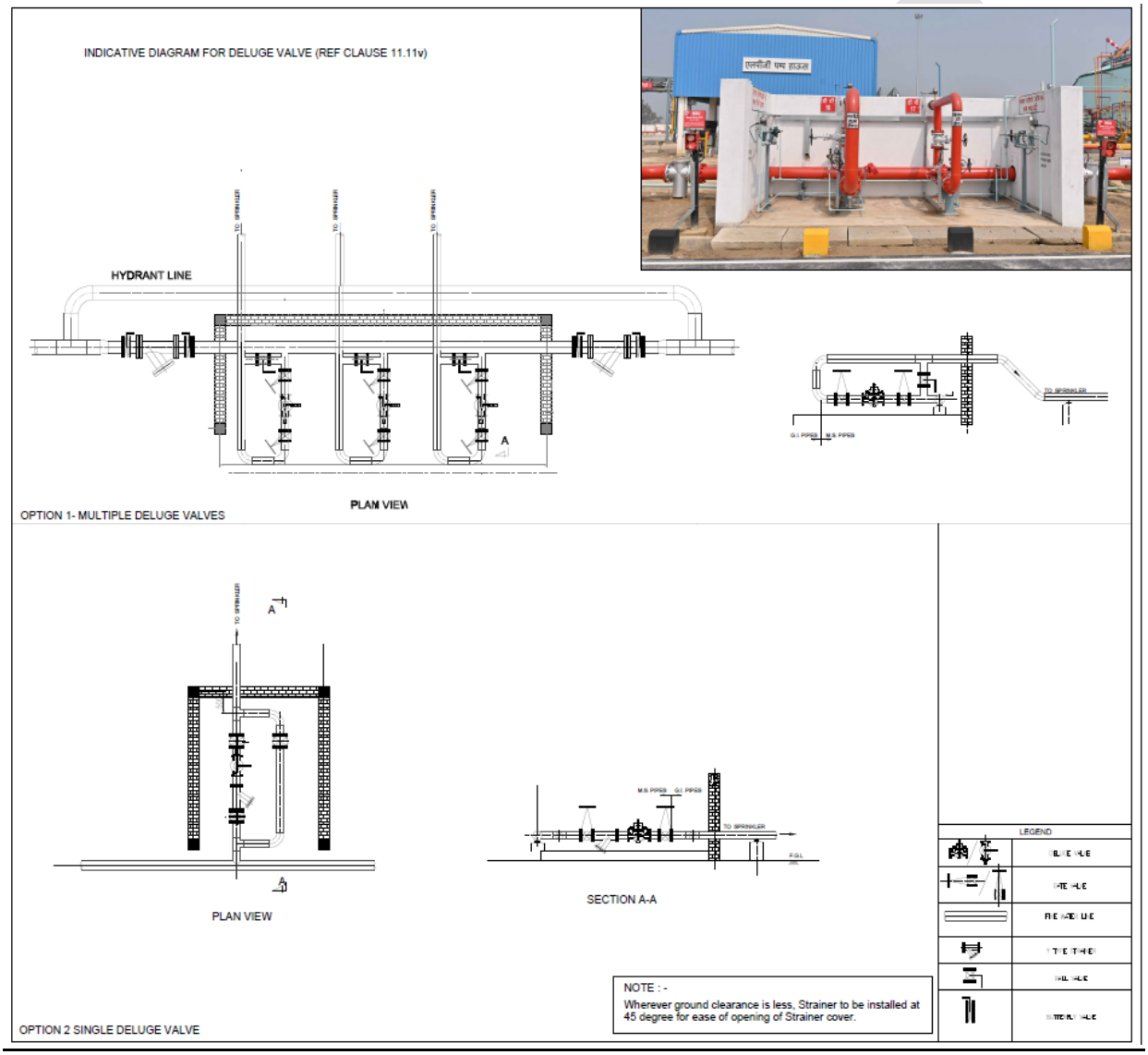
DRAFT 2

1	Single Largest Risk Area (Facility Name: HS2 & HS3 Zones: 12,13, 14 for HS2 and 13,14,15 for HS3) Considering a single risk at a time, either at HS2 or HS3	3054.9	SQM	
a.	Water Required per Hr. for the Single Largest Area	1869.6	KL/HR	
b.	Add 288 Kl/Hr for Hydrants	288	KL/HR	
c.	Total Water Required for Maximum Risk Area	2157.6	KL/HR	
2	Fire Pump Requirement: -			
e.	Say the Capacity of each main fire pump: -	616	KL/HR	
f.	Total number of Main Fire Pumps required (c/e)	3.5	No	
	Total number of Main Fire Pumps required	4.0	No	
g.	Total number of Standby Fire Pumps required	2	no	
3	Fire water storage requirement			
h.	**Total water storage required for 4 hrs. running of all main pumps (in KL) (Above the suction level of the water tank)**	4 nos x 616 kl/hr x 4 hr = 9856 KL	KL	

DRAFT 2

Annexure-11-II

INDICATIVE DIAGRAM FOR DELUGE VALVE



Annexure-14-I

DAILY SAFETY CHECK FOR LPG INSTALLATION

LOCATION _____

AUDIT/ INSPECTION BY _____

S. No.	Item	Observation	Action Plan	Target Date
A. SAFETY/ FIRE PROTECTION SYSTEM				
A-01	The water level in the static water tank Required water qty in KL Available water qty in KL			
A-02	Record Pressure of the fire water ring main.			
A-03	Fire water pumps/engines a) Check and sign logbook b) Are the diesel tanks full c) Start one pump in auto mode, record pressure developed. Mention pump operated.			
A-04	Check by operating the following: a) Manual siren (at least one) b) Page phone (at least one)			
A-05	The Automatic Gas Detection system is operating. Check randomly at one place by releasing LPG. Mention the sensor number.			
A-06	Record any unsafe conditions of the Safety & Fire Protection system (Specify in brief under the remarks column.)			
A-07	Check CCTV footage to see whether frisking is being done at the gate.			
A-08	Observation on the use of PPE			
B. BULK STORAGE AREA				
B-01	The earthing connection apparently sound.			

DRAFT 2

	Check at random. Mention area			
B-02	Whether the level instruments are working.	Yes/ No		
B-03	Portable fire extinguishers in position	Yes/ No		
B-04	Inlet and Outlet valves of all Deluge Valves is in open condition and DV is in fire readiness.			
B-05	Housekeeping is in order	Yes/ No		
B-06	Any sign of LPG leakage. If yes, mention the location and action taken	Yes/ No		
B-07	Record the condition of the Water seal. Mention the location of the water seal.			
B-08	Operation of the ROV of one storage vessel (not in operation) to check and record the ROV number.			
C. TANK WAGON UNLOADING				
C-01	Any visible sign of LPG leakage. If yes, mention the location and action taken.			
C-02	The earthing connection apparently sound. Check at Random. Mention area			
C-03	a) Whether the physical condition of the hose is proper. Record if any action is to be taken. b) Are these properly plugged if not in use.			
C-04	Housekeeping in order	Yes/ No		
C-05	Portable fire Extinguishers in position.	Yes/ No		
C-06	Inlet and Outlet valves of all Deluge Valves is in open condition and DV is in fire readiness.			
C-07	Record the condition of the water seal	~~Yes/ No~~		
C-08	ROVs in remote mode	Yes/ No		
D. BULK LPG TANK TRUCK GANTRY				

DRAFT 2

D-01	Are all vehicles allowed in the licensed area are permanently fitted with PESO approved spark arrestors on the exhaust Pipe (Random check. Record TT No)	Yes/ No		
D-02	Is the vehicle engine and master switch being shut off during Unloading bulk LPG (Random check. Record TT No)	Yes/ No		
D-03	Are the earthing connections properly made before connecting the unloading arms and hoses to the tank trucks.	Yes/ No		
D-04	Any visible sign of LPG leakage. If yes, mention the location and action taken.	Yes/ No		
D-05	Safety check of the Tank truck carried out. (Random check. Record TT No.) a) 2 Nos. of fire extinguishers available in good condition/ easily accessible. (Physical check). 1 no. of DCP fire extinguisher available in driver's cabin b) Safety instruction booklet, TREMCARD available. Record any abnormality/ discrepancies observed. c) Wheel choke used d) Whether Crash guard provided	Yes/ No		
D-06	Portable extinguishers are available in designated places.	Yes/ No		
D-07	Inlet and Outlet valves of all Deluge Valves is in open condition and DV is in fire readiness.			
D-08	Record observations on Housekeeping	~~Yes/ No~~		
E. PUMP HOUSE/ COMPRESSOR ROOM				
E-01	Check the condition of Electrical connections and record any deviation	Yes/ No		
E-02	Gauges on pumps/ compressors in working condition	Yes/ No		
E-03	Safety guards in position	Yes/ No		

DRAFT 2

E-04	Record any Abnormal vibration/ excessive noise	~~Yes/ No~~		
E-05	Portable fire extinguisher in position	Yes/ No		
E-06	Inlet and Outlet valves of all Deluge Valve is in open condition and DV is in fire readiness.			
E-07	Any visible sign of LPG leak or any Unsafe conditions observed (record under remarks column	Yes/ No		
E-08	Record observation on Housekeeping	~~Yes/ No~~		
F. FILLING SHED				
F-01	Any horizontal rolling of LPG cylinders is observed in the shed. Trolley is being used for shifting of cylinders.	Yes/ No		
F-02	Observation on earthing connection of the carousel guard rail and filling machines	~~Yes/ No~~		
F-03	Water in the test bath is clean and maintained up to the required level.	Yes/ No		
F-04	Any sign of leakage or unsafe condition. If yes, mention the location and action taken.	~~Yes/ No~~		
F-05	Are all the fire extinguishers available in an accessible location	Yes/ No		
F-06	Inlet and Outlet valves of all Deluge Valve is in open condition and DV is in fire readiness.			
F-07	Record observation on Housekeeping	Yes/ No		
F-08	Condition of Rotary equipment/ coupling and coupling guard etc. Record any deviation			
F-09	Check interlock of Vapour Extraction Unit with filling system			
G. CYLINDER STORAGE/ HANDLING				
G-01	Are cylinders stacked properly in	Yes/ No		

DRAFT 2

	demarcated areas?			
G-02	Any leaky, overfilled cylinder is kept at the end of the day. Record No. if any of the leaky un-evacuated cylinders	Yes/ No		
G-03	Safety caps fixed on cylinders in stock	Yes/ No		
G-04	Inlet and Outlet valves of all Deluge Valves are in open condition and DV is in fire readiness.	Yes/ No		
G-05	Are all the fire extinguishers available in an accessible location	Yes/ No		
G-06	Record observation on Housekeeping			
H. TT Parking Area				
H-01	Observation on the parking of the Bulk LPG tank trucks & packed trucks: a) Number of Bulk LPG tank trucks & packed trucks parked. b) Whether trucks are parked in the slot/marking in drive out condition c) Is cleaning of the cabin of the truck observed? If yes, action taken			
H-02	Record observation on Housekeeping of the parking area and the Driver's rest room			
H-03	Availability of fire protection equipment (Water monitor, hydrant hoses and nozzle)			
*** Put specific observation wherever the conditions are not in line with specified requirement levels.**				

DRAFT 2

Annexure-14-II

CHECKLIST FOR ELECTRICAL AUDIT

LOCATION:

DATED:

AUDIT/ INSPECTION BY:

Sr No	Location	Particular Details
1	General and Statutory	Is the record of the designated persons properly made and kept up to date and duly attested? Check the Supervisor/ wireman license and check the validity
2	General and Statutory	Whether any unauthorised temporary installation exists?
3	General and Statutory	Give a report on the condition of service lines, cables, wires, apparatus, and such other fittings placed by the supplier or owner of the premises. If not satisfactory, give details.
4	General and Statutory	Whether a suitable firefighting system as per the regulation has been provided?
5	General and Statutory	Designated Electrical Safety Officer available?
6	General and Statutory	(i) Work permit being followed? (ii) Electrical isolation - Lockout/ Energization Line clearance being taken? (iii) Earth excavation clearance being obtained?
7	General and Statutory	Are values observed are within the prescribed range in line with OISD-STD-137 for Electrical tests (Insulation Resistance/ Earth Resistance/ Earth continuity/ Transformer Tan Delta Test/ Transformer oil test/ Dissolved gas analysis)
8	General and Statutory	Failure analysis is being done for the following: High voltage equipment Repetitive failure of all kinds Power system failure affecting Plant operation Failure due to arcing or sparking, which caused unsafe condition/ accidents Failure to newly installed or repaired systems Critical equipment and systems

DRAFT 2

9	General and Statutory	History cards are being maintained for following Rotary equipment Stationary equipment Protective system Distribution network Other critical equipment
10	General and Statutory	The plant is having identification of zones as Z0, Z1 and Z2, and the same is in line with OISD-STD-113
11	General and Statutory	Electrical Caution Notice/ First Aid Training in case of HT installations - Every medium, high voltage installation shall affix permanently in a conspicuous position a caution notice (in English or Hindi and the local language of the district, and where Hindi is the local language, in English and Hindi).
12	General and Statutory	Provided that where it is not possible to affix such notices on any generator, motor, transformer, or other apparatus, they shall be affixed as near as possible thereto: Provided further that where the generator, motor, transformer or other apparatus is within an enclosure, one notice affixed to the said enclosure shall be sufficient for the purpose of this rule.
13	General and Statutory	In the case of generating and enclosed stations have fire buckets, filled with clean, dry sand, have been conspicuously marked and kept in convenient situations in addition to fire extinguishers suitable for dealing with electric fires?
14	General and Statutory	State if First Aid Boxes or Cupboards are conspicuously marked and properly equipped are provided and maintained.
15	General and Statutory	Is some staff trained in First Aid Treatment?
16	General and Statutory	Is a 30 m Amp RCCB/ ELCB provided in the admin building/ canteen area and other buildings, and are in working condition?
17	General and Statutory	Is the Lux as per standard for proper lighting in the admin building?
18	General and Statutory	Whether C02 fire extinguisher is provided in the admin building for electrical fire fighting?
19	General and Statutory	Are the buried underground earth conductors getting exposed at any location?

DRAFT 2

20	Metering and VCB Room	The following also shall be checked/ ensured for HT/ LT Breaker: i) Breaker interlock mechanisms are operative. ii) All releases are provided and connected properly as per the design drawings. iii) Alignment of fixed busbar contacts has been properly done with the draw-out contacts of breakers. iv) Breaker contact resistance by millivolt drop test is within recommended values. v) Whether checking of VCB bottles for healthiness is done every three years as per OISD-STD-137.
21	Metering and VCB Room	SWITCHGEAR PANEL i) Check for proper clearance and electrical isolation. ii) Check the initial IR values of the three phases and record them. iii) Check the vermin proof-ness and cleanliness of the panel. iv) Check the condition of the cable end termination. v) Check for any hot spots/ flash on the conductor or insulator. vi) Check for any cracks on the insulator and replace if necessary. vii) Check for tightness of connections. viii) Check the condition of joints between copper and aluminium and ensure that there is no corrosion due to bimetallic reaction. ix) Check for any wear on the main bus male contacts and replace if necessary. x) Check the condition and alignment of shutters in the panel. xi) Check the condition of the rails on which the breaker moves. xii) Check the alignment and condition of carriage limit switches. xiii) Check the tightness of control wiring and soldering of the multipoint socket. Check the IR value of the control cables. xiv) Check the condition of the indicating lamps and Panel meters xv) Check the safety interlock for proper functioning. xvi) Check the final IR value of the bus. xvii) Check the protection system by the Secondary Injection test. xviii) Check the IR value of the power cable of the panel before charging. xix) Check corrosion on the panel and paint if necessary. xx) Check the condition of the explosion vent for

DRAFT 2

		cleanliness. xxi) Check the space heater for its proper functioning.
22	Metering and VCB Room	(Whether instructions (in English or Hindi and the local language of the district, and where Hindi is the local language, in English and Hindi) for the resuscitation of persons suffering from electric shock have been affixed in a “conspicuous place”?
23	Metering and VCB Room	Whether adequate precautions, e.g., Bakelite, etc., are taken to ensure that no live parts are so exposed as to cause danger?
24	Metering and VCB Room	Whether the space behind the switchboard exceeds 75 cm in width or is less than 20 cm?
25	Metering and VCB Room	Whether all conductors and apparatus, including live parts thereof, are inaccessible
26	Metering and VCB Room	Whether trenches inside the substation containing cables are filled with non-inflammable material or completely covered with non-inflammable slabs?
27	Substation	Is a painted line diagram of power/ lighting distribution provided?

DRAFT 2

28	Substation	MCC Panel condition: a) Doors are in order and closed. not openable in switched ON condition. b) Dust/soot/ cobwebs do not exist c) Lugs existing at each cable, termination not loose. d) Visual condition of switch, contact, fuse, etc. e) Control wiring condition satisfactory. f) Jumper condition between the bus and distribution is in order. g) Grounding in order h) Alarm/ trip circuit working.
29	Substation	No temporary electrical connection exists.
30	Substation	Rubber mats (ISI15652) are provided
31	Substation	a) Protective relays set at recommended values. b) Whether relays and cables are tested, and records are maintained.
32	Substation	All fuses are of the proper type and rating as specified in the single line diagram.
33	Substation	Fire extinguishers are of the proper type and are in working condition.
34	Substation	Lighting condition satisfactory (fixtures/ tubes/ switches, etc.)
35	Substation	All energized equipment is provided with caution boards.
36	Substation	The following also shall be checked/ ensured for HT/ LT/ Panel: i) The panel is firmly bolted to the base plate, which should be grouped properly. ii) Panel is earthed with the required number and size of earthing conductors. iii) Panels are properly levelled and aligned in all directions. iv) The tag numbers are prominently displayed on each panel on both front and rear sides, and they match. v) Whether the doors of the MCC panel are not openable when the panel is in the switched-on condition? vi) Whether the MCC panel are vermin proof vii) Whether thermography of the panels has been carried out and a record maintained.
37	Substation	The following also shall be checked/ ensured for HT/LT/Panel: i) Danger board has been provided. ii) All panel doors fixing bolts are in position and tightened.

DRAFT 2

		iii) Bus bar connecting links are firmly connected with suitable size and type of bolts. iv) All connections are tight.
38	Substation	The following also shall be checked/ ensured for HT/LT/Panel: i) Bus bar insulation, wherever provided, is in good condition. ii) Bus bar support insulators and bushings do not have cracks/ damage. iii) All the openings have been sealed. iv) All gaskets are in position.
39	Substation	The following also shall be checked/ ensured for HT/ LT/ Panel: i) Fire extinguishers of the correct type and capacity have been provided, in the required number. ii) The exhaust fan/ pressurisation system, wherever provided, is in working order. iii) Bus bar terminal shutters are in order. iv) Cables entering the panel are provided with cable glands iv) Unused holes are plugged properly.
40	Substation	Is there any visible signs of overloading in respect of any apparatus?
41	Substation	Whether flexible cables used for portable or transportable equipment covered under the Regulation are heavily insulated and adequately protected from mechanical injury?
42	Substation	Whether instructions in English or Hindi and the local language of the district, and where Hindi is the local language, in English and Hindi, for the resuscitation of persons suffering from electric shock have been affixed in a “conspicuous place”?
43	Substation	(ii) Are the persons mentioned in this regulation able to apply instructions for resuscitation of persons suffering from electric shock?
44	Substation	Whether a suitable linked switch, or a circuit breaker, or an emergency tripping device is placed near the point of commencement of supply so as to be readily accessible and capable of being easily operated to completely isolate the supply?
45	Substation	Whether adequate precautions are taken to ensure that no live parts are so exposed as to cause danger?

DRAFT 2

46	Substation	i) Whether a clear space of 100 cm is provided in front of the main switchboard?
47	Substation	(ii) Whether the space behind the switchboard exceeds 75 cm in width or is less than 20 cm?
48	Substation	In case the clear space behind the switchboard exceeds 75 cm, state whether a passageway from either end of the switchboard to a height of 1.80 m is provided.
49	Substation	Whether all conductors and apparatus, including live parts thereof are inaccessible
50	Substation	Whether trenches inside the substation containing cables are filled with non-inflammable material or completely covered with non-inflammable slabs?
51	Substation	Are conductors and apparatus so arranged that they may be made dead in sections for carrying out work thereon?
52	Substation	Whether protections and interlocks have been provided? Give the details of the protection schemes and their settings. 1) Interlock of Mains with DG 2) Interlock of the Transformer with VCB
53	Substation	Have all non-current carrying metal parts associated with the installation been effectively earthed with the earthing system or mat by two separate and distinct connections?
54	Substation	Test records are available for the following Earth Resistance test Insulation Resistance test Cathodic protection parameters Transformer oil test Calibration tests of meters/ instruments Simulation test of protective relay circuits Relay test record Test records on the battery Other records
55	Substation	In the case of generating stations and enclosed substations, whether fire buckets filled with clean, dry sand have been conspicuously marked and kept in a convenient location in addition to fire-extinguishers suitable for dealing with electric fires?
56	Sub Station	Are any unused cable entries left open?

DRAFT 2

57	Sub Station	Whether the bus is provided with distinct colour identification (Red, Yellow, Blue and Black)
58	Sub Station	**LV Transformer/ Lighting Transformer**
59	Sub Station	Cable terminations/ horn gap/ CTPT/ main circuit breakers, etc., are in good condition.
60	Sub Station	The foundation is in good condition.
61	Sub Station	The transformer is positioned properly.
62	Sub Station	No oil leak**/** sweating observed.
63	Sub Station	Silica gel is in good condition.
64	Sub Station	Oil / winding temperature is in order.
65	Sub Station	Terminal Box earthing is in good condition.
66	Sub Station	Oil level in transformer(s) and circuit breakers(s) found in topped up condition.
67	Sub Station	Periodic / scheduled oil quality test/ oil change conducted, and dielectric strength test results are found acceptable.
68	Sub Station	Whether the general condition of the transformer yard/ shed, the flooring surface, painting of the transformer/ shed/ fencing/ gates, and overall cleanliness/ housekeeping, etc. is in good condition? Whether dry vegetation/ grass found in the yard?
69	Sub Station	Whether there is a locking arrangement for the Transformer yard to prevent entry by unauthorized persons? If so, is it being followed?
70	Sub Station	Are the buried underground earth conductors getting exposed at any location?
71	Sub Station	3 phase Motors
72	Sub Station	Is the motor provided with double earthing?
73	Sub Station	Is the motor provided with fan guard and coupling guard?
74	Sub Station	Are the junction boxes/push button/ lighting fixtures provided with the double earthing?

DRAFT 2

75	Switchyard	Whether the location/ type of transformer is in conformity with the area classification drawings.
76	Switchyard	Condition of transformer: a) Grounding b) Oil level c) Terminal box earthing d) Oil/ winding temp. e) Periodic oil quality test conducted.
77	Switchyard	The following shall be checked/ensured for the Transformer: i) The name plate details conform to specifications. ii) All the accessories have been fixed properly. iii) Proper earthing as per IS: 3043 IS:10028 is provided for the transformer body, neutral.
78	Switchyard	The following also shall be checked/ ensured for the Transformer: i) Oil filtering and conditioning is to be done as per IS:10028. ii) The oil levels in the transformer conservator tank and all the bushings are up to the marked point. iii) Dielectric strength and acidity of the oil, taking three samples (minimum), (if required, filter the oil) iv) There is no oil leakage.
79	Switchyard	The following also shall be checked/ensured for the Transformer: i) The silica gel is in reactivated condition. ii) The breather pipe is clear of any blocking and contains oil up to the proper level. iii) The explosion vent diaphragm does not have any dents, accumulation, or any oil and air has been released. iv) The Buchholz is mounted correctly, there is no friction in the movement, and the floats are free. There should be no air bubbles inside the Buchholz relay.
80	Switchyard	The following shall be checked/ ensured for the Lightning Arrestor: a) The voltage and current rating of the arrestor match with the approved drawings. b) Location and spacing between arrestors is as per approved drawings. c) Height of grading ring above ground is as per approved drawing. d) Explosion vent is directed away from adjacent equipment. e) Porcelain surfaces are clean and not damaged.

DRAFT 2

		f) Connections to earth do not pass through any metal pipe. g) Earth connection from arrestor to counter is insulated from earth. h) Separate earth pit has been provided for arrestor. i) Insulation resistance value of arrestor is in order. j) Line connection is made as recommended.
81	Switchyard	Whether the transformer oil samples are being tested periodically and the results recorded in a register? State value of dielectric strength of oil.
82	Switchyard	Whether all circuits and apparatus are so arranged that there is no danger of any part(s) becoming accidentally charged to any voltage beyond the limits of voltage for which it/ they is/ are intended?
83	Switchyard	(iii) Has the neutral point at the transformer and generator been earthed by two separate and distinct connections with earth?
84	Switchyard	(i) Is the outdoor (except pole type) substation efficiently protected by fencing not less than 1.8 m in height?
85	Switchyard	Whether the earth pits are properly maintained, and tests are carried out once in six months.
86	Switchyard	Whether two separate earth electrodes of proper size are provided properly for the lighting arrestors?
87	Switchyard	Whether there is a locking arrangement for the yard area to prevent entry to the yard by unauthorised persons? If so, whether is it being followed and keys readily available?
88	Switchyard	Are the buried underground earth conductors getting exposed at any location?
89	Earthing System	Check/ Ensure: The earthing installation has been done in accordance with IS: 3043 approved earthing drawing and specifications.
90	Earthing System	Check/ Ensure: The main earth loop is laid at a depth of 500 mm below grade level. Wherever cable trenches are available, the earth lead is laid in the trenches and firmly cleated to the sidewall of concrete lined trenches. The earthing strip is protected against mechanical damage in process unit areas; the earthing conductor is run along

DRAFT 2

		cable trays wherever possible. The earthing conductor is suitably cleated and electrically bonded to the cable tray at regular intervals. The earthing for equipment is tapped from the main earth conductor and not from the cable tray structure.
91	Earthing System	Check/ Ensure: Earth pit locations are identified and displayed
92	Earthing System	Check/ Ensure: All earth electrodes are tested for earth resistance by means of a standard earth test megger.
93	Earthing System	Check/ Ensure: i) A disconnect facility is provided for the individual earth pits to check their earth resistance periodically ii) In case joints are through connectors/ lugs, ensure no gap is available between the surface of stripes.
94	Earthing System	(ii) General visible condition of the earthing arrangement.
95	Earthing System	(ii) Is the earth wire free from any mechanical damage?
96	Earthing System	(v) Whether earthing has been properly executed and has been tested. If yes, give the value of Earth's resistance.
97	Earthing System	Are the values observed is within range as per IS:3043 and other relevant OISD standard
98	Motor	Is a double independent earthing connection for all motors provided?
99	Motor	Is the grounding wire in good condition (not loose).
100	Motor	a) Is the terminal box and gland flame-proofness maintained and not tampered with? b) Existing cable terminations are proper
101	Motor	a) Is the push button earthed properly, and flame proof condition not tampered. b) General condition of push button good.
102	Motor	Ground resistance at motor/ push button was measured and found in order.
103	Motor	No motor abnormality - Vibration/ bearing condition, faulty guard, fan condition, excessive noise, high body temperature.

DRAFT 4

104	Motor	a) Carousel machine frame to filling gun earthing continuity maintained. b) Carousel machine frame to central column earthing continuity is in good condition.
105	Motor	Conveyor system bonding and earthing are maintained
106	Motor	The following shall be checked/ensured: i) Name plate details of motors and suitability to the specified area classification. ii) The insulation resistance between phase and earth and between phases (wherever provisions exist) of motors is measured before and after connection of the power cables. iii) No unauthorised site modification for flameproof equipment is done. iv) Double earthing with proper earth conductor size is provided as per IS:3043, and earth resistance is within the specified limit.
107	Motor	The following also shall be checked/ensured: i) The direction of rotation is correct. ii) All the terminal boxes are tight with proper gaskets and fasteners. iii) Ensure the coupling guard has been provided. iv) Ventilation and cooling system. v) Condition of bearing grease/ oil
108	Motor	(iii) Whether the motors and controlling equipment are being overhauled periodically, and a record kept of the same in a register?
109	DG set	Whether DGs is provided with two independent body earthing?
110	DG set	Whether the generator panels are made vermin proof?
111	DG set	Wherever more than one generator is provided, whether provision of individual breakers is kept for a paralleling arrangement?
112	DG set	Whether danger notices are provided on equipment/ panels in the generator room?
113	DG set	Whether the generator room is having adequate ventilation?
114	DG set	Whether control panel provided include all protection devices like short circuit, over current, and earth fault relays? Whether calibration of these devices carried out annually?

DRAFT 2

115	solar installations	(1) The following general safety requirements for solar installations shall be ensured, namely: –
116	solar installations	(i) clear pathways of a minimum of seventy five centimetres in width with handrails for roof access and emergency exit shall be provided for the rooftop system;
117	solar installations	(ii) there shall be clear pathways, walkways between the rows or columns of solar panels, which are necessary for cleaning and maintenance;
118	solar installations	(iii) cables shall be laid in trenches for ground based photovoltaic installation;
119	solar installations	(iv) ground mounted solar installations shall be protected by fencing or other means not less than 1.8 m in height so as to prevent unauthorised entry;
120	solar installations	v) Disconnection switches or circuit breakers provided in combiner boxes to disconnect the photovoltaic system from all other conductors of the system shall be located at a readily accessible location;
121	solar installations	(vi) three phases on the alternating current side, and positive and negative conductors on the direct current side shall be marked and identified with different colours:
122	solar installations	(vii) Inverter unit for solar photovoltaics shall be installed in the periphery of the building and as near as the solar panel: Provided that the direct current cable shall be ultraviolet protected or routed through ultraviolet protected pipe;
123	solar installations	viii) there shall be a manual disconnection switch to isolate the system from the grid, and it shall be situated outside the alternating current combiner box; and
124	solar installations	(ix) protection shall be deployed (for both input and output) on site for overload, surge current, surge voltage, short circuit, high temperature, over voltage, under voltage and over frequency, under frequency, reverse polarity, and lightning.
125	solar installations	(2) The following earthing requirements for solar installations shall be ensured, namely:
126	solar installations	(i) Solar earthing shall be as per the relevant standard.
127	solar installations	(ii) the frame of the inverter cabinet shall be connected with the earthing bus bar through the earthing terminals using

DRAFT 2

		flexible braided copper wire;
128	solar installations	iii) all metal casing and shielding of the Plant, each array structure of the photovoltaic yard, equipment, inverters and control systems shall be earthed through proper earthing;
129	solar installations	(iv) the earthing system shall connect all non-current carrying metal receptacles, electrical boxes, appliance frames, chassis and photovoltaic module mounting structures in one long run, and the earth strips shall be interconnected by proper welding and shall not be bolted;
130	solar installations	(v) There shall be adequate number of interconnected earth pits provided in each location, and a minimum required gap shall be provided in between earth pits as per the relevant standard.
131	solar installations	(3) The following protection, testing and interlocking requirements for solar installations shall be ensured; namely: –
132	solar installations	(i) the solar photovoltaic power Plant shall be provided with lightning and over voltage protection by deploying the required number of lightning arresters as per the relevant standards;
133	solar installations	ii) Every combiner box shall be provided with a suitable surge protective device with arc extinguishing capability as per the relevant standards to avoid any risk of fire;
134	solar installations	(iii) the input circuits of the combiner box shall be provided with overcurrent protection as per the relevant standards;
135	solar installations	iv) The output circuits of the combiner box shall be provided with isolation protection;
136	solar installations	(v) Earth fault protection and insulation monitoring for photovoltaic arrays and inverters shall be provided;
Additional		
137	General and Statutory	**Central Control Room (CCR - Not applicable to monitoring room)**
138	General and Statutory	Is 30 m Amp RCCB/ ELCB provided, and is it in working condition?
139	General and Statutory	Are all the air conditioners provided with earthing?

DRAFT 2

140	General and Statutory	Are all the electronic equipment that stores energy in CCR provided with earthing?
141	General and Statutory	What is type of battery is used for UPS?
142	General and Statutory	Whether the battery Room is clean and well ventilated?
143	General and Statutory	Is the cell voltage of the cell being checked and recorded as per approved schedules?
144	General and Statutory	Is there any corrosion on terminals/ connectors?
145	General and Statutory	Whether all the safety equipment (rubber apron, googles, acid proof hand gloves etc.) are available?
146	General and Statutory	Is the operation of UPS satisfactory? (Test for critical service)?
147	General and Statutory	Whether trickle charger working and checked properly?
148	General and Statutory	Is the Lux value as per standard for proper lighting in CCR?
149	General and Statutory	Is Co2 fire extinguisher provided for electrical fire fighting?
150	Sub Station	**DC Battery Charger**
151	Sub Station	What is type of battery used?
152	Sub Station	Whether the battery Room is clean and well ventilated
153	Sub Station	Are the floors acid resistant?
154	Sub Station	Whether exhaust fans provided?
155	Sub Station	Is the cell voltage of the cell being checked and recorded as per approved schedules?
155	Sub Station	Is there any corrosion on terminals/ connectors?
156	Sub Station	Whether safety equipment like rubber apron, googles, acid proof hand gloves, etc., are available?
157	Sub Station	Is the operation of the Battery Charger satisfactory?
158	Sub Station	Whether the trickle charger is working and is it checked properly?

DRAFT 2

Annexure-14-III

CHECKLIST FOR INSPECTION & MAINTENANCE SYSTEM AUDIT

LOCATION:

DATED:

AUDIT/ INSPECTION BY:

FILLING SHED EQUIPMENT - CAROUSELS/ UFM of 2/ 5/ 10/ 47.5/ 425 kg etc. Cylinder Filling	
S. No.	**Check Point**
1	LPG filling carousel, manual filling machine, Drive unit, central column, running rails, Introduction/ ejection devices, and unit filling machines, etc., daily checked, preventive maintenance schedules prepared, carried out and records maintained
2	Whether a 5 kg / 10 kg composite Filling system is provided and functioning properly. Check availability of water spray/water mist facility, purging facility, availability of cage, bonding and earthing of filling gun and filling station, etc.
3	Whether 47.5 kg Filling system is provided and functioning properly. Check availability of Purging, WCU, post Quality Checks, etc...
4	Whether 425 kg. Filling system is provided and functioning properly. Check availability of Purging, WCU, post Quality Checks etc.
5	The remote Carousel data monitoring system is functioning for all the equipment
6	Condition of belt and belt tension/ drive wheel and tyre is satisfactory Oil level of carousel DU maintained.
7	Earthing continuity maintained for: a) Carousel/ machine frame to filling gun, b) Carousel frame to central column, c) The condition of carbon bushes is satisfactory.
8	The earthing interlock of carousel is provided and functioning
9	Filling machine hoses are regularly tested, and records maintained
10	Condition of filling guns - nil LPG leaks
11	LPG return line valves kept fully open - DP valve adjusted for required pressure.
12	Whether the post filling quality checking facilities are available for 5 kg/ 10 kg/ 47.5 kg/ 425 kg filling facilities
13	a) Whether negative and positive tare weight setting is maintained in electronic carousel

DRAFT 2

	and Test cylinders for Positive Tare and Negative Tare are available.
14	Introduction/ Ejection devices and filling machines are checked daily, and scheduled maintenance/ overhaul work carried out.
15	Whether the checking of pin travel is done regularly, and records are maintained?
CVT / VLD / OLD	
S. No.	**Check Point**
1	Calibrations of CVT/ VLD/ OLD are being done regularly, and maintenance schedules as per OEM are followed.
2	A comparator for the valve leak tester and an approved calibration chart is available from the manufacturer.
3	The machine is functioning properly, and rejection is proper
4	Whether calibrated test cylinders are available for checking the accuracy/ calibration of the valve leak detection machine to ensure the accurate performance of the machine? (Requirement: One cylinder per carousel)
5	Whether test cylinders (one cylinder without O and another with a cut “O”-ring) are available and kept in the rejection loop conveyor to ensure the performance of the machine continuously?
Miscellaneous Filling Shed equipment	
1	Whether a trolley for handling cylinders is available and is functional.
2	Whether the Scissor lift up and down mechanism is of Deadman Type/ Inching Type.
CHECK SCALES: single static/ double static/ dynamic check scales.	
S. No.	**Check Point**
1	Check scales are well maintained, and calibrations carried out on a daily basis with standard weights
2	The lifting table mechanism is operating satisfactorily
3	Dynamic check scale triplex chain tension, sprockets condition is satisfactory.
FILLING SHED EQUIPMENT - WATER BATH	
S. No.	**Check Point**
1	Movement of cylinders is smooth. Water level is maintained up to Bung of cylinder.

DRAFT 2

2	Is the water clean and without foul smell.
3	Safety grating/ handrail around the water bath is in place
FILLING SHED EQUIPMENT - CONVEYORS	
S. No.	**Check Point**
1	Noise level is satisfactory and smooth running, tension rollers and conveyor chains tension is satisfactory and no jerks
2	Condition of chain and channels/ liners is satisfactory (not worn out, corroded, broken, etc.)
3	Condition of soap solution is satisfactory (level/ leakage/ dirty/ concentration etc)
4	Conveyer railing is in place
5	Deadman control for the Telescopic conveyers is provided/ is in working condition.
FILLING SHED EQUIPMENT - CONVEYOR DRIVE UNITS	
S. No.	**Check Point**
1	Oil level in gear boxes is satisfactory, Breathers/ sight glass / dipstick is satisfactory and no oil spillage
2	Condition of sprockets and chains is satisfactory.
3	Couplings/ alignment is satisfactory, all coupling guards and motor guards are in position
	Coupling guard, drive chain cover is in place
4	All foot bolts / base frame bolts are tight. Gear box operation is smooth,
5	Condition of Bearing housing/ Counterweights etc is satisfactory.
FILLING SHED EQUIPMENT - PNEUMATICS (Actuators; FRLs; tubing; pneumatic instruments; etc.)	
S. No.	**Check Point**
1	Condition of Filter Regulator is in satisfactory condition. No water/dust in filters / condensate drained
2	No air leaks from lines, ferrules, valves and Pneumatic dressing is satisfactory
3	Air pressure at buffer tanks in sheds is satisfactory and condensate drained from all vessels

DRAFT 2

FILLING SHED EQUIPMENT - AUXILIARY UNITS (Purging; Evacuation; Wash/Dry; Vapour Extraction.; Sealing; etc.)	
S. No.	**Check Point**
1	Purging unit in good working order (permanent LPG piping provided / vacuum / LPG pressure/ pneumatics is satisfactory). The cylinder purging system cut-off shall be based on vacuum pressure. Note: Check the working of the purging unit by operating the machine for a batch of 20 cylinders. The machine shall work without any manual intervention for all 20 cylinders.
2	Whether a purging unit is provided for all types of LPG cylinders, i.e., 2 kg, 5 kg, 10 kg, 14.2 kg, 19 kg. 35 kg. 47.5 kg. 425 kg.
3	Whether an evacuation unit is provided for all types of LPG cylinders, i.e.,2 kg, 5 kg, 10 kg, 14.2 kg, 19 kg, 35 kg, 47.5 kg, 425 kg.
4	Evacuation unit is functional / condition of guns is satisfactory / no leakages from guns / hoses/ flanges/ valves/ strainer/ swivel joints.
5	Evacuation unit vessels: HLA is operational; Auto changeover of valves between vessels is provided and functional.
6	The valve changing machine is in working condition, and the torque setting is within range, with the availability of a torque checking bench.
7	Valve Change Without Evacuation is in working order? a) There are no LPG leaks from the sealing bell b) The control for lifting of sealing bell is in working condition. c) Vents are not choked. d) Torque setting is within range, and the torque checking bench is available. e) The arm is available for replacing SC valves and is in working condition. f) Weight correction and valve leak check are done after the valve change g) The control for lowering the wrench head is in working condition.Note: Check whether all the controls of the valve change machine are dead man type?
8	Whether the degassing unit is in working order and has been provided with a closed-loop gun?
9	Whether a degassing unit is provided for all types of LPG cylinders, i.e.,2 kg, 5 kg, 10 kg, 14.2 kg, 19 kg, 35 kg, 47.5 kg, 425 kg.
10	De pressurisation unit is in working order
11	Whether a depressurising unit is provided for all types of LPG cylinders, i.e.,2 kg, 5 kg, 10kg, 14.2 kg, 19 kg, 35 kg, 47.5 kg, and 425 kg.

DRAFT 2

12	Vapour extraction unit functional/ preventive maintenance of blowers carried out/ no noise/vibrations in blowers/ foundations, and ducts are satisfactory. Carousel (including filling machines of 5 kg, 10 kg, 14.2 kg, 425 kg and 47.5 kg and 425 kg) and at all places where LPG is likely to leak – Blower interlock available and working.
13	Whether the suction of the Vapour extraction system is effective.
14	The cylinder washing/ drying unit is in good working order/ no water spillage around. Washing unit pump properly maintained
15	The Inline Hot air sealing unit and portable hot air sealing units are in good working condition

IN-HOUSE STATUTORY TESTING AND PAINTING

S. No.	Check Point
1	Whether rejection criteria are displayed for all types of cylinder defects (for dents, cuts, gouges, bulges cracks, laminations or punctures, bottom shell thickness, foot ring bottom clearance etc.) in line with Table 1, 2 and 3 of IS:16054.
2	Whether the bottom shell thickness is measured for all cylinders and rejected if not found is satisfactory and records maintained thereof.
3	Whether vacuum pump is in proper working condition and cylinders are being cleaned from inside.
4	Whether magnetic rod is available and being used to remove ferrous particles from bottom of cylinder surface.
5	Whether low voltage lamp is available, and cylinders are being visually inspected.
6	Whether check scale of Least Count of not more than 10 gms is available and cylinders are being checked for tare weight after external and internal cleaning.
7	Whether soft wire brush and $3/4^{th}$ 14 NGT thread is available and used for cleaning tap. Whether L-1, L-9 gauges are available to check valve pad thread.
8	Is the hydro testing machine is in proper working condition with rotary movement and cylinder being tested at the test pressure of 2.45 MPa (25 kg/cm^2). Whether the auto cut-off arrangement is available
9	Whether all pipework, flexible tubing, valves, fittings, and other components forming the pressure system of the test equipment has been designed to withstand a pressure of 1.5 times of the maximum test pressure of cylinder to be tested. Records to be checked.
10	Whether dedicated pressure gauges for each cylinder to read the cylinder test pressure for hydro test machine.

DRAFT 2

11	Whether two numbers of main gauges are provided on the inlet line of the equipment after the isolation valve of hydro test machine.
12	Whether gauges used are calibrated using a master gauge at a minimum interval of one month.
13	Recalibration of the master gauge is being done
14	Whether a suitable device (SRV, additional pressure switch, etc.) is fitted in the hydrotest machine to ensure that no cylinder is subjected to a pressure in excess of its test pressure by more than 10%
15	Whether the paint booth blower is in working condition with an adequate water curtain.
16	Whether both the primer coating and enamel paint application on the external surface of cylinder being done by using a spraying gun.
17	Whether an auto cut-off mechanism is available for the valve fixing machine, and the machine will stop automatically after reaching the set torque of 20 +/- 2 kg.
23	Whether the pneumatic testing machine is in working condition and is able to develop a pressure of 12 kg/cm^2.
24	Whether facility available to fully immerse the cylinder during pneumatic test and whether this facility has provision for rotation of cylinder manually during the test.
26	Whether Air drier of suitable capacity is provided
27	Whether Pressure gauge of suitable capacity is provided at the air filling point to check the pressure of air in the cylinder prior to starting the test.
28	Whether the following tools are available for quality/ non-conformity testing.
	a) Elcometer
	b) Dial gauge
	c) Vernier calliper
	d) Bulge ring
	e) L1 and L9 gauges
	f) Bung screw tap
	g) Low voltage lamp insertable into cylinder
	h) Magnet insertable into cylinder for cleaning metallic particles.

DRAFT 2

	i) Weighing machine with least count 10mg
	j) Torque wrench
	k) Ultrasonic Thickness gauge. LC of 0.01mm.
	l) Pneumatic air flaring unit
TANK TRUCK GANTRY/TANK WAGON GANTRY/ WEIGHBRIDGE	
S. No.	**Check Point**
1	Hoses used are as per OISD-STD-135 or Unloading arms are free and easy to operate, quick couplings in good working condition
2	Loading hoses/ arms when not in use are stored properly and open end kept blinded [O] / Female coupling kept covered with plastic/rubber caps.
3	Loading / unloading hoses do not have kinks / cracks.
4	Hoses periodically tested as per OISD-STD-135 and test records available / Unloading Arms tested, and test records maintained
5	Studs and gaskets used for connecting with tankers are of acceptable quality. Whether foundation bolts and other bolts of unloading arms are properly tightened
6	All quick shutoff valves functioning properly
7	Interlocking system of female coupling with male coupling is satisfactory in all unloading arms - not tied with ropes/wires.
8	Check for LPG leakage from ends of loading arm, swivel joints etc.,
9	Gantry structural in good condition, painting in good condition.
10	Weighbridge in working order / calibration not overdue
11	Entry/ exit ramps at level with WB platform and condition is satisfactory
12	Railway track insulation checked and found is satisfactory.
13	Whether function tests of Breakaway couplings are carried out as per OEM recommendations and records available.
14	Whether the function test of Breakaway couplings is carried out using a trolley, simulating the movement of a tank truck. (The function test shall never be carried out with Bulk LPG tank trucks) Check the availability of the manual trolley at the Plant for simulating the truck movement.

DRAFT 2

15	Whether the earthing interlock system is provided and working? Note: Check that the system should not work when the crocodile clip is connected to the TLD structure. The system shall work only if the crocodile clip is connected to the Tank Truck vessel and chassis.
HORTON SPHERES / BULLETS/MOUNDED BULLETS	
S. No.	**Check Point**
1	Five-year inspection / Hydrostatic tests as per SMPV Rules,2016, for all pressure vessels carried out and records maintained
2	All SRVs are tested annually / tests are not overdue [O]. SRVs are set at different pressures as per guidelines.
3	Two independent level gauges are provided and in working condition
4	Whether Level gauges of LPG Vessels are calibrated once in 5 years during NDT Testing.
5	HLA - ROV interlocks with the audio-visual alarm working properly.
6	Fireproof coating/ insulation periodically checked for cracks, bulging and suspect areas are further inspected for presence of corrosion, etc., and corrective steps taken
7	Bullet pedestals in good condition - proper slope on top / no voids between pedestal and supports
8	Condition of paint of vessel acceptable.
9	Condition of kerb wall, water seal in shallow sump is satisfactory. There shall not be any leaks / cracks in kerb wall. Whether flooring inside kerb wall is free of vegetation.
10	Condition of mound is satisfactory and no cracks on the wall. Nil water seepage observed from the walls and inside tunnel area.
11	Whether tunnel inside area is free from cobwebs, honeycombs and any unwanted material and Whether water is not allowed to enter the tunnel from the outside surroundings and whether the tunnel is sloped away from the vessel
MSV CP SYSTEM AND SETTLEMENT	
S. No.	**Check Point**
1	a) Whether all the reference electrodes are in working condition. b) Whether Instant OFF PSP 2nd reading within 1 second is in the range of -0.85v to -1.2v for Cu-CuSO4 and +250mV to -100 mV for Zinc reference. c) Whether high impedance voltmeter >10 Mega ohm is available at location.

DRAFT 2

2	a) Whether voltage of reference electrode located at 6 O'clock position of each bullet in the MSV is being monitored fortnightly at TR unit and records maintained. b) Whether PSP reading at the test lead points (monitoring junction box) for entire vessel is being taken once in a quarter and records maintained.
3	a) Whether "OFF PSP" readings at the 12 O'clock (top) and bottom quadrant of the vessel (where monitoring pipes are installed) with Cu/ CuS04 portable half-cell is being monitored quarterly. b) Whether one portable reference electrode (half-cell) is maintained at location for measuring PSP and proper verification/calibration of half-cell is ensured for accuracy
4	a) Whether anode current is being monitored quarterly by measuring millivolt drop across the shunt and records maintained. b) Whether protective current is monitored quarterly and is being maintained less than the design current. c) In case protective current is beyond the design criteria whether the same is being analysed for deterioration of coating, proper electrical isolation of the vessel and interference effect due to Cathodic Protection of underground cross-country pipelines.
5	a) Whether monolithic Insulating joints are being inspected quarterly by ON/OFF PSP test method and using Insulation Joint Tester once in a year and records maintained. b) Whether isolation of vessels from foreign structures is quarterly tested by checking On and Instant Off PSP on protected and unprotected side of Monolithic Isolation Joint and records maintained.
6	a) Whether the DC De-coupler is being checked quarterly using the ON/ OFF PSP test method, and the potential across the protected side and unprotected side and records are maintained. b) If the DC De-coupler is of electrolytic type, the liquid level (KOH) is being replenished quarterly or early if required.
7	a) Whether a single TR unit is provided for MSV. If multiple TRUs are provided for an MSV, whether the same is based on TR unit rating, coating breakdown factor, CP protection criteria, stray current interference between the vessels and whether the same is documented in design documents. b) Whether minimum one reference cell (Cu/CuSO4) at 6 O'clock position of each vessel is connected to TRU.
8	a) Whether Location of anode and permanent reference electrodes with numbering is clearly identified in the layout drawing and the anode/ reference electrode to be installed in the same sequence at EJB. b) Whether minimum three number of reference cells of each vessel are connected to TRU and PSP in TRU is being monitored fortnightly.
9	a) Whether Transformer Rectifier is provided with digital meters to read the output voltage, current and PSP readings. b) Whether current interrupters are synchronized while measuring Off PSP wherever multiple TR units are provided for a mound.

DRAFT 2

10	a) Whether the CP system design document and operations manual are available. b) Whether the ATR report of the CP system AMC report are available. c) Whether design documents mention current drawn at the initial and end of the life of coating.
11	a) Whether all the anode-to-anode tail cable field joints are outside the mound area for future maintenance. b) Whether Surge diverters installed across Monolithic Insulation Joint are tested yearly using a 500 V AC or DC megger. c) Whether surge diverter cable leads connected across the Monolithic Isolation Joint are not tampered. Size and length of the cable leads shall be as per manufacturer original supply.
12	a) If cathodic protection criteria ((-) 0.85 to (-) 1.2 V) is not achieved for MSV, then whether appropriate measures are taken to identify the root causes and their mitigation so as to achieve criteria (-) 0.85 to (-) 1.2 V with respect to Copper/ Copper Sulphate half-cell at all measurement points on Mounded Storage Vessel. b) Whether the Root cause and mitigations have been recorded.
13	Whether electrical isolation using epoxy wrapped laminate, circumferentially wrapped around the pipe with a minimum thickness of 5 mm and width covering the entire contact area, is carried out to achieve electrical isolation.
14	Whether an access pit is provided for inserting the portable reference Cu/CuSo4 electrode near the Liquid line ROV for each vessel, for checking the integrity of the Insulation Joint
15	Settlement analysis of mounded vessels is carried out with respect to the initial reference before hydrotest, and records available with respect to Mean Sea Level, and the half-yearly reports are available.
16	Initial Settlement readings before hydro testing of Mounded vessels are available. The settlement readings are compared with initial readings before hydro testing and the same are within the allowable settlements as per design documents/ PESO approved drawings.
17	a) Whether a minimum of two permanent benchmarks are provided away from the MSVs and are earmarked. b) Whether cross verification/ correctness of the established permanent benchmark levels with each other is being carried out every six months to ascertain nil settlement or deviation in the existing permanent benchmarks.
18	a) Whether a minimum of three reference points are provided for each vessel to ascertain uniform/ differential settlement and identify possible vessel bending. b) Whether the maximum spacing of these reference points is approximately twice the vessel diameter. Note 1.
19	Whether the maximum spacing of these reference points is approximately twice the

DRAFT 2

	vessel diameter. Note: 1
20	Whether the settlement analysis report consists of the following 1. Overall settlement. 2. Differential settlement, which causes bending of the vessel. 3. Differential settlement, which causes sloping of the vessel.
21	a) Whether the document consists of a permissible overall settlement available. b) Whether the document consists of permissible differential settlement available.
22	Whether Spring supports are provided for liquid lines, and the readings of the same are recorded along with the settlement readings of MSVs
23	a) Whether the settlement of spring support of the Liquid inlet/ outline pipeline in the tunnel is recorded half-yearly and monitored. b) Whether Stress analysis of the liquid inlet/ outlet pipeline is carried out if spring support is compressed more than 60% of the full-scale reading, since the initial hydro test of the MSV.
24	Whether the fasteners/ nuts provided on the studs of spring supports are not in a tightened condition to enabling free movement.
PIPELINES / VALVES: PIPELINES A/G & U/G (LPG & UTILITIES)	
S. No.	**Check Point**
1	Inspection records of underground/ aboveground LPG lines are available
2	Whether inspection data as well as thickness data of newly constructed pipelines is collected to function as base for the establishment of corrosion rates.
3	Whether details of the thickness survey of LPG pipelines are maintained on an isometric sketch and inspection format as provided in line with Annexure-V of OISD-STD-130.
4	Pipe supports are not damaged Pedestals are provided with point contact and are spaced at less than 6 m. Whether Pipelines at any area have been dislodged from its support such that the weight of the pipe is distributed unevenly on the hangers or the saddles.
5	Whether pipeline shall be laid at a height of 300 mm to 400 mm above finished ground level and no vegetation growth below pipelines observed or concreting/ brick pitching done below pipelines to prevent unwanted vegetation.
6	Whether Pipelines crossing roads/ dykes are wrapped and coated - visible portions are not damaged. Pipe surfaces are free from deep pitting corrosion and large pitted areas.
7	Whether all pop-action valves are tested (properly set and not passing) and provided with isolation valves and are kept locked open.

DRAFT 2

8	Flange joints have proper studs/ nuts, gaskets and are leak free.
9	F/W monitors/ hydrants brick pitched and provided with platforms.
10	FW pipelines are in healthy condition and no leakages observed
11	Standard Pipeline colour coding maintained.
12	Valves: Whether leaks observed in Valve Gland/ Gasket. Whether maintenance (painting / Greasing of valve stems) of valves being carried out. Overall condition of valves (No bent/ damaged valve stems, missing wheels). Valves on Drain line and vent are capped.
SERVICE / SECURITY AIR & WATER LINES, BOREWELL	
S. No.	**Check Point**
1	Bore wells are in working condition, flowmeters are provided and are in working condition, and Daily records of water drawn are recorded for each bore well
2	No leaks observed in service/ security air and service water lines
LPG P/H: COMPRESSORS and PUMPS - LPG	
S. No.	**Check Point**
1	Stuffing box leakage is not excessive
2	No oil carryover into the distance piece
3	The condition of belts/ pulleys and alignment is satisfactory
4	Coupling guards in position - rigid
5	Whether double valves have been provided for all the vents of the LPG Compressor.
6	Oil level in sump is sufficient, and quality is satisfactory (no frothing) .Lubricating oil pressure is proper. No oil spillage observed.
7	All tripping devices are healthy.
8	All cooling water passages, including gland cooling, are clear with proper flow
9	No broken/missing foot bolts/ foundation bolts in compressors/ motors / CW pumps.
10	Suction/ discharge bottles are in good condition and tested
LPG P/H: PUMPS / MOTORS - LPG	

DRAFT 2

S. No.	Check Point
1	No leaks from Mech. Seals. API Plan 52 (buffer fluid) systems/ gland packings in good working condition
2	Coupling guards in position
3	Coupling is locked on shafts with Allen / grub screws only (including Cooling Water pumps). No oversized bolts used
4	Constant-level oilers/ grease nipples are in working order
5	Regular schedule for oil change/ greasing followed
6	No broken/missing foot bolts/ foundation bolts in motors/ LPG/ CW pumps
7	All foundation bolts have proper washers/ nuts and are fully tightened. No spacers / packings / earth connections used under main nuts
8	Whether double valves have been provided for all the vents of LPG pumps.
9	Shims / packings used for alignment are of Brass / SS only and of proper shape and size
10	All tripping devices are healthy.
AIR COMPRESSORS (Reciprocating / Screw)	
S. No.	**Check Point**
1	Stuffing box leakage not excessive
2	Condition of belts/ pulleys and alignment is satisfactory
3	Coupling guards in position - rigid
4	Oil level in sump is sufficient, and quality is satisfactory (no frothing)
5	Discharge air temperatures within limits, lubricating oil pressure is proper
6	All Cooling water passages including gland cooling are clear with proper flow
7	No broken/missing foot bolts/ foundation bolts in compressors / motors.
8	All foundation bolts have proper washers/nuts and are fully tightened. No spacers / packings / earth connections used under main nut
9	Aftercoolers are in good condition - no overheating of shell
10	Screw air compressor (filters/ VFD) are in good working condition

DRAFT 2

11	Frequency of starting of Security Air Compressor is more than 30 minutes
12	No abnormal noise from Screw Air Compressor
13	No frothing observed in Air Oil Separator of Screw Air Compressor.
14	Acoustic enclosure of Screw Air Compressor is properly maintained, all doors are locked, locks intact and all the cooling fans and blower are in working condition.
AIR DRYER	
S. No.	**Check Point**
1	Air dryer in proper working condition. Air pressure at inlet / outlet is satisfactory
2	Dew point test done in last 6 months and record available
3	Buffer vessels drained of condensate/ Auto drain system is in working order
4	Air receiver vessels are in good condition - no corrosion/ proper paint and supports/ testing not overdue
5	Whether an online hygrometer/ dew point meter is installed on the outlet of the dryer and is in working condition?
PUMPS / ENGINES / MOTORS - WATER	
S. No.	**Check Point**
1	No excessive leaks from glands - packings in good working condition
2	Gland plates are not overtightened - no smoke from glands during pump running
3	All bearing housing bolts are tight - no bolts missing
4	Constant-level oilers / grease nipples are in working order
5	Regular schedule for oil change / greasing followed
6	Pump suction strainers /engine air filters cleaned as per schedule
7	All gauges in the engines are working accurately
8	All audio-visual alarms of the fire engines are operating
9	Breathers / Vents of bearing housings are as per requirement and kept clean
10	All drive belts are in good condition - proper tension / no cracks / not worn out
11	Coupling hubs are locked on shafts with Allen/ grub screws (CW/ Jockey pumps). No

DRAFT 2

	oversized bolts used
12	Bearing / lube oil temperatures are within limits.
13	Fire Engine exhaust lines are properly supported. Cladding done.
14	Fire engines batteries and charger in good working condition
15	Diesel tanks are satisfactory (breather, LG, covers, etc.), piping and hoses is satisfactory
16	Cooling water / coolant flow in engines is satisfactory and temperatures normal
17	Frequency of starting of Jockey pump is _____ minutes
18	Cooling water system is provided as per manufacturer's recommendations.
FIREWATER TANKS	
S. No.	**Check Point**
1	Foundation-tank pad: No cracks, no excessive settlement; proper slope
2	No voids between tank pad and annular plates and no grass growth from below the plates
3	Annular plate is exposed and painted and not covered by debris/ vegetation
4	Inlet and outlet lines properly supported
5	No excessive undulations/ buckling on the shell/ roof
6	Roof and shell properly painted and condition acceptable
7	Level gauges in working condition
8	Tank outlet valve in good working condition
9	Roof vents (goosenecks) are available and covered with wire mesh
10	Thickness survey carried out on _____. Report available, compared with original thickness readings, accepted by LPG ~~Bottling Plant~~ Installation.
11	Internal and External Inspections of Water tanks are carried out as per OISD-STD-129.
12	Whether settlement analysis of fire water tank has been carried out during comprehensive testing of FWT's.
DG SETS	

DRAFT 2

S. No.	Check Point
1	All trips / alarms (high water temp, low lube oil pressure, over speed) are healthy.
2	Coupling guards in position
3	Insulation provided up to silencer.
4	Condition of radiators/ heat exchangers and no excessive fin damage/ leakages/ scaling
5	Diesel tanks are satisfactory (breather, LG, covers, etc.), piping and hoses is satisfactory
6	Coolant in engines is satisfactory, and temperatures are normal
7	The condition of diesel and coolant hoses is good
8	All drive belts are in good condition - proper tension / no cracks / not worn out
9	No excessive smoke in the exhaust
10	Exhaust lines are properly supported
11	Base frames and vibration dampers/ foot bolts in good condition
12	Engines batteries and charger in good working condition
13	Panels are well maintained. Engine and panel meters are in working condition.
DG ROOM / PMCC: DG SETS	
S. No.	**Check Point**
1.a	Earthing is as per norms for: Equipment earthing
1.b	Earthing is as per norms for: Structure earthing
1.c	Earthing is as per norms for: storage vessels
1.d	Earthing is as per norms for: Lightening
1.d	Earthing is as per norms for: Neutral earthing (Independent/ Separate grid for each equipment i.e., transformer, DGs, etc...)
2	Earth strips connected to electrodes by electro- galvanized bolts/ bolts not rusted. Earthing pits' condition is satisfactory
3.a	a) Panel condition: Doors in position and closed. b) Doors not opening in switched ON condition.

DRAFT 2

3.b	MCC Panel condition: Dust/ soot/ cobwebs not existing
3.c	MCC Panel condition: Visual condition of switch contact, fuse, etc., is satisfactory
3.d	MCC Panel condition: Control wiring condition satisfactory
3.e	MCC Panel condition: Earthing in order
4.a	Condition of cable/duct: Cables are on rack
4.b	Condition of cable/duct: Cable ducts are clean
4.c	Condition of cable/duct: Sealing of ducts/ panels is satisfactory
4.d	Condition of cable/duct: Cable identification tags are fixed
5.a	Condition of transformer: Earthing is satisfactory
5.b	Condition of transformer: No oil leaks
5.c	Condition of transformer: Silica Gel is satisfactory
5.d	Condition of transformer: Oil level is satisfactory
5.e	Condition of transformer: Terminal box earthing is satisfactory
5.f	Condition of transformer: Oil/ winding temperatures is satisfactory
5.g	Condition of transformer: Periodic oil quality test conducted as per IS-1866
5.h	Condition of transformer: Tripping are in healthy condition.
5. i	Earthing to fencing including gate of transformer has been provided
6.a	Earthing of fixtures proper
6.b	Condition of wiring good
6.c	Water does not enter and accumulate in fixtures/ distribution boxes etc.
7.a	FLP property of fixtures: No Allen bolts missing, and all are tight
7.b	FLP property of fixtures: Cable gland proper
7.c	FLP property of fixtures: No unplugged openings observed
8	Lighting DBs, switches, MCBs, etc., well maintained
9.a	Tower/ Mast Lights (Area Lighting): Earthing of mast is satisfactory

DRAFT 2

9.b	Tower/ Mast Lights (Area Lighting): Condition of JBs, switch boards, fuses, etc., is satisfactory
9.c	Tower/ Mast Lights (Area Lighting): Approach to electrical installation and ladders good and safe
9.d	Whether Wire ropes and winch of high mast are tested as per norms of State Factories Act and certificates are available
10	Ground resistance at motor/ push buttons measured and found is satisfactory.
11.a	Terminal boxes/ glands/ push buttons, etc., and their FLP properties is satisfactory.
11.b	Existing cable terminations are proper
12	Whether protective devices are provided in ACB/ fuse of PMCC/ Lighting panels for FWPH, Control Room, Canteen, Admn Building etc. From where single cables are laid to the panels available at FWPH, Control Room, Canteen, Admn Building etc., respectively?
13	All fuses are of proper type and rating as specified in single line diagram (SLD) and SLD is displayed
14	Emergency lighting system in order
15	Condition of flameproof/ Ex -equipment in licensed area is good
16	Cable insulation resistance between phases and phases to earth checked (yearly)
17	Servicing of motors done as per schedule (megger test/ PM jobs)
18	Calibration of protective/ overload relays was done as per schedule, and the record was maintained
19	Substation wiring and terminations are neatly done. No loose wires. Panels sealed. Cable glands and tags is satisfactory
20	All light points are in working condition; glass covers are clean
21	The condition of the capacitor bank and the power factor controller is satisfactory
22	UPS/ batteries/ chargers in working condition
23	Interlocking is provided between the Main breaker and the DG breaker and is in working condition.
24	Earth pits are tested, and resistance values are within the acceptable limit.
25	Average Power Factor for the last 12 months (to be calculated from the SEB Power Bills)

DRAFT 2

BUILDINGS / SHEDS	
S. No.	**Check Point**
1	Periodic painting/ distempering of all buildings is done
2	General maint. and cleanliness of buildings/ structural/ sheds/ DV chambers, etc., is good
3	Canteen building maintenance and facilities are satisfactory
4	Shed, structural, ladders, platforms are in good condition
5	Spaceframe/ roof supporting structures (including columns) in sheds in good condition – no sagging/ bending/ broken structures/ bolts. Whether Plant has obtained safety/ stability certificate from competent person as per State Factory Rules?
6	Sheds roofing sheets are in good condition & no damages/ leaks
7	Mastic flooring in sheds/ columns is satisfactory/ no gaps/ discontinuities
8	Exhaust fans working in battery room/ urinals/ lavatories/ bathrooms/ PMCC/ Canteen
9	General cleanliness in pump houses, filling/ other sheds, and other buildings satisfactory
10	Office equipment (fans, lights, PCs, ACs, MCB, ELCB etc.) are in good working condition.
11	Condition of building structure: whether free from cracks/ damage etc.
12	Check for seepage marks in the building, condition of roof drainpipes, water seepage marks on window chajjas, toilet walls etc.
13	Whether columns of LPG Bottling sheds are made of RCC up to full height/ if made of structural steel, then painted with fireproof painting as per LPG Eng specs and relevant documents are available at Plant for the paint applied with repaint schedule on steel columns.
STORE MANAGEMENT	
S. No.	**Check Point**
1	Stores material properly stacked/ stored; neatness maintained.
2	Whether the stacking pattern in Stores allow FIFO (First in First out) while consuming the spares/ LPG Consumables/ LPG Equipment.
3	Segregation, numbering and labelling of materials and respective racks and bins.

DRAFT 2

	Maintenance of records and Bin cards.
4	Codification of all materials, Individual item-wise stock data and Valuation of items in SAP/ any other system.
5	Equipment for handling of material in the store is provided.
6	Provisioning of firefighting equipment in stores. Display of slogans in stores.
7	Whether critical spares for various equipment are maintained at the Plant
BOUNDARY WALLS AND GATES	
S. No.	**Check Point**
1	Condition of boundary walls: a) free from any damage/ cracks/ gaps b) Condition of painting, etc. c) Weep holes provided in the boundary wall are clear d) Condition of a grill at the entry/ exit of the stormwater drain.
2	Condition of Concertina fencing above the boundary wall/ painting etc. (free from any discontinuity, spacing of coils is uniform)
3	Condition of gates/ barriers and fencing: Visual impact, painting, condition, safe operation and stability of gates and barriers.
4	Condition of Emergency Gate and Emergency Exit for Bulk LPG tank trucks and packed trucks: Check the condition of emergency roads and gates for safe passage of trucks by sending out one Bulk LPG tank trucks and packed trucks through designated emergency gate.
5	Whether sliding gates if provided have the smooth movement?
6	Whether sliding gates if provided are motorised?
7	Whether sliding gates if provided have the stopping arrangements to avoid the movement of sliding panels beyond the desired position by way of: a. Wheel stopper welded to wheel rail. b. Stopping arrangement at the rear side of the sliding panel with corresponding arrangement in the wall/ robust structure to stop the sliding panel to move beyond desired position while closing and opening. c. Stopping arrangement on top of the sliding panel in the form of cleat welded with corresponding cleat provided in the wall/ robust independent structure to stop the movement of sliding panels while opening/ closing.
8	Whether the rail provided is straight and is free of dust/ dirt to allow free movement of sliding panel of the sliding gate.
9	Whether all the hinge type gates are easily operable by single person and the gate

DRAFT 2

	columns and hinges are free of cracks?
ROADS / DRAINS/ VAPOUR TRAPS	
S. No.	**Check Point**
1	Roads/ hard-surfaced areas are free from potholes and maintained well
2	Whether pedestrian paths/ pavements/ foot paths are in good condition and marked with night glow paints?
3	Emergency exit route is properly displayed and is clear of vegetation / debris
4	Entry/ Exit barriers and gates are maintained well and in operating condition
5	Surface/ storm water drains are clean and free from any debris
6	Vapour traps provided as per approved drawings, permanent water connections along with a metallic float available and water levels maintained
7	Trench drains in sheds are clean & no choking/ accumulation of water
8	Whether a motorable road is available in the buffer area/ green belt, and is it maintained in good condition?
9	Check whether the camber is maintained for all roads to avoid the accumulation of water on roads.
10	Condition of the approach road to the main entry/ Emergency gate - Free from Potholes, accumulation of water and obstruction for the free movement of vehicles.
11	Maintenance of road signage/ road markers/ proper speed breakers, with zebra marking, etc.
12	Whether the TT Parking Area is maintained well. Shall be free from cigarette/ bidi butts, bottles, plastic/ paper rags, etc.
13	Whether NO Bulk LPG tank trucks and packed trucks are parked on the approach road of the Plant.
HOUSEKEEPING	
S. No.	**Check Point**
1	Grass cutting is done regularly, and licensed area is maintained well and no excessive wild growth
2	No excessive surplus/ scrap material lying around in the Plant
3	General cleanliness and hygiene are maintained at all places in the Plant

DRAFT 2

4	All Plant equipment is well cleaned, no oil, water spillage
5	All equipment painting is satisfactory and are corrosion free
6	Scrap yard: defined area/ display name plate/ cleanliness/ proper stacking/ disposal action etc.
7	Whether TT driver's waiting area near the entry gate is well maintained.
8	Entity sign board and mandatory statutory boards at main entrance is available and maintained
9	Whether Rolling plan for painting is available and adhered to?
GENERAL	
S. No.	**Check Point**
1	All other SRV's / TSV/ PAVs are tested and sealed. Isolation valves are kept locked in open condition. Test records available
2	All pressure gauges / temp gauges are in working condition and are calibrated.
3	All LPG strainers cleaned on quarterly basis and records maintained
4	Condition of Lightning Protection System
5	Cooling water pumps maintained properly: no abnormal noise/vibration/corrosion
6	Cooling towers are well maintained, and effective cooling achieved
7	Interlocking of cooling water pump with Air/ LPG compressor and LPG pump (if applicable) provided and in working condition.
8	All foundation bolts have proper washers/nuts and are fully tightened. No spacers/ packings/ earth connections used under main nut
9	Whether blind turns inside the Plant are provided with convex mirrors?
10	Whether periodic maintenance is being carried out for the LPG compressor and allied equipment of emergency rescue vehicle and records available.
DOCUMENTATION / DRAWINGS / EQUIPMENT DATA / HISTORY	
S. No.	**Check Point**
1	Layout plan/ P&ID incorporating facilities modifications and new additions available.
2	Layout drawings of cabling, lighting, and earthing are available and displayed

DRAFT 2

3	Equipment History Cards for critical equipment are maintained
4	Equipment running hours meters (totalizers) maintained in working order, and hours are logged: DG sets
5	Equipment running hours meters (totalizers) maintained in working order, and hours are logged: Fire engines (electrical totalizers)
6	Equipment running hours meters (totalizers) maintained in working order, and hours are logged: LPG compressors
7	Equipment running hours meters (totalizers) maintained in working order, and hours are logged: LPG pumps
8	Equipment running hours meters (totalizers) maintained in working order, and hours are logged: Air compressors
PLANT MAINTENANCE MODULE IMPLEMENTATION	
S. No.	**Check Point**
1	All equipment is covered in the Maintenance Module.
2	PM task lists prepared for all equipment as per guidelines.
3	PM scheduling is being carried out
4	Equipment data maintained
5	Preventive maintenance of equipment is being done.
6	All types of Maintenance details (breakdown/ shutdown/ normal, etc.) are being recorded
VIBRATION MONITORING	
S. No.	**Check Point**
1	Safe base vibration limits identified for major rotary equipment (Fire engine, Air compressor, LPG Compressor, LPG pump, Jockey pump and DG set) (vibration analysis done once in year)
2	Vibration monitoring schedule prepared and adhered to
3	Corrective actions taken as per recommendations of vibration analysis
BREAKDOWNS	
S. No.	**Check Point**

DRAFT 2

1	No major equipment breakdowns in last 1 year
2	AMC for all critical/major equipment available (Viz Carousels, Fire engines, DG sets, weigh bridge, CP System)

DRAFT 2

Annexure-14-IV

SELF-AUDIT BY PLANT IN CHARGE

LOCATION:

DATED:

AUDIT/ INSPECTION BY:

checklist is indicative and can be further supplemented and modified suitably to meet the specific requirements of the individual Plant.

Obsv ID	Sub Grp Code	Particular Detail	Observation
1	HSE MANAGEMENT	Whether health monitoring is done for all personnel as per the Factory Act (OSH Code/ Rules)/the entity's policy, and records maintained.	
2	HSE MANAGEMENT	Whether any suggestions scheme to improve safety management a) Available b) If yes, is any reward/ incentive scheme available c) Details of response to the above reward/ incentive scheme in the last two years.	
3	SAFETY COMMITTEE	Safety Committee: a) Whether a Safety Committee exists at the location as per OISD-GDN-206. b) Whether sufficient representation from non-management cadre exists in the committee as per Factories Act (OSH Code/ Rules)/ OISD-GDN-206 c) What is the frequency of meetings of the committee. (at least quarterly) d) Whether Safety Committee meetings are chaired by the Occupier or his authorised representative. In case the Occupier does not chair the meeting, whether minutes of the safety committee meeting are marked to him. e) Whether the functions and duties of safety committee members are documented. Interaction with Safety Committee members is to be made to ascertain their awareness level.	

DRAFT 2

4	SAFETY COMMITTEE	Whether the safety committee reviews the following critical aspects, and a remedial action plan is prepared and monitored for implementation: a) The compliance of MOM of the previous meeting(s). b) Findings of audits. c) Unsafe acts, unsafe conditions, near misses, off the job and on the job incidents. d) Shortcoming in fire/ emergency drills e) Incident investigation/ analysis reports/ recommendations. f) Long pending safety concerns, escalation of concerns (if required)?	
5	MANPOWER / TRAINING	a) Whether orientation/ refresher trainings are imparted to the following category of personnel as per OISD-STD-154 and OISD-STD-178, and a record is maintained. Mention the number of personnel/ covered/ pending for each category in the last one year. i) Head of the Departments ii) Fresh Entrant Officers, transfer-in personnel from other functions/ disciplines and Supervisors iii) Officers and Supervisors (In Service) iv) Workmen v) Security Personnel vi) Drivers and Truck crew vii) Contract workmen and a supervisor b) Whether refresher courses are carried out after initial training, once in 3 years/ earlier as per ~~company~~ training policy of entity (OISD-STD-154)? c) Details of personnel are not covered in the training as per a) and b) above. d) Whether location specific HSE training cum awareness program has been provided to the officer (in case of location change due to transfer and having same functional responsibilities) as per clause 6.0 (e) of OISD-STD-154	

DRAFT 2

6	EMERGENCY MANAGEMENT SYSTEM	QRA and Hazop Study: i) Whether Hazop Study and Quantitative Risk Analysis have been conducted for the location. ii) Whether recommendations thereof have been complied with, and a record of compliance is available. iii) Whether the same is revised in 5 years or with any major change/ modification. iv) Whether the QRA study of the Plant has covered various hazards associated with the adjoining pipeline receiving station. v) Whether Hazop and QRA have been accepted/ approved by the location.	
7	EMERGENCY MANAGEMENT SYSTEM	a) Whether a well-defined ERDMP Plan, certified by accredited ~~by~~ third party empanelled by PNGRB, is available. b) Whether the same has been prepared based on a Quantitative Risk-Analysis study. c) Whether the same is revised in 5 years or with any major change/ modification. d) Whether the ERDMP Plan of the Plant has covered various hazards associated with adjoining pipeline receiving station.	
8	EMERGENCY MANAGEMENT SYSTEM	Mock Drills: a) Whether Mock Drills are conducted regularly. b) What is the frequency of Mock Drills. c) All scenarios as per ERDMP are covered. d) Whether Mock drills are conducted in the night and on holidays. e) Lessons learnt/ actions taken are recorded in mock drill reports.	
9	EMERGENCY MANAGEMENT SYSTEM	a) Whether the Mutual Aid Agreement (valid for a maximum of 3 years) is prepared and signed by all mutual aid members? b) Whether the Plan includes details of a minimum number of firefighting equipment, manpower and quantity of consumables which to be shared among the mutual aid members. c) Whether the contact details of the officials of the mutual aid members are updated. d) Whether minutes of the meeting are recorded and reviewed. e) Whether data on available resources for disaster management has been uploaded on	

DRAFT 2

		IDRN website.	
10	EMERGENCY MANAGEMENT SYSTEM	a) Whether On-Site (Level-II)/Off-Site (Level-III) Mock Drills for Disaster Management Plan are organized. What is the frequency? b) Comments on participation of District Administration and Mutual Aid members etc. c) Whether Crisis Management Scheme for Bomb Threat exists, what is the frequency followed.	
11	EMERGENCY MANAGEMENT SYSTEM	Are records pertaining to maintenance of fire extinguishers for routine checks and hydro test available (Refer OISD-STD-142).	
12	EMERGENCY MANAGEMENT SYSTEM	a) Whether the accident-reporting system, including “Near Miss”, is available. b) Whether lessons learnt are recorded/ investigation is carried out for all near miss incidents. c) Whether unsafe acts and unsafe conditions are reported? d) How many unsafe acts, unsafe conditions and near-miss incidents have been reported over the last 2 years?	
13	EMERGENCY MANAGEMENT SYSTEM	a) A list of incidents that have happened within LPG Plants of the entity is available in last five years. b) Whether applicable recommendations of internal investigation of above accidents implemented and record of the same maintained.	
14	SAFETY AUDIT	a) Internal Safety Audit i) Frequency/ last audit on ii) Formation of team (multi-disciplinary) iii) Implementation status b) Electrical Safety Audits. i) Inspecting Agency ii) Frequency/ last audit on.	

DRAFT 2

		iii) Implementation status c) Maintenance and Inspection Audit. i) Frequency/ last audit on. ii) Implementation Status d) Whether any system available to review of compliance of the above audits? If yes, whether it is effective or not.	
15	LAYOUT & FACILITIES	a) Whether the updated PESO approved Plant layout showing all facilities is available. What is the last date of updation. b) Whether the updated PESO approved chain conveyor layout, showing all facilities, is available. What is the last date of updation.	
16	LAYOUT & FACILITIES	Whether the updated Factory Department approved Plant layout is available. What is the last date of updation.	
17	LAYOUT & FACILITIES	a) Whether inter distances are as per OISD-STD-144/ 150 (attach Annexure showing actual inter-distances). b) Comment on deviation, if any.	
18	LAYOUT & FACILITIES	a) Whether updated drawing showing fixed firefighting system is available. What is the last date of updation. b) Whether updated drawing showing positioning of all type of fire extinguishers are available. What is the last date of updation.	
19	LAYOUT & FACILITIES	Whether the updated P&I diagram is available. What is the last date of updation.	
20	LAYOUT & FACILITIES	Whether the latest updated as-built drawing is used for the preparation of various plan drawings, i.e., hydrant layout, firefighting layout, cable route diagram, earth pit diagram etc.	
21	OPERATIONAL ACTIVITIES	a) Whether checklists for T/T, T/W and Packed Trucks are available (as per latest OISD-STD-144). b) Whether a record of checking and deviation is available.	

DRAFT 2

22	OPERATIONAL ACTIVITIES	a) When the last NDT inspection of bulk storage vessels was carried out. b) Whether the results of the NDT test are compared with the original thickness and the last tested. c) Comment on deviation, if any.	
23	OPERATIONAL ACTIVITIES	a) Whether statutory test certificates for knockout drums/ air vessels/ evacuation vessels are available and are valid. b) Whether the Test Date and next Due Date are displayed.	
24	OPERATIONAL ACTIVITIES	a) Whether Certificates/ records for annual testing for SRVs are available and valid. b) Whether the relief valves are set at different values not exceeding 110% of the design pressure of the vessel. In case the vessel is designed for propane and LPG is stored, then the SRV set pressure is ensured as per the design pressure of LPG. c) Whether the Test Date and next Due Date are displayed. d) Whether a record of total PAV/ TSV/ SRVs is available in the Plant. e) Whether the record of SRV testing of 425 kg capacity cylinders is available.	
25	OPERATIONAL ACTIVITIES	a) Whether Work permits are issued for all maintenance jobs, and other than normal operating work. b) Whether Job Safety Analysis is being carried out for all work permits. c) Toolbox talk and LOTO system to be implemented as per clauses 6.3.2 and 6.3.3 of OISD-STD-105. d) The list of authorized personnel involved in the work permit system, with their roles and responsibilities shall be available at the location as per clause 6.1.5 and 6.4.1 vii of OISD-STD-105. e) Whether provision to display the real-time status of critical work permit as per clause 6.4.5 of OISD-STD-105. f) CCTV and GMS are being utilized to monitor site conditions continuously for critical jobs as per clause 6.5.1(iii) of OISD-STD-105.	

DRAFT 2

26	OPERATIONAL ACTIVITIES	a) Whether Cold work permits are issued as per OISD-STD-105, and relevant details, including completion details, are recorded. b) Whether Hot work permits are issued as per OISD-STD-105 and relevant details, including completion, are recorded. c) Whether Electrical Isolation/ Energisation Permit are issued as per OISD-STD-105 and relevant details, including completion, is recorded. d) Whether Electrical isolation and energisation permits are taken along with cold/ hot work permits wherever required, as per the nature of the job. e) Whether Height work permits are issued as per OISD-STD-105/ 192 and relevant details, including completion details, are recorded.	
27	OPERATIONAL ACTIVITIES	a) Whether organisation specific MOC procedure and approval matrix are available for different types of changes as per clause 6.0 of OISD-STD-178. b) Whether Management of Change is being followed in line with OISD-STD-178, and records are maintained. c) Whether MOC closure is being intimated to the approver as per clause 5.1(x) of OISD-STD-178.	
28	M & I	a) Whether the preventive maintenance schedule followed in the location is as per OISD-STD-144/ Equipment Manufacturer's recommendations. b) Whether records pertaining to the preventative schedule vis-a–vis actual work are available. c) Whether the master list, along with serial number, is available for Plant equipment like LPG hose, Fire hose, earth pits, pressure gauge, temperature gauge, TSV, SRV, etc.	
29	M & I	Whether Log Sheets/ records are maintained for daily activities, breakdown, and maintenance for the following areas, and the same are being reviewed by the concerned officer: a) Filling Shed b) Fire Fighting system c) PMCC	

DRAFT 2

		d) LPG Pump House.	
30	M & I	Whether history cards for pressure vessels/ Fire Pumps/ LPG Pumps/ LPG Compressors/ Air Compressors, etc. are available.	
31	M & I	a) Whether any major breakdown had taken place in the last one year on LPG equipment/ firefighting facilities. b) Comment on Breakdown Analysis and present status.	
32	M & I	Whether testing of LPG pipelines (including unloading arms/ carousel pipeline, etc.) has been carried out as per clause no. 9.5.3 of OISD-STD-144: a) Comprehensive inspection in line with OISD-STD-130 and statutory requirement as applicable. b) External on-stream inspection, including ultrasonic thickness testing, once in 3 years. c) Whether the results of the NDT test are compared with the original thickness and the last tested, and comments are recorded. d) Whether details of the thickness survey are maintained on an isometric sketch. e) Whether the document for the retiring thickness calculation is available, or the retiring thickness is mentioned in the NDT report. f) Piping in coastal installations and corrosive environments is visually inspected once a year.	
33	M & I	a) Whether pressure gauges/ other gauges are checked regularly with the master gauge and marked with an identification number. b) Whether the calibration certificate of the master gauge used is available and attached to various test records. c) Whether all AMC reports/ test certificates issued in-house/ by a third party have been reviewed and signed by the Plant Officer and/ or location In-Charge. d) In case, recommendation has been given in the AMC report, has the Plant-initiated action.	

DRAFT 2

34	M & I	a) Are LPG filling hoses as per OISD-STD-135. b) Whether a record of testing of LPG hoses/ filling machine hoses, evacuation hoses is available. c) Whether a unique identification number with the date of put in use is permanently marked on the hoses.	
35	M & I	Whether a rolling plan for painting of various Plant facilities is available, including the frequency of painting of various structures/ pipelines, and actual implementation of the plan is monitored.	
36	LICENCE & DOCUMENTATION	Whether following PESO licenses is available: a) Bulk LPG storage b) Cylinder filling c) Filled Cylinder storage d) Permission for degassing e) Permission for operating location and dispatching cylinders between sunset and sunrise f) Permission for bulk TT unloading/ loading/ PLT is available between sunset and sunrise g) Bulk storage of HSD for own use h) PESO License for statutory testing and painting of cylinders.	
37	LICENCE & DOCUMENTATION	a) Factory license b) Prohibited Area declaration	
38	LICENCE & DOCUMENTATION	Whether following consent from PCB is available: a) Consent to operate - Air (Prevention and Control of Pollution) Act b) Consent to operate - Water (Prevention and Control of Pollution) Act c) Authorization for handling Hazardous Waste	
39	LICENCE & DOCUMENTATION	a) Whether a mechanism for paint sludge disposal is available, and records are maintained. b) Whether the mechanism of used oil disposal is available, and records are maintained. c) Water potability test d) ETP discharge water test.	
40	LICENCE & DOCUMENTATION	a) Availability of VHF/ UHF License for all sets provided at the location and its validity.	

		b) Weights & Measures approval/ certificates.	
41	LICENCE & DOCUMENTATION	Whether electrical installations were inspected by an Electrical Inspector in the last 5 years.	
42	STATUS ON IMPLEMENTATION OF RECOMMENDATIONS	a) Year of previous OISD audits (ESA/ SSA/ PCSA). b) Whether a copy of the last OISD audit and a signed copy of the compliance status is available. c) No. of pending points of previous OISD Audits. (Attach Annexure for the same) d) Whether recommendations of the last ESA/ SSA/ PCSA shown complied are pending/ repetitive.	
43	BULK STORAGE AREA	Whether the water drain line has two valves, with a suitable distance piece between them, provided between the ROV and the first isolation valve. The first drain valve from the vessel should be of quick shut-off type, while the second should be a anti-freeze throttling type (Globe Valve/ dead man). In case of refinery/ pipeline fed locations dead man valve is to be provided in place of a throttle valve. Checking the water draining from one vessel per cluster.	
44	BULK STORAGE AREA	a) Are safety instructions for draining operations displayed prominently. b) Whether water draining is done under supervision.	
45	BULK STORAGE AREA	Safety Relief Valves: a) Whether isolation valves are kept in open and locked conditions. b) Whether the Test Date and next Due Date are displayed.	
46	BULK STORAGE AREA	a) Whether each storage vessel has at least two level gauges. In addition, one independent high level switch is also provided. b) Whether the daily recording of both level gauges is done. c) Whether high-level alarm of storage vessels is physically checked by raising the bulk level	

DRAFT 2

		up to 85% capacity of the vessel once a year.	
47	BULK STORAGE AREA	a) Is a sprinkler system provided on all vessels and in working condition. SRV, the first valve of the vessel (ROV) and the valve manifold are adequately covered with sprinklers. (CL. 11.11, OISD-STD-144) b) Whether Deluge Valves are remotely operated and are located at a safe distance (15 m) from storage vessels. (CL. 11.11, OISD-STD-144) c) Whether the area is adequately covered by Hydrant/ Monitors, including positioning of hoses/ nozzles. Is the throw of the monitor reaching the SRVs. d) Whether hydrant/ monitors are positioned at a safe distance (Minimum 15 m). (Cl. 11.8 (v), OISD-STD-144). e) Whether only one type and size of spray nozzles are used in a particular facility. (Cl. 11.10.iii, OISD-STD-144)	
48	BULK STORAGE AREA	a) Whether an adequate number of crossovers is provided (Cl. 5.6.9, OISD-STD-144). b) Whether foundation/ anchor bolts, ladders, stairways, platforms/structures are checked periodically for any corrosion/ damage, etc.	
49	BULK STORAGE AREA	Check the following for Mounded Storage Vessels	
50	BULK STORAGE AREA	a) Whether the cathodically protected surface is isolated from the unprotected structures/ surfaces by installing monolithic joints: on liquid line and vapour lines. b) Whether suitable isolation is also provided on utility pipelines such as air line to ROV, metallic structures, instrument lines, etc. (Annexure-I Cl. 7.0, OISD-STD-150).	

DRAFT 2

51	BULK STORAGE AREA	Whether to prevent drainage of protective CP current at the pipe support locations, the pipe is electrically isolated using epoxy wrapped laminate, circumferentially wrapped around the pipe with a minimum thickness of 5 mm and width covering the entire contact area (Annexure-I, Cl 7.0, OISD-STD-150). (Wherever in existing MSVs, provision of epoxy wrapped laminate is not feasible, alternatively minimum 5 mm thick Teflon pad should be provided)	
52	BULK STORAGE AREA	a) Whether provision of a minimum of three reference points (One each near the vessel ends and one in the middle) has been made for monitoring the settlement of the mound. (Cl. 5.2.15, OISD-STD-150) b) Whether a minimum of two permanent benchmarks have been provided away from the MSVs and earmarked. (Cl. 5.2.15, OISD-STD-150) c) Whether the level of this benchmark is traceable w.r.t. to the original benchmark from which it was transferred to the MSV site. (Cl. 5.2.15, OISD-STD-150) d) Comment on the condition of the reference points and benchmark.	
53	BULK STORAGE AREA	a) Whether a signed document for initial settlement, reading before initial hydro-testing, is available. b) Whether the design document for permissible total and differential settlement is available. c) Whether the settlement of the vessel is monitored on a half-yearly basis, and a record is maintained since the beginning, initial hydro testing. d) Whether total and differential settlement is within the permissible limit as per design. (Cl. 9.4, OISD-STD-150).	
54	BULK STORAGE AREA	a) Whether spring support is provided for LPG liquid inlet/ outlet pipeline in the tunnel, and reading is being monitored/ recorded on half half-yearly basis. (Cl. 5.2.15, OISD-STD-150) b) Whether stress analysis of the liquid inlet/ outlet pipeline is carried out if spring support is compressed more than 60% of the full-scale	

DRAFT 2

		reading, since the initial hydro test of the MSV. (Cl. 5.2.15, OISD-STD-150).	
55	BULK STORAGE AREA	a) Whether one transformer rectifier unit (TRU) for one mound is provided. b) In case of multiple TRUs for one mound, whether stray current interference is positive potential swing, less than 50 mV (as per IS 8062:2006) measured for all Cu-CuSo4 reference electrodes. (Annexure-I, Cl 2.2.2.iv, OISD-STD-150) c) Whether reference cells (Cu/CuSO4) connected to TRU are identified, and minimum one reference cell (Cu/CuSO4) at the 6 O'clock position of each vessel is connected to TRU. (Annexure-I, Cl 8.0, OISD-STD-150)	
56	BULK STORAGE AREA	a) Whether the Transformer Rectifier unit is fitted with a current interrupter. (Annexure-I, Cl 8.0, OISD-STD-150) b) Whether the current interrupters are synchronized while measuring Off PSP wherever multiple TR units are provided for a mound. (Annexure-I, Cl 8.0, OISD-STD-150) c) If remote monitoring unit is provided for monitoring of CP system, whether customized report with provision of '"OFF PSP" readings and same is being monitored.	
57	BULK STORAGE AREA	Whether insulation joints are checked quarterly by the ON/OFF PSP test method and using the Insulation Joint Tester once in a year and records are maintained. (Cl. 9.2(b), OISD-STD-150)	
58	BULK STORAGE AREA	a) Whether "OFF PSP" Pipe to Soil Potential (PSP) readings are in the range of (-) 0.85 to (-) 1.2 volts. (Cl. 9.2.a.ii, OISD-STD-150) b) Whether "OFF PSP" readings are recorded on a fortnightly basis at TRU. (Cl. 9.2.a.i, OISD-STD-150) c) Whether "OFF PSP" reading at the test lead points for the entire vessel is being taken once in a quarter. (Cl 9.2.a.ii, OISD-STD-150) d) Whether the protective current is monitored quarterly, and whether the same is in line with	

DRAFT 2

		the design document. (Cl 9.2.a.iii, OISD-STD-150)	
59	BULK STORAGE AREA	Whether the Solid-state de-coupler/ Kirk Cell has been provided, checked periodically and records maintained. (Cl 9.2.c, OISD-STD-150)	
60	BULK STORAGE AREA	Are gas detectors provided near ROVs, Inspection tunnel, inside the nozzle box enclosure (if provided) or dome connection, near water draining/ sampling points. (Cl 8.2, OISD-STD-150)	
61	BULK STORAGE AREA	Whether any water seepage is observed from the sides and inspection tunnel of the Mounded storage.	
62	BULK STORAGE AREA	Wherever PCC/Tiles have been provided on top of the mound, provisioning have been made to prevent the possibility of gas accumulation inside the mound. (Cl 5.2.9, OISD-STD-150)	
63	BULK STORAGE AREA	Whether a kerb wall and vapour trap are provided for the manifold area of the MSV.	
64	BULK STORAGE AREA	Check the following for bulk LPG receipt through pipeline transfer	
65	BULK STORAGE AREA	a) Whether Critical control parameters, i.e., storage vessel level, pressure, temperature, flow, quantity, ROV and MOV status are shared/ made available and monitored in the control rooms of Supplier's and Consumer's company. (OISD-STD-214, clause 5.16.3). b) The following indications shall be incorporated in the control rooms of both the Pipeline Operator and the consumer company.(clause 5.16.4 of OISD-STD-214). (i) Close indication of LPG storage ROV. (ii) Pressure Transmitters/ Switch High-High of LPG storage. (iii) Level Transmitters/ Switch High-High of LPG storage.	

DRAFT 2

		(iv) Pressure Transmitters/ Switch High-High of LPG receiving pipeline system. (v) Pressure Transmitters/ Switch High-High of main LPG pipeline. (vi) Any fire/emergency in the installation.	
66	BULK STORAGE AREA	In case of emergency, whether a) pipeline ROV is closing earlier than the storage vessel ROV (Cl. 5.16.3, OISD-STD-214) b) Supplier end ROV is closing earlier than the consumer end ROV. (Cl., 5.16.2, OISD-STD-214)	
67	BULK STORAGE AREA	Whether records in line with OISD-STD-178 are maintained in case of any interlock being compromised between the supplier and consumer, with adequate precautions taken.	
68	BULK STORAGE AREA	Whether the Battery limit of the pipeline receipt line between the supplier and receiver is identified and documented/shown in SCADA.	
69	PUMP/ COMPRESSOR AREA	a) Whether any abnormal sound and vibration are noticed from the motor and bearings, and their condition. b) Whether the coupling guards are in position. c) Whether periodic vibration analysis is conducted for critical equipment like Fire engine, Air compressor, LPG Compressor, LPG pump, Jockey pump and DG set etc.	
70	PUMP/ COMPRESSOR AREA	Whether double valves are provided on LPG drain points.	
71	PUMP/ COMPRESSOR AREA	a) Is sprinkler system provided in the pump house and on the valve manifold are in working condition. (Cl. 11.11, OISD-STD-144) b) Whether density observed during operation of sprinkler system is 20.4 lpm/m^2. (Cl. 11.4 (c), OISD-STD-144)	

DRAFT 2

		c) Whether only one type and size of spray nozzles are used in a particular facility. (Cl. 11.10.iii, OISD-STD-144)	
72	PUMP/ COMPRESSOR AREA	a) Whether safety features for pressure, temp, cooling water, and lube oil are provided with audiovisual indications. (Cl. 6.6 and 6.7, OISD-STD-144) b) Whether safety tripping is periodically checked, and records are maintained. c) Whether the safety trip set values are displayed.	
73	CYL FILLING STORAGE AREA	Is a proper purging facility provided for new/ retested cylinders (all capacities of cylinders being filled in the Plant). (Cl. 6.13, OISD-STD-144)	
74	CYL FILLING STORAGE AREA	Whether visual inspection of cylinders is carried out for spurious/ defective cylinders, and records are maintained for segregated cylinders. (Cl. 8.2.2.1, OISD-STD-144)	
75	CYL FILLING STORAGE AREA	Is segregated storage of empty cylinders marked for the following: a) Spurious b) Repairable c) Due for statutory testing d) Other defectives (Cl. 8.2.2.1, OISD-STD-144)	
76	CYL FILLING STORAGE AREA	Whether a gross weight filling facility with Auto-cut off is provided in manual filling machines. (Cl 8.2.2.2 (c), OISD-STD-144)	
77	CYL FILLING STORAGE AREA	a) Whether the vapour extraction system provided in the filling shed and system is working satisfactorily and is interlocked with the filling machine or ROV on the liquid line at the filling shed entry. (Cl. 6.4 (xiv), OISD-STD-144) b) Whether jumpers are provided in the joints of the vapour extraction ducts.	
78	CYL FILLING STORAGE AREA	a) Whether the earthing system is maintained properly. b) Whether continuity of earthing for the carousel is maintained, including the filling gun.	

DRAFT 2

79	CYL FILLING STORAGE AREA	Whether a 100% post-filling quality checking facility, for all capacities of cylinders being filled in the Plant, is provided and in use. (Cl. 8.2.2.3, OISD-STD-144)	
80	CYL FILLING STORAGE AREA	Whether the accuracy of filling/ check scales is being checked daily, and records are maintained.	
81	CYL FILLING STORAGE AREA	a) Whether negative and positive tare weight setting is maintained in an electronic carousel. b) Whether SOP for handling of Positive Tare (PT)/ Negative Tare (NT) segregated cylinder is in place and displayed. (Cl. 8.2.2.2, OISD-STD-144)	
82	CYL FILLING STORAGE AREA	a) Whether valve/ "O"-ring defects are being checked for all capacities of cylinders filled in the Plant. b) Whether a calibration/ verification mechanism is available for the valve leak detector/ "O"-ring leak detector. c) Whether manual CVT is checked through the master cylinder/ verifier. (Cl. 8.2.2.3.3(f), OISD-STD-144) d) Whether a customised report is available for the accuracy of carousel/ OLD rejection/ VLD rejection, and the report is being monitored, and action is being taken.	
83	CYL FILLING STORAGE AREA	Whether a Weight correction arrangement is provided for underweight/ overweight cylinders for all capacities of cylinders filled in the Plant. (Cl. 8.2.2.3.2, OISD-STD-144).	
84	CYL FILLING STORAGE AREA	Whether water test bath or electronic leak detectors are provided to detect body and bung leak in cylinders for all capacities of cylinders being filled in the Plant. (Cl. 8.2.2.3.7, OISD-STD-144) Note: Refer clause 8.2.5 for 425 kg cyl	
85	CYL FILLING STORAGE AREA	Is an effective capping and pilfer proof sealing facility is provided. (Cl. 8.2.2.3.6, OISD-STD-144)	

DRAFT 2

86	CYL FILLING STORAGE AREA	a) Whether sample quality control checks (SQC) are being done as per the entity's policy. b) Whether the record is maintained, including details of defective cylinders observed/ action taken during SQC.	
87	CYL FILLING STORAGE AREA	Sample (SQC) for min. 32 cylinders to be checked as per Rule 19 (5th Schedule) Legal Metrology (Packaged Commodities) Rules 2011.	
88	CYL FILLING STORAGE AREA	Whether proper facilities are provided for the evacuation of defective (of all capacities of cylinders being filled in the Plant) cylinders and are working satisfactorily. (cl 8.2.2.3.9, OISD-STD-144)	
89	CYL FILLING STORAGE AREA	a) Whether depressurization facility provided for evacuated cylinders. (Cl. 8.2.4.3, OISD-STD-144) b) Whether degassing facility is provided with closed loop system for venting. (Cl. 5.6.4(e), OISD-STD-144)	
90	CYL FILLING STORAGE AREA	a) Whether valve change/ Online valve replacement machine's torque setting is being checked every day. (Cl. 8.2.2.3.3(d), OISD-STD-144) b) Comment on the LPG leak from the sealing bell of VCWE. c) Whether valve-defective cylinders are evacuated immediately. (Cl. 8.2.2.3.9, OISD-STD-144)	
91	CYL FILLING STORAGE AREA	Whether mastic flooring is provided on the floor and on structures up to 1.5 m height to avoid accidental sparks, and fireproofing of structural columns is provided. (Cl. 5.6.3, OISD-STD-144)	
92	CYL FILLING STORAGE AREA	a) Whether stacking zones are painted on floor for empty cum filled cylinders for all capacity of cylinders being handled. b) Whether cylinders are stacked in a vertical position as per the stacking norms, with safety caps on. (Cl.8.2.3.3, OISD-STD-144)	

DRAFT 2

93	CYL FILLING STORAGE AREA	a) Is a sprinkler system provided in the area, and is it in working condition? (Cl. 11.4 (b), OISD-STD-144) b) Comments on periodic testing and performance test (Cl. 11.15.1(v) of OISD-STD-144) c) Whether only one type and size of spray nozzles are used in a particular facility. (Cl. 11.10.(iii) OISD-STD-144)	
94	CYL FILLING STORAGE AREA	Whether the emergency tripping system of LPG handling facilities is operative.	
95	CYL FILLING STORAGE AREA	Whether the Differential Pressure Valve is in working condition, and the LPG filling pressure is restricted to 16.87 kg/cm2. (Cl. 8.2.2.2(f), OISD-STD-144)	
96	CYL FILLING STORAGE AREA	Whether a fire safe ROV is provided in the LPG liquid line entering the shed. Is a QB detector and water sprinkler nozzle provided for ROV if provided within 15 m of the facility? (Cl. 6.2.2(c), OISD-STD-144)	
97	CYL FILLING STORAGE AREA	Whether the Statutory Testing of Cylinders facility is provided in line with Clause 10, OISD-STD-144/ IS 16054 for all capacities of cylinders being filled in the Plant.	
98	CYL FILLING STORAGE AREA	Whether a Flow Chart is displayed for all activities involved. (Annexure 10-I, OISD-STD-144)	
99	CYL FILLING STORAGE AREA	Whether any Personnel meeting the requirements of the manager, Supervisor and Operators have been deployed for inspection and approvals. (Cl. 10.0, OISD-STD-144)	
100	CYL FILLING STORAGE AREA	Whether ETP is provided as per PCB approval and in working condition.	

DRAFT 2

101	CYL FILLING STORAGE AREA	Check the following for a few of the packed trucks: a) A spark arrestor or any spark elimination system duly approved by CCEs is welded to the exhaust. Exhaust shall be fitted on the front side of the vehicle in line with OISD-STD-159 b) 2 nos. 9 kg. DCP extinguishers are in an easily accessible and removable position with the Truck number, the last date of checking and charging painted on it. c) Air brakes are not bypassed d) Double pole master cut-off switch of minimum 300 amp rating e) Whether TREM card is available and whether "112" - Emergency response system of GOI - the Single Emergency number, for the people in distress to call 24h/7 in all 36 states/ UTs of India to get immediate assistance from the fire brigade, a medical team or the police is available on TREM card. f) Driver's Training as per CMV Rule 9/ OISD-STD-154 to TT Crew and valid authorization on the driving license by RTO for carrying hazardous goods as per MV Act. g) The voltage step-up device is not available in the cabin.	
102	CYL FILLING STORAGE AREA	Whether an adequate number of crossovers is provided (Cl. 5.6.9, OISD-STD-144).	
103	BULK LPG LOADING / UNLOADING FACILITY	Whether high vents are located at a safe height of 1.5 m above the eaves of the roof, and provision for dissipation of static charge is provided. (Cl. 7.2.1, OISD-STD-144)	
104	BULK LPG LOADING / UNLOADING FACILITY	a) Is a sprinkler system provided at least in three tiers on all bays with effective coverage of tires, manifold, and dish ends, and is in working condition. (Cl. 11.4 (d), OISD-STD-144). b) Whether only one type and size of spray nozzles are used in a particular facility. (Cl. 11.10.(iii), OISD-STD-144).	
105	BULK LPG LOADING / UNLOADING FACILITY	Check the following for a few of the bulk LPG tank trucks: a) Whether a Safety Checklist is being issued by the location for the decanted TTs, and deficiencies are recorded for compliance.	

DRAFT 2

		b) Explosive license, Safety Certificate, Fitness Certificate. c) Whether TREM card and Route Map are available and whether “112" - Emergency response system of GOI - the Single Emergency number, for the people in distress to call 24h/7 in all 36 states/ UTs of India to get immediate assistance from the fire brigade, a medical team or the police is available on TREM card. d) Valid authorization on the driving license by the RTO for carrying hazardous goods as per the MV Act. e) Driver’s training as per CMV Rules 9/ OISD-STD-154 to TT Crew.	
106	BULK LPG LOADING / UNLOADING FACILITY	Check the following for a few of the bulk LPG tank trucks: a) 2 nos. 9 kg DCP extinguishers in an easily accessible and removable position with Truck No., the last date of checking and charging painted on it. b) A spark arrestor or any spark elimination system duly approved by CCEs is welded to the exhaust. Exhaust shall be fitted on the front side of the vehicle in line with OISD-STD-159 c) Is internal EFCV provided on TTs d) Whether a crash guard is provided to protect the valves/ manifold in line with OISD-STD-159. e) Whether Emergency Tool kit as per OISD-GDN-161, Annexure - III is available with Bulk TT. f) Comment on the awareness level of drivers on the use of the emergency kit. g) Double pole master cut-off switch of minimum 300 amp rating. h) The voltage step-up device is not available in the cabin.	
107	BULK LPG LOADING / UNLOADING FACILITY	Is EFCV provided on the pipeline (verify direction). (Cl. 6.3, OISD-STD-144) During the next hydro testing of the pipeline, direction marking on EFCV (toward TT side) should be ensured in top position.	
108	TANK WAGON LOADING / UNLODING	a) Whether the condition of loading/ unloading the hose/ arms is proper. b) Are loading/ unloading hose/ arms properly	

DRAFT 2

		plugged when not in use.	
109	TANK WAGON LOADING / UNLODING	Whether Electrical Insulation Block is provided for siding, and a record of checking is maintained.	
110	TANK WAGON LOADING / UNLODING	Is a vent provided 1.5 m above eve of the roof.	
111	TANK WAGON LOADING / UNLODING	a) What is the heat detection system for sprinkler operation? (QB/EP Detector). b) Whether automatic fire protection is provided. (Cl.11.11.v, OISD-STD-144) c) Is sprinkler system provided at least in three tiers to uniformly cool the top, middle and the bottom of the tank wagons. (Cl. 11.4 (e), OISD-STD-144) d) Whether only one type and size of spray nozzles are used in a particular facility. (Cl. 11.10.iii, OISD-STD-144)	
112	TANK WAGON LOADING / UNLODING	Whether the lower portion of the flapper bridge at the wagon side is fitted with rubber or wooden padding. (Cl. 7.3.1 (6), OISD-STD-144)	
113	TANK WAGON LOADING / UNLODING	Whether for connecting and disconnecting arms or hoses, only non-sparking tools are used. (Cl. 7.3.1 (4), OISD-STD-144)	
114	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Is a line diagram of power/ lighting distribution displayed, and the last date of updation. b) Whether IS:15652 marked insulating mats are provided in electrical switch rooms, and their condition is satisfactory. c) Whether IS:13774marked electrical hand gloves for LT/ HT are available.	
115	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	Whether the earthing system is maintained properly in the area below: a) Bulk storage area b) Pumps/ Compressors area c) Cylinder Filling storage area d) Bulk LPG loading/ unloading facility e) Tank Wagon loading/ unloading	
116	ELECTRICAL SYSTEM/ SUBSTATION /	Is double independent earthing connection for all motors provided.	

DRAFT 2

	SWITCH ROOM		
117	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Whether test records for earth resistance/ continuity are available. b) Whether the earth pit/ grid resistance value is within the limit as per Cl. 7.0 (II)-(b), OISD-STD-137. c) Are earthing pits are as per the latest IS 3043 and are in good condition. d) Whether the updated drawing showing earthing grids and electric cable route is available, and when it was updated last. e) Whether the main earth loop is laid at a depth of 500 mm below ground level (Cl 27.2.2, IS:3043 and cl 6.17(vii), OISD-STD-144). f) In case joints are through connector/ lugs, ensure no gap between the surface of stripes by a minimum of two or more bolts.	
118	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Whether the electrical supply of air compressor/ jockey water pump/ siren/ barriers/ yard lighting/ bore wells/ CCTV is through a lighting distribution panel, which will not trip in case of emergency. (Cl. 6.5(c), OISD-STD-144) b) SOP to be displayed for disabling of auto start of DG in case of LPG leakage.	
119	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Are cables laid properly. b) Whether cable route markers are provided. c) Whether identification tags are provided at terminals.	
120	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Comment on the condition of insulation on DG Exhaust. b) Whether Earmuffs/ earplugs are in use in the DG room (where acoustic enclosure of DG is not provided)/ the Fire Water Pump House.	

DRAFT

121	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Is the transformer periodically checked for oil level and dielectric strength, etc. and records maintained. (Cl. 6.2.2, OISD-STD-137) b) Whether calibration of Transformer: Winding Temperature Indicator (WTI) and Oil Temperature Indicator (OTI) relays are carried out as per clause 6.2.2 B vii of OISD-STD-137. c) Whether Tan Delta Testing of Power and Distribution Transformers is carried out every 3 years as per clause 6.2.2 B xiii. of OISD-STD-137. d) Whether a circuit breaker is provided to isolate the transformer in case of an emergency. e) Whether relays and cables are tested, and records are maintained. (Cl. 6.3.1B and Cl.9.2.6, OISD-STD-137) f) Whether thermography of the panels has been carried out and a record maintained. (Cl. 6.2.1A, OISD-STD-137) g) Whether checking of VCB bottles for healthiness is done every three years as per clause 6.2.1 B xi. of OISD-STD-137.	
122	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Whether the doors of the MCC panel are not openable when the panel is in the switched ON condition in line with clause 47(v) of CEA regulation 2023. b) Whether live phases of the busbar inside the MCC panel are protected with a Bakelite sheet. c) Whether the MCC panel is vermin proof in line with clause 9.2.1 XV of OISD-STD-137	
123	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Whether all electrical fittings are flame proof and no loose/ missing screws or over size/ improper glands in hazardous area. (OISD-STD-113) b) Whether extra cable entry openings of FLP fittings are plugged using FLP plugs of matching thread.	
124	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Whether 30 milliamp ELCB is provided in building electrical circuits. (Cl. 6.3.8, OISD-STD-137 and IS 12640 - Part I) b) Whether the Lock Out Tag Out (LOTO) provision for panel isolation is provided and in practice. (Cl. 4.5 (iii), OISD-STD-137)	

DRAFT 2

125	ELECTRICAL SYSTEM/ SUBSTATION / SWITCH ROOM	a) Emergency Lights (DC based) are provided in all critical areas like Sub-Station, DG Room, Control Room, Security cabin(s), cylinder sheds, Fire Water Pump House, for CCTV, etc. (Cl. 6.5 (c), OISD-STD-144) b) Whether FLP torches are available. Cl 11.12.3(v), OISD-STD-144).	
126	SAFETY / FIRE PROTECTION SYSTEM	a) Whether above ground hydrant pipeline is at a height of 300 mm to 400 mm above finished ground level. (Cl. 11.8 (i) a, OISD-STD-144) b) A thickness survey is being carried out in line with cl.6.2.2, OISD-STD-142.	
127	SAFETY / FIRE PROTECTION SYSTEM	Whether the fire water network layout is displayed prominently.	
128	SAFETY / FIRE PROTECTION SYSTEM	a) Whether a pressure gauge is provided at the farthest end of the hydrant line. b) What is the pressure at the farthest end (min. 7 kg/cm^2). (Cl. 11.7(ii), OISD-STD-144)	
129	SAFETY / FIRE PROTECTION SYSTEM	a) Is any water leakage observed in the hydrant and monitor point. b) Whether hydrant hoses are being tested at 1.5 times of the working pressure and records are maintained. (Annexure G, IS-636)	
130	SAFETY / FIRE PROTECTION SYSTEM	Whether strainers are provided on deluge/ sprinkler water supply main piping, cleaned periodically and records maintained.	
131	SAFETY / FIRE PROTECTION SYSTEM	Comments on fire water pumps:	
132	SAFETY / FIRE PROTECTION SYSTEM	Adequacy of capacity, nos. and discharge pressure, etc.	
133	SAFETY / FIRE PROTECTION SYSTEM	a) Whether each fire water pump is being test run twice a week for half an hour at the rated RPM without compromising the auto mode of the firefighting system, and records are maintained. (Cl. 11.15.1 (i), OISD-STD-144) b) Whether the shut-off pressure of the fire pump is checked once a month, and a record is maintained. (Cl. 11.15.1(ii), OISD-STD-144)	

DRAFT 2

134	SAFETY / FIRE PROTECTION SYSTEM	a) Whether each fire water pump is being test run continuously for four hours once a year at the rated RPM and flow using the circulation line of the fire water storage tank, and records/ observations logged. (Cl. 11.15.1 (iii), OISD-STD-144) b) Whether each fire water pump is being tested for its performance once in six months by opening the required nos. of hydrants/ monitors, depending on the capacity of the pump, to verify the rated parameters as per OEM guidelines and records/ observations logged. (Cl. 11.15.1 (v), OISD-STD-144). c) Whether the design details of the fire hydrant system, Deluge Valve and MV Sprinkler system, including design pressure, operating pressure, test pressure and hydraulic calculations, are available at the Plant. (Cl. 11.8.iv OISD-STD-144)	
135	SAFETY / FIRE PROTECTION SYSTEM	Whether Fire water pumps have provision for high coolant temperature and low lube oil pressure alarms, and same are in working condition. (Cl. 11.7 (vii), OISD-STD-144)	
136	SAFETY / FIRE PROTECTION SYSTEM	Whether fire water pumps have been provided with auto auto-starting facility and are auto-starting based on their preset pressure to supplement/ to maintain the fire water pressure in the ring main. (Cl. 11.7 (x) , OISD-STD-144)	
137	SAFETY / FIRE PROTECTION SYSTEM	Working of the Jockey Pump and maintenance. (Cl. 11.15.1 (vi), OISD-STD-144)	
138	SAFETY / FIRE PROTECTION SYSTEM	Whether water storage tanks provided are of adequate capacity	
139	SAFETY / FIRE PROTECTION SYSTEM	a) Whether a proper source of water supply is available. (Cl. 11.16, OISD-STD-144) b) If a bore well is provided, whether state authorities' approval has been taken for the same. c) Is the yield recorded on a quarterly basis for each bore well. (Cl. 11.16, OISD-STD-144)	
140	SAFETY / FIRE PROTECTION SYSTEM	What is the water level in storage tanks.	

DRAFT 2

141	SAFETY / FIRE PROTECTION SYSTEM	Comments on the following protective clothing: a) Availability of the Fire Proximity suit b) Availability of low temperature protective clothing c) Breathing apparatus and its upkeep. d) Water gel blankets. (Cl. 11.12.3 (vi), OISD-STD-144)	
142	SAFETY / FIRE PROTECTION SYSTEM	a) Whether Gas Monitoring System (GMS) sensors are provided in line with OISD-STD-144 and OISD-STD-150. b) Whether a mimic/ repeater panel is provided at 24 hrs. manned location. (Cl. 12.1(ii), OISD-STD-144) c) Whether the height of the detector is not more than 0.3 m from the mounting level (Cl. 12.5 i, OISD-STD-144)	
143	SAFETY / FIRE PROTECTION SYSTEM	Whether the Gas Monitoring system is working satisfactorily in the following areas- a) Bulk Storage Area. b) LPG Cylinder sheds c) LPG Pump House d) Tanker Unloading/ Loading area e) Tank Wagon Gantry (Cl. 12.5 (v), OISD-STD-144)	
144	SAFETY / FIRE PROTECTION SYSTEM	a) Whether GMS are regularly tested and calibrated. (Cl. 12.6, OISD-STD-144) b) Whether GMS alarms are being monitored daily, and the action taken is recorded. (Cl. 12.3 (iii), OISD-STD-144) c) Whether a minimum 2 nos. of spare gas sensors for each system maintained in the Plant is available. (Cl.12.6(iv), OISD-STD-144)	
145	SAFETY / FIRE PROTECTION SYSTEM	How many portable explosive meters are provided. Are they calibrated for use in LPG service. (Cl. 11.12.3 (vi), OISD-STD-144)	
146	SAFETY / FIRE PROTECTION SYSTEM	Whether the audiovisual alarm system is maintained and is in working condition for the following (individually): - a) Deluge Valves (Cl. 11.11(v) b, OISD-STD-144) b) Manual Call Points (Cl. 11.13.1, OISD-STD-144) c) Emergency Shut Down (cl 11.13.2, OISD-	

DRAFT 2

		STD-144).	
147	SAFETY / FIRE PROTECTION SYSTEM	a) Whether an audiovisual alarm indicating the affected zone at the Fire Water Pump House and, manned control panel are provided (Cl. 11.11.v, OISD-STD-144). b) Whether the control panel has status indications for Deluge Valves with facility for actuation (Cl. 11.11.v, OISD-STD-144). c) Whether systems like push buttons and/or air release valves, etc. provided for each Deluge Valve in the field at 15 m from the facility protected. (Cl. 11.11, OISD-STD-144) d) In case PLC based firefighting system is provided, whether logging of cause and effect events for ESD/ MCP is available and a customized report is available for effective review (cl 11.11(xii), OISD-STD-144)	
148	SAFETY / FIRE PROTECTION SYSTEM	a) Whether First aid boxes with requisite medicines as per the Factory Act/ OISD-STD-227 are available? (cl 11.12.3 (vii), OISD-STD-144) b) Whether First Aid boxes are checked periodically, and records maintained. (Cl. 11.12.3 (v), OISD-STD-144) c) Whether first aid personnel are available during working hours, and a list of such personnel is displayed at a prominent place? (? (cl 11.12.3 (vii), OISD-STD-144) d) Whether the electrical shock restoration chart (in English or Hindi and the local language of the district, and where Hindi is the local language, in English and Hindi) is displayed in generator/ electrical rooms as per Regulation 30 of CEA MSES Regulations 2023 and its amendments thereof? (cl 11.12.3 (vii), OISD-STD-144)	
149	SAFETY / FIRE PROTECTION SYSTEM	Whether a suitable pressure relief mechanism is provided in the hydrant line to ensure that the line pressure never exceeds its design pressure. (Cl. 11.5 (v), OISD-STD-144)	
150	SAFETY / FIRE PROTECTION SYSTEM	Whether isolation valves provided on the fire water network have a visible indication for open	

DRAFT 2

		and closed conditions. (Cl. 11.5 (iv), OISD-STD-144)	
151	SAFETY / FIRE PROTECTION SYSTEM	Is testing of Fire Water Tanks carried out as per OISD-STD-129: a) External Inspection of FWT b) Internal Inspection and painting c) Whether the fabrication drawing of FWT is available d) Whether the results of the NDT test are compared with the original thickness, and comments are recorded.	
152	LAYOUT & GENERAL FEATURES	Is a proper industry type boundary wall at least 3.0 m high with 0.6 m barbed wire on top provided around the installation. (Cl. 5.7.3, OISD-STD-144)	
153	LAYOUT & GENERAL FEATURES	Whether the parking area provided is adequate to accommodate (minimum) following requirement (Cl. 5.5.11, OISD-STD-144): a) In case of bulk unloading, bulk tankers required for 8 hours of maximum rated capacity of the Plant. b) In case of bulk loading location, bulk tankers required for 8 hours of maximum rated bulk loading capacity of the Plant. c) Packed trucks required for 4 hours of maximum rated capacity of the Plant bottling.	
154	LAYOUT & GENERAL FEATURES	Whether parking area provided for packed trucks and bulk LPG tank trucks is under real time visual surveillance (security guard/ CCTV cameras). (Cl. 5.5.9, OISD-STD-144)	
155	LAYOUT & GENERAL FEATURES	Whether parking slotting/marking is done for safe parking of packed trucks and bulk LPG tank trucks. (Cl. 5.5.9, OISD-STD-144)	
156	LAYOUT & GENERAL FEATURES	a) Whether adequate coverage is provided through hydrant/ monitor, for all the packed trucks and bulk LPG tank trucks parked in parking area. (Cl. 5.5.9, OISD-STD-144) b) Whether protection is provided to avoid hitting of TTs to fire hydrant line of the Parking area. c) Whether no obstruction to hydrant/monitor for effective firefighting coverage of TT's is	

DRAFT 2

		ensured. Cl. 5.5.10, OISD-STD-144)	
157	LAYOUT & GENERAL FEATURES	Is water seal provided for (Cl. 5.4(c), OISD-STD-144): a) Bulk Storage area, b) TLD/ TLF, c) Filling shed, and d) Drain discharging into the storm drain going outside the Plant. Are the above water seal provided with permanent water connection and metallic float for auto replenishment of water.	
158	LAYOUT & GENERAL FEATURES	a) In case provision for green belt is provided, whether the same is segregated from hazardous area by a 1 m high wall/ chain link fencing. (Cl. 5.7.3, OISD-STD-144) b) Comments on the maintenance of the green belt.	
159	LAYOUT & GENERAL FEATURES	a) Whether proper identification and frisking of visitors/ contractors is made at the gate and entries are controlled through strict security checks to ensure the identity of each personnel inside hazardous and non-hazardous areas on real time basis. b) In case of a biometric/ face/ card reader type access control system is provided, a customised monitoring report is available to monitor in/ out of each personnel, including multiple in/ out of the same person and the same is monitored at day end.	
160	LAYOUT & GENERAL FEATURES	Whether the gate pass system is as per the security manual for hazardous and non-hazardous areas.	
161	LAYOUT & GENERAL FEATURES	a) Whether security personnel are trained on the hazardous nature of LPG and handling of emergency/ firefighting equipment. b) Are security staff conversant with the communication system. c) Whether patrolling of the hazardous and peripheral area is being done by the security	

DRAFT 2

		personnel, and records are maintained. d) Whether the outer perimeter wall is visible for monitoring through CCTV and/ or patrolling the pathway.	
162	LAYOUT & GENERAL FEATURES	a) Whether important telephone numbers of State Fire Services, Police, Hospital, District Admin./ other offices, etc. displayed as per ERDMP of the location. (Cl. 13.1 (i), OISD-STD-144) b) Whether the role during an emergency is displayed at workstation. (cl 13.2.5 (a), OISD-STD-144)	
163	LAYOUT & GENERAL FEATURES	Comments on the general housekeeping level in various areas - a) Bulk Storage Area. b) LPG Cylinder sheds. c) LPG Pump House d) Tanker Unloading/ Loading area e) Tank Wagon Gantry f) Electrical Areas g) Parking area h) Any other area	
164	LAYOUT & GENERAL FEATURES	Whether pipelines are colour coded as per entity policy, and flow directions are marked. (cl 9.5.13, OISD-STD-144)	
165	LAYOUT & GENERAL FEATURES	a) Whether proper roads around facilities are provided for smooth access of the fire tender, etc., in such a way that all facilities are accessed at least from two sides in case of emergency. (Cl. 5.7.1, OISD-STD-144) b) Whether an emergency exit with a proper gate is provided in the earmarked zone, and the same is not blocked. (Cl. 5.7.5, OISD-STD-144)	
166	LAYOUT & GENERAL FEATURES	a) Which is the nearest Plant where the emergency rescue vehicle is positioned. (Cl. 4.1, OISD-GDN-161) b) If provided in the Plant, whether the same is operational. Comment on maintenance.	
167	LAYOUT & GENERAL FEATURES	Is lux level in all areas maintained as per Table in OISD-STD-144. Checks are done once a	

DRAFT 2

		year, and record are maintained.	
168	LAYOUT & GENERAL FEATURES	a) Is a CCTV system with 24-hour monitoring provided. b) In case the size of the monitor is not adequate to accommodate all the cameras simultaneously, whether auto changeover of CCTV monitoring screen for continuous surveillance is provided. c) Who and how does the location monitor the area surveillance through the CCTV system, and is the record of any abnormality, unsafe act, or unsafe condition maintained for the same. d) Whether a record of random review by an officer of the CCTV recording is maintained. e) Whether the UPS of CCTV is in working condition and has a battery backup of 4 hrs. f) Whether CCTV data is stored as per the entity's security policy. g) Comments on any unsafe acts/ conditions observed during the audit upon review of CCTV recording. (Cl 16.0, OISD-STD-144)	
169	LAYOUT & GENERAL FEATURES	Comments on monitoring of Vehicle Tracking System.Cl 8.2.1(e) & cl 7.2.2, OISD-STD-144)	
170	LAYOUT & GENERAL FEATURES	LPG Cylinder Bank, if provided, shall meet the requirements of OISD-STD-162 for the following: i) 9 kg DCP FE, ii) Double Wire mesh made of Copper or non-corroding metal of not less than 11 mesh per linear cm available in line with clause 4.1.8 of OISD-STD-162. iii) isolation valves outside the canteen/cylinder bank and closing/ opening sign displayed in line with clause 4.1.9 of OISD-STD-162. iv) Whether LPG Cylinder manifold at canteen is tested in line with clause no 09 of OISD-STD-162.	
171	LAYOUT & GENERAL FEATURES	a) Whether PPE policy has been created in line with OISD-STD-155. b) Whether same is being followed in the location.	

DRAFT 2

Annexure-15

MAJOR CHANGES IN THE REVISED EDITION

1. The following definitions are added.

a. Deadman Valve
b. Emergency shut-off valve.
c. Excess flow check valve
d. Flameproof enclosure
e. Personal Protective Equipment
f. Safety Relief Valve
g. Thermal Safety Valve

2. Clause 5.4: LPG Storage Facilities

There shall be a minimum of 3 no's vessels for storage of LPG in the installation.

3. Clause 5.4 (d): Grouping

Where LPG Pump House and Storage Vessel are situated as an integrated block (where piping manifold of the LPG Pump House and the vessel is between the vessel and the LPG Pump House), the roads from all four sides shall be provided for the combined facility of LPG Pump House and storage vessel. However, adequacy of firefighting to be ensured.

4. Clause 5.4 (e): Level elevation

Documentary evidence to authenticate vessels in different groups are in the same elevation shall be available in the Plant.

5. Clause 5.5: LPG bulk loading/ unloading facilities – Bulk LPG tank truck.

- Number of tank truck gantries in a premise shall not exceed to two.
- Suitable earthing interlock arrangement shall be provided to ensure that unloading/ loading operation does not commence without earthing. The system should not work when the earthing connection is made to the TLD structure or ground. The system shall work only if the earthing connection clip is connected to the Tank Truck bullet and chassis.

6. Clause 5.6: LPG Bottling Facilities

Suitable earthing interlock arrangement with carousel rail shall be provided so as to ensure that the filling operation does not commence without earthing. Whenever earthing connections are removed for maintenance of filling guns/ filling machines, the electrical continuity of such filling guns/ filling machines should be checked, ensured, and recorded appropriately before re-commissioning.

7. Clause 6.2: Inter-distance Table:

Inter distance table is elaborated.

8. Clause 6.2.2: Fittings

- ROVs provided on the vessel inlet/ outlet liquid line and vapour line, on the liquid line and vapour line at the entry of Tank truck Gantry and Tank Wagon Gantry and on the entry of the liquid line at LPG filling shed shall be treated as Safety ROV. Location shall define safety and operational ROVs.
- Provision should be made in the bulk LPG receipt header or in the water drain line for filling of water in the vessel in case of any exigencies. Water filling line to have an isolation valve with NRV and male coupling for attaching the hydrant hose, so that in

DRAFT 2

case of emergency, only the hose has to be connected. Refer to Annexure-6-I for the Water drain provision and water filling arrangement.

9. Clause 6.3: LPG loading/ Unloading facilities.

QCDC (Quick Connect Dry Coupling) shall be provided at the end of filling liquid and vapour return lines for connecting to the tank truck vessel or Tank Wagon.

10. Clause 6.5: Emergency power

- Location of Emergency Lighting has been defined.

  During a power failure, the battery bank shall be used to provide power to critical emergency lighting. Emergency lighting shall be provided at critical places such as Tank Truck Gantry, Tank Wagon Gantry, LPG shed, LPG Pump/ Compressor House, Bulk Storage Area, Fire Water Pump House, PMCC room, DG room, and security cabin.

- For CCTV, a UPS backup for 30 min. operation shall be provided.

11. Clause 6.9: Thermal Pressure Relief System

- The pressure relief devices shall be capable of preventing the pressure from rising more than 20% above the design pressure of the pipeline. ASME Section VIII Div. 1 and ASME B31.3 shall be referred to.

12. Clause 6.16: Installation Lighting
Lux Level Table defined.

13. **Clause 7.2.2:**
For Bulk LPG Tank Truck, requirement of replacement of SRV and IEFCV every ten years is introduced.

14. **Clause 7.3.3: Degassing of Sick/ Leaky Tank Wagon**
The water displacement technique for the degassing of Tank Wagon is defined.

15. **Clause 7.3.3: Purging of Tank Wagon.**
Purging of the Tank Wagon is defined.

16. **Clause 8.2.5 and 8.2.6: Operation and Handling of 425 kg and Composite cylinders**
Operation and Handling of 425 kg and Composite Cylinder is defined.

17. **Clause 8.2.7: Handling of Mercaptan**
Handling of Mercaptan is defined in detail.

18. **Clause 9.5.3: Inspection and Testing of Pipelines**

Hydro testing of pipeline during a comprehensive inspection under the following conditions, as applicable:

- After any major alteration/ re-rating / replacement
- Inaccessible piping and complicated manifold
- System after chemical cleaning
- When the piping is out of service for more than six months

19. **Clause 9.5.14: Drawings**
Various drawings to be maintained at the location is detailed.

20. **Clause 11.13.1: Fire Alarm System**

DRAFT 2

- Provision of Emergency shutdown and its location is introduced.

Emergency Shut Down (ESD) provision near to Filling shed, Bulk storage area, Tank truck gantry, Tank wagon gantry, LPG Pump/ compressor House, Pipeline transfer battery limit, etc. shall be provided. Operation of these ESD points shall trip the power supply, barring the emergency power and raise the audio-visual alarm in the control panel with identification of the alarm location.

21. **Clause 11.17: Combine POL and LPG Facility**

Introduced Clause for sharing of the water storage facility for firefighting purposes between the POL Installation and the LPG Plant.

22. **Clause 11.18: Data Monitoring system**

Introduced a clause on the data monitoring system.

23. **Clause 15.1: Training**

Introduced clause on Training.

24. **Clause 15.2: Designated Safety Officer**

Introduced clause on the Designated Safety Officer.

25. **Clause 16.0: CCTV System**

Introduced a clause on the CCTV system.

26. **Clause 16.0: Access Control System**
Introduced clause on Access Control System.

27. Annexure-5-I on “Typical Design for construction of Water Seal” is added.
28. Annexure-5-II on “Typical Design for Closed-loop degassing system” is added.
29. Annexure-6-I on “typical pipeline connection for water drain arrangement in LPG storage vessel” is added.
30. Annexure-6-VII on “degassing and gas-in procedures of bulk LPG storage vessels” is added.
31. Annexure-7-I-A and 7-I-B on “checklist for bulk LPG tank trucks at unloading and loading location” is modified.
32. Annexure-8-I on “checklist for Safety fitting of Packed Cylinder Truck” is modified.
33. Annexure-9-I to 9-III to on “Maintenance schedule for Compressor, Pump, Filling shed Equipment is modified.
34. Annexure-9V on “Typical Inspection Frequency” is added.
35. Annexure-10-II on “Rejection Criteria of cylinder during statutory testing” is added.
36. Annexure-10-III on “Route Card of cylinder” is added.
37. Annexure-10-V on “Certificate of periodic testing of LPG cylinder” is added.
38. Annexure-10-VI on “Quality Control Checks on Re-tested cylinder” is added.
39. Annexure-11-I on “Sample Water Calculation” is added.
40. Annexure-11-II on “Indicative diagram for Deluge Valve” is added.
41. Annexure-14-I on the checklist for daily safety check of LPG installation is modified.
42. Annexure-14-II on the checklist for Electrical audit is modified.
43. Annexure-14-I on the checklist for the Inspection and Maintenance system audit is modified.
44. Annexure-14-IV on the checklist for Self-audit by Plant In charge is modified.

DRAFT 2

ANNEXUE-16

FUNCTIONAL COMMITTEE

(Second Edition - October 2005)

NAME	ORGANISATION
LEADER Sh. R. Krishnan	Hindustan Petroleum Corporation Ltd., Mumbai
MEMBERS	
Sh. A.N. Biswas	Deptt. of Explosives, Nagpur
Sh. N. Rajkhowa,	Engineers India Limited, New Delhi
Sh. B.S. Giridhar	Indian Oil Corporation Ltd., - HQ, Mumbai
Sh. N. Dasgupta	Bharat Petroleum Corporation Ltd., Mumbai
Sh. T. Koteeswaran	Chennai Petroleum Corporation Ltd., Chennai
Sh. B.J. Phukan	Numaligarh Refinery Ltd., Assam
Sh. P. Jaidevan	Indian Oil Corporation Ltd., Calicut
Sh. S. Ramesh	Bharat Petroleum Corporation Ltd., Noida
MEMBER COORDINATOR Sh. S. C. Gupta	Oil Industry Safety Directorate, New Delhi

DRAFT 2

FUNCTIONAL COMMITTEE MEMBERS

(First Edition – April 1994)

NAME	ORGANISATION
LEADER Sh. S.C. Tandon	Indian Oil Corporation Ltd., (Ref.)
MEMBERS	
Sh. H.G. Bhinde	Indian Oil Corporation Ltd., (Mkt.)
Sh. A.C. Kelkar	Indian Oil Corporation Ltd., (Mkt.)
Sh. V. Narayana Rao	Bharat Petroleum Corporation Ltd., (Mkt.)
Sh. P. Kumaraswamy	Bharat Petroleum Corporation Ltd., (Ref.)
Sh. P.P. Nadkarni	Hindustan Petroleum Corporation Ltd., (Mkt.)
Sh. S. Mathur	Engineers India Limited, New Delhi
MEMBER COORDINATOR Sh. S. C. Gupta	Oil Industry Safety Directorate, New Delhi

DRAFT 2